

Getting to Know Vivado

Course Workbook

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About this Workbook

This workbook is designed to be used in conjunction with the Getting to Know Vivado course.

The contents of this workbook are created by Aduvo Engineering & Training, Ltd.

If you have any questions about the contents, or need assistance, please contact Adam Taylor at adam@adiuvoengineering.com.

Pre-Lab

Workshop Pre-requisites

Required Hardware

There is no required hardware for this course.

Downloads and Installations

Step 1 – Download and install the following at least one day prior to the workshop. This may take a significant amount of time and drive space.

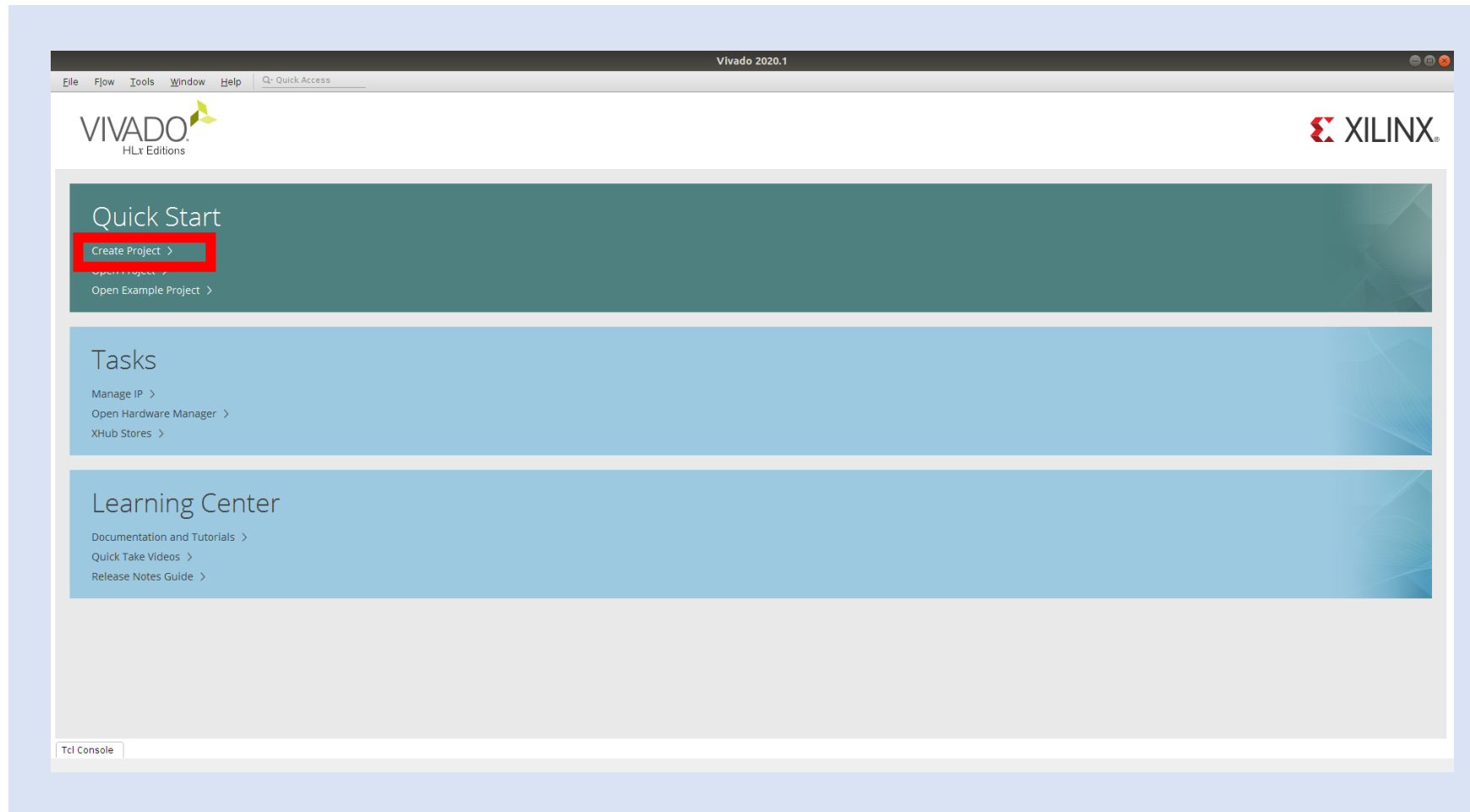
Vivado 2020.1	Download
Source Project Files	Download

Lab 1

Overview and Introduction to Vivado

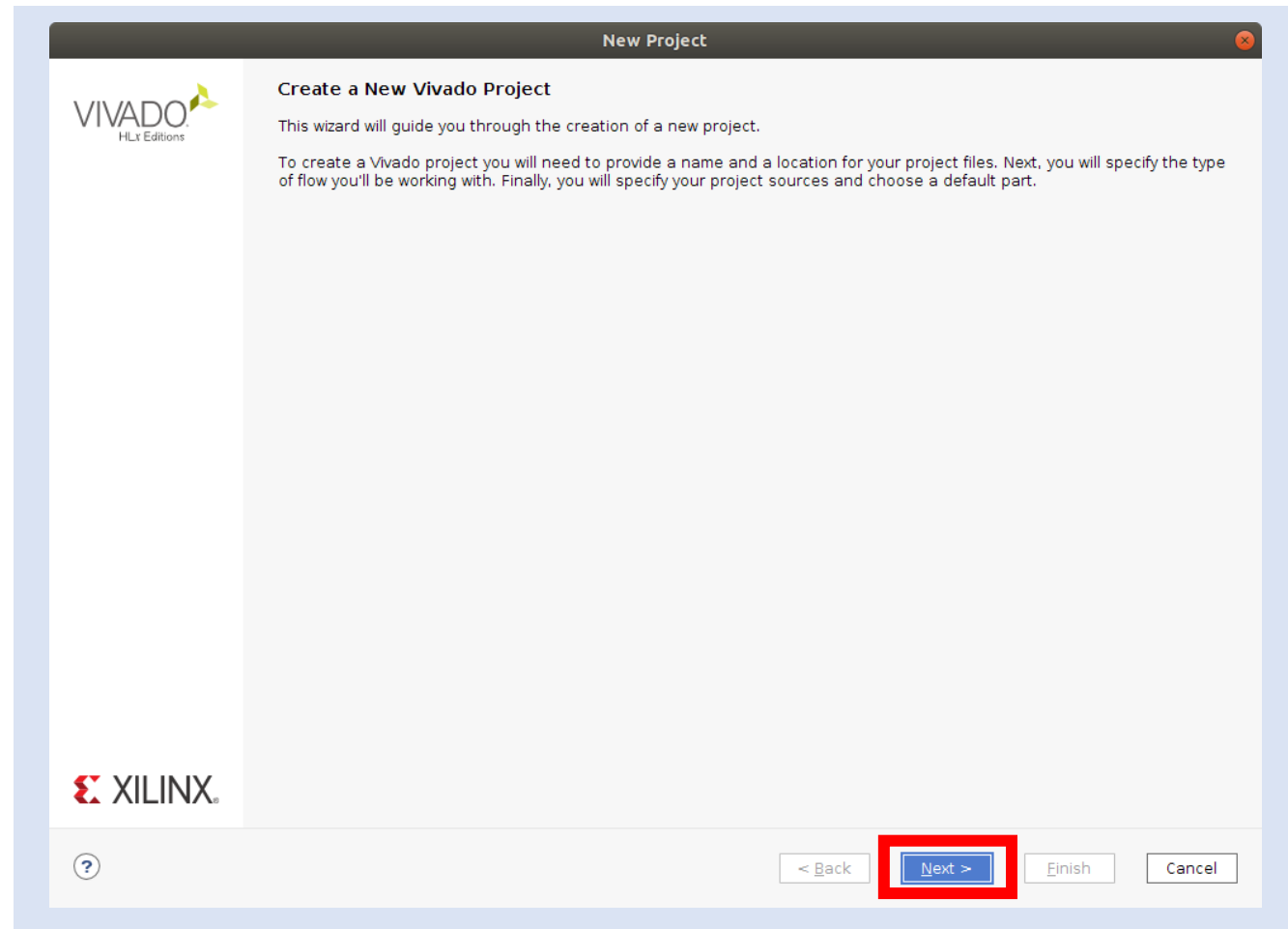
Lab 1: Overview and Introduction to Vivado

Step 1 – Open Vivado 2020.1.



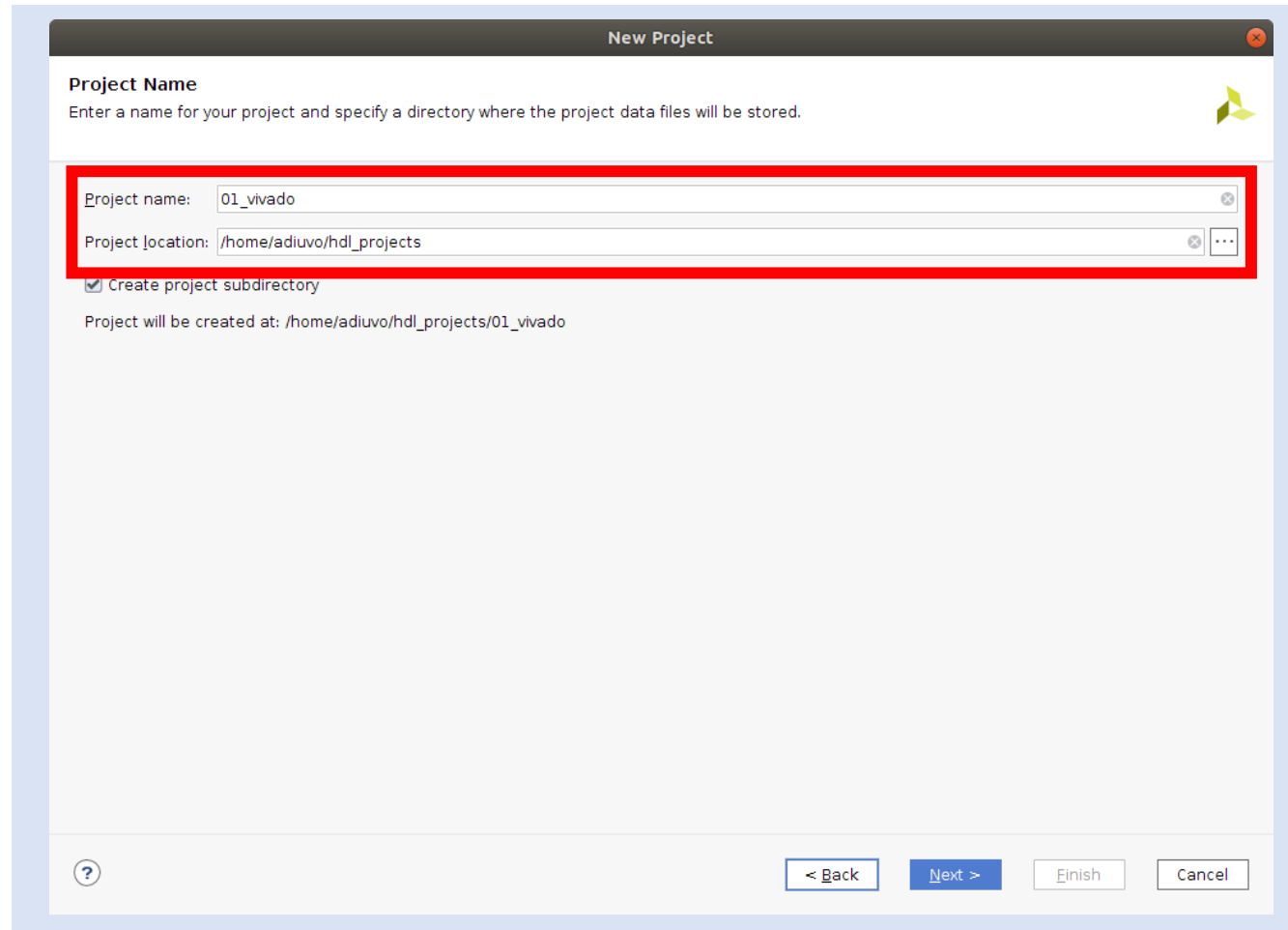
Lab 1: Overview and Introduction to Vivado

Step 2 – Click on **Create Project** – This will open the New Project Wizard – Click **Next**.



Lab 1: Overview and Introduction to Vivado

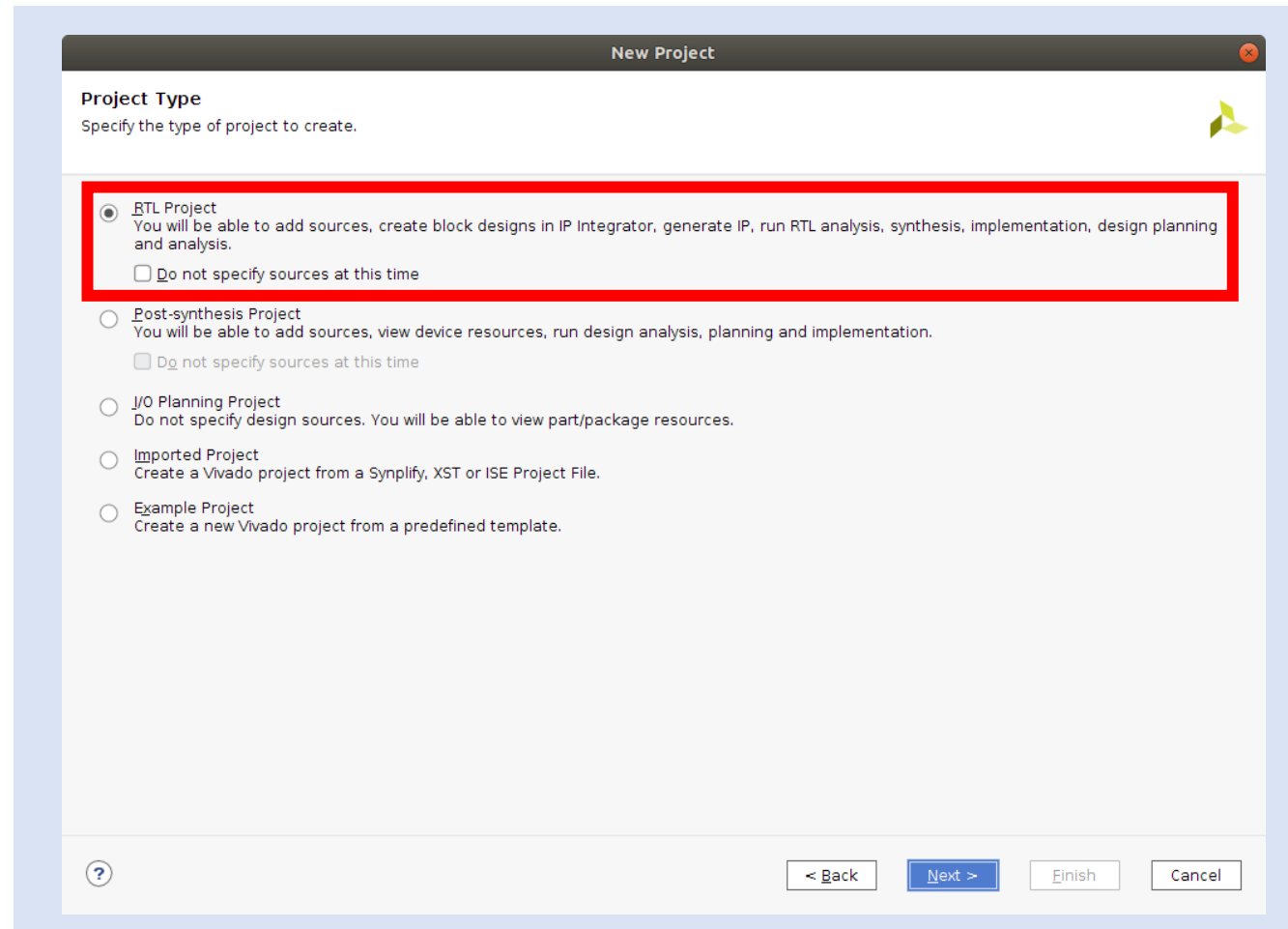
Step 3 – Enter the project name of “**01_Vivado**” and select the location you want to save the project.



The screenshot shows the 'New Project' dialog box in Vivado. The 'Project Name' field is set to '01_vivado' and the 'Project location' is set to '/home/adiuvo/hdl_projects'. The 'Create project subdirectory' checkbox is checked. The project will be created at '/home/adiuvo/hdl_projects/01_vivado'. The 'Next >' button is highlighted.

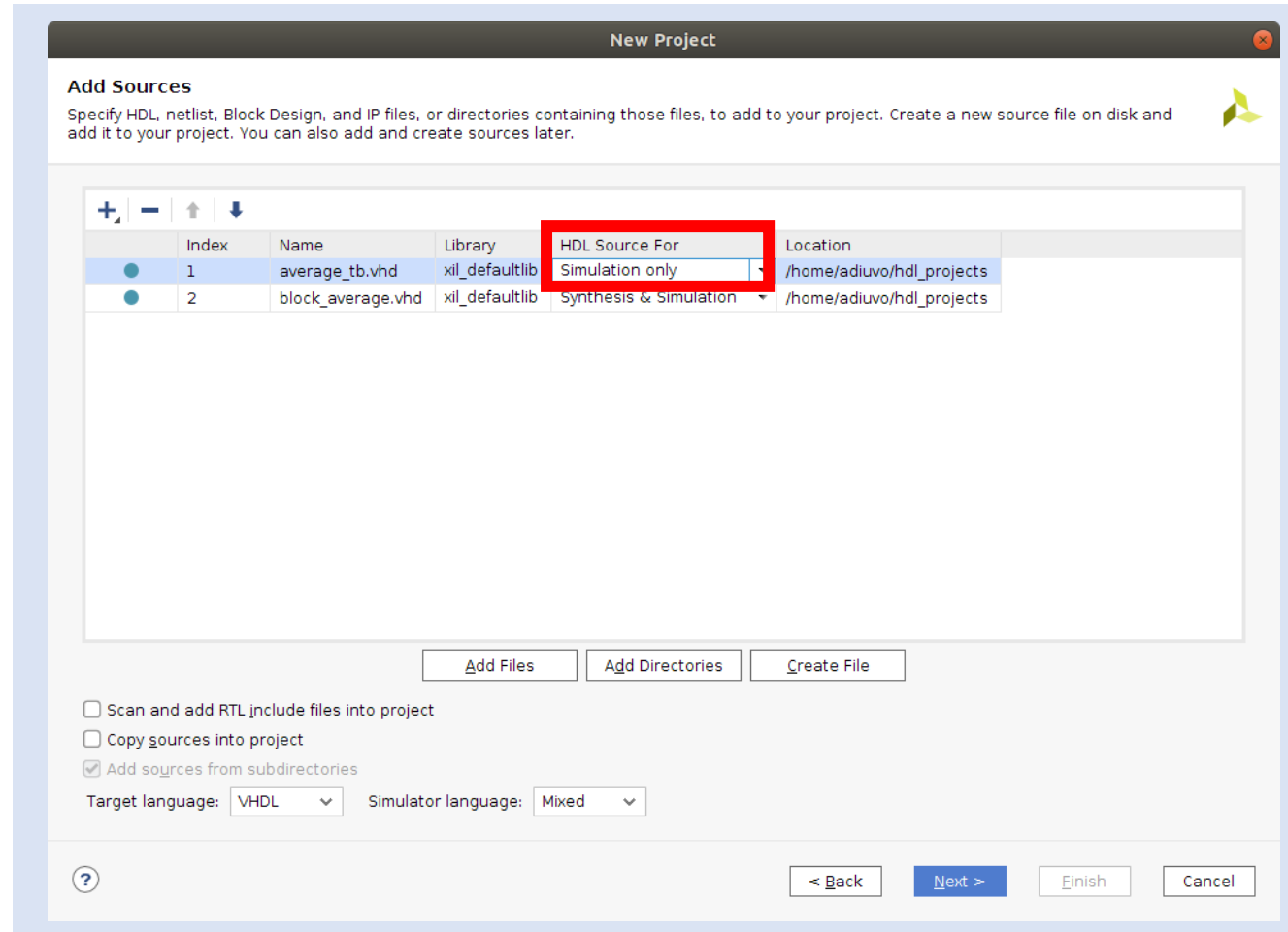
Lab 1: Overview and Introduction to Vivado

Step 4 – Select RTL Project.



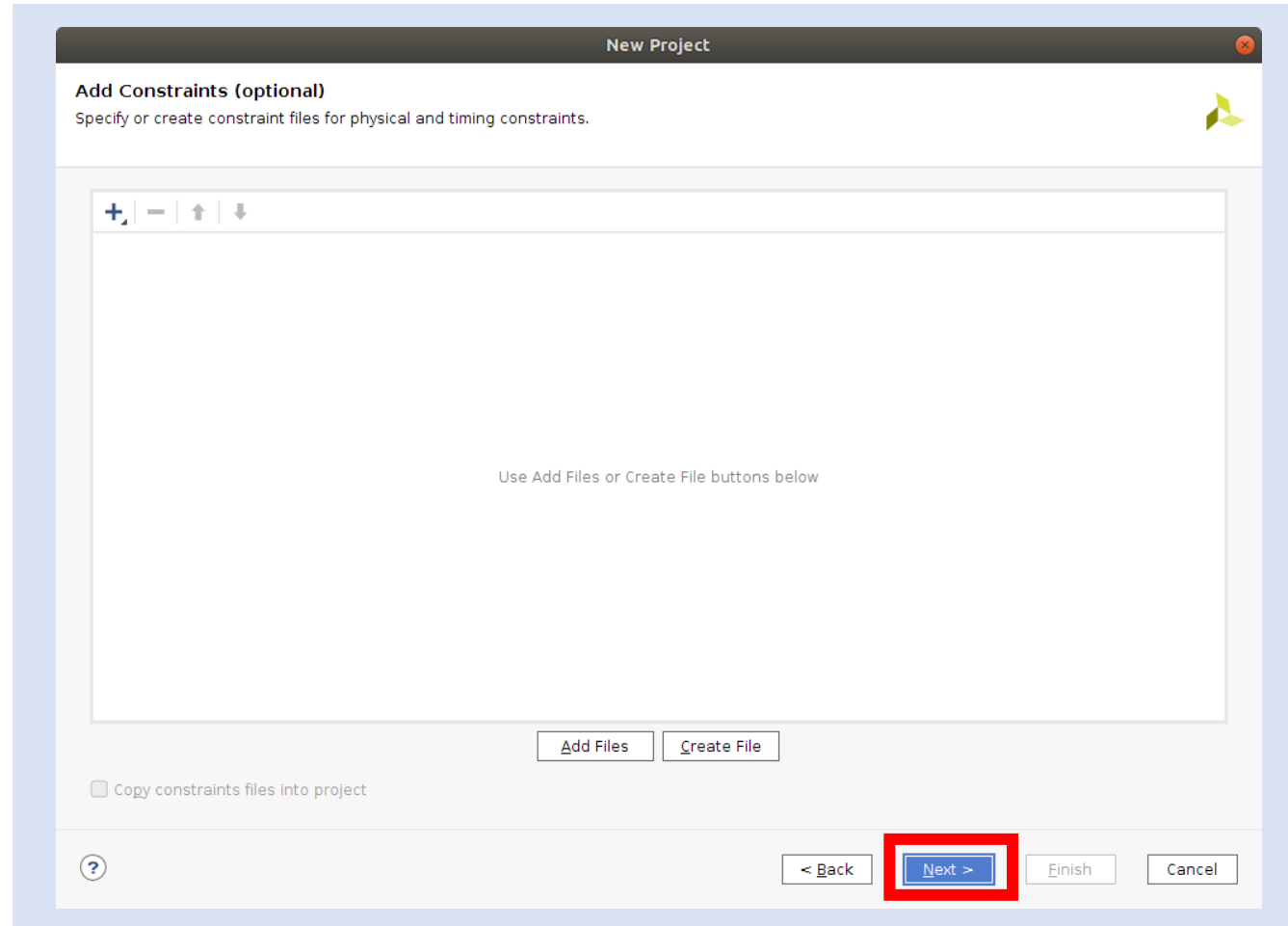
Lab 1: Overview and Introduction to Vivado

Step 5 – Click **ADD FILES** and select the two files downloaded from Github. For the file `average_tb.vhd`, change the HDL source to **Simulation Only**.



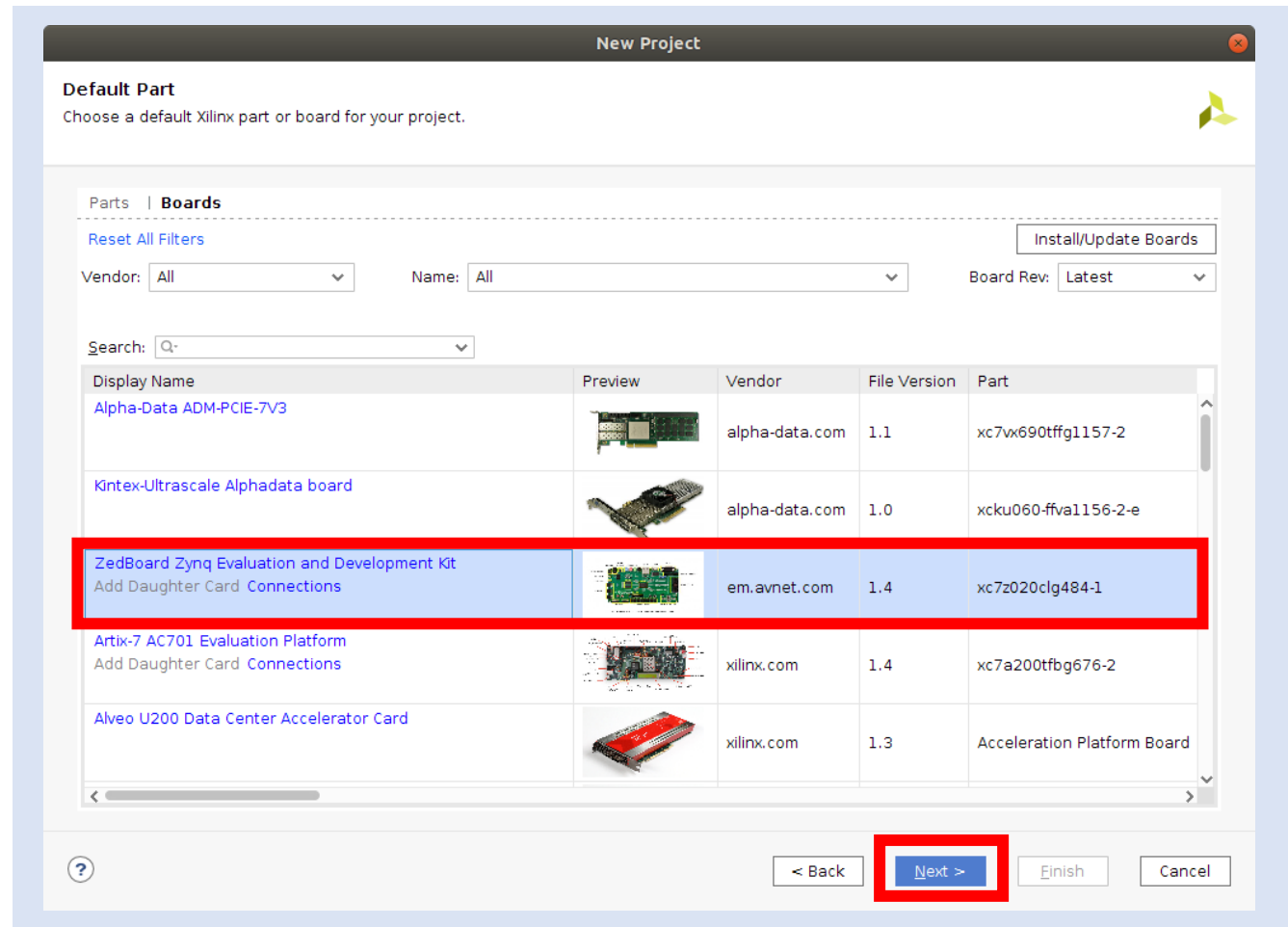
Lab 1: Overview and Introduction to Vivado

Step 6 – At this time we do not want to add any constraints files. Click **Next**.



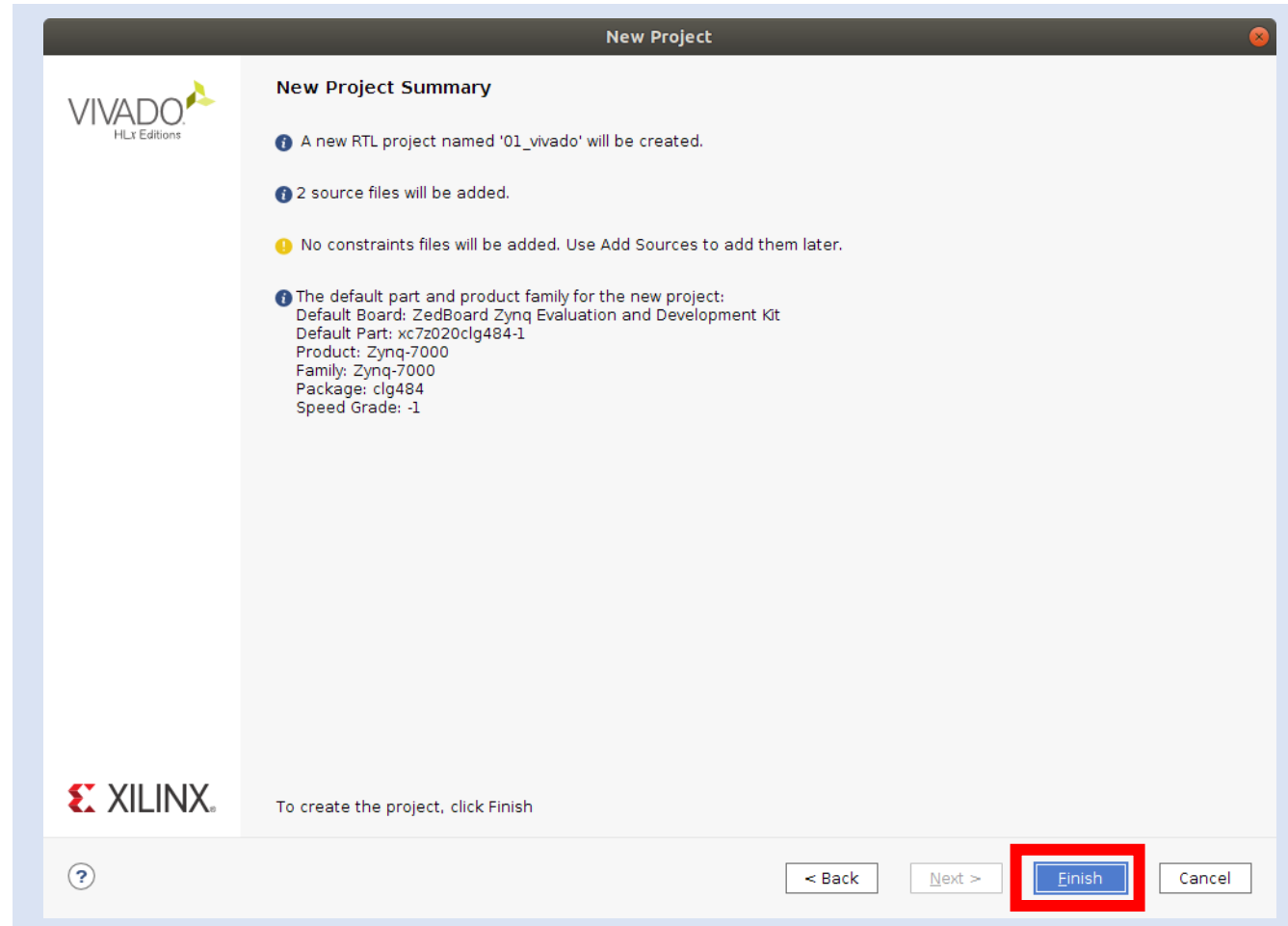
Lab 1: Overview and Introduction to Vivado

Step 7 – Select the ZedBoard and click Next.



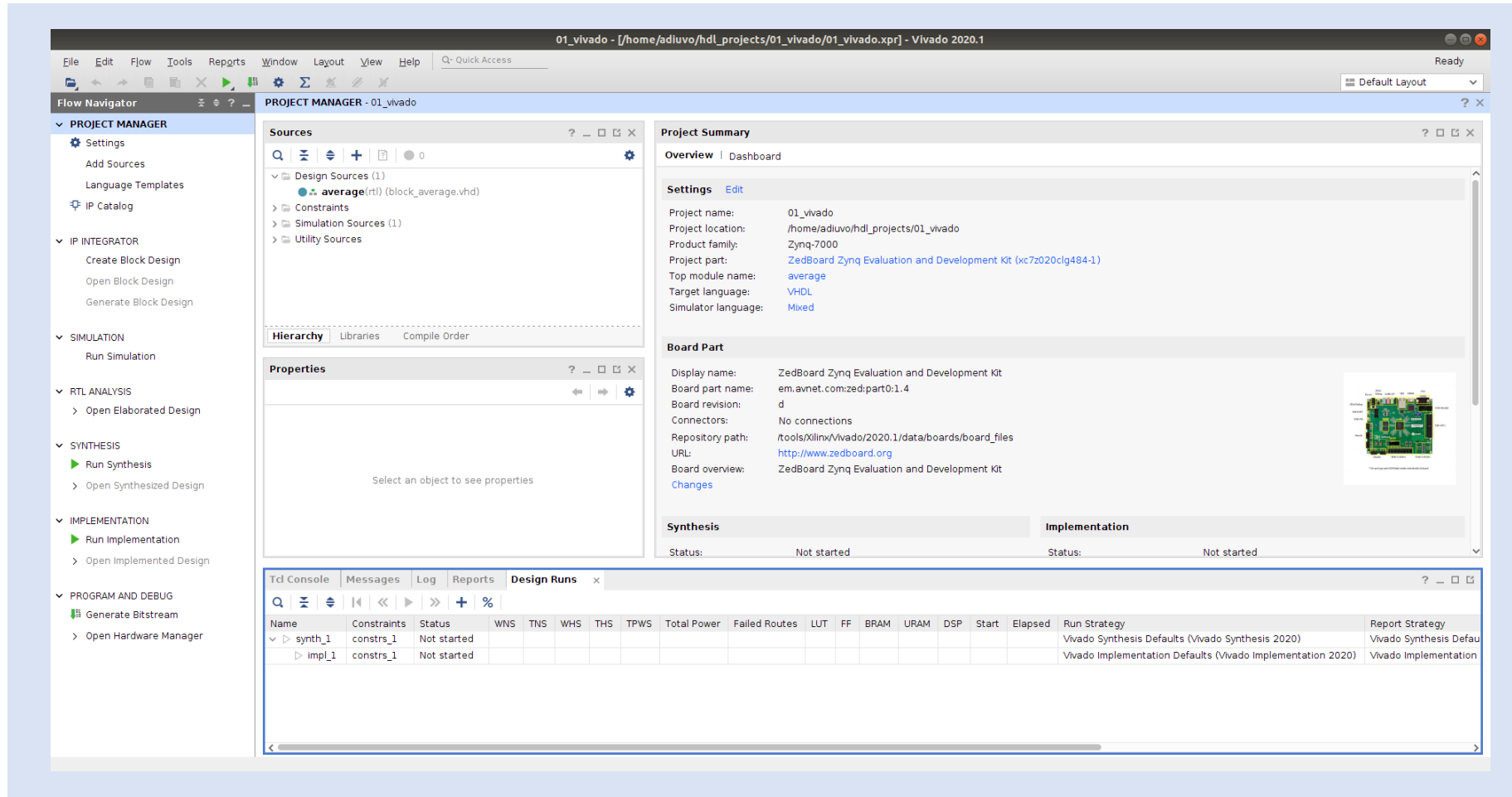
Lab 1: Overview and Introduction to Vivado

Step 8 – On the project summary tab, select **Finish**.



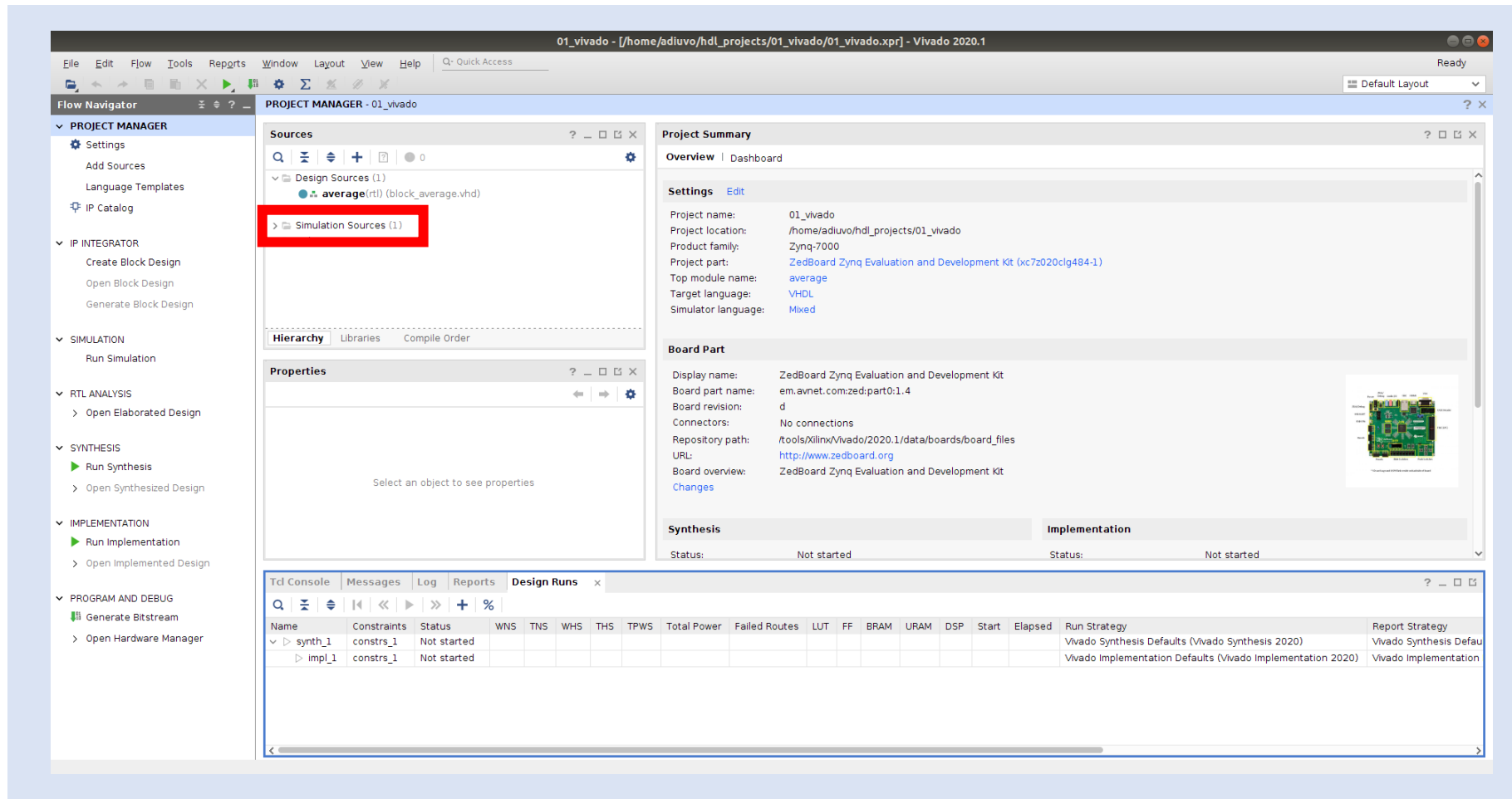
Lab 1: Overview and Introduction to Vivado

Step 9 – This will open Vivado in the project manager view.



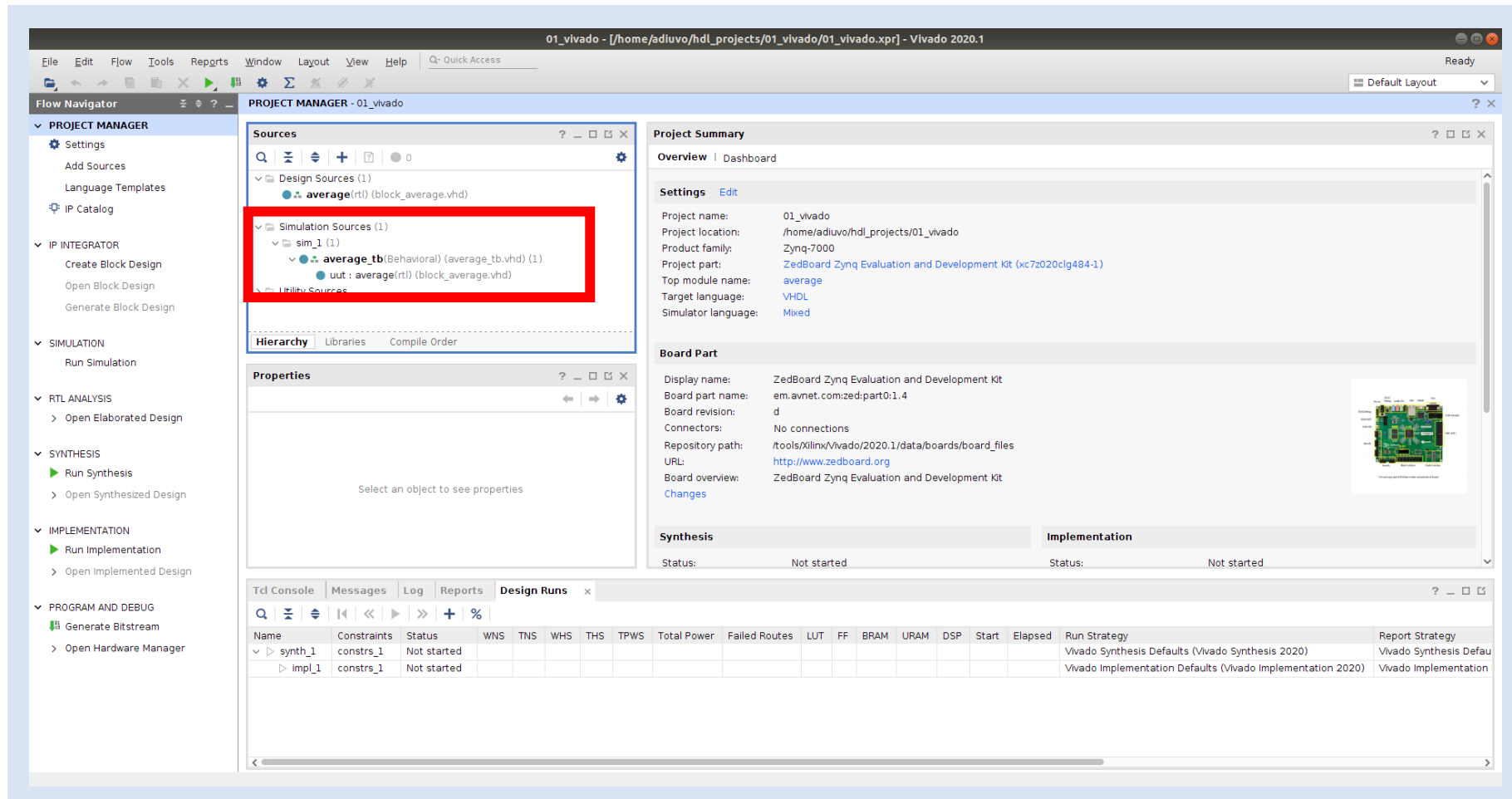
Lab 1: Overview and Introduction to Vivado

Step 10 – Expand the Simulation Sources.



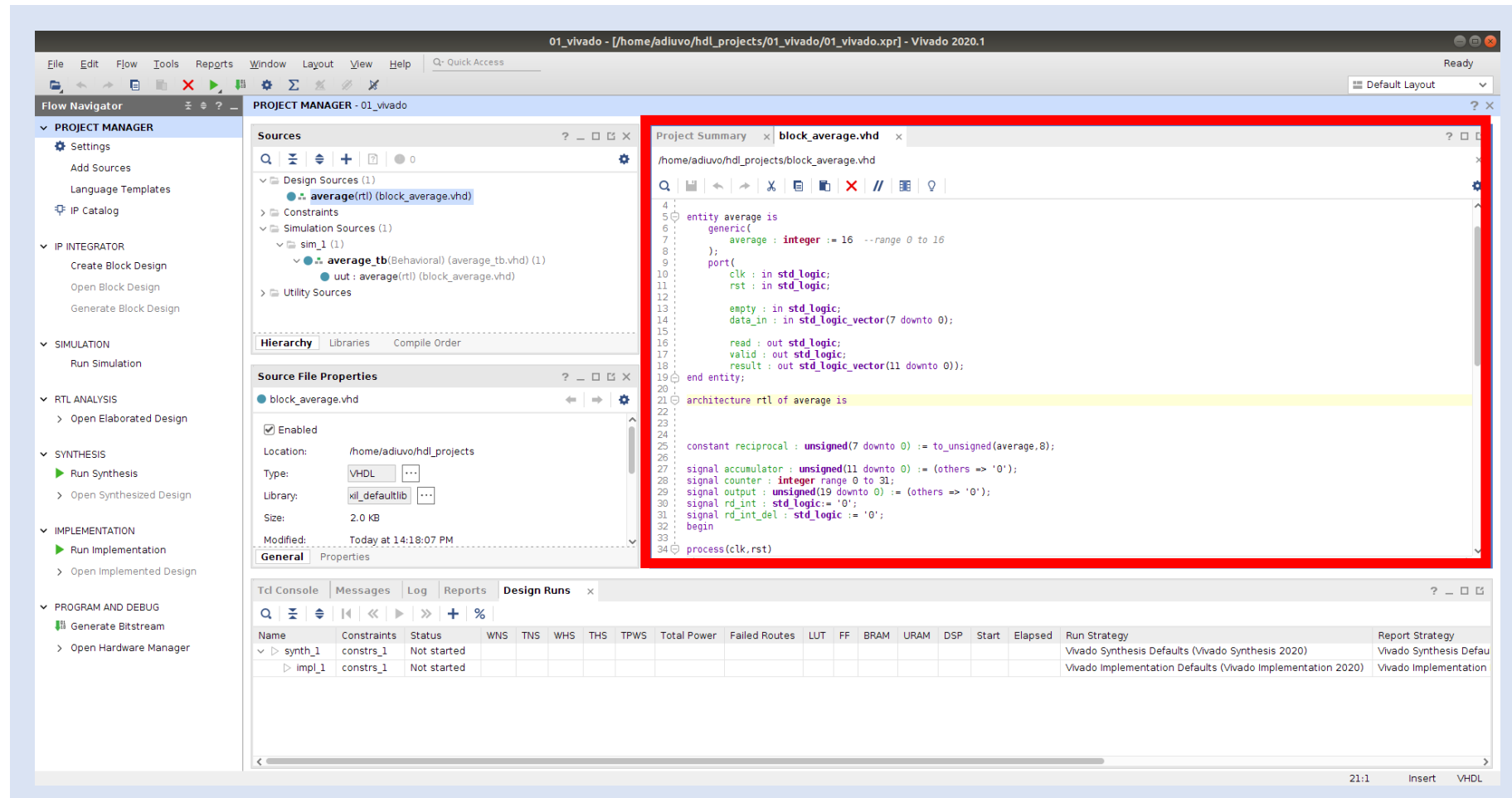
Lab 1: Overview and Introduction to Vivado

Step 11 – This will show the test bench and the design source to be simulated.



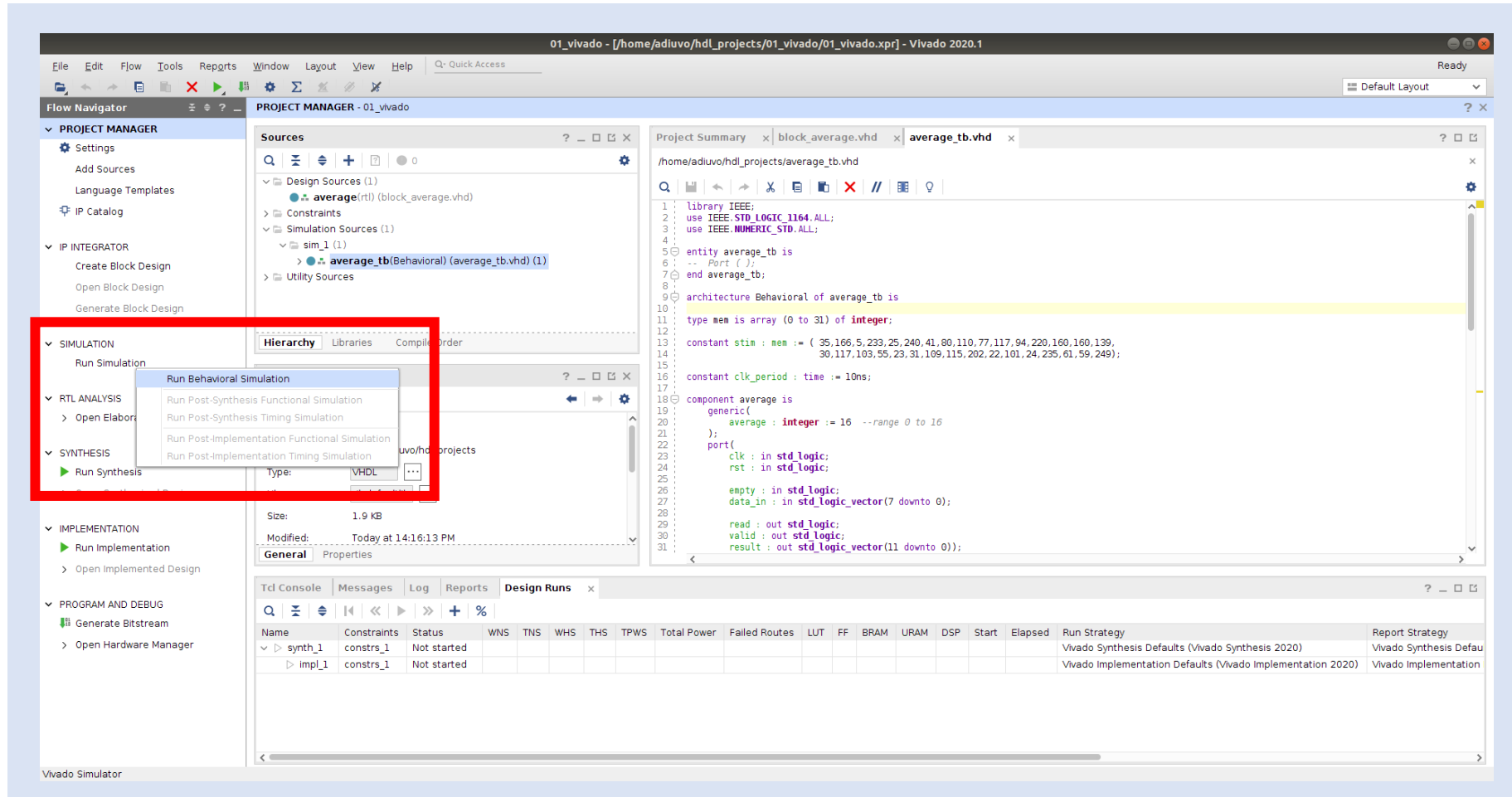
Lab 1: Overview and Introduction to Vivado

Step 12 – Double clicking on the VHDL files will open the source for inspection.



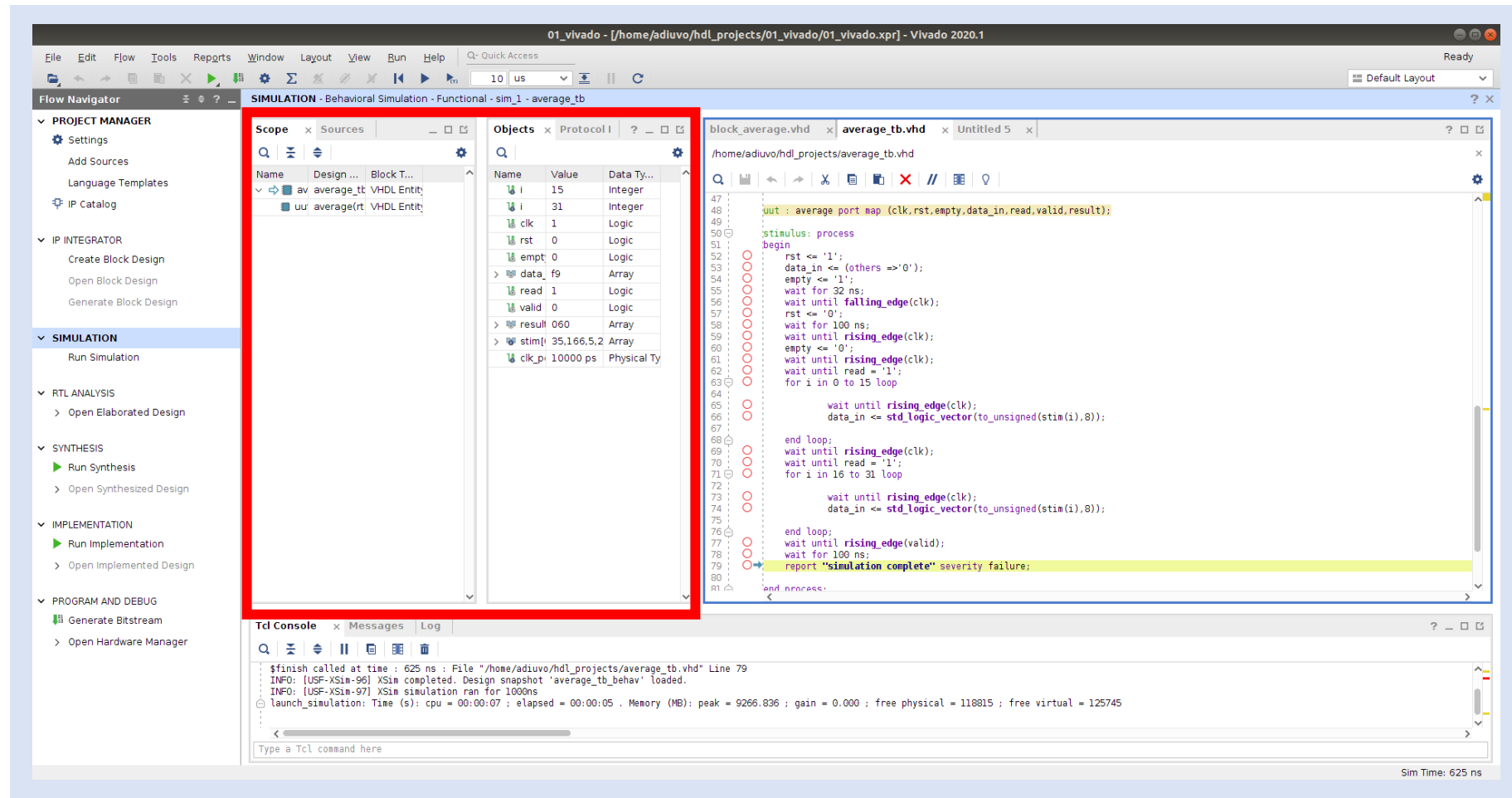
Lab 1: Overview and Introduction to Vivado

Step 13 – To run a simulation, click on **Run Simulation** and select **Run Behavioral Simulation**.



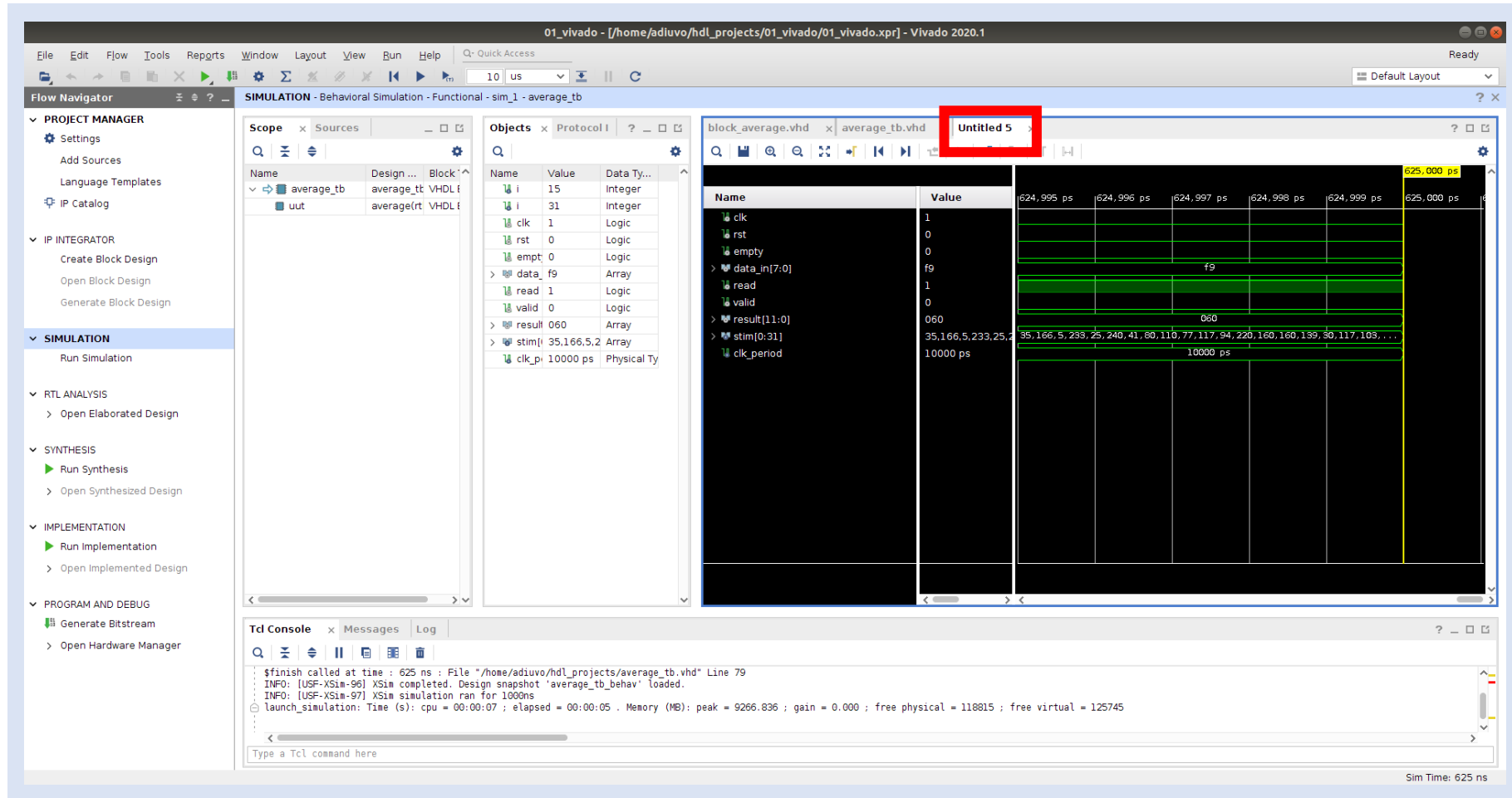
Lab 1: Overview and Introduction to Vivado

Step 14 – This will open the behavioral simulation view. Note the scope and objects.



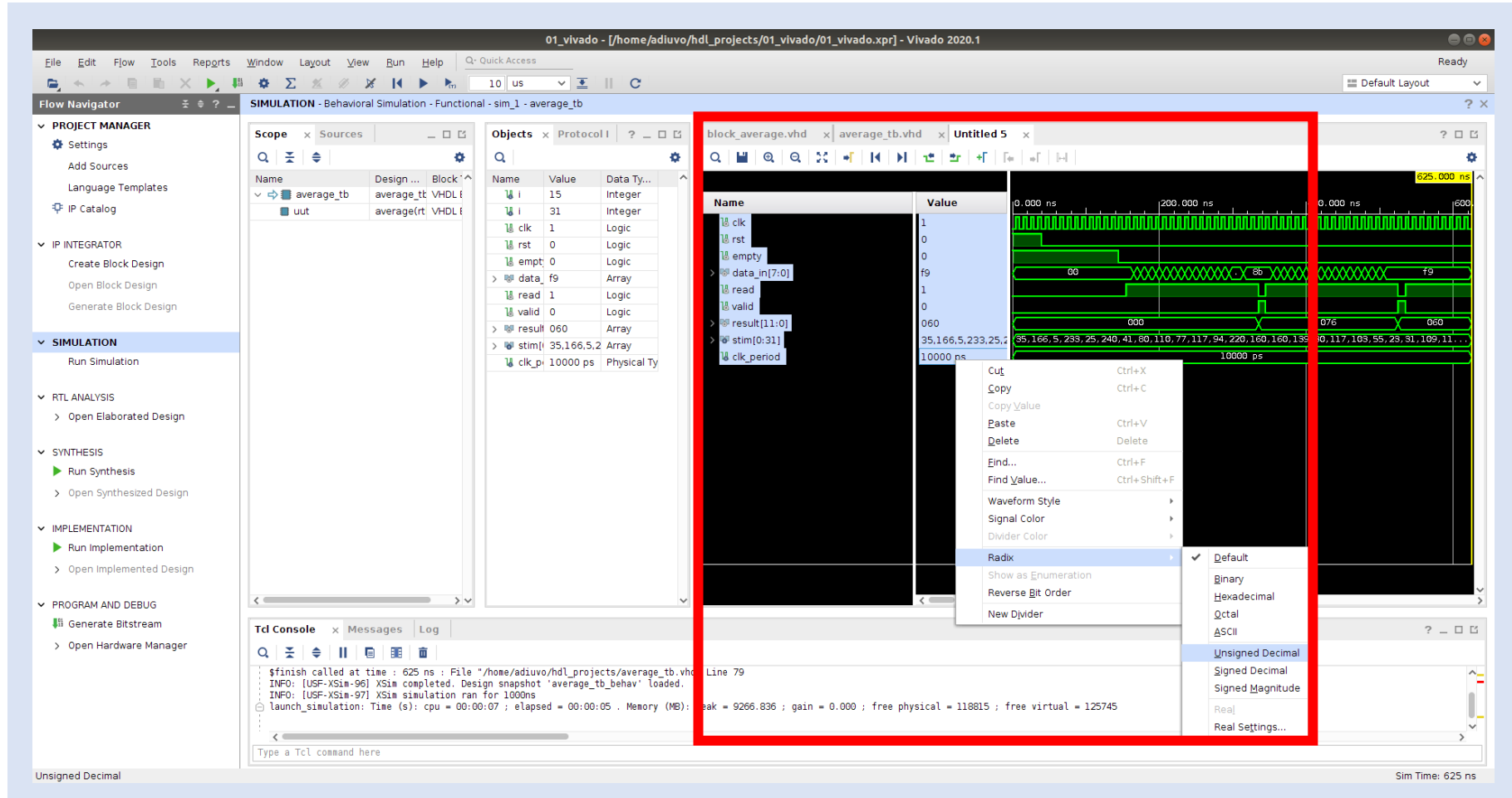
Lab 1: Overview and Introduction to Vivado

Step 15 – Click on the **Untitled*** tab to see the waveform of the simulation. Note this view shows the signals defined within the test bench only, not the Unit Under Test.



Lab 1: Overview and Introduction to Vivado

Step 16 – At the moment, the results are in hexadecimal but they make more sense in decimal. Select all the signals, right click, and select **unsigned decimal** from the **radix**.



Lab 1: Overview and Introduction to Vivado

Step 17 – This will change the results to decimal. Correct operation has result showing 118 then 96. This is the block average of 16 input values.

The screenshot displays the Vivado 2020.1 interface during a behavioral simulation. The 'Scope' window shows the 'result[11:0]' signal, which is highlighted in blue. The 'Objects' window shows the 'result' signal as an array of 16 elements. The 'Tcl Console' window shows the simulation results, including the final value of 96.

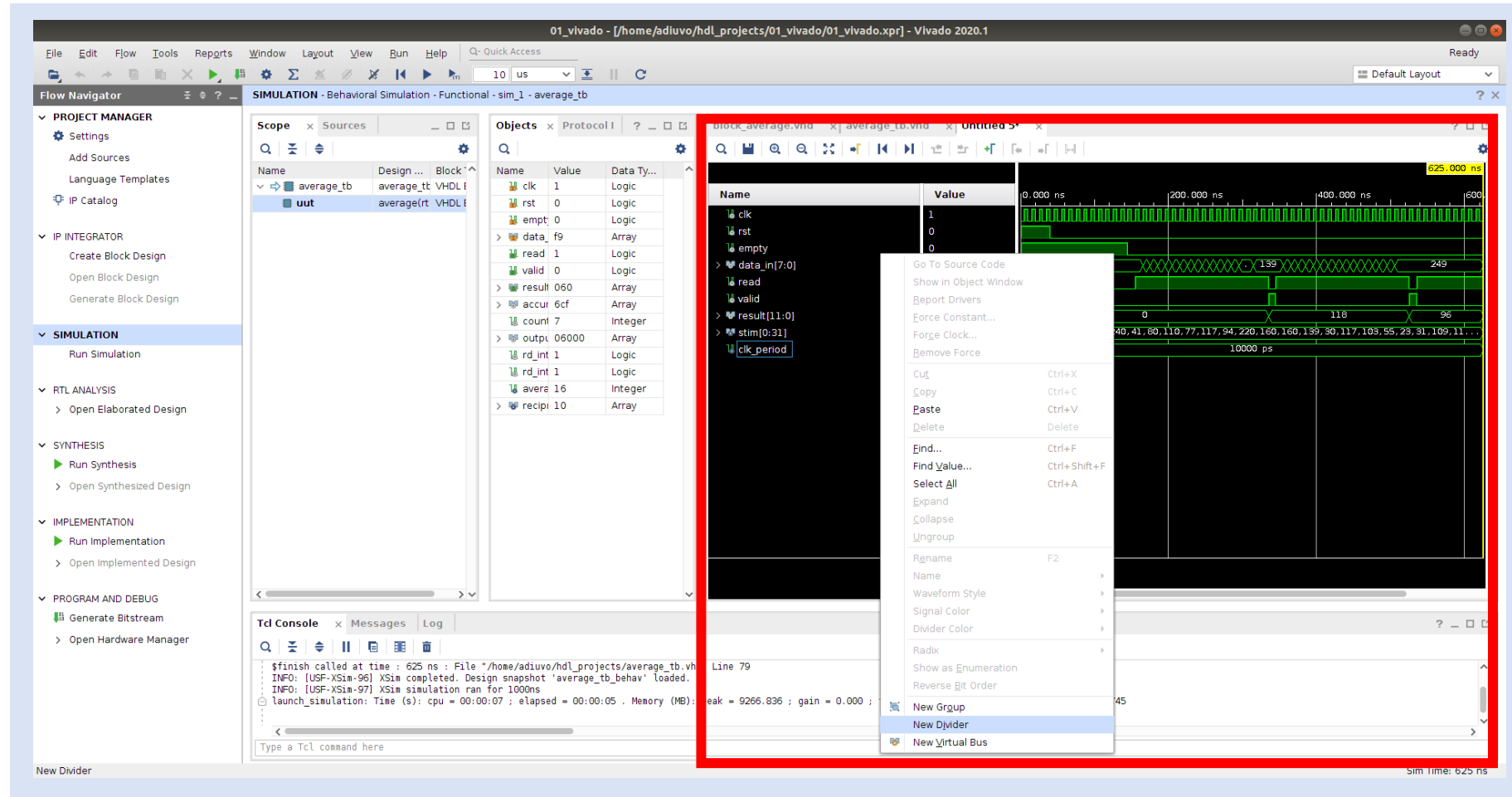
Name	Value
clk	1
rst	0
empty	0
data_in[7:0]	249
read	1
valid	0
result[11:0]	96
stim[0:31]	35,166,5,233,25,2
clk_pi	10000 ps

The 'Tcl Console' window shows the following output:

```
$finish called at time : 625 ns : File "/home/adiuvo/hdl_projects/average_tb.vhd" Line 79
INFO: [USF-XSim-96] XSim completed. Design snapshot 'average_tb_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:07 ; elapsed = 00:00:05 . Memory (MB): peak = 9266.836 ; gain = 0.000 ; free physical = 118815 ; free virtual = 125745
```

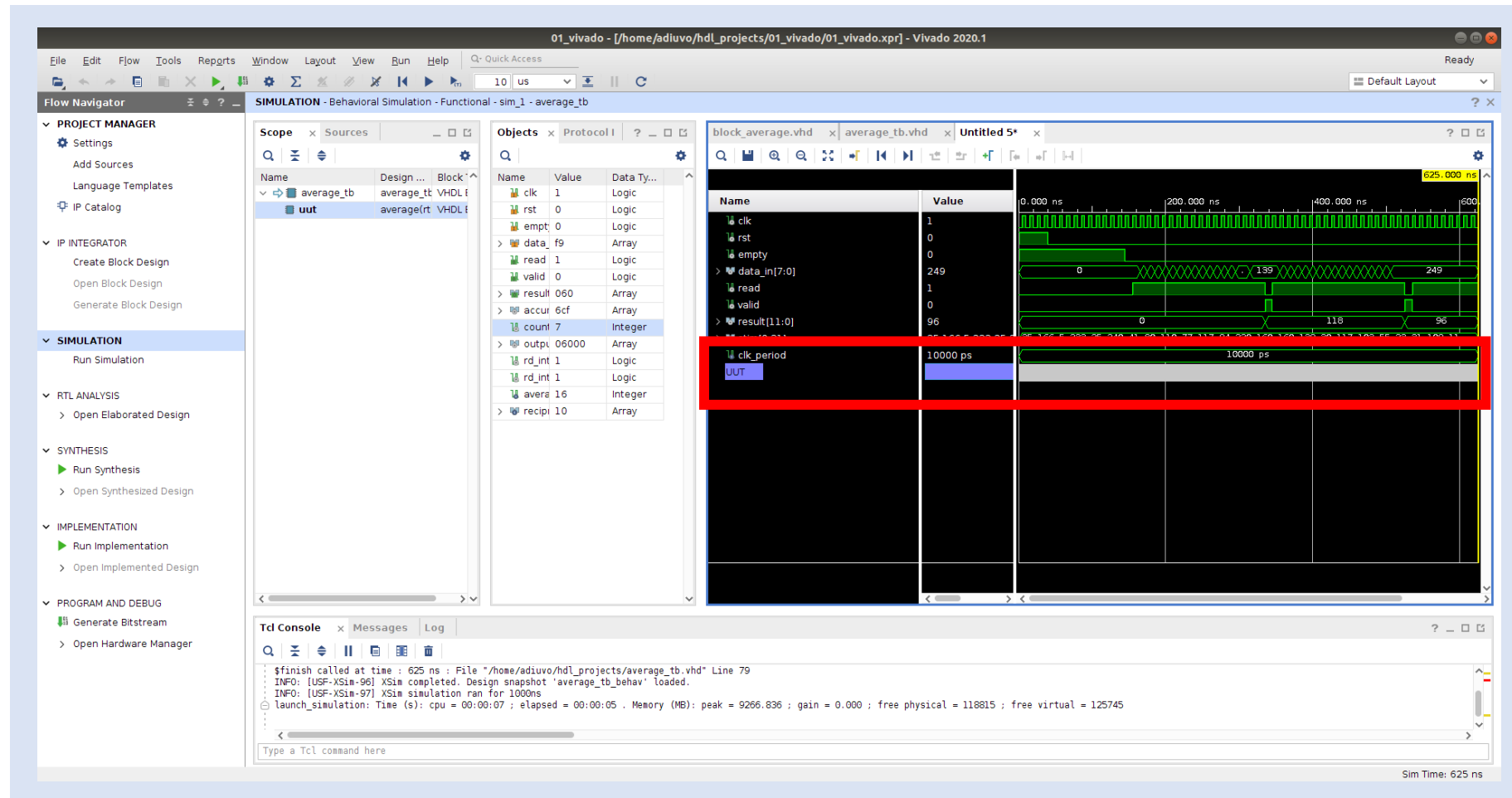
Lab 1: Overview and Introduction to Vivado

Step 18 – Often we want to be able to see the signals in the UUT. To do this, first let's insert a divider. Right click on the **bottom signal** and select **New Divider**.



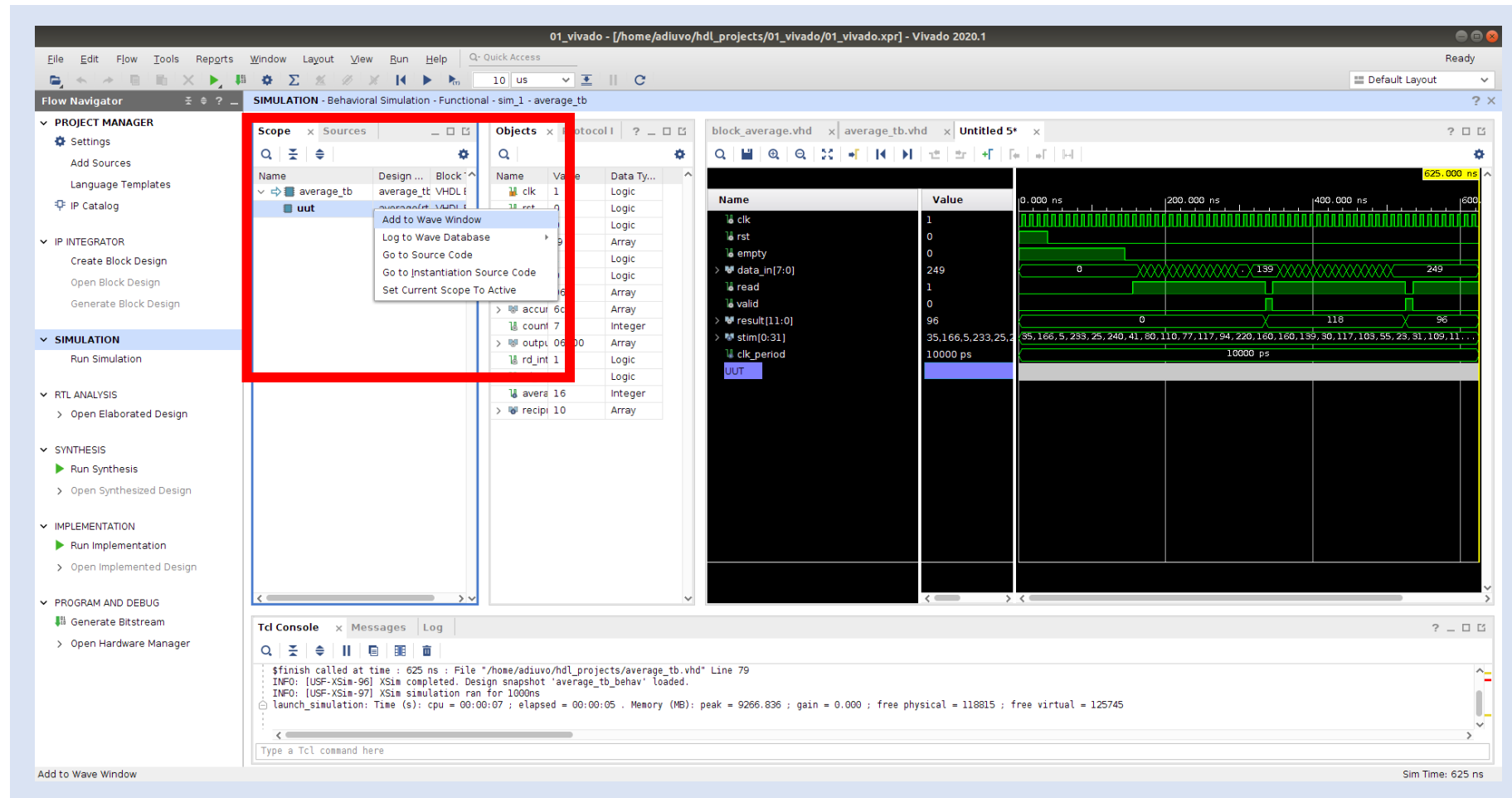
Lab 1: Overview and Introduction to Vivado

Step 19 – When prompted, enter the name **UUT** and you will see the new divider in the waveform.



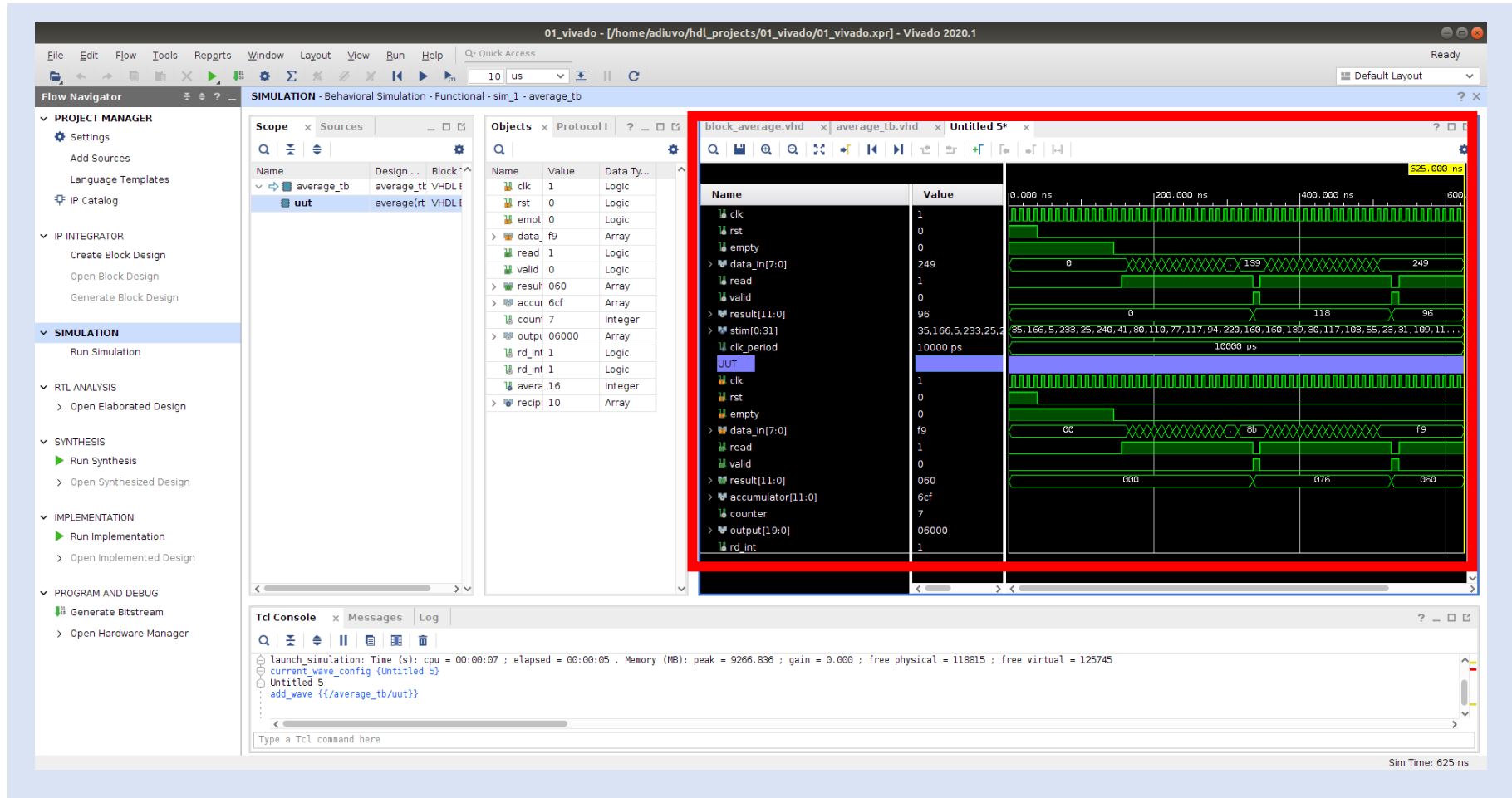
Lab 1: Overview and Introduction to Vivado

Step 20 – To add in the UUT, right click on the **UUT** under the scope and select **Add to Wave Window**.



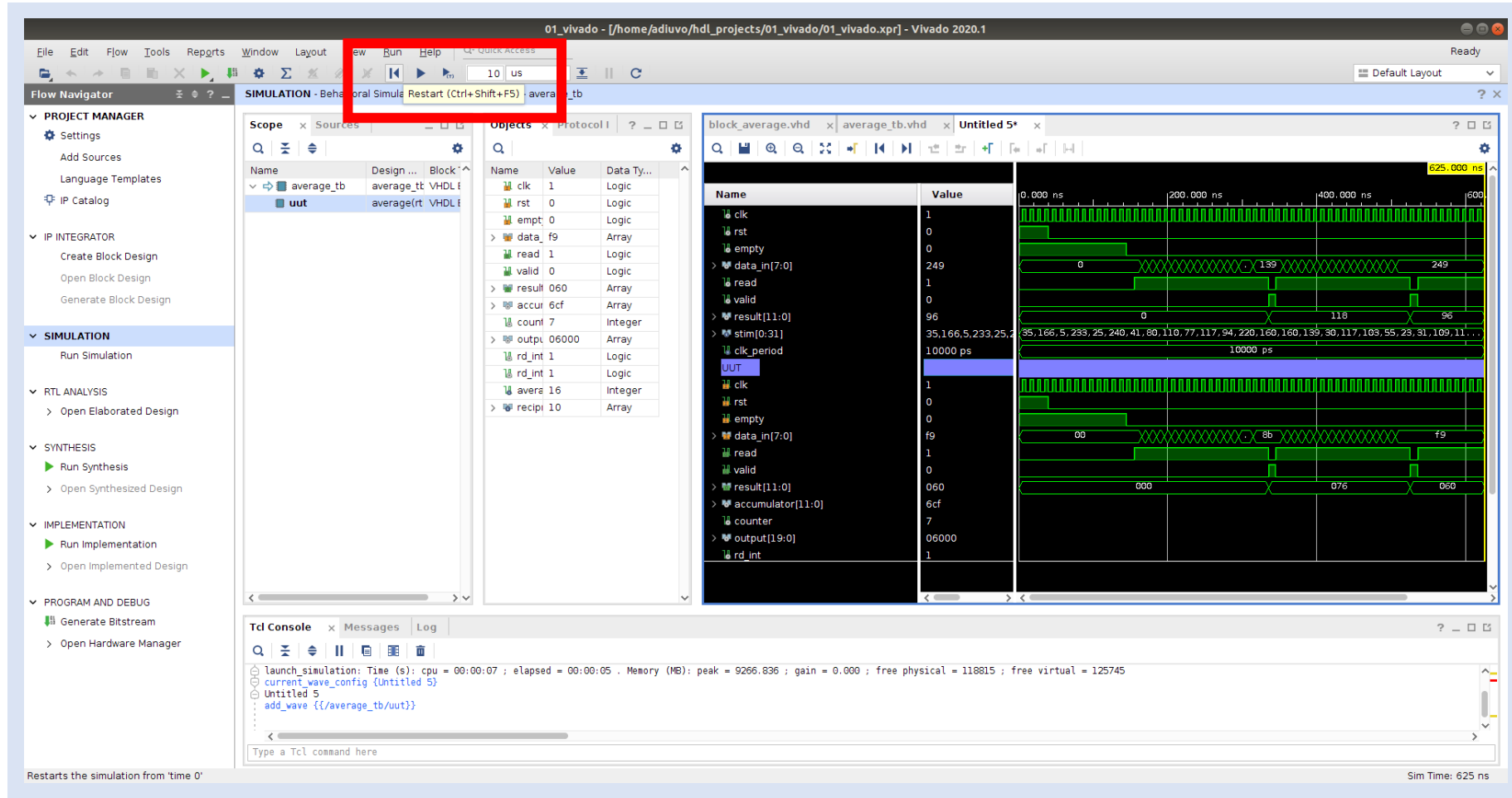
Lab 1: Overview and Introduction to Vivado

Step 21 – This will add in the UUT signals, however, some information may be missing as it was not saved during the simulation.



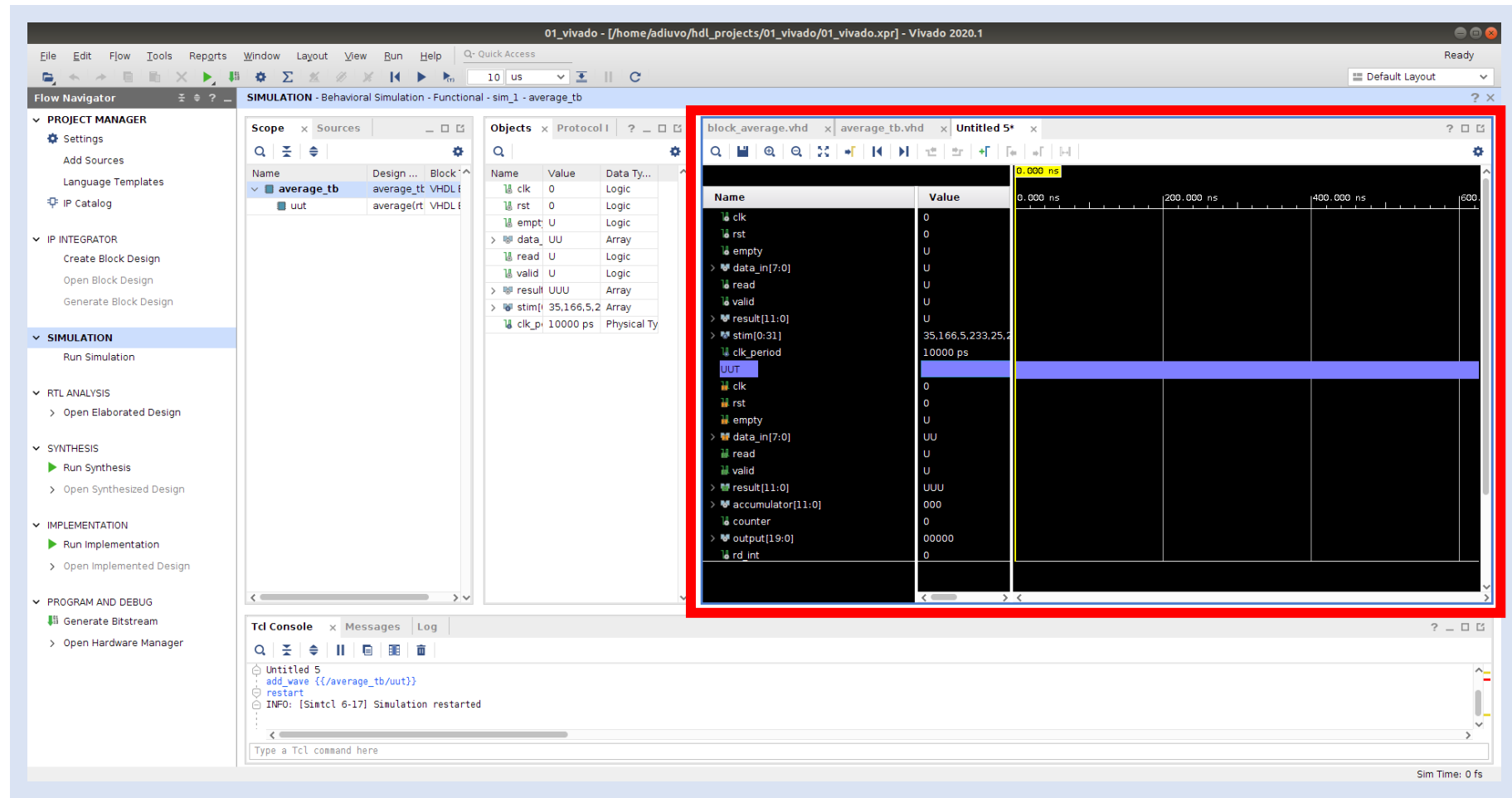
Lab 1: Overview and Introduction to Vivado

Step 22 – To add in the missing waveform, we need to restart the simulation. Select **Restart** from menu bar.



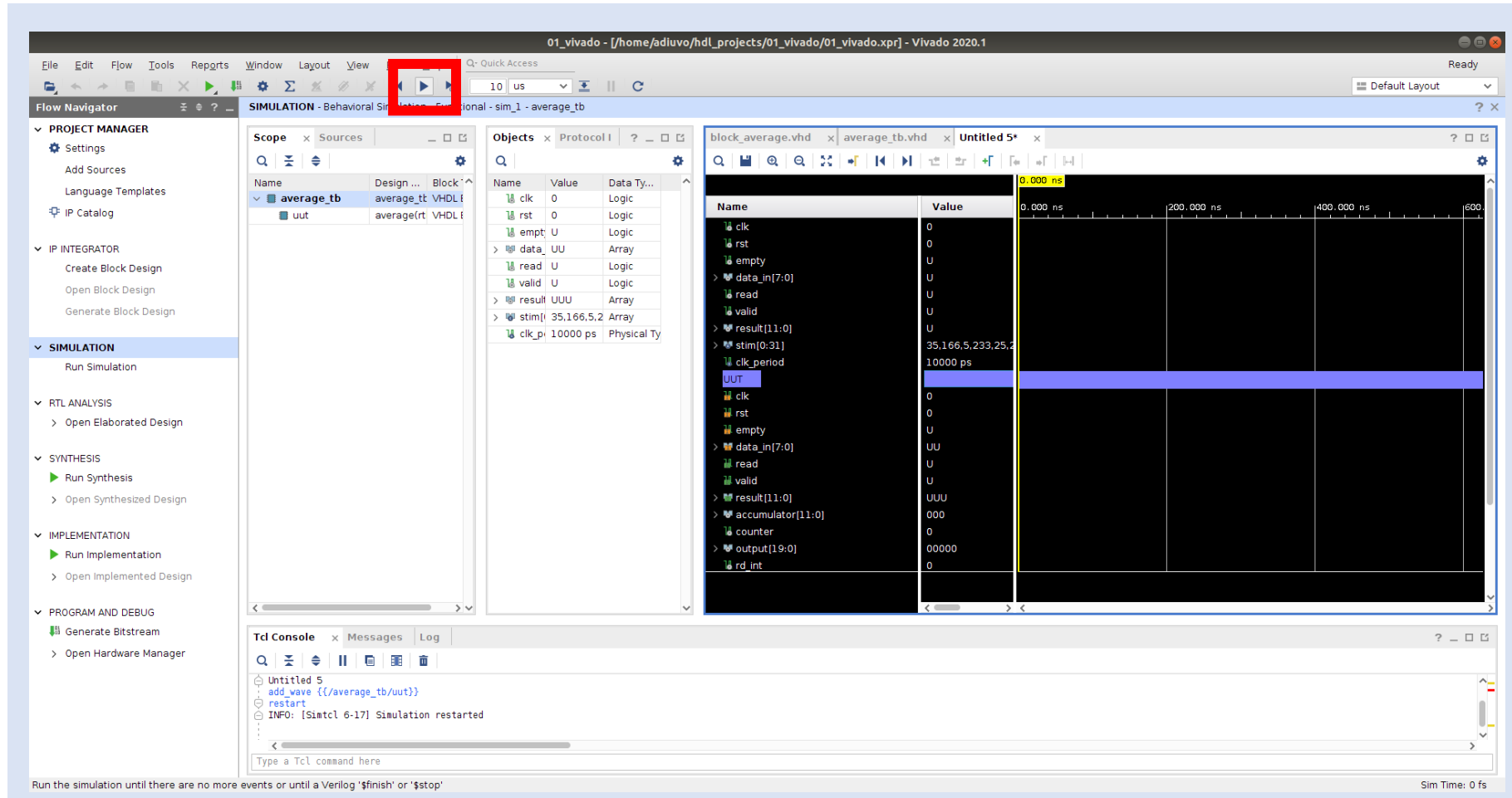
Lab 1: Overview and Introduction to Vivado

Step 23 – This will clear all waveform data and restart the simulation.



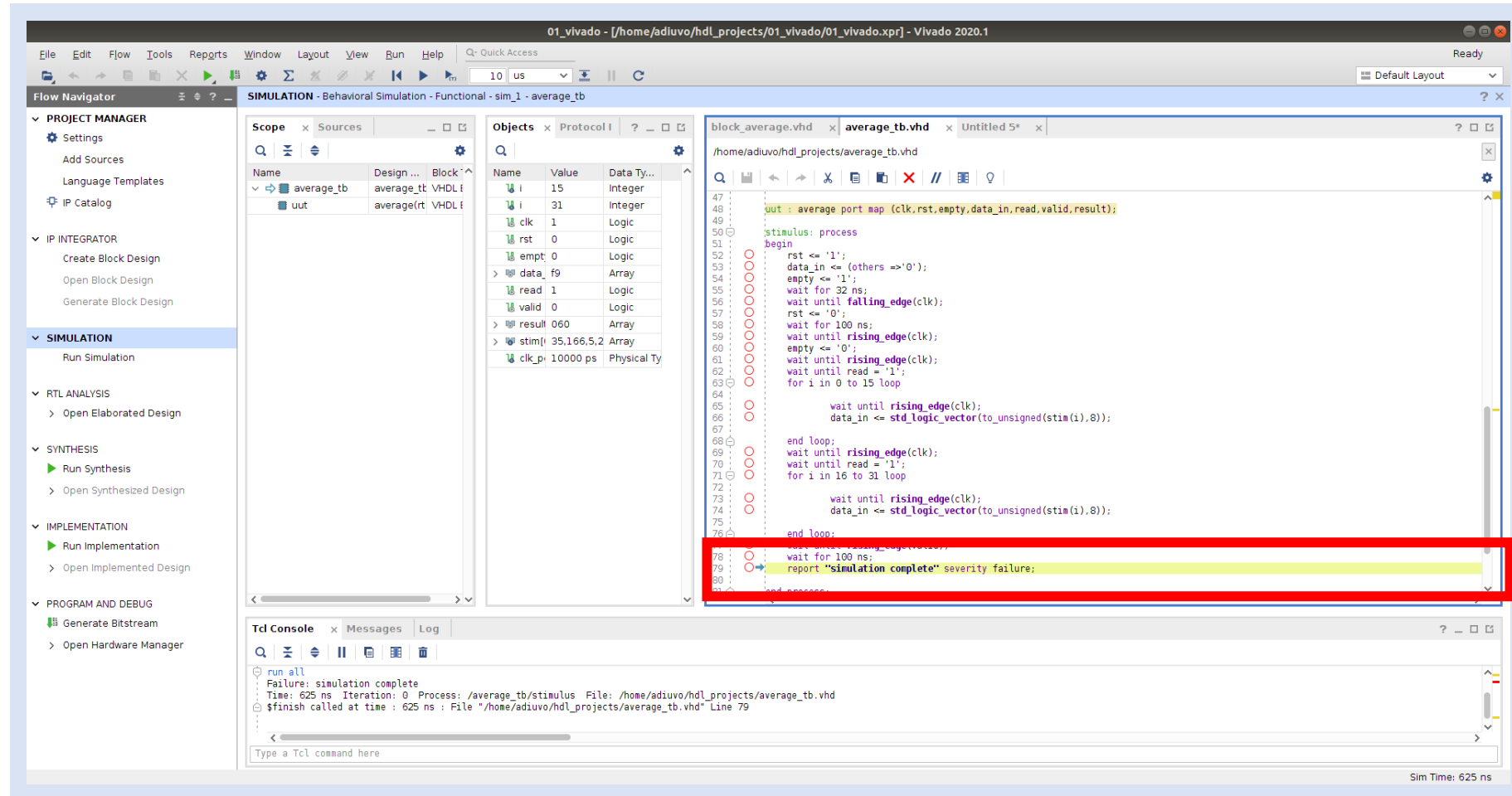
Lab 1: Overview and Introduction to Vivado

Step 24 – To rerun the simulation, select the **Run** button on the menu. The simulation will stop automatically.



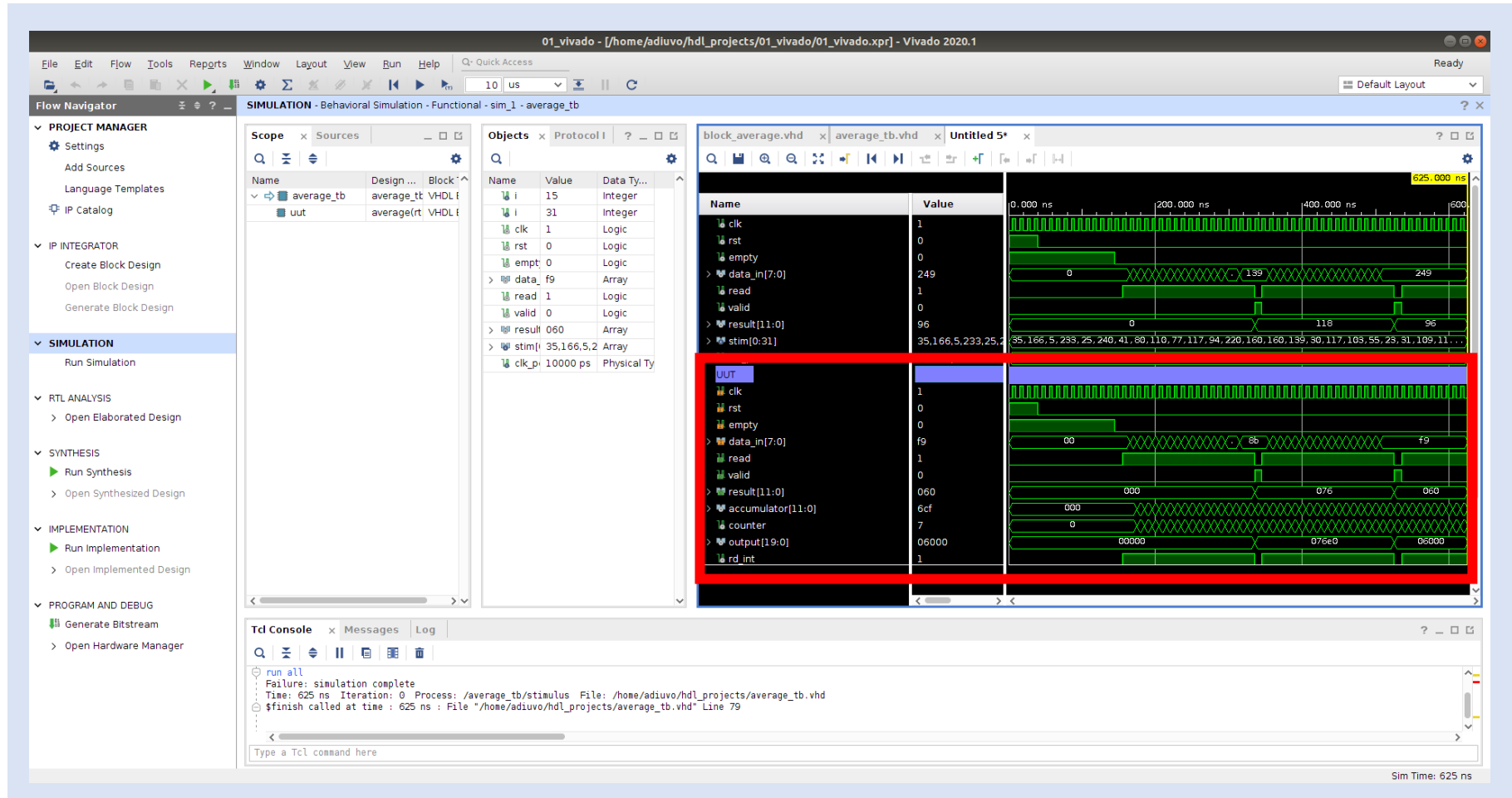
Lab 1: Overview and Introduction to Vivado

Step 25 – When the simulation completes, you will see the highlighted line in the test bench.



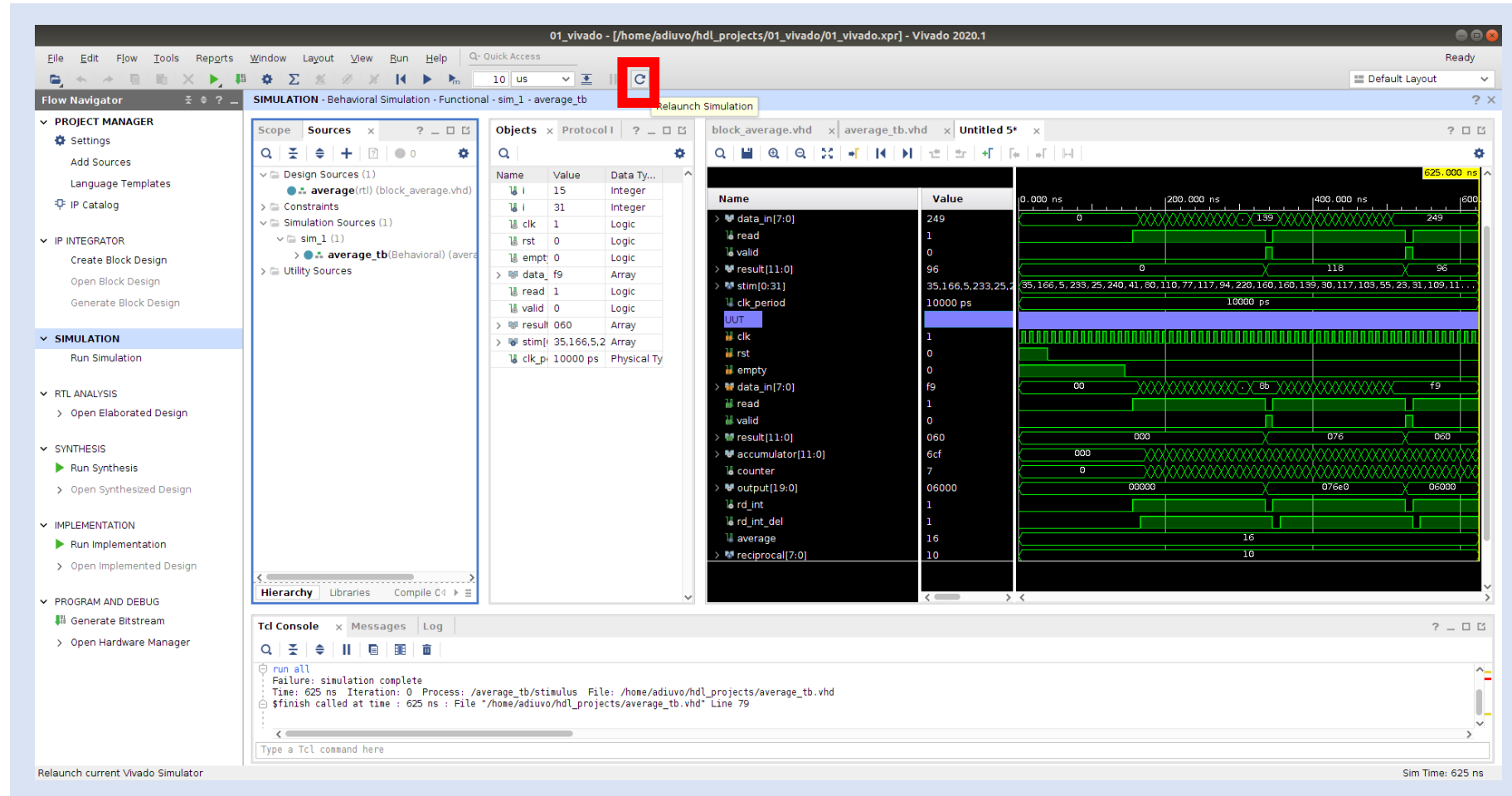
Lab 1: Overview and Introduction to Vivado

Step 26 – Selecting the **waveform tab** again will show all the signals for the UUT.



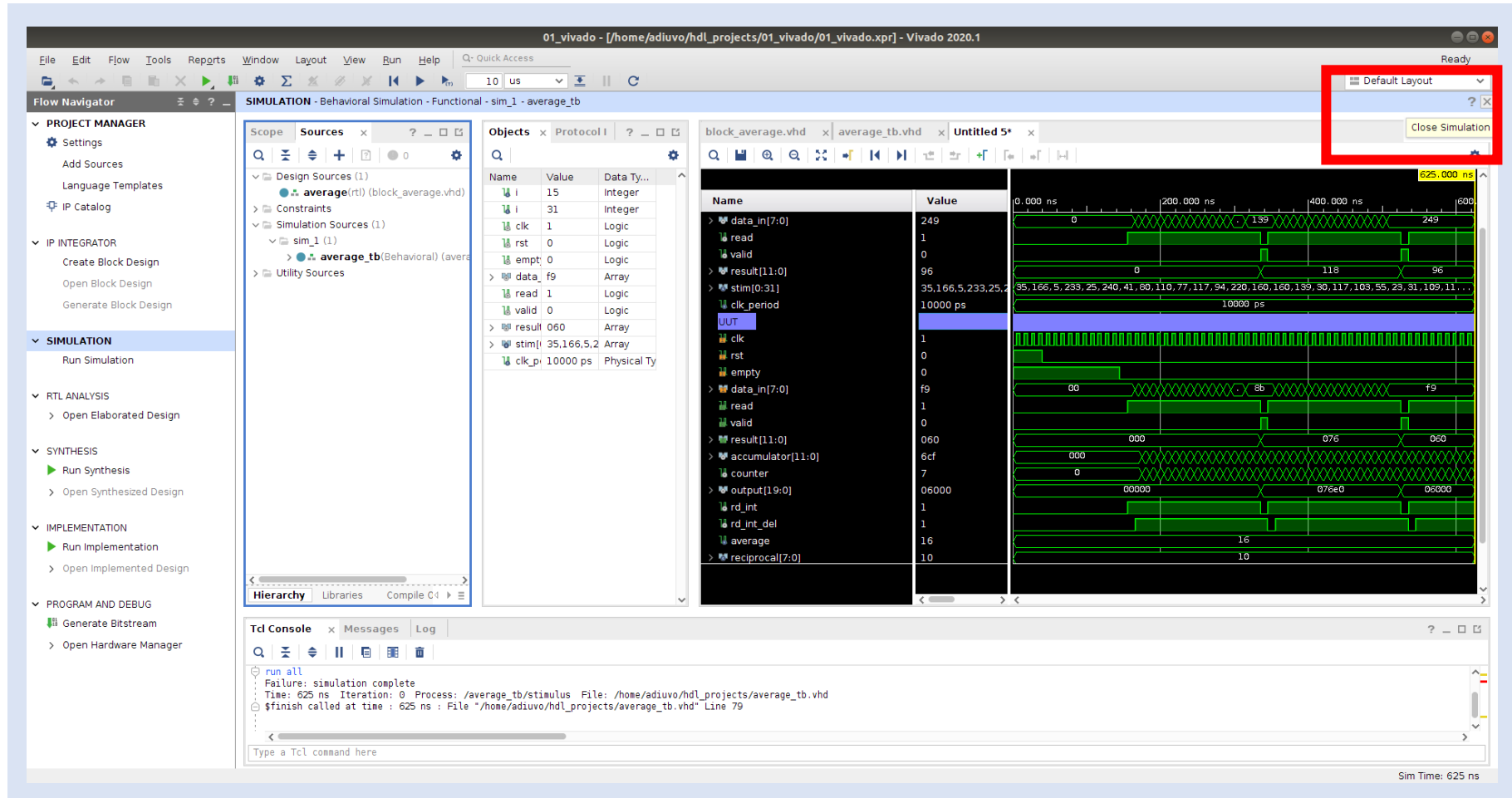
Lab 1: Overview and Introduction to Vivado

Step 27 – If you make changes to the source code, you need to relaunch the simulation. This can be achieved using the relaunch button on the menu.



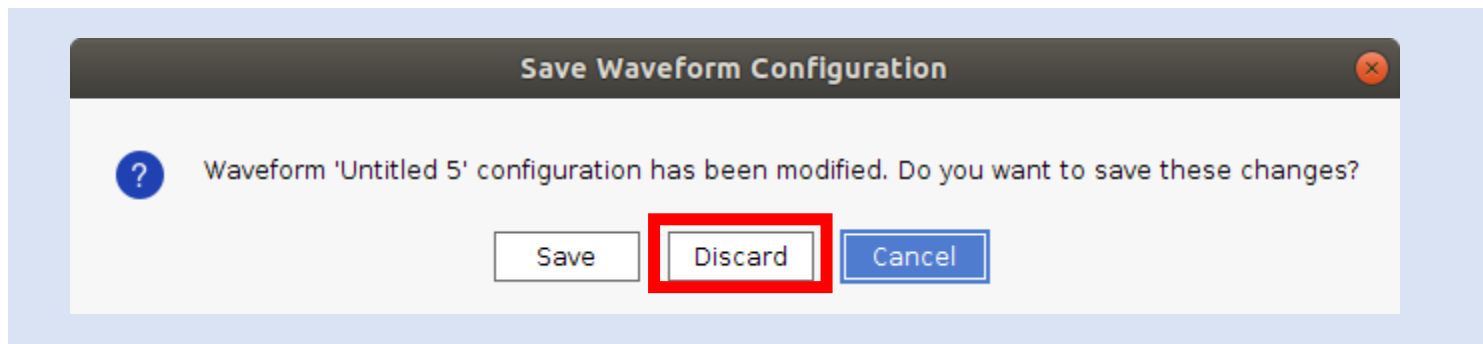
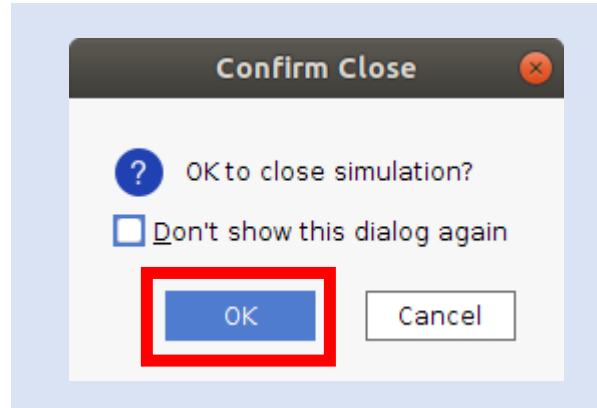
Lab 1: Overview and Introduction to Vivado

Step 28 – With the simulation complete, we are now ready to implement the design. Close the simulation view.



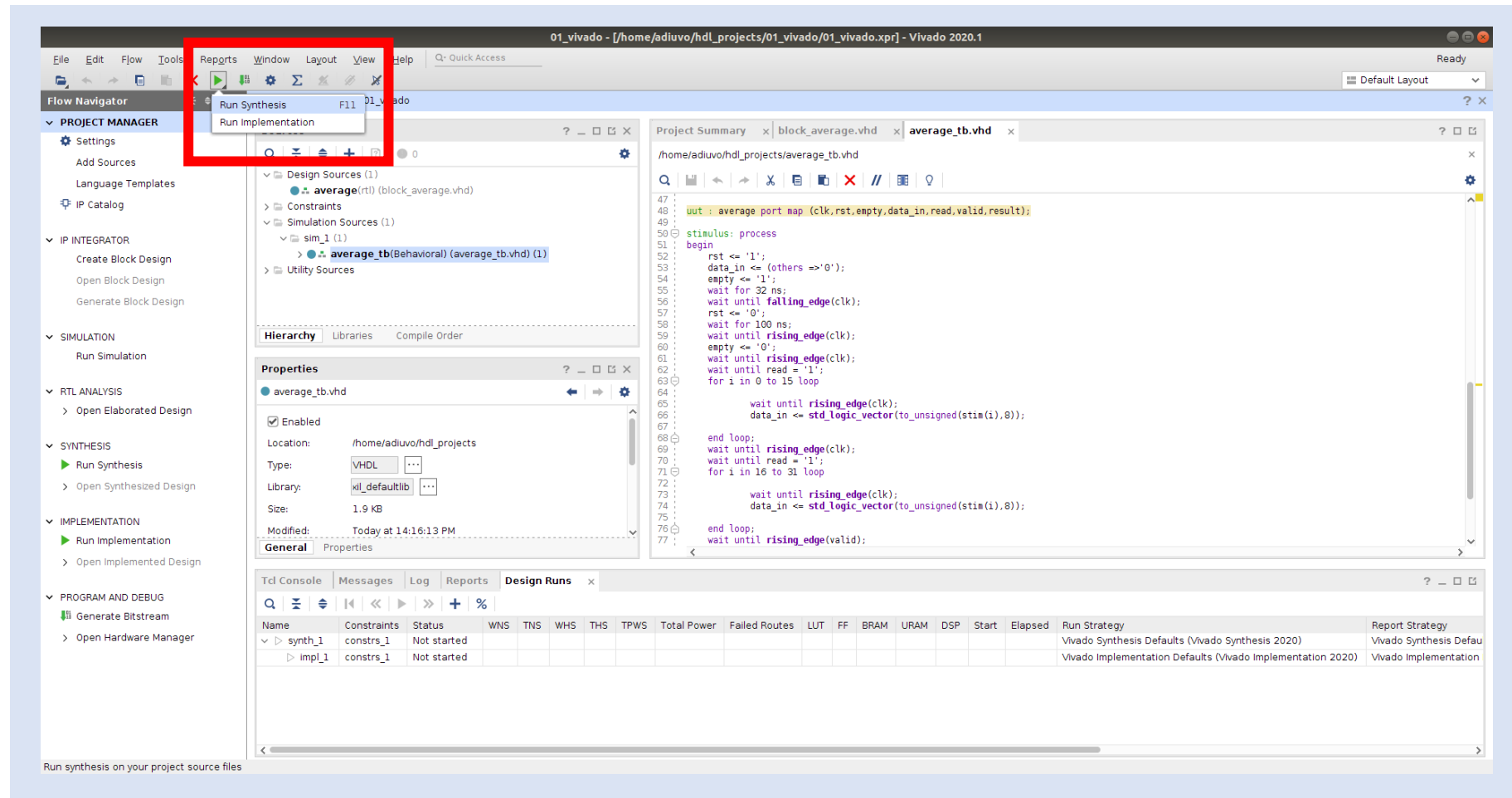
Lab 1: Overview and Introduction to Vivado

Step 29 – When asked to confirm, click **OK**. If a save waveform dialog pops up, select **Discard**.



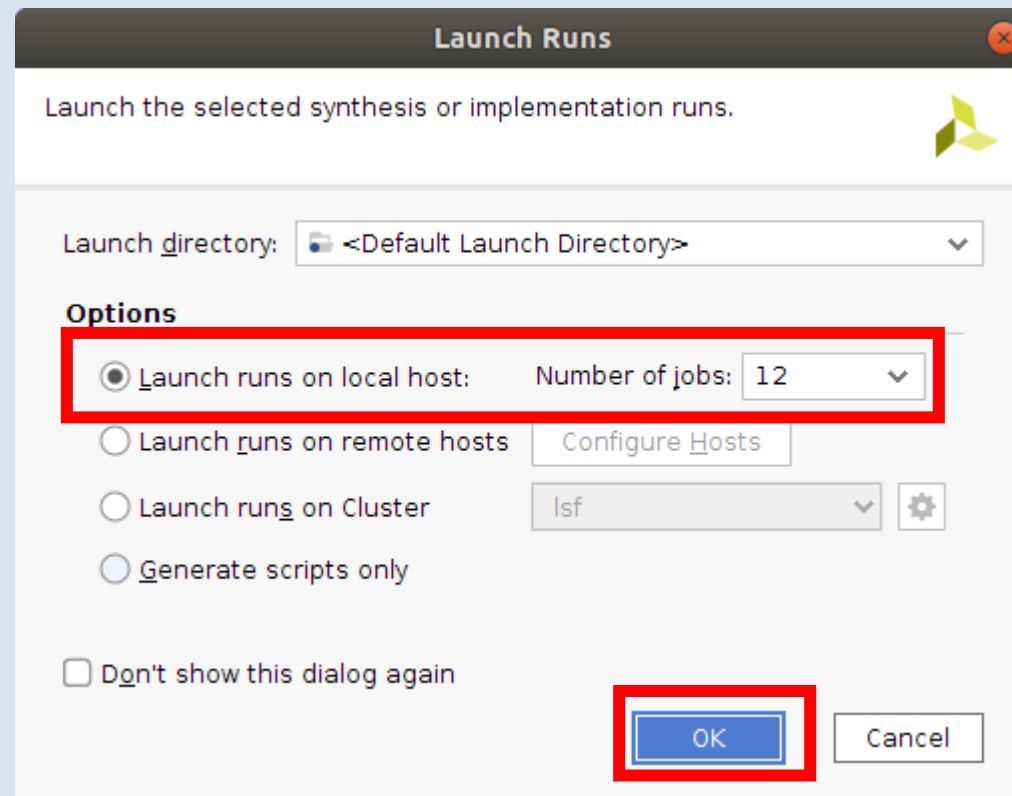
Lab 1: Overview and Introduction to Vivado

Step 30 – To synthesize the design, click the **green run arrow** and select **Run Synthesis**.



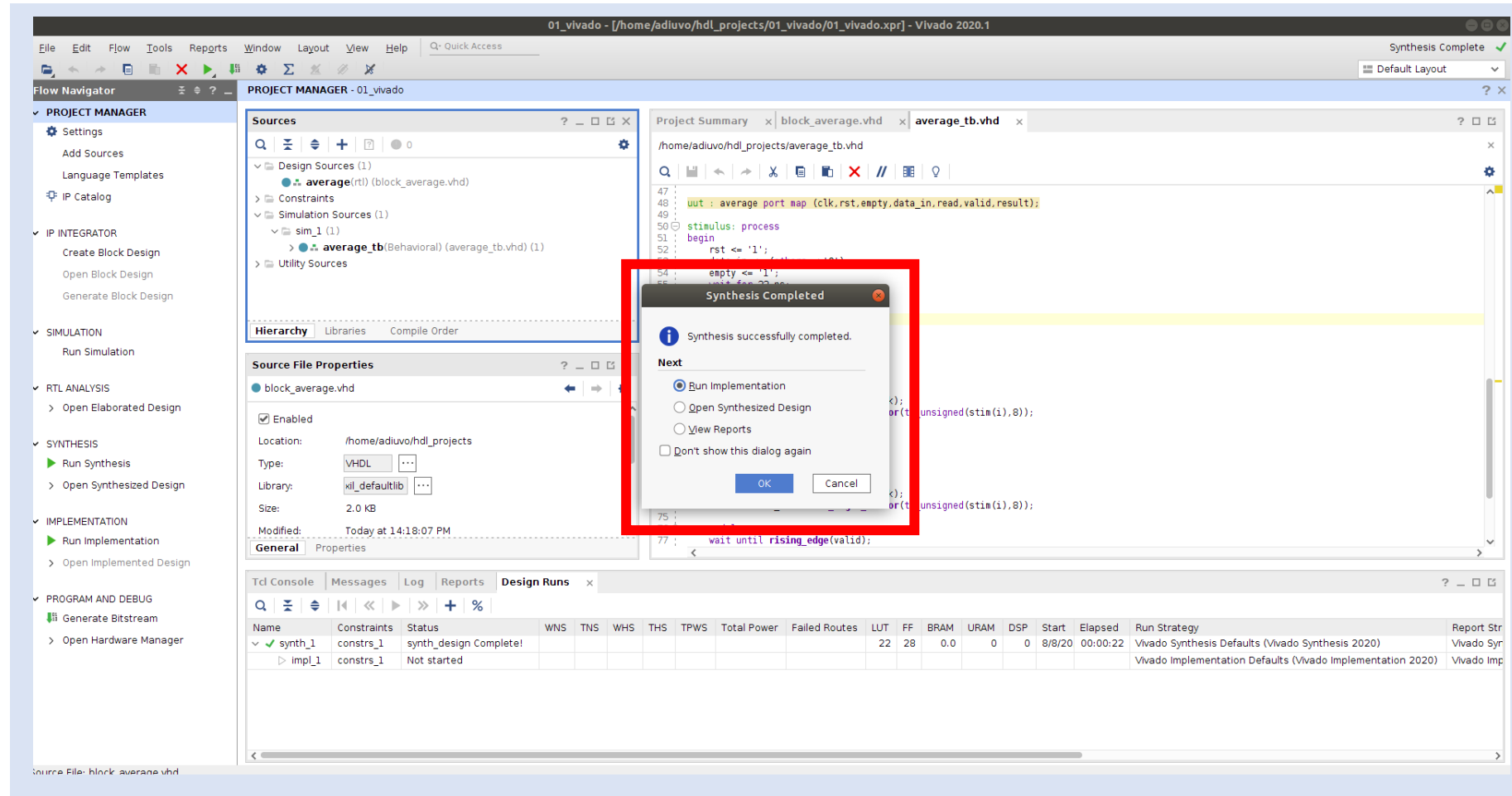
Lab 1: Overview and Introduction to Vivado

Step 31 – On the Launch Runs dialog, select the number of jobs you want to run on your system and click **OK**.



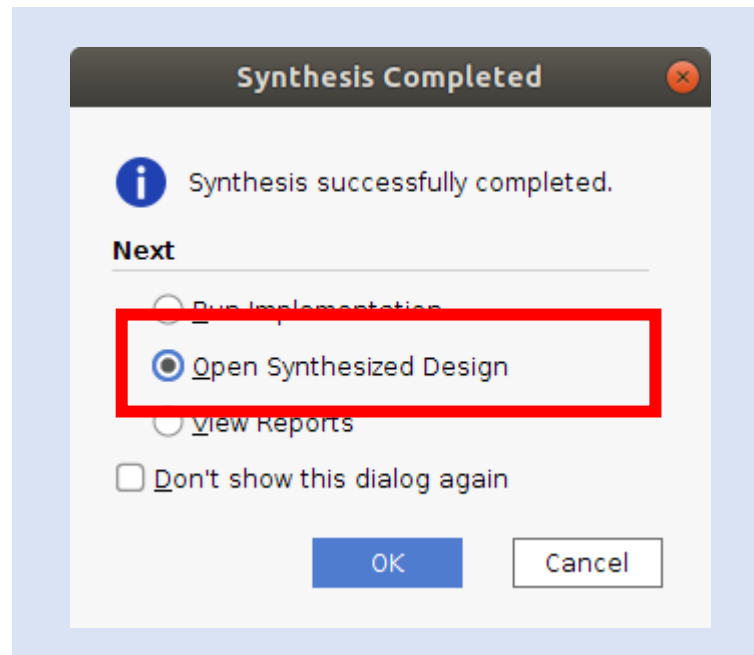
Lab 1: Overview and Introduction to Vivado

Step 32 – When synthesis is complete, you will see a dialog box appear.



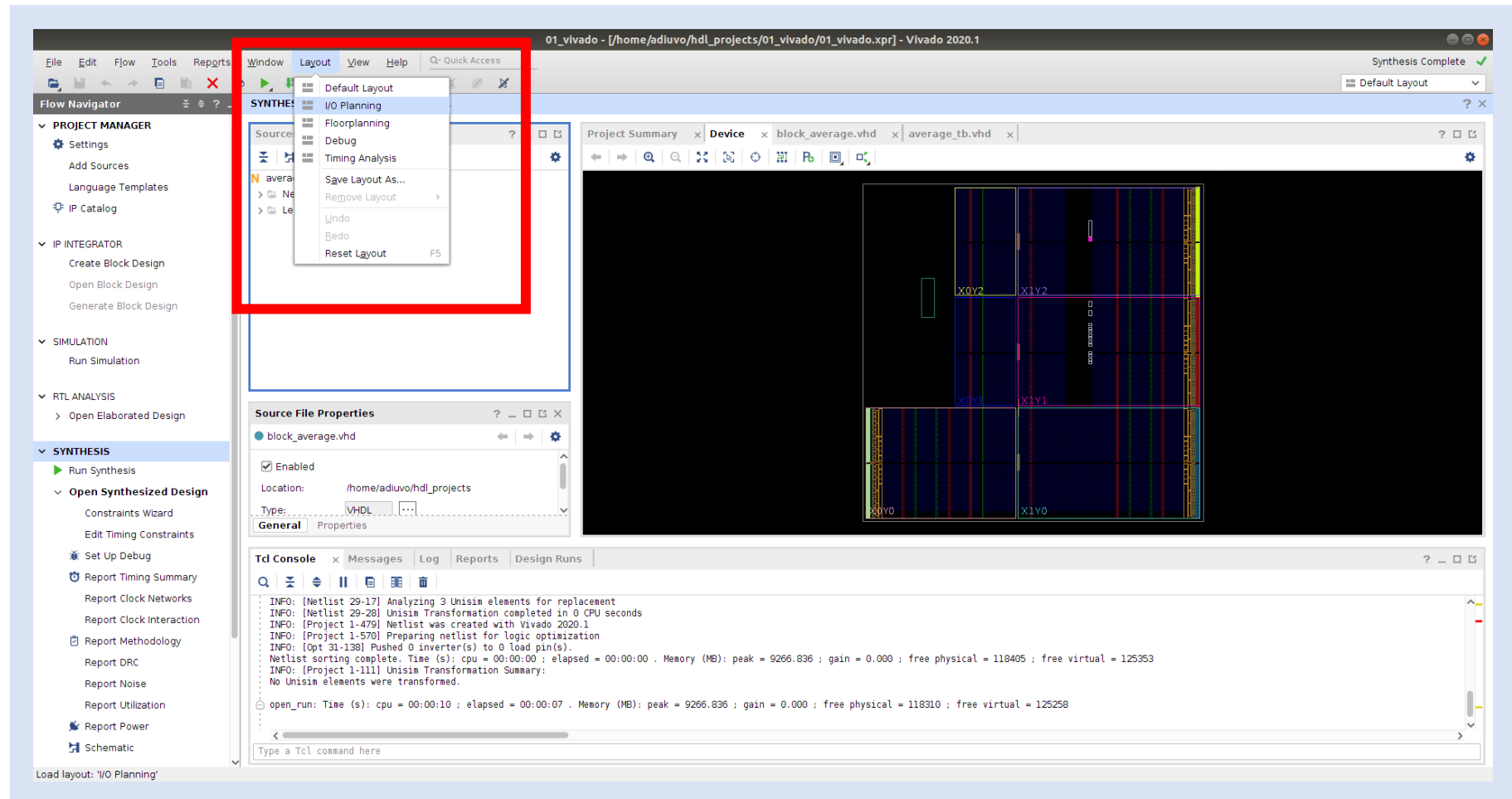
Lab 1: Overview and Introduction to Vivado

Step 33 – Select Open Synthesized Design.



Lab 1: Overview and Introduction to Vivado

Step 34 – This will open the synthesis view. From the menu layout, select **I/O Planning**.



Lab 1: Overview and Introduction to Vivado

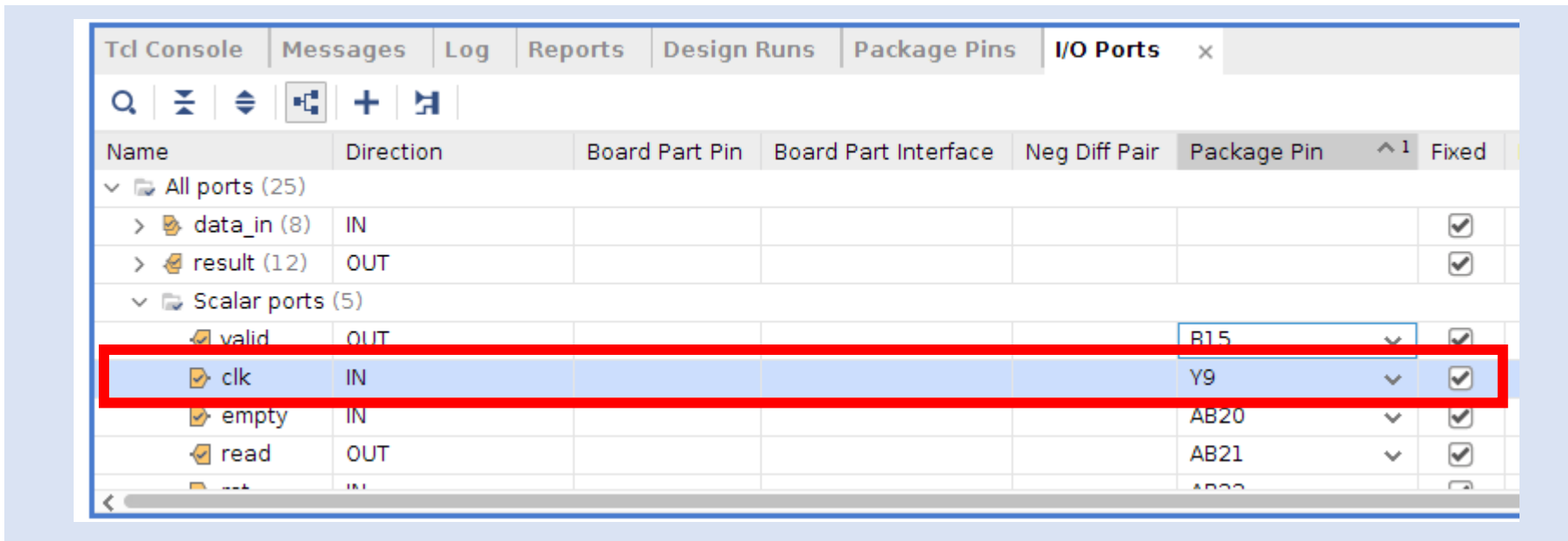
Step 35 – Expand the Data_In, Result and Scalar Ports and assign them to pins. All EXCEPT the clock pin can be assigned to any pin.

The screenshot shows the Vivado 2020.1 interface. The left sidebar contains the Project Manager, IP Integrator, Simulation, RTL Analysis, and Synthesis sections. The main window displays the Synthesized Design, showing the Sources, Netlist, and Package tabs. The I/O Port Properties window is open, showing the properties for 'data_in[7]'. The I/O Ports window is also open, showing a table of ports and their assignments to package pins.

Name	Direction	Board Part Pin	Board Part Interface	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type	Pull Type
data_in (8)	IN							(Multiple)	default (LVCMOS18)	1.800			NONE
data_in[7]	IN				A16	✓		35	default (LVCMOS18)	1.800			NONE
data_in[6]	IN				A17	✓		35	default (LVCMOS18)	1.800			NONE
data_in[5]	IN				A18	✓		35	default (LVCMOS18)	1.800			NONE
data_in[4]	IN				A19	✓		35	default (LVCMOS18)	1.800			NONE
data_in[3]	IN				A21	✓		35	default (LVCMOS18)	1.800			NONE
data_in[2]	IN				A22	✓		35	default (LVCMOS18)	1.800			NONE

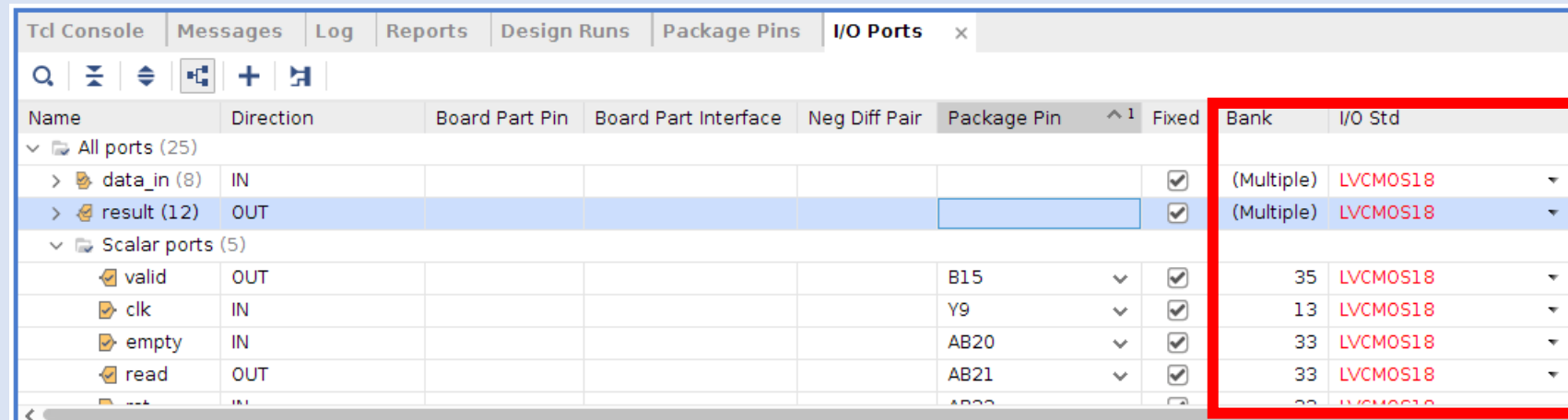
Lab 1: Overview and Introduction to Vivado

Step 36 – Assign the **clock pin to Y9**. Clocks have to be assigned to clock capable pins.



Lab 1: Overview and Introduction to Vivado

Step 37 – Ensure the **IO Standard** is set to **LVC MOS18**. Do not leave it as default because this will lead to a failure to implement and generate a bitstream.

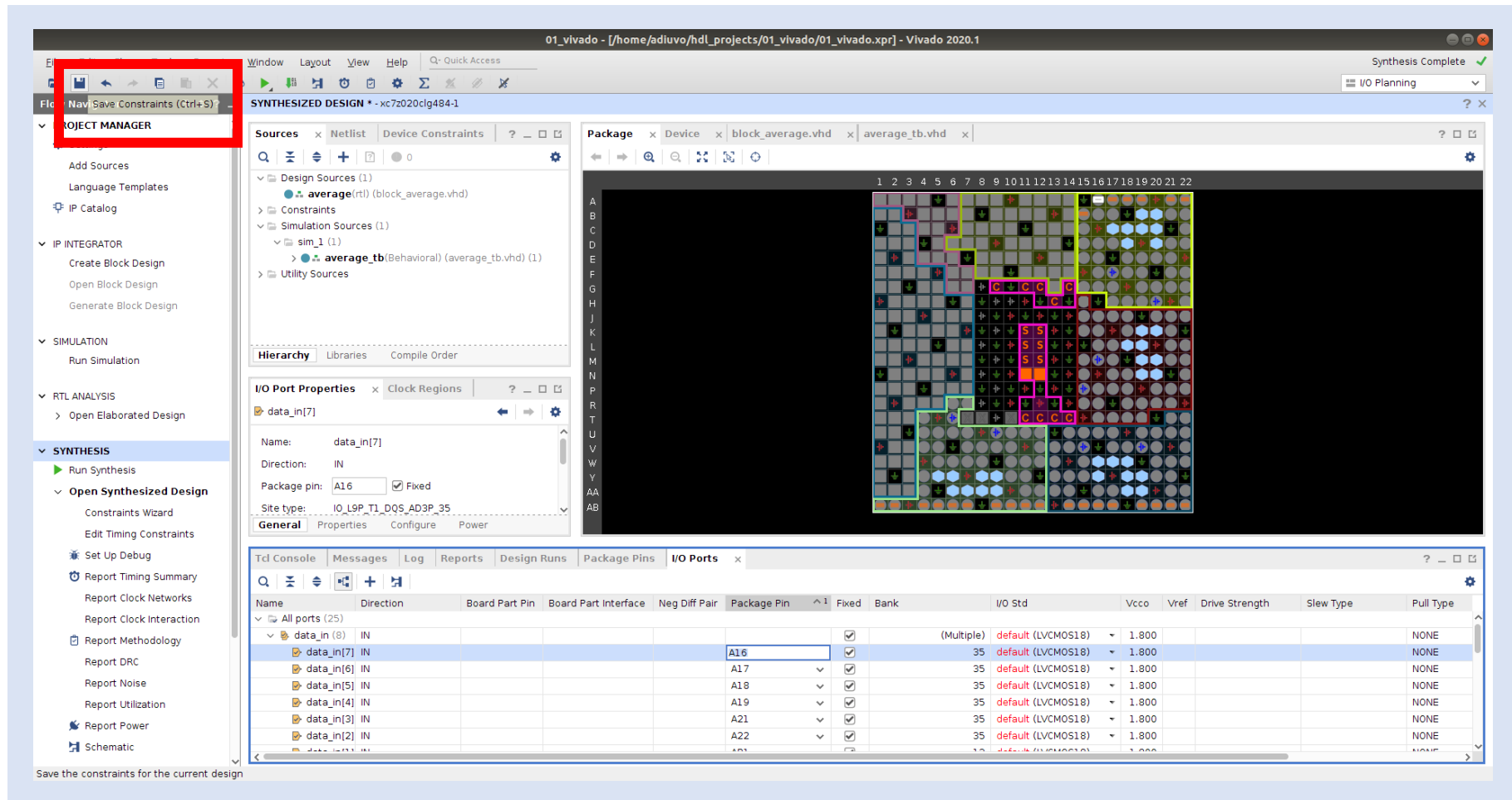


The screenshot shows the 'I/O Ports' window in Vivado. The table lists various ports and their configurations. A red box highlights the 'Bank' and 'I/O Std' columns, showing that all ports are configured with 'LVC MOS18' as the I/O standard.

Name	Direction	Board Part Pin	Board Part Interface	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std
All ports (25)								
> data_in (8)	IN					☒	(Multiple)	LVC MOS18
> result (12)	OUT					☒	(Multiple)	LVC MOS18
Scalar ports (5)								
valid	OUT				B15	☒	35	LVC MOS18
clk	IN				Y9	☒	13	LVC MOS18
empty	IN				AB20	☒	33	LVC MOS18
read	OUT				AB21	☒	33	LVC MOS18

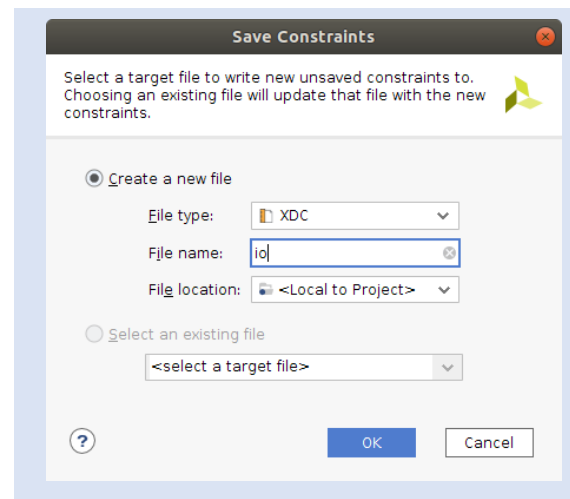
Lab 1: Overview and Introduction to Vivado

Step 38 – Save the Constraints we just edited.



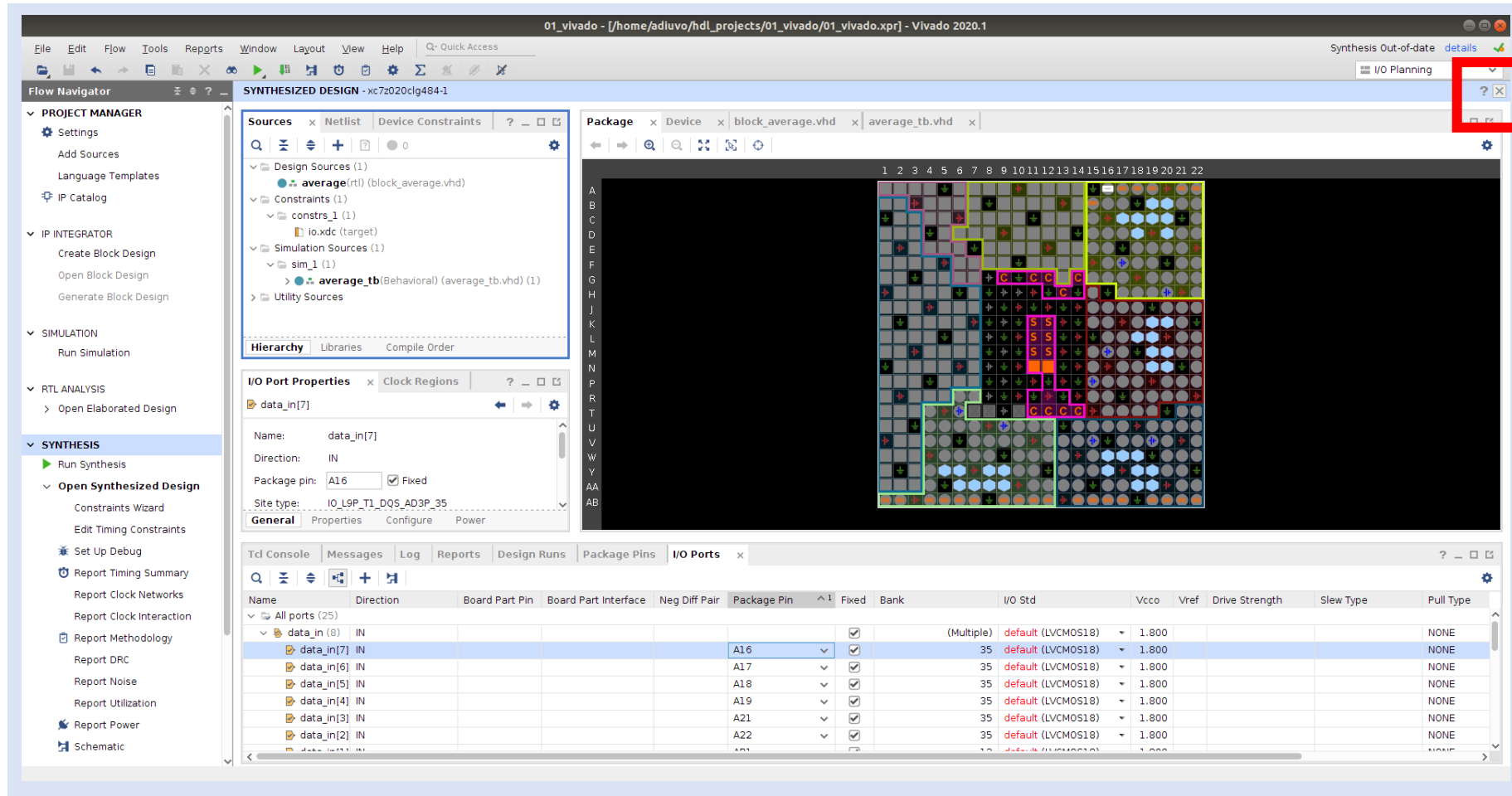
Lab 1: Overview and Introduction to Vivado

Step 39 – This will present two new dialogs. Click **OK** on the first and enter a **file name** for the second.



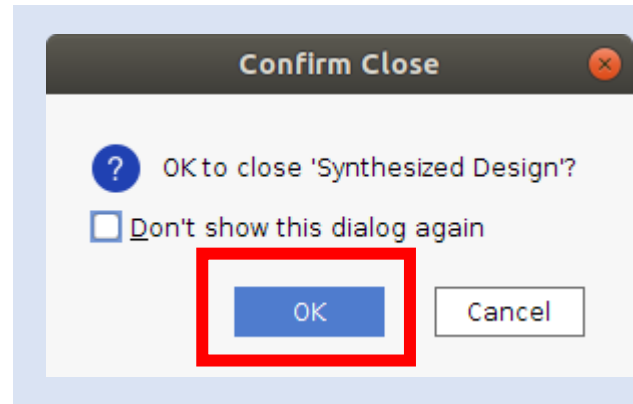
Lab 1: Overview and Introduction to Vivado

Step 40 – Close the Synthesis View.



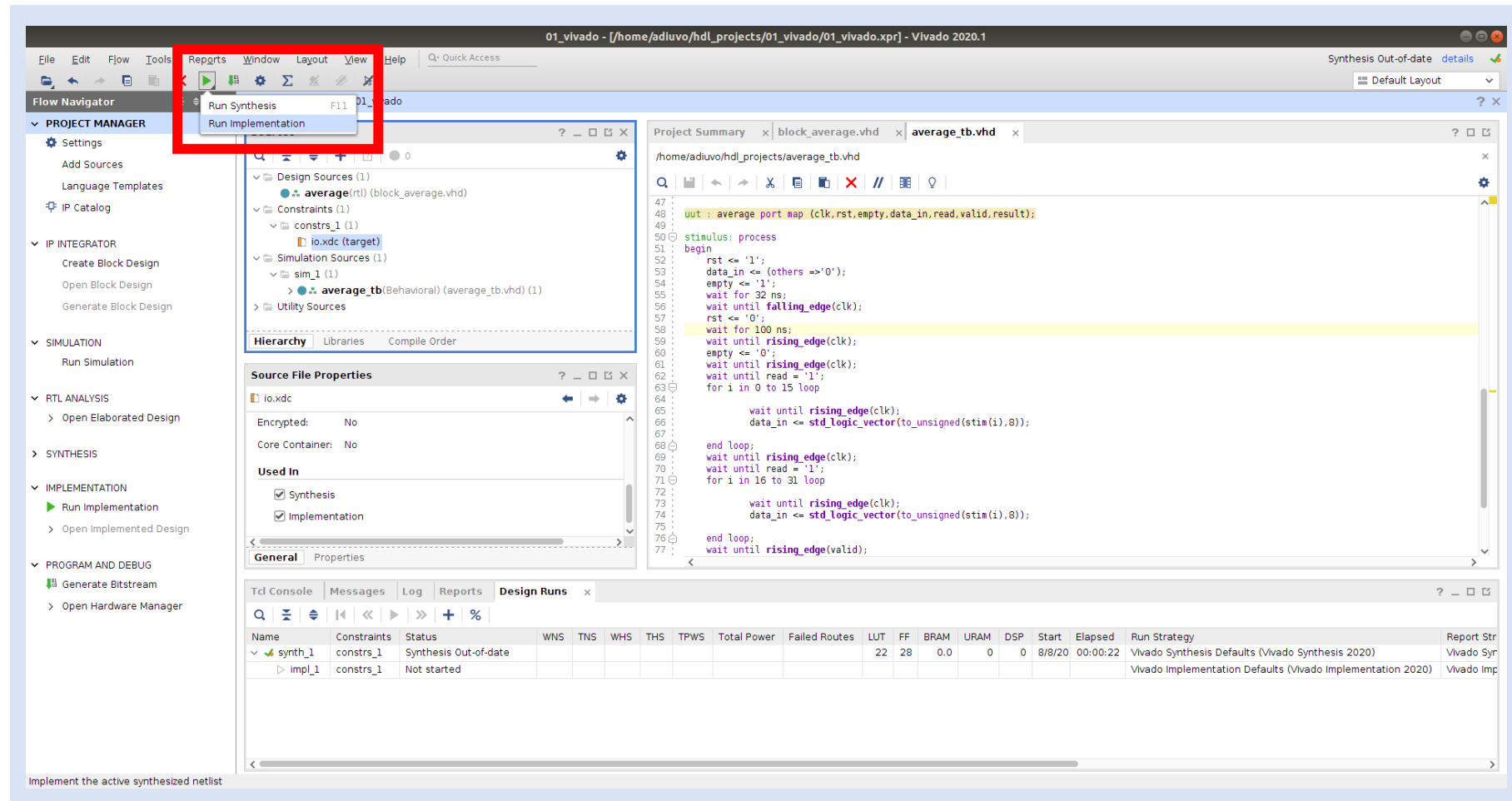
Lab 1: Overview and Introduction to Vivado

Step 41 – Confirm the decision to close by clicking **OK**. This will take us back to the project manager view.



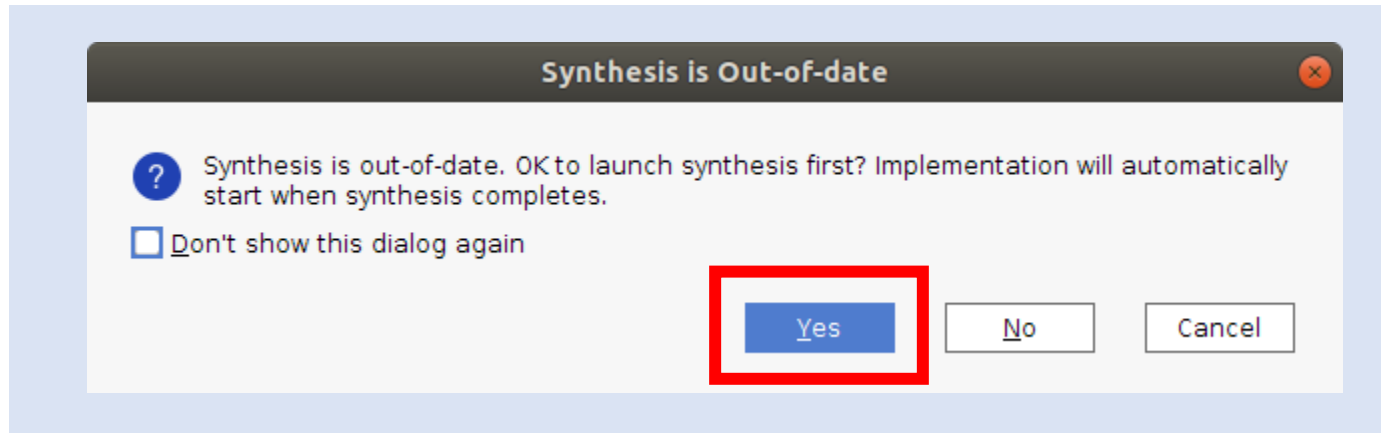
Lab 1: Overview and Introduction to Vivado

Step 42 – We are now ready to Run Implementation.



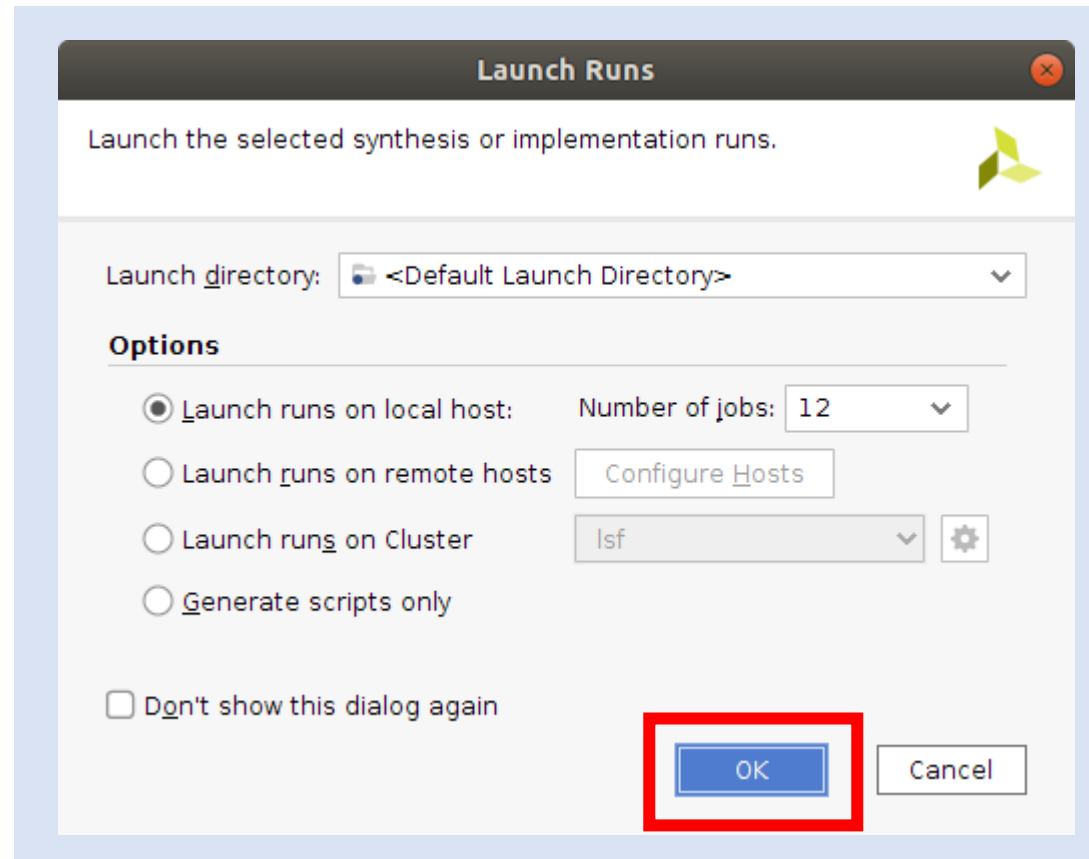
Lab 1: Overview and Introduction to Vivado

Step 43 – Click **Yes** when the Synthesis Out-of-Date dialog pops up.



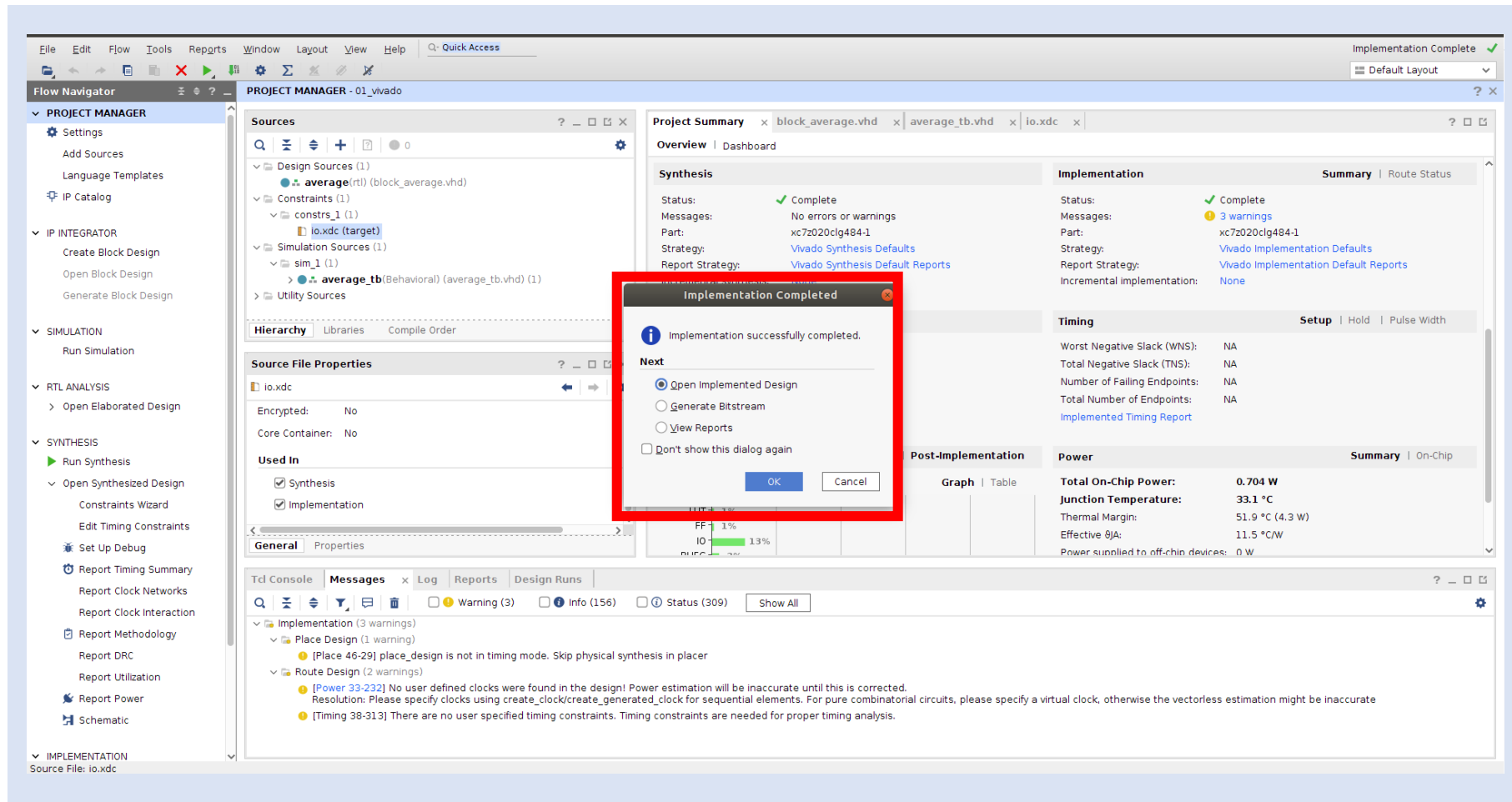
Lab 1: Overview and Introduction to Vivado

Step 44 – Select **OK** on the Launch Runs dialog.



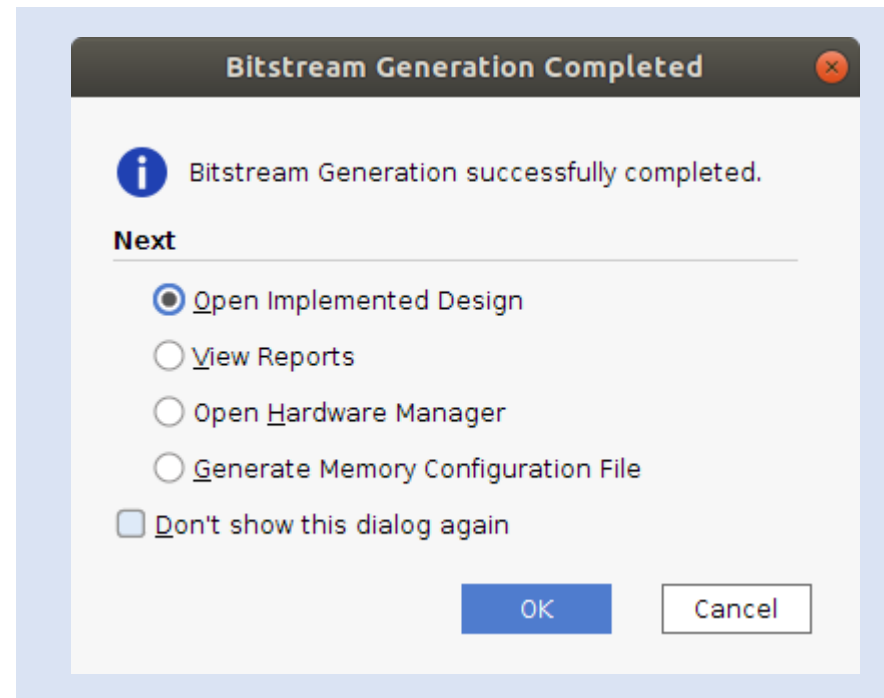
Lab 1: Overview and Introduction to Vivado

Step 45 – When the implementation completes, you will see a dialog appear. Select **Generate Bitstream**.



Lab 1: Overview and Introduction to Vivado

Step 46 – A dialog box will appear when the bitstream generates. Congratulations you have completed your first Vivado FPGA implementation.

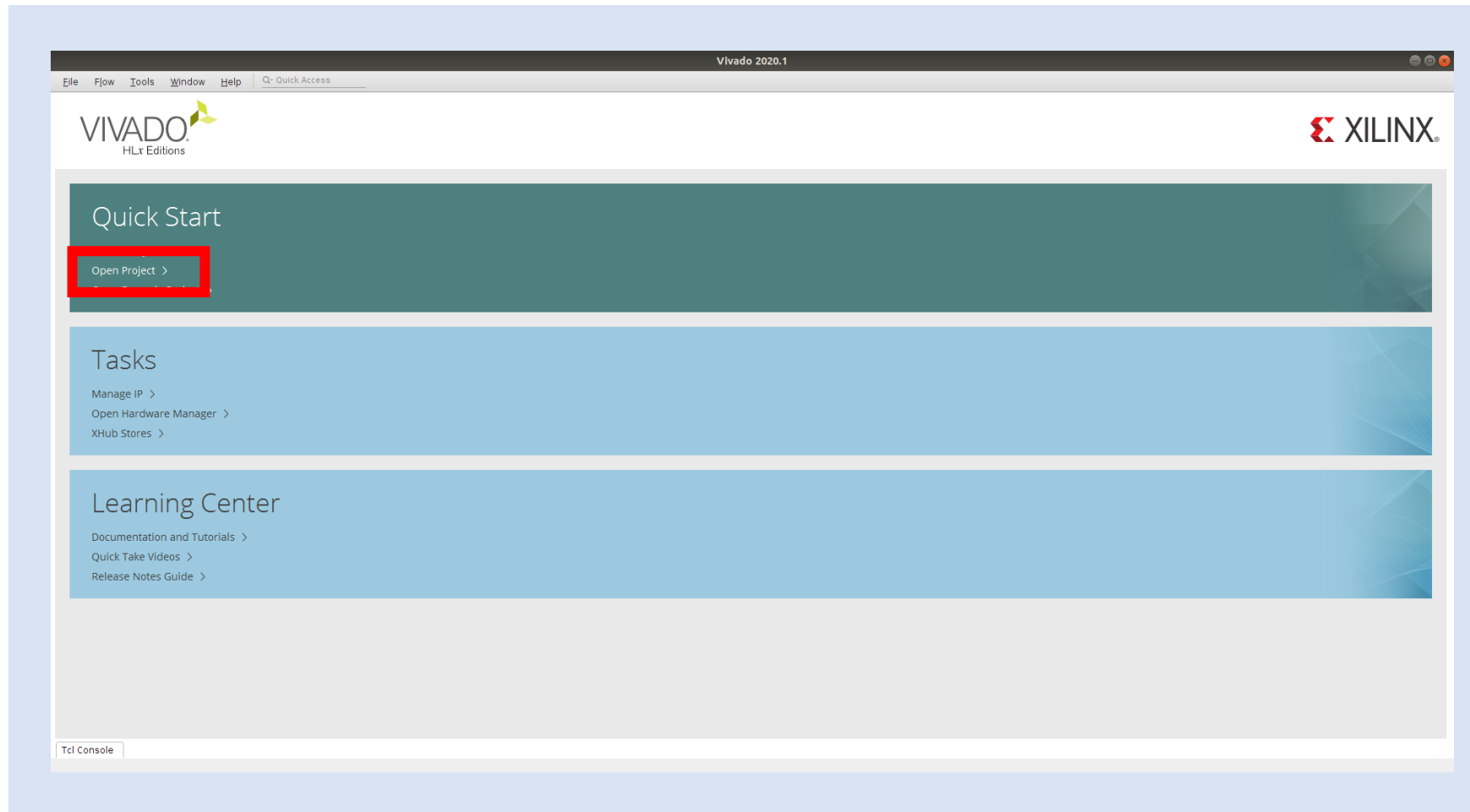


Lab 2

Intermediate Vivado

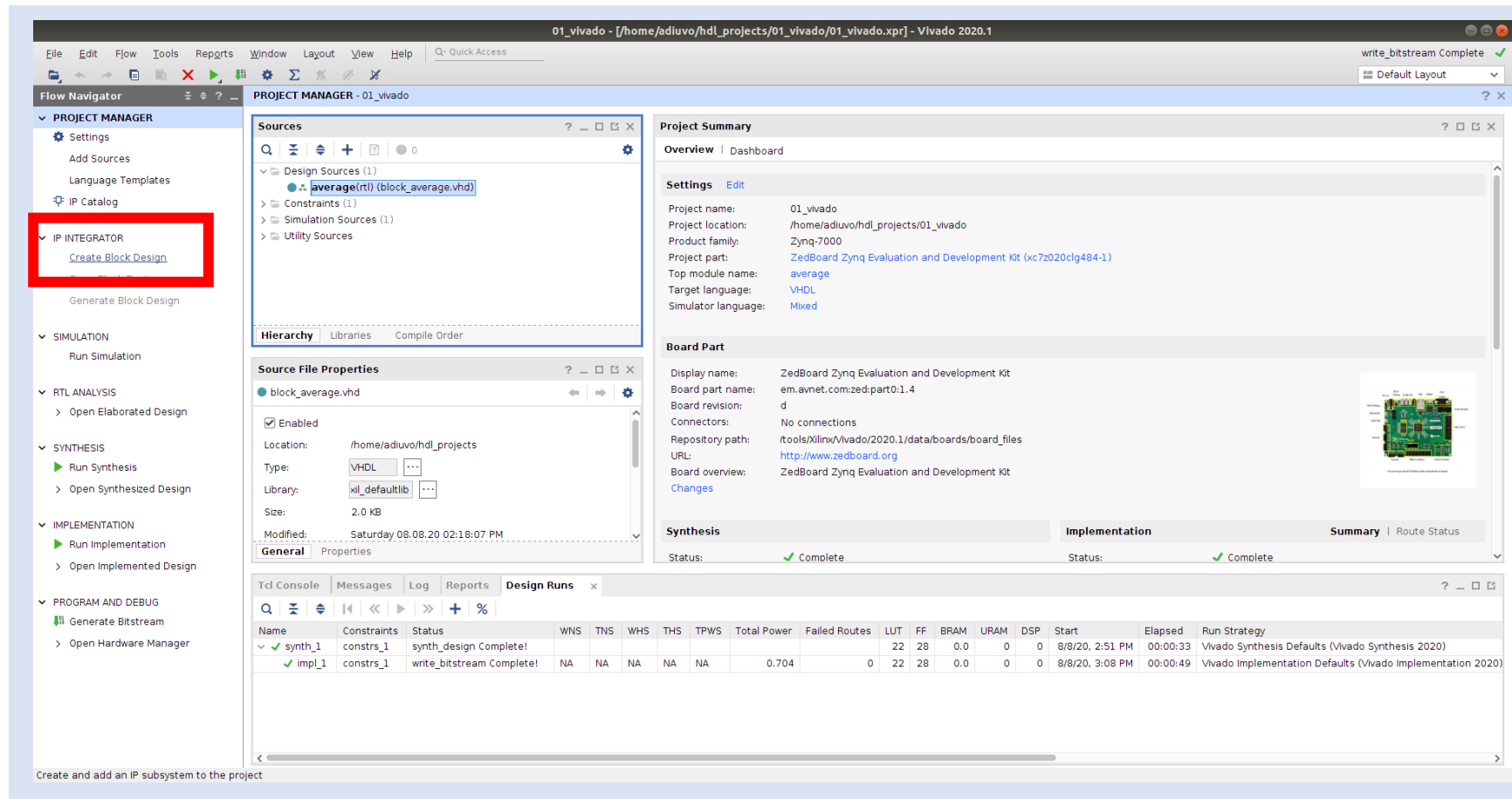
Lab 2: Intermediate Vivado

Step 1 – Open the project created in part one.



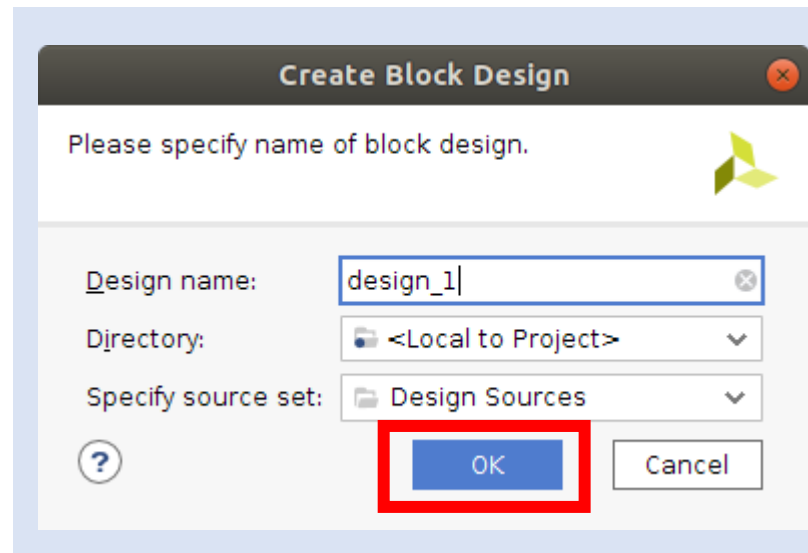
Lab 2: Intermediate Vivado

Step 2 – This will open in project management view. Click on **Create Block Diagram**.



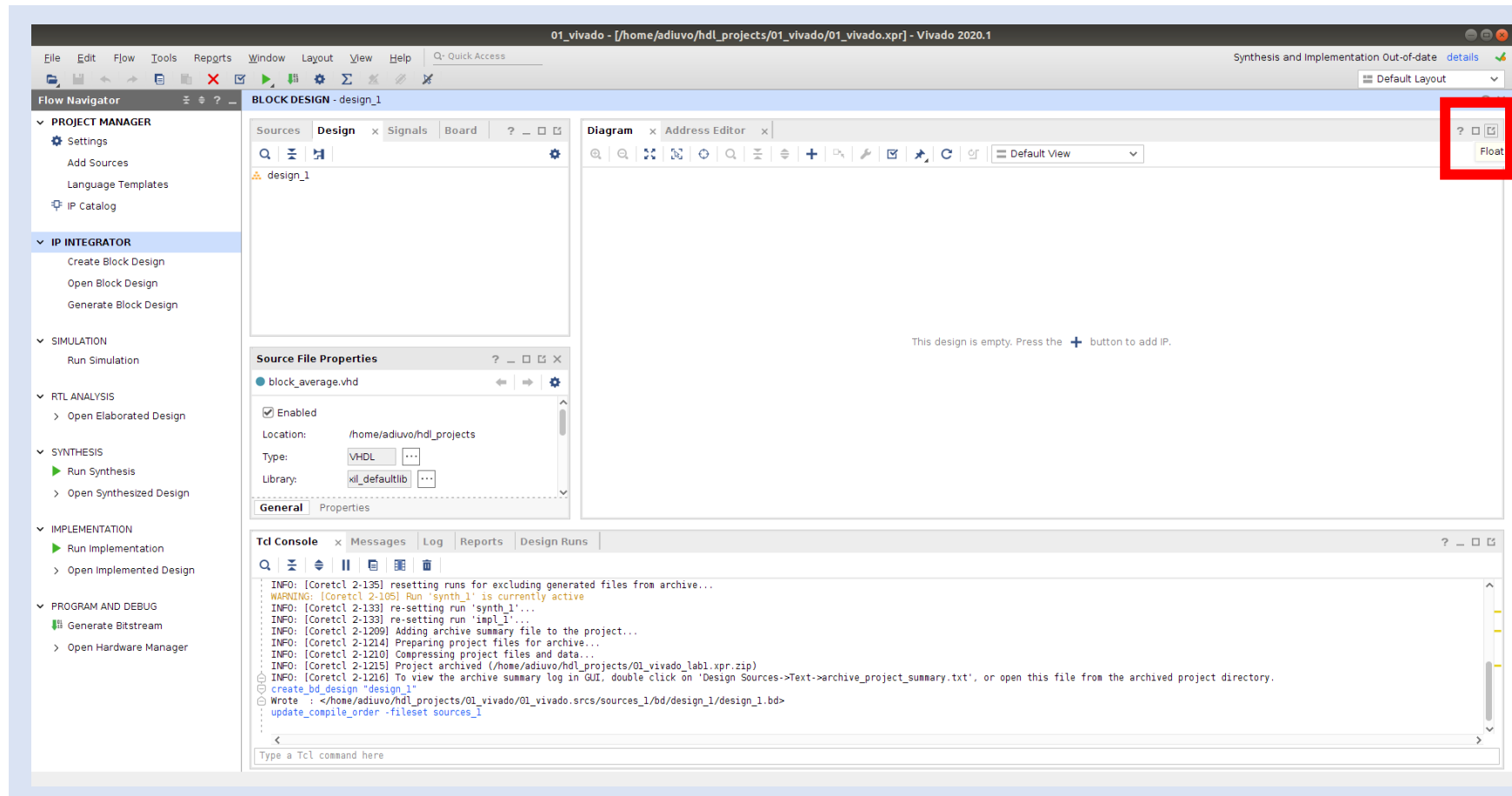
Lab 2: Intermediate Vivado

Step 3 – Leave the predefined name and locations unchanged and click **OK**.



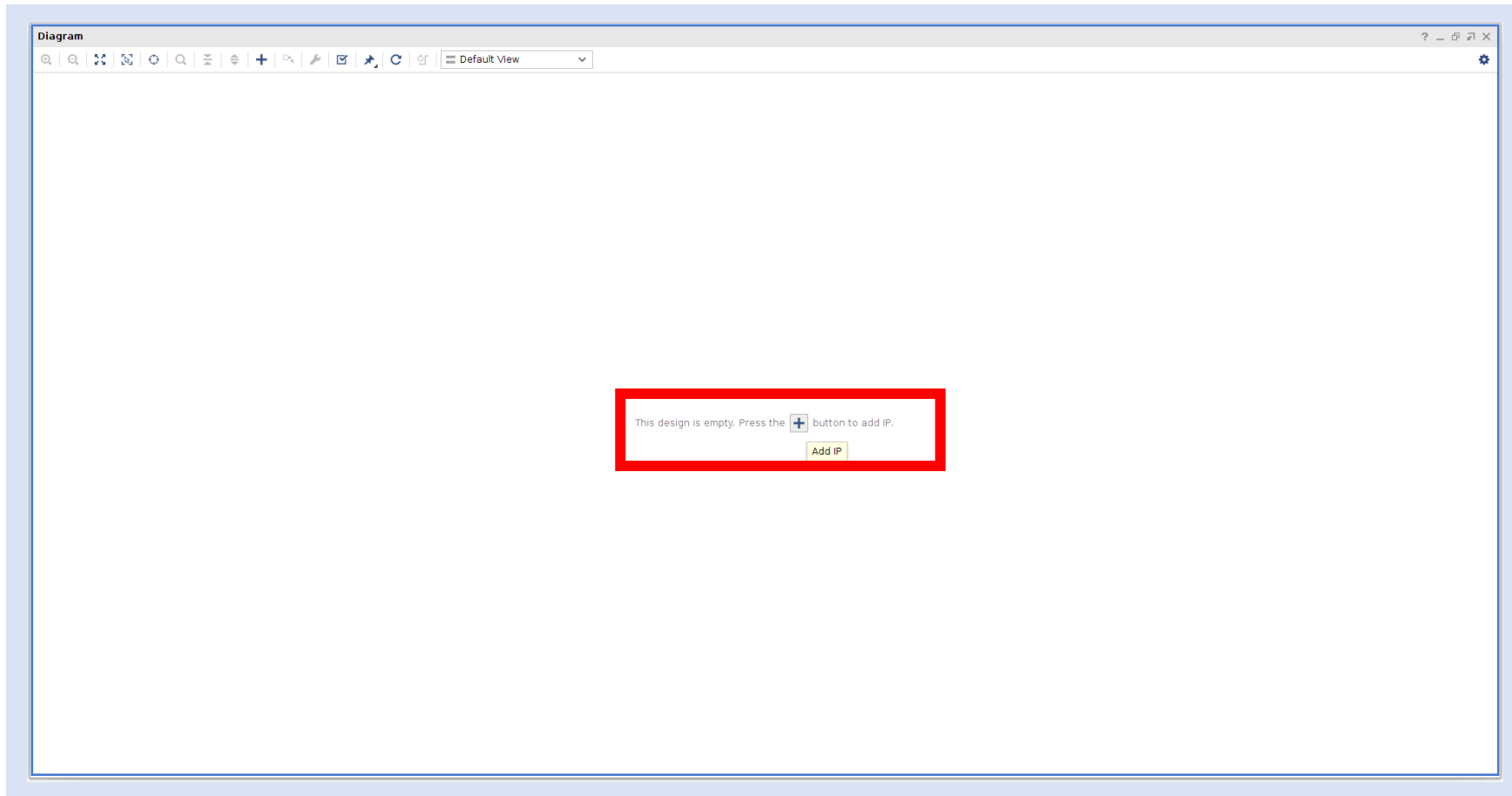
Lab 2: Intermediate Vivado

Step 4 – Undock the block diagram window and maximize it.



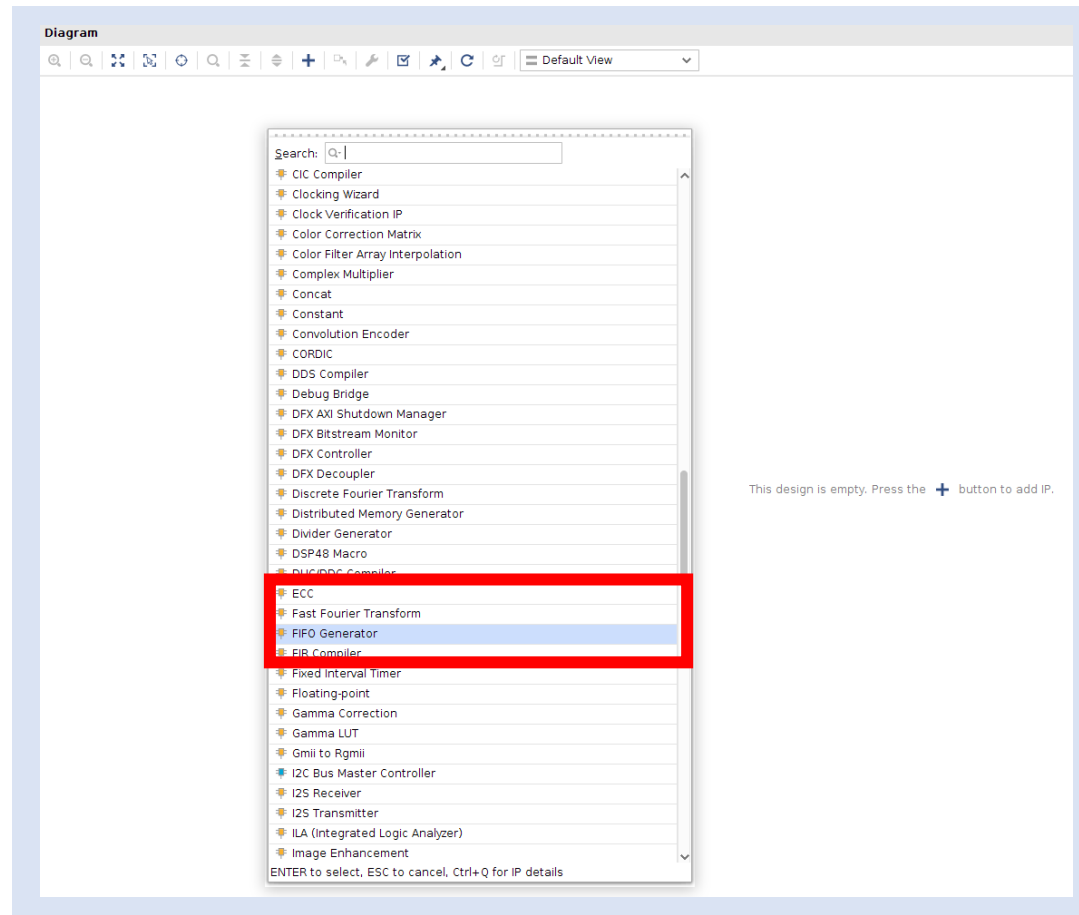
Lab 2: Intermediate Vivado

Step 5 – We are going to add in new IP. Click on the **+** button.



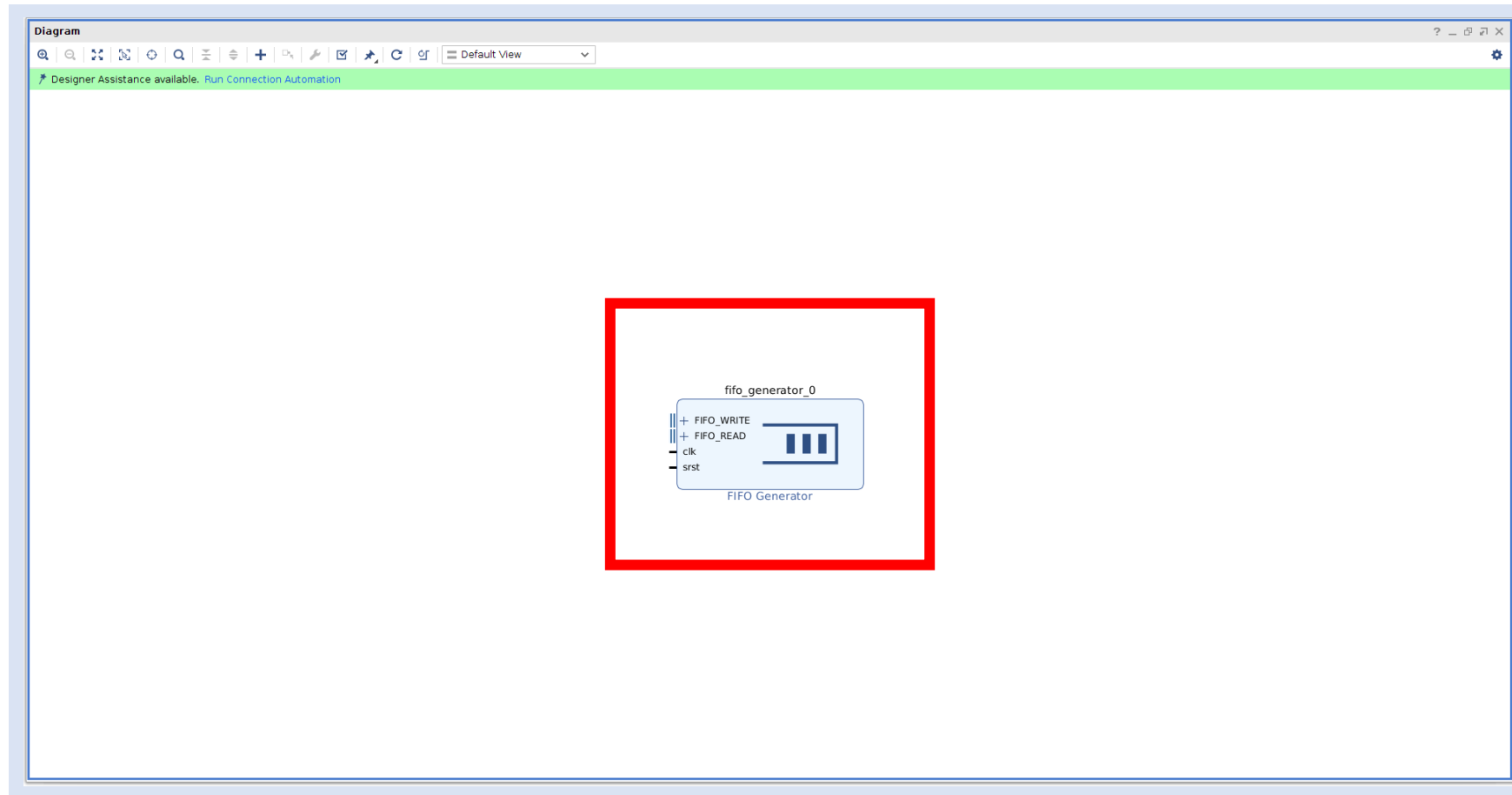
Lab 2: Intermediate Vivado

Step 6 – Select the FIFO Generator. This will add a FIFO to the block diagram.



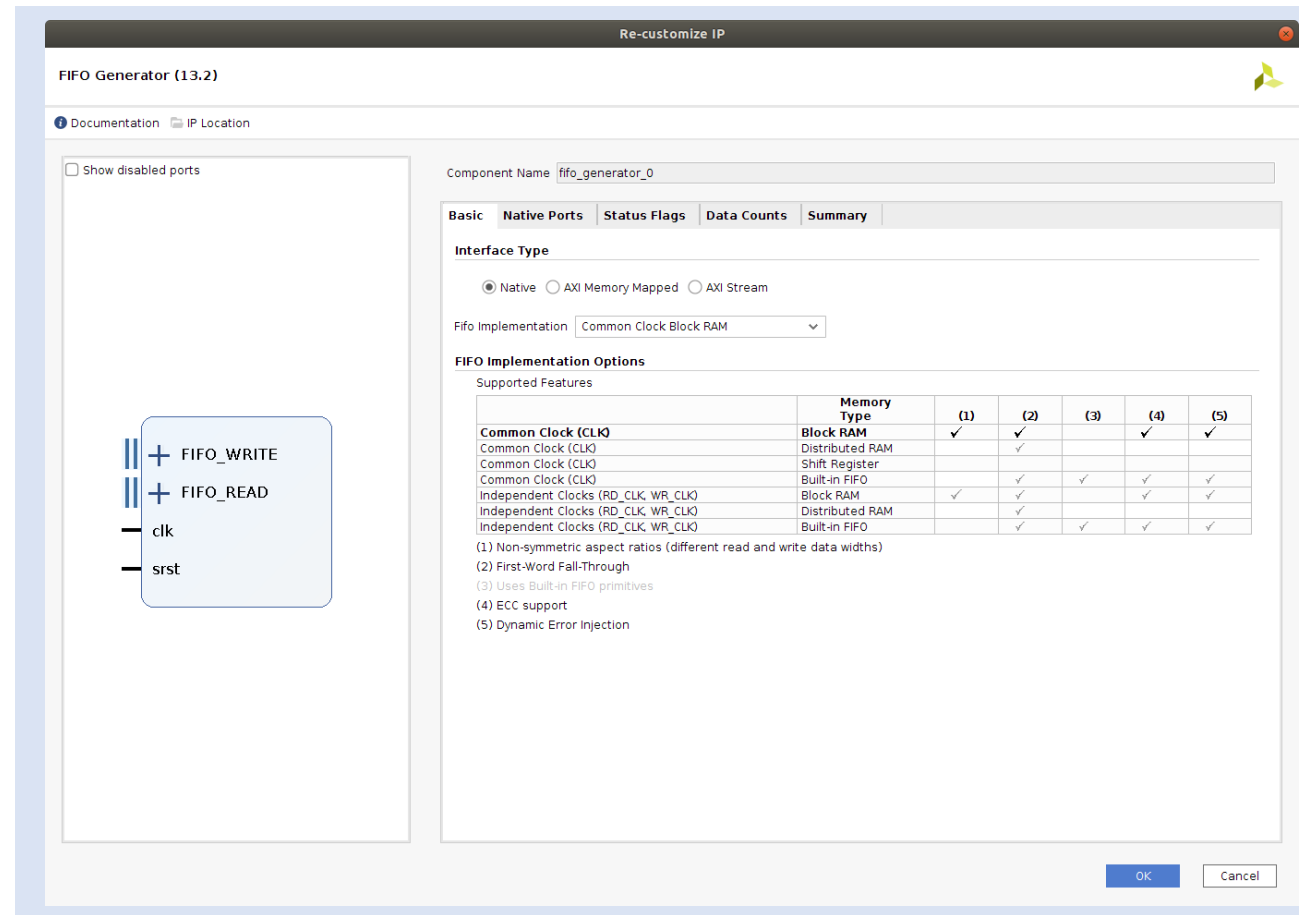
Lab 2: Intermediate Vivado

Step 7 – Double click on the **FIFO Generator** to customize it.



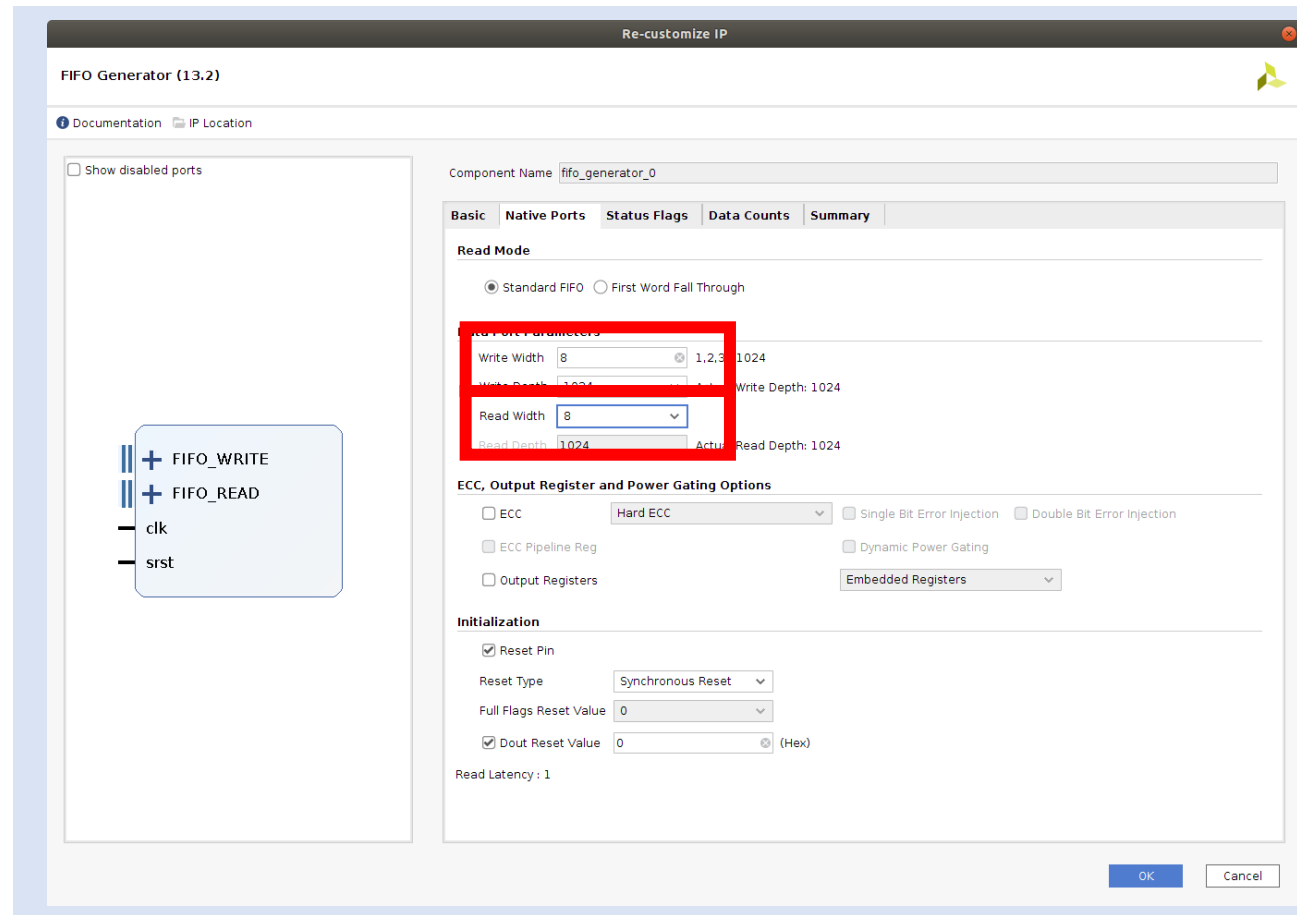
Lab 2: Intermediate Vivado

Step 8 – Leave the first page unchanged.



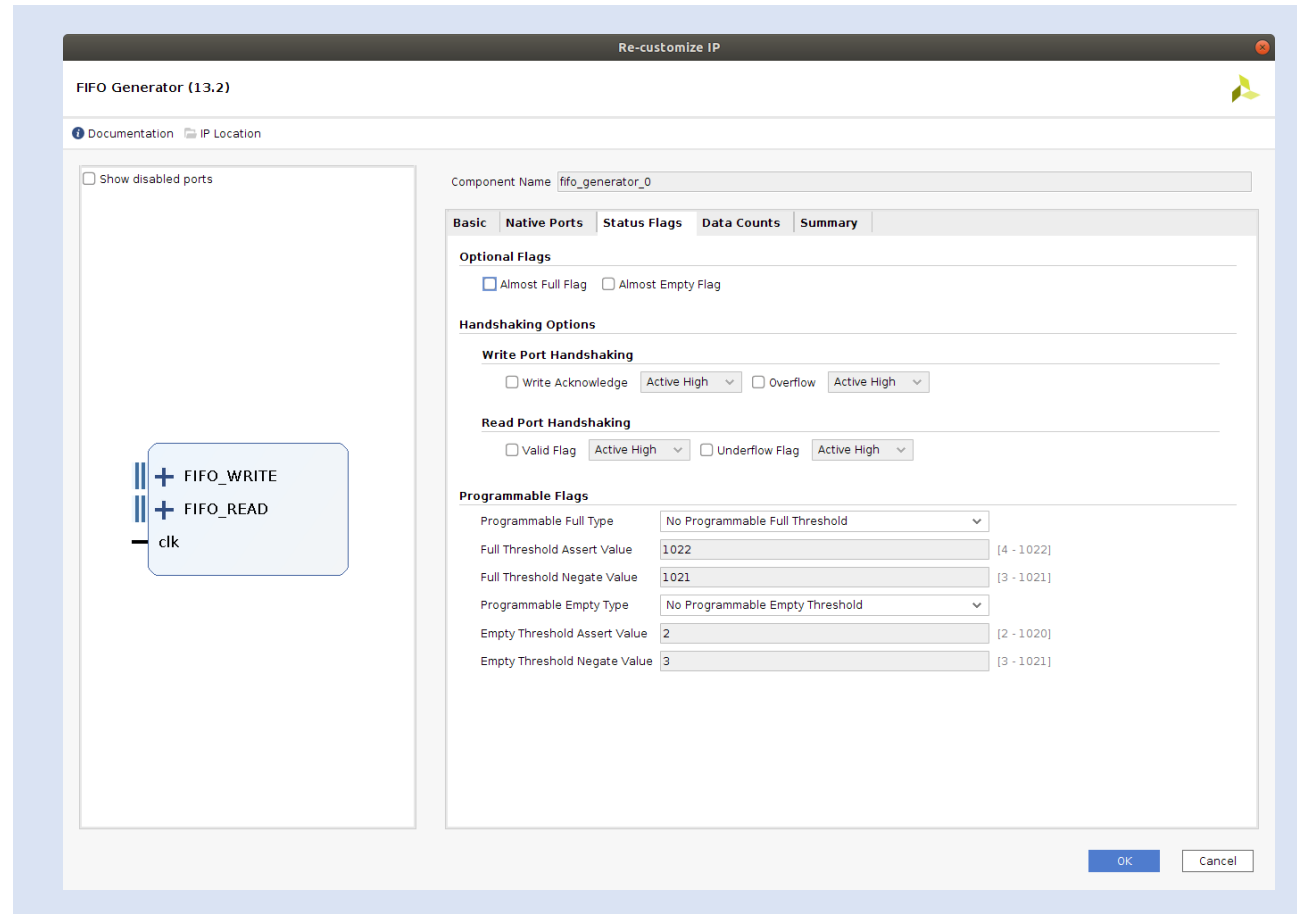
Lab 2: Intermediate Vivado

Step 9 – Change the Write and Read Width to be **8 bits**.



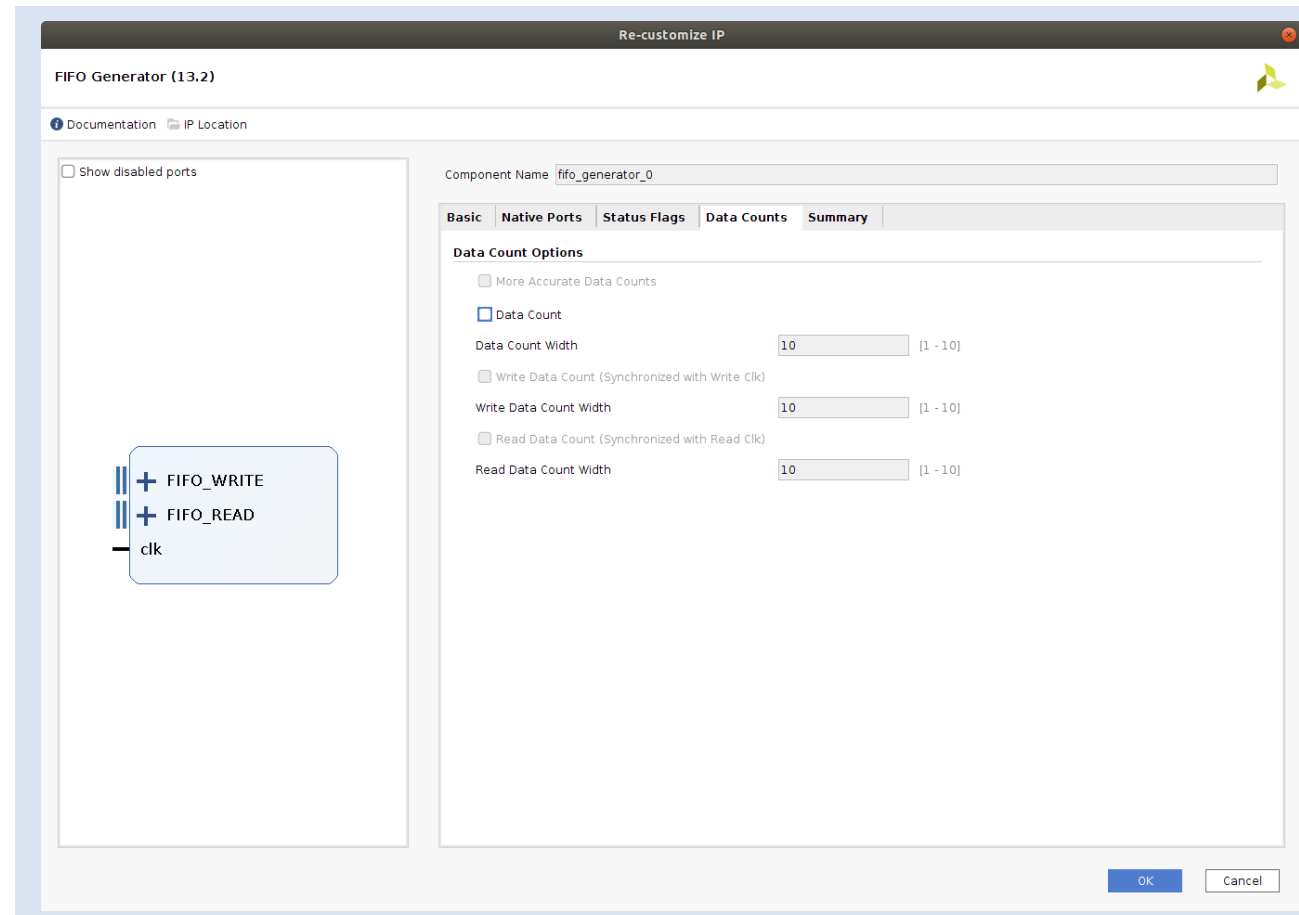
Lab 2: Intermediate Vivado

Step 10 – Leave the third page unchanged.



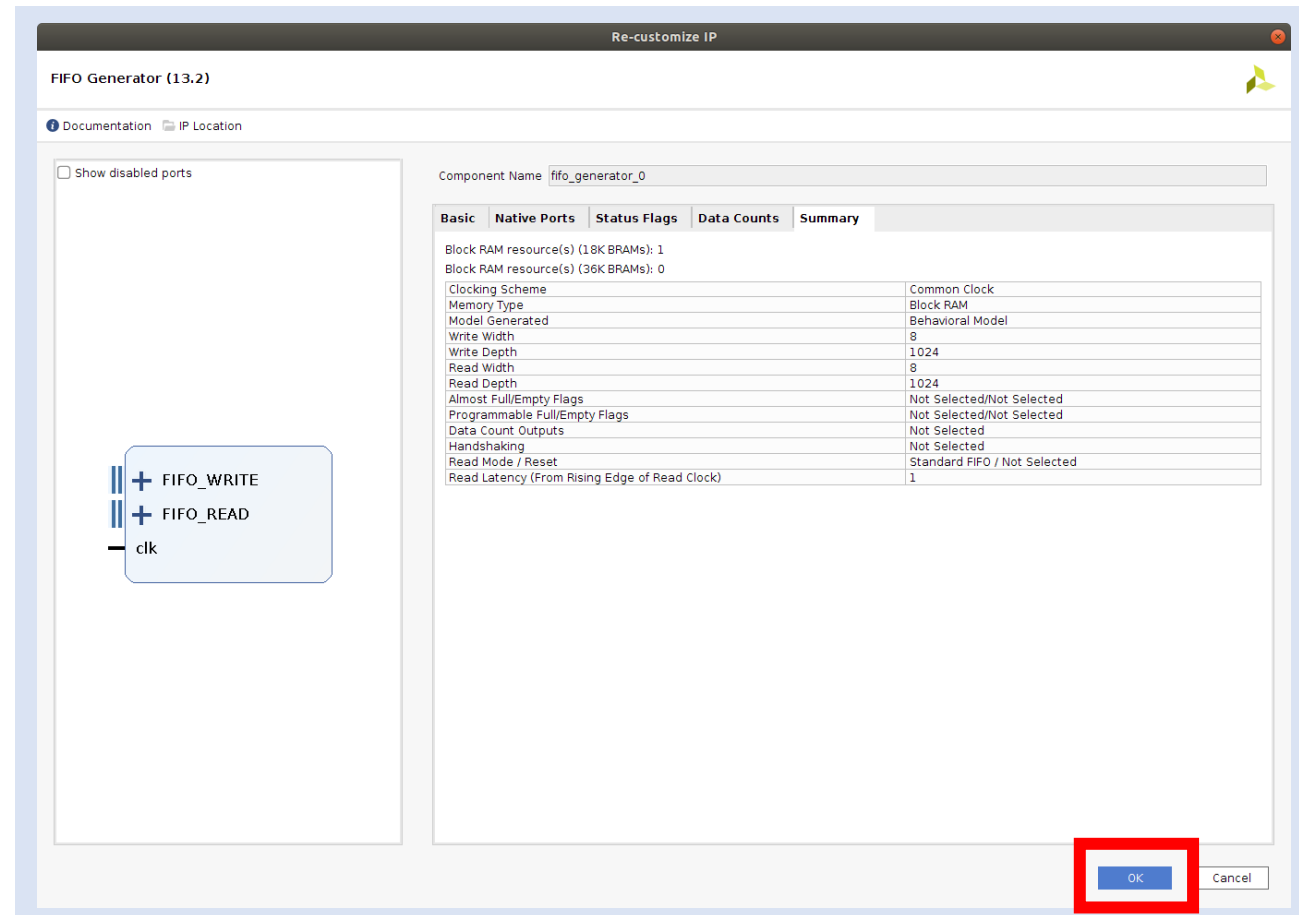
Lab 2: Intermediate Vivado

Step 11 – Leave the fourth page unchanged.



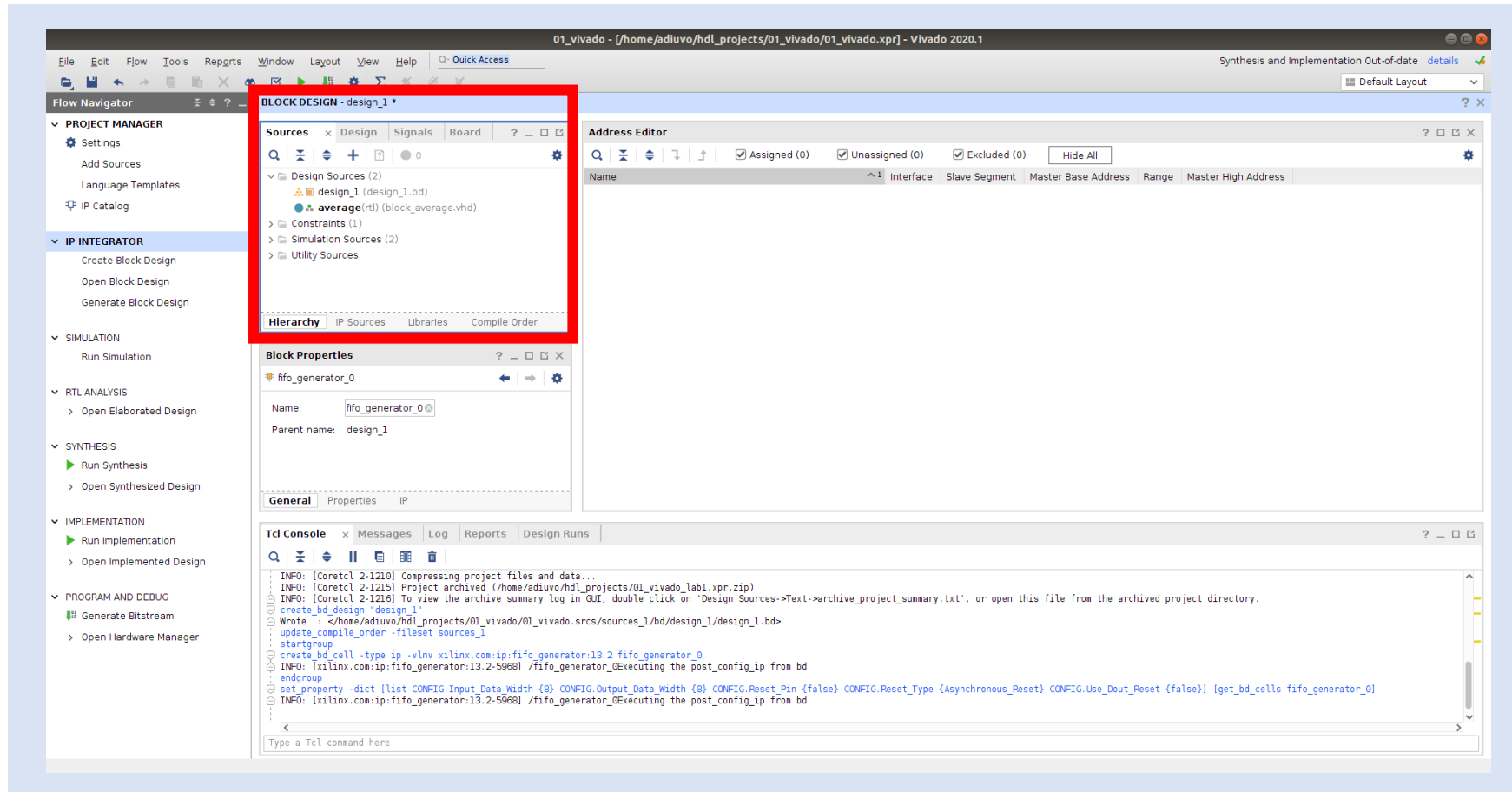
Lab 2: Intermediate Vivado

Step 12 – Leave the final page unchanged. Note that the FIFO OP are unregistered. Click **OK**.



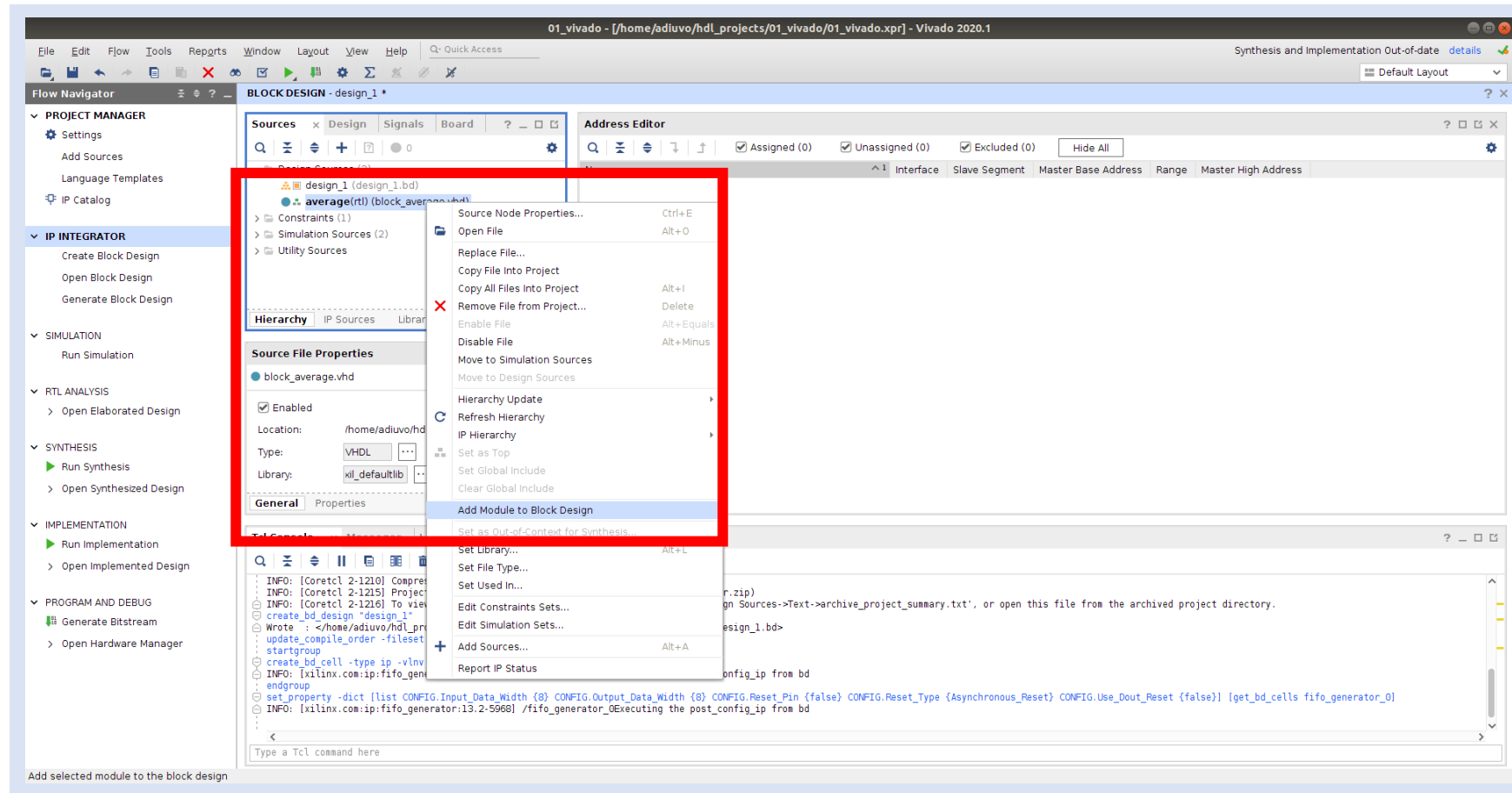
Lab 2: Intermediate Vivado

Step 13 – Click back on the Vivado Project Management view you will see the block diagram under the design sources.



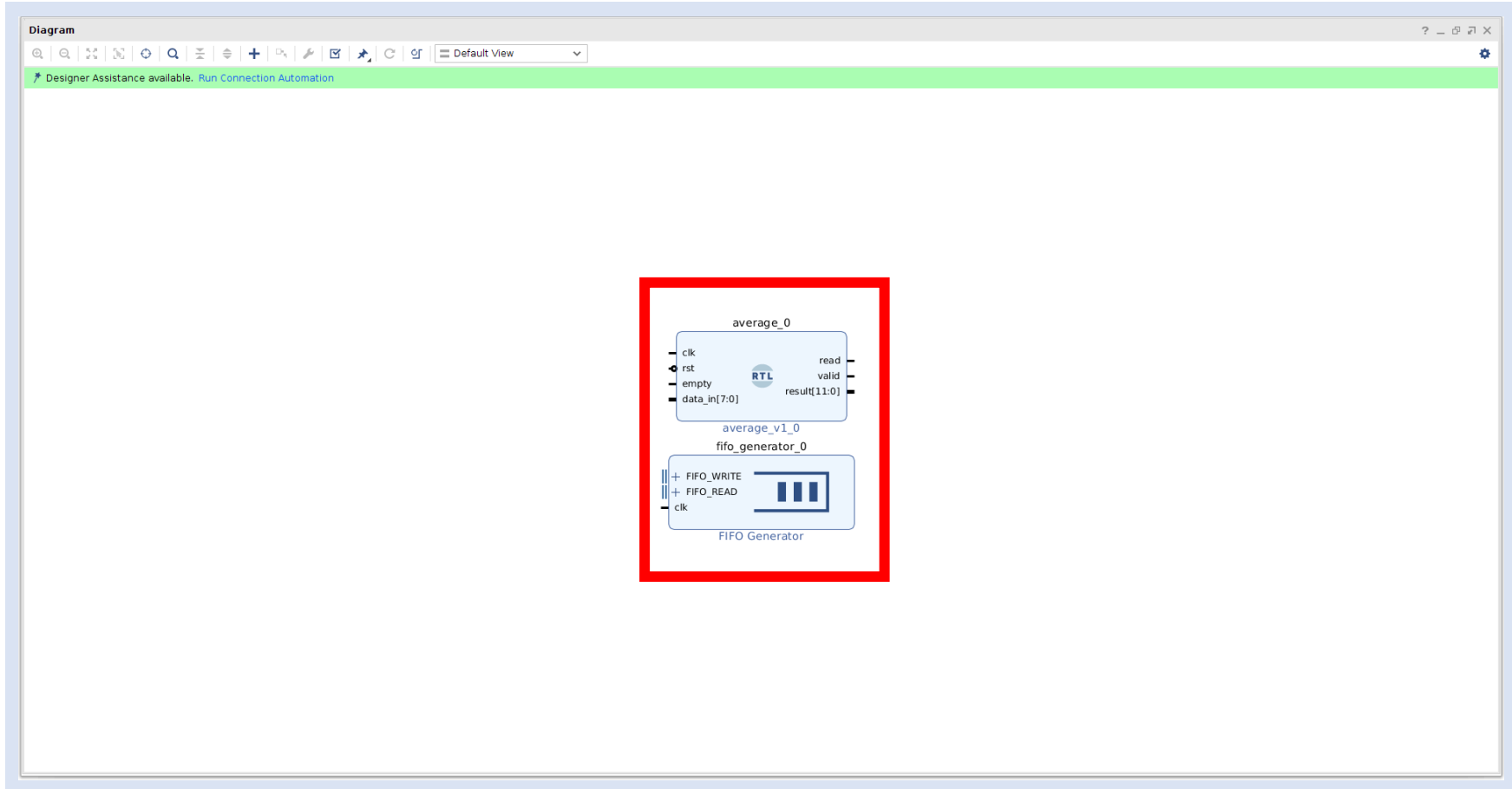
Lab 2: Intermediate Vivado

Step 14 – Right click on the average RTL block and select **Add Module to Block Design.**



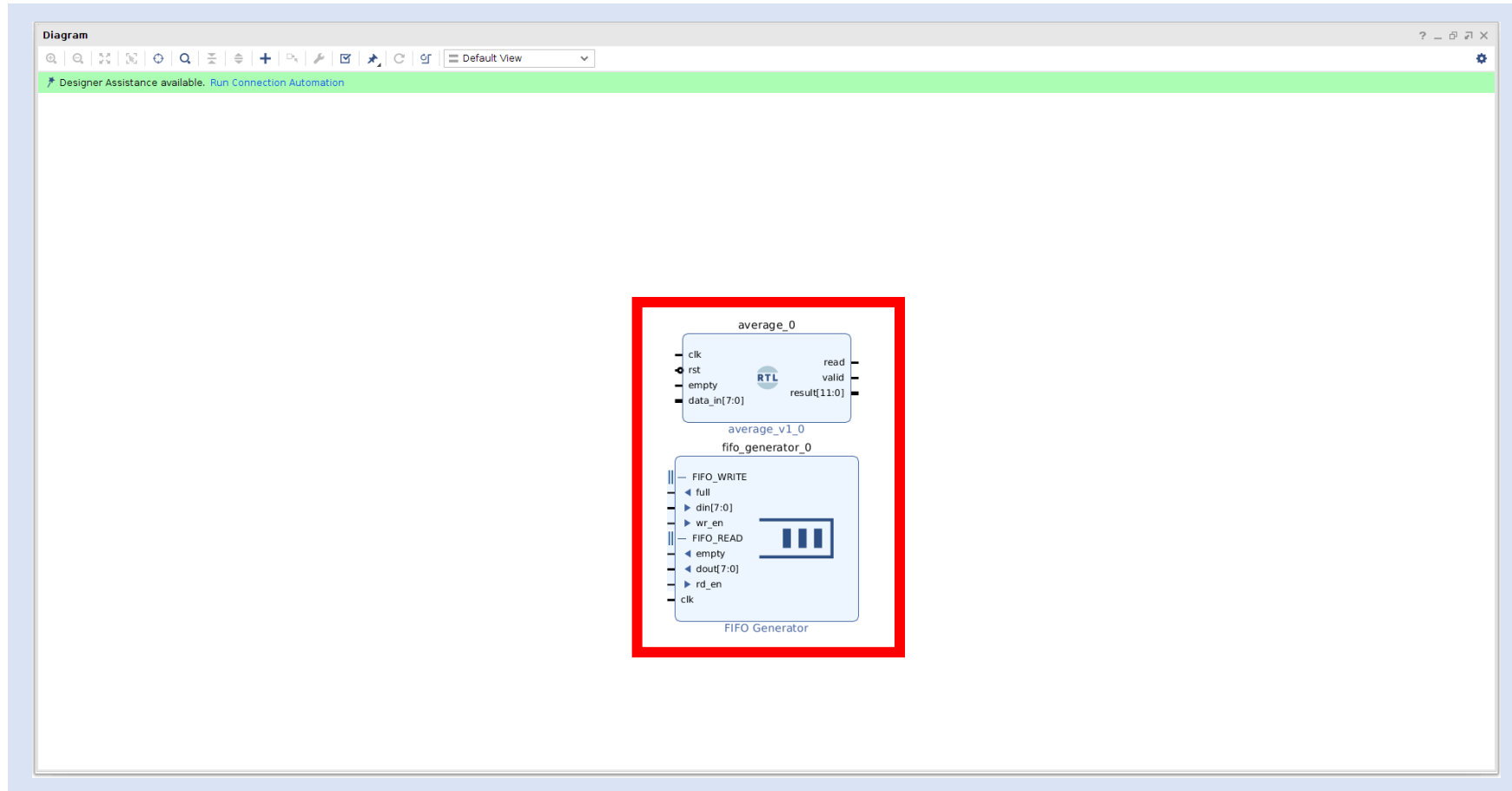
Lab 2: Intermediate Vivado

Step 15 – This will add the average block to the block diagram.



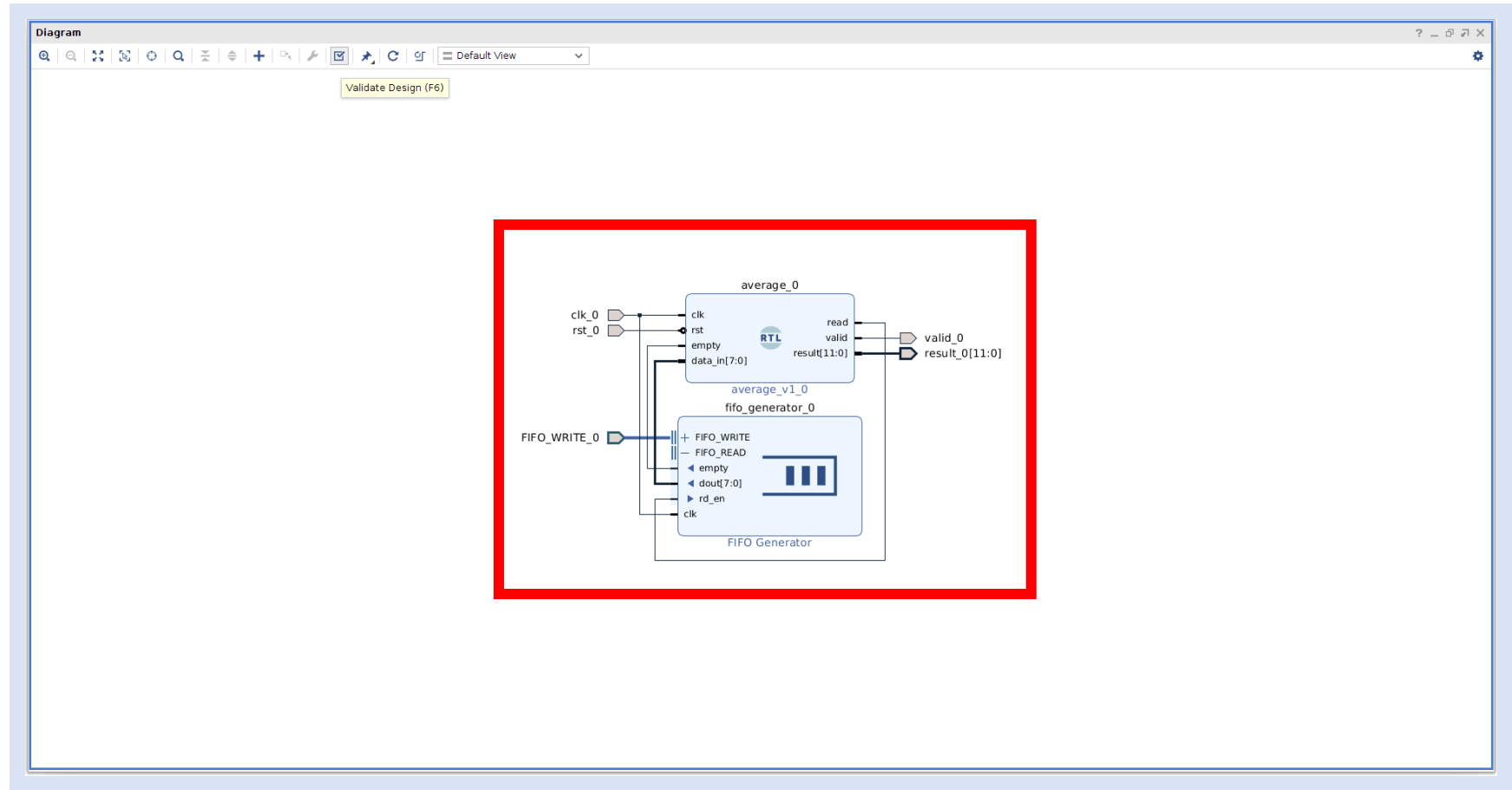
Lab 2: Intermediate Vivado

Step 16 – Expand the FIFO Write and Read Interfaces.



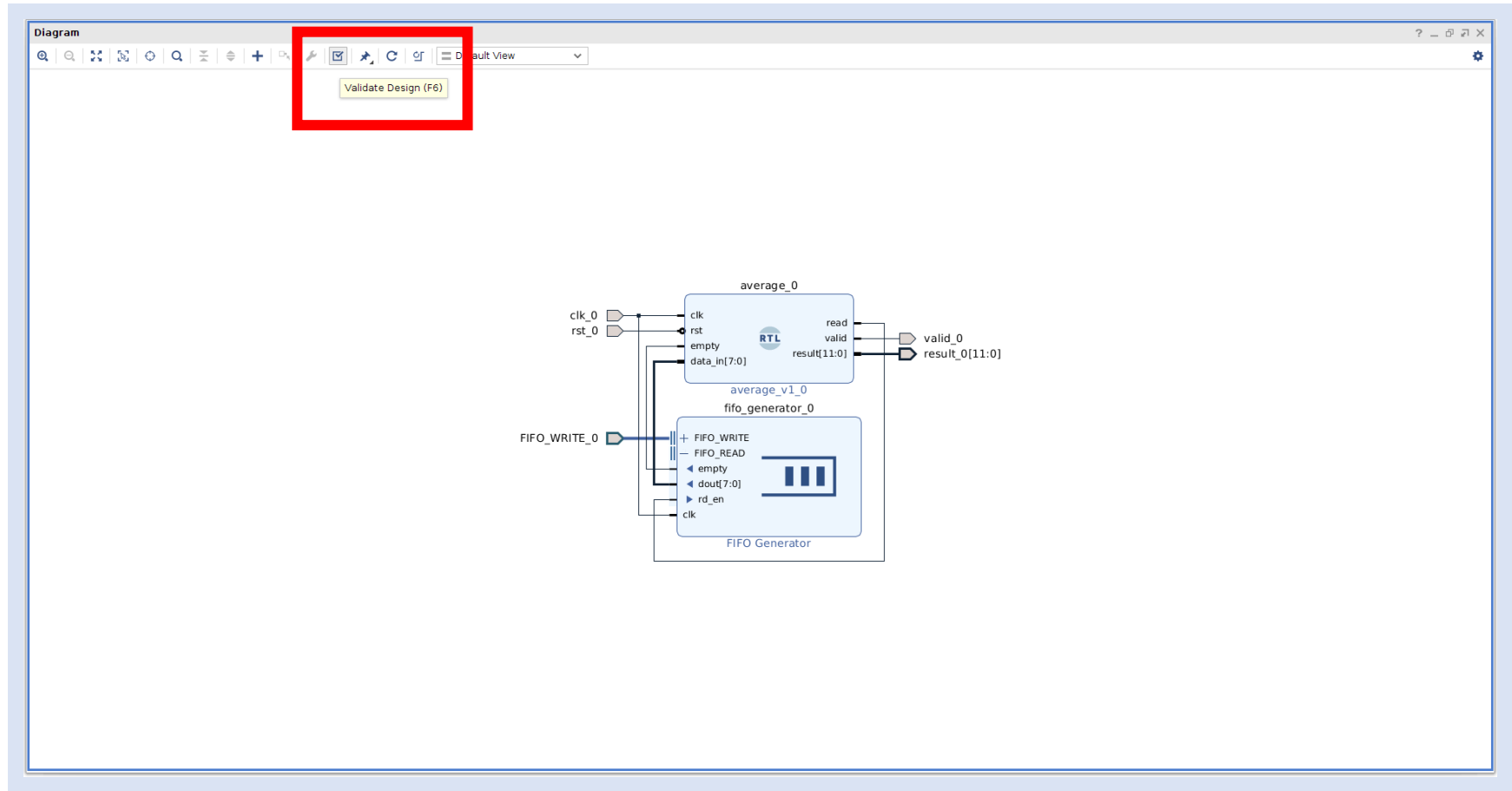
Lab 2: Intermediate Vivado

Step 17 – Make the Clk, Reset, Result, Valid and FIFO Write interfaces external by right-clicking on each pin and selecting **Make External**. Connect the remaining interfaces as below.



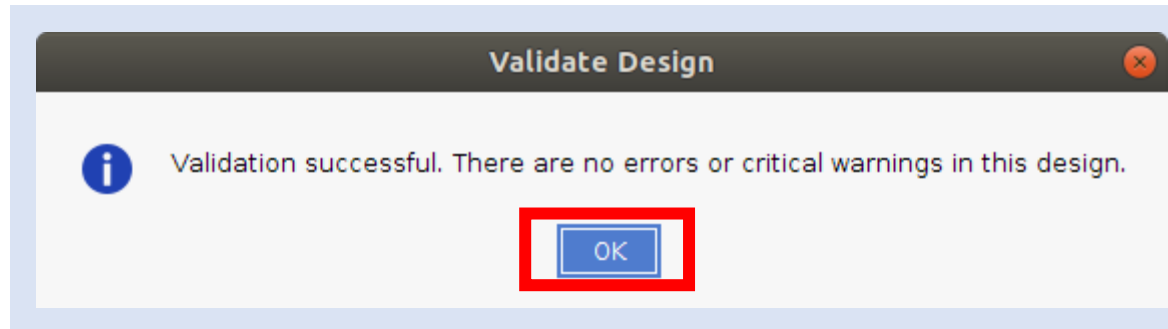
Lab 2: Intermediate Vivado

Step 18 – Click on **Validate Design.**



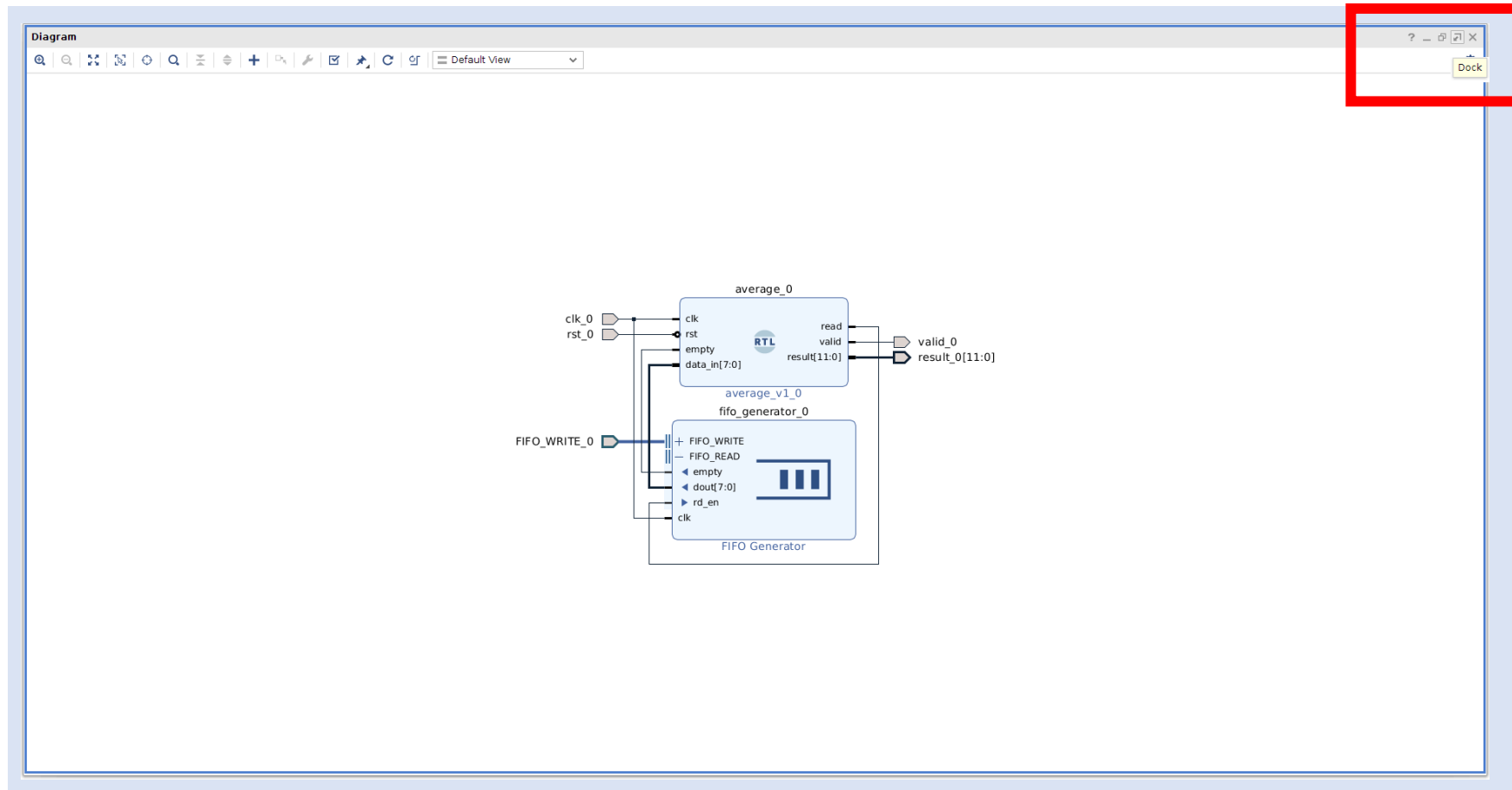
Lab 2: Intermediate Vivado

Step 19 – The validated design should result in no error or critical warnings. Click **OK**.



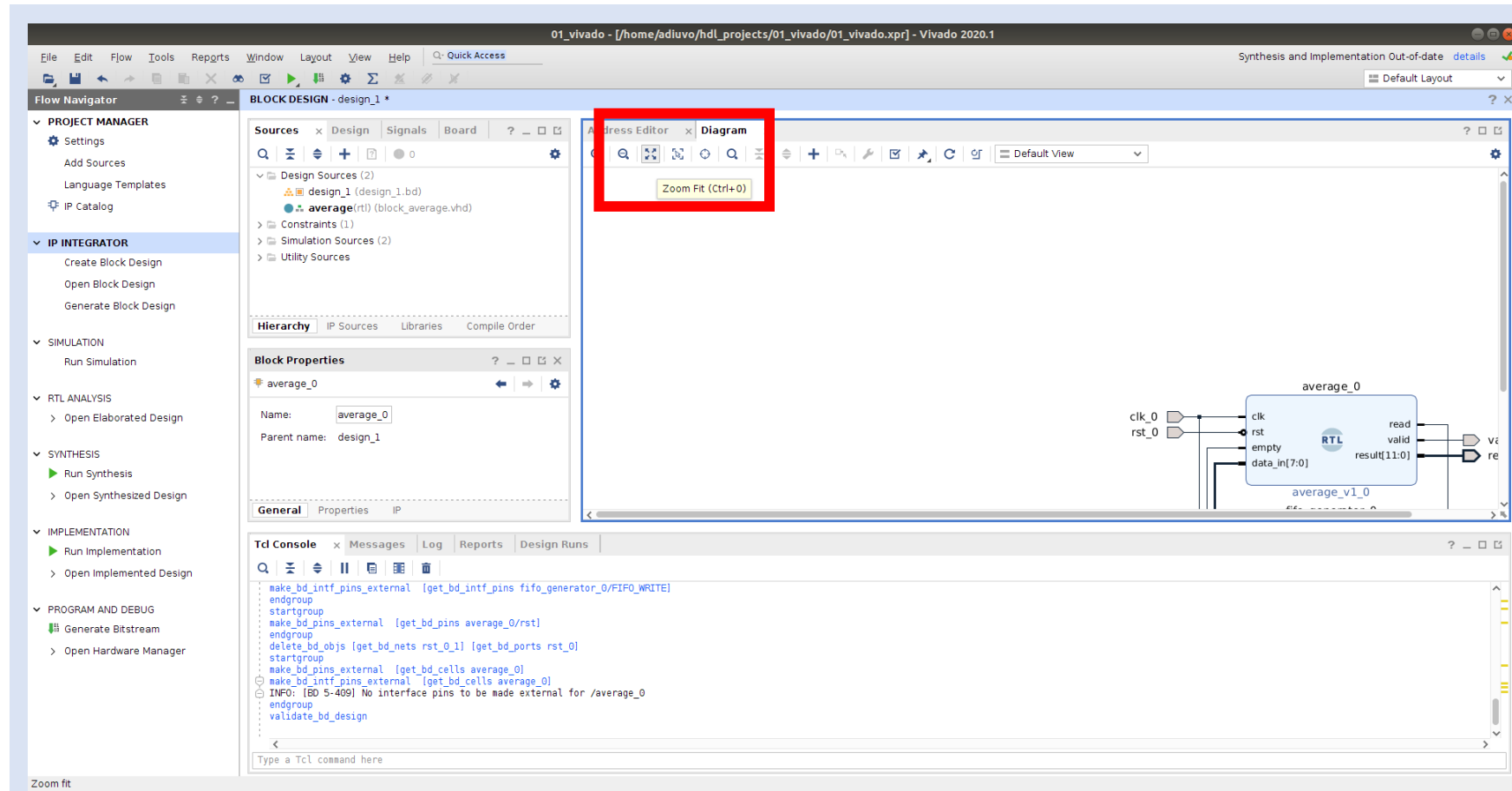
Lab 2: Intermediate Vivado

Step 20 – Re-dock the block diagram window into the Vivado Project Manager.



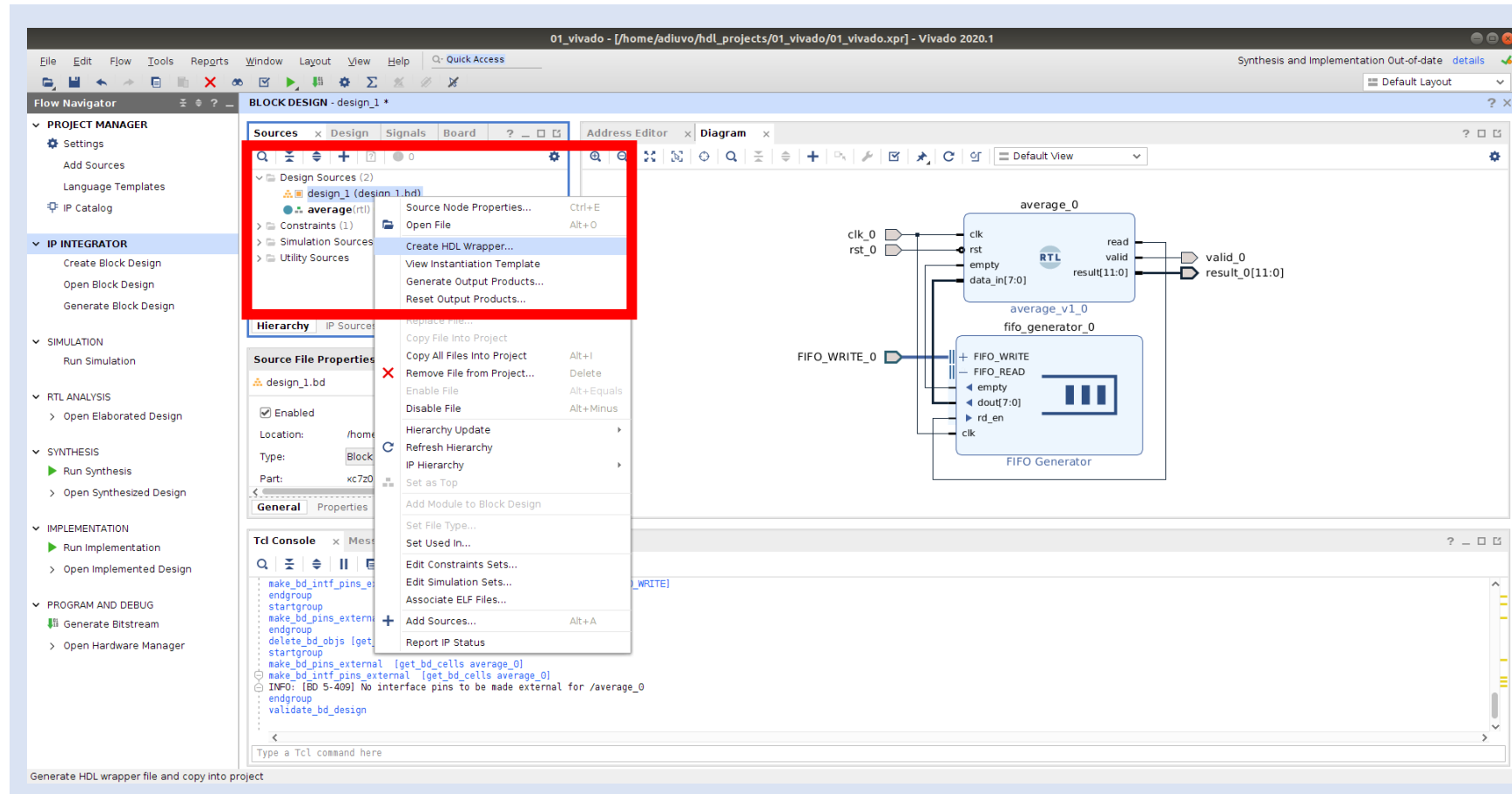
Lab 2: Intermediate Vivado

Step 21 – Click on the **Zoom Fit** button to fit the design to the window.



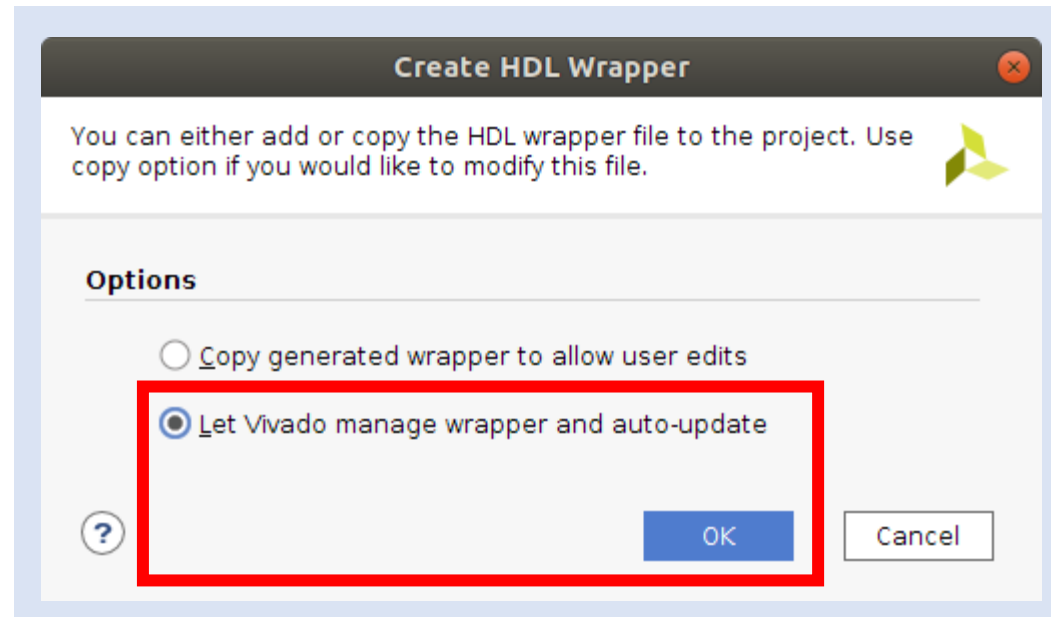
Lab 2: Intermediate Vivado

Step 22 – Right click on the block diagram design under the Design Sources tab and select **Create HDL Wrapper.**



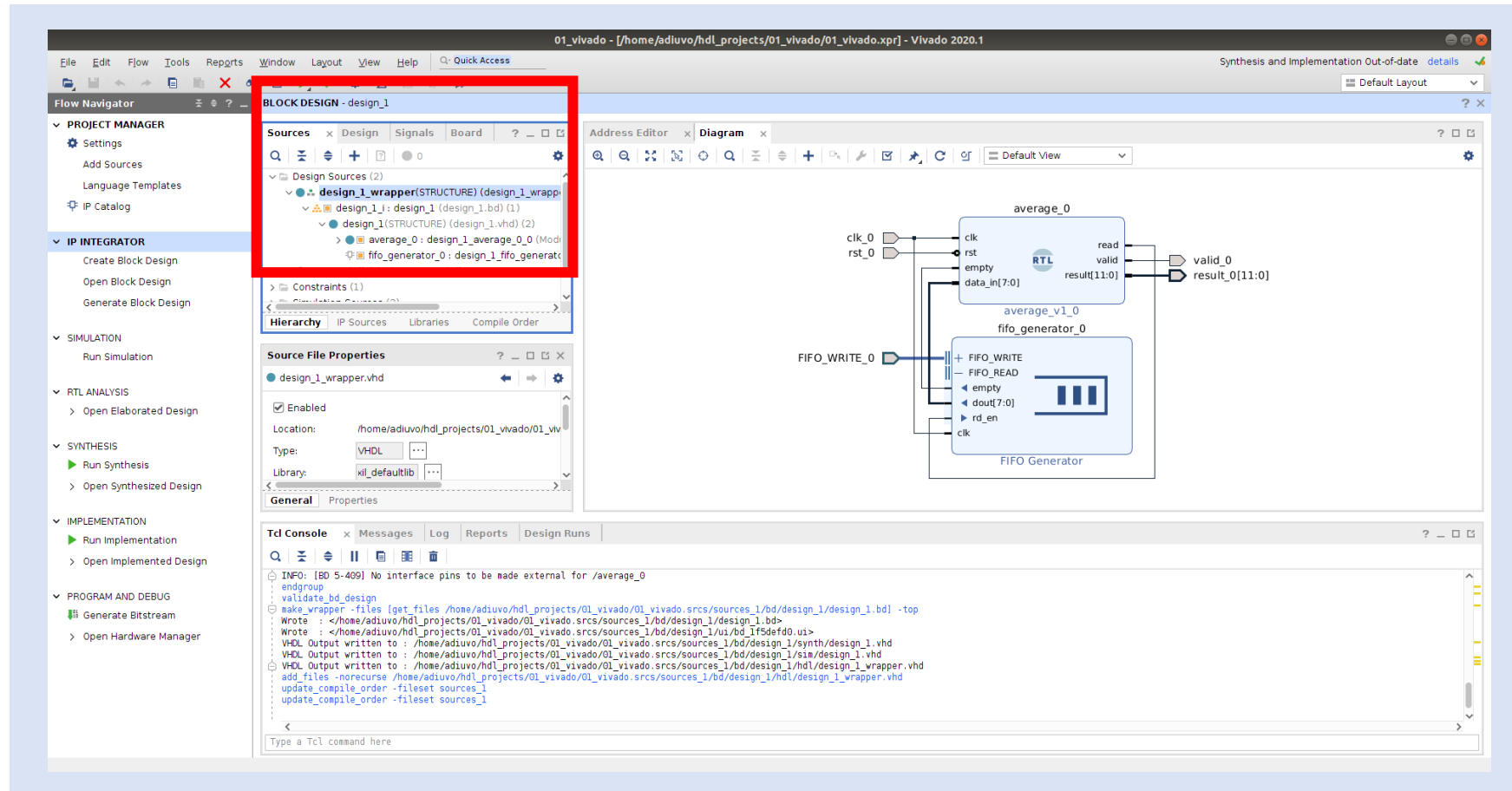
Lab 2: Intermediate Vivado

Step 23 – Allow Vivado to manage the wrapper and click **OK**.



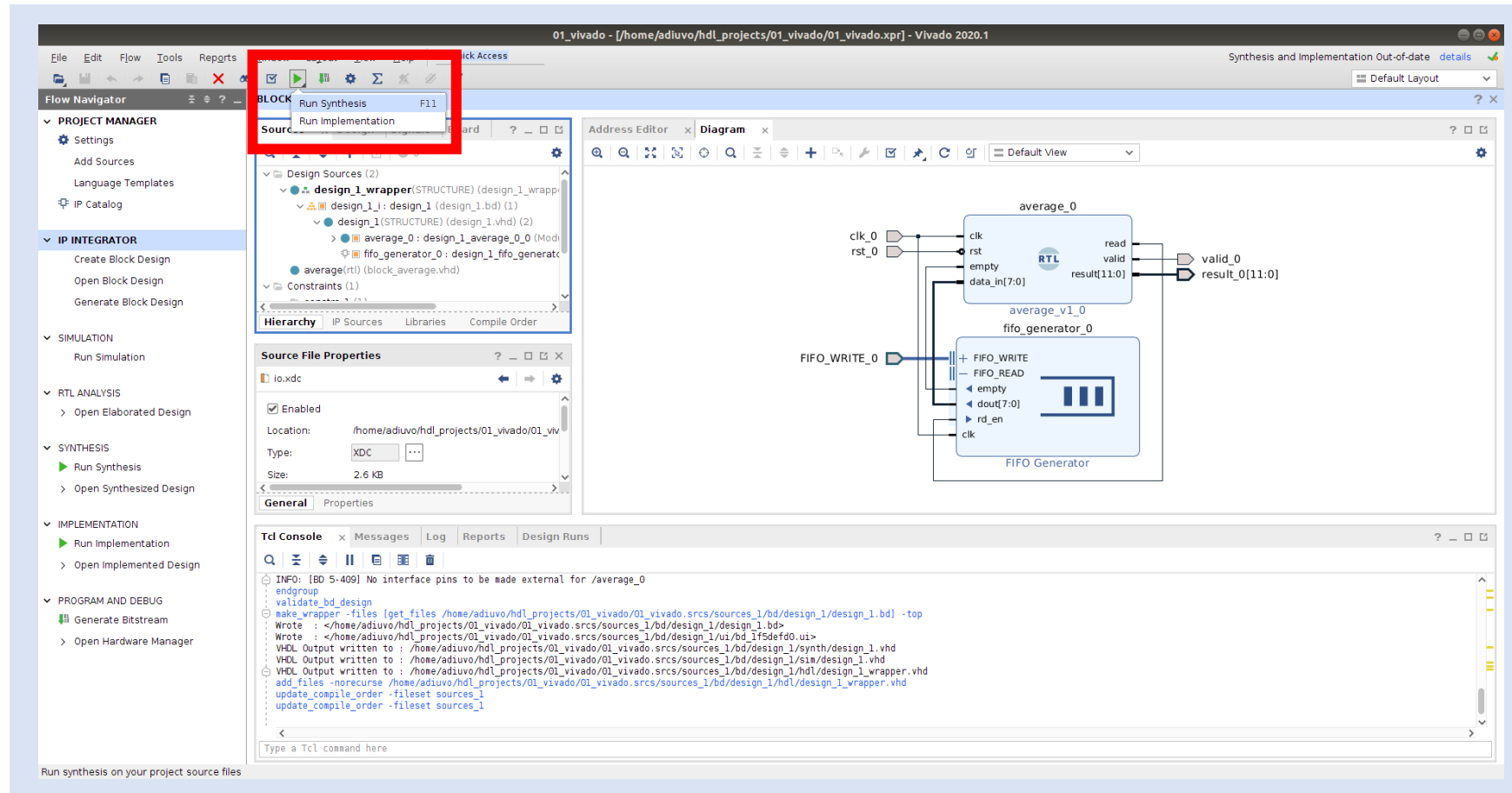
Lab 2: Intermediate Vivado

Step 24 – Expand the newly created wrapper and you will see the entire design.



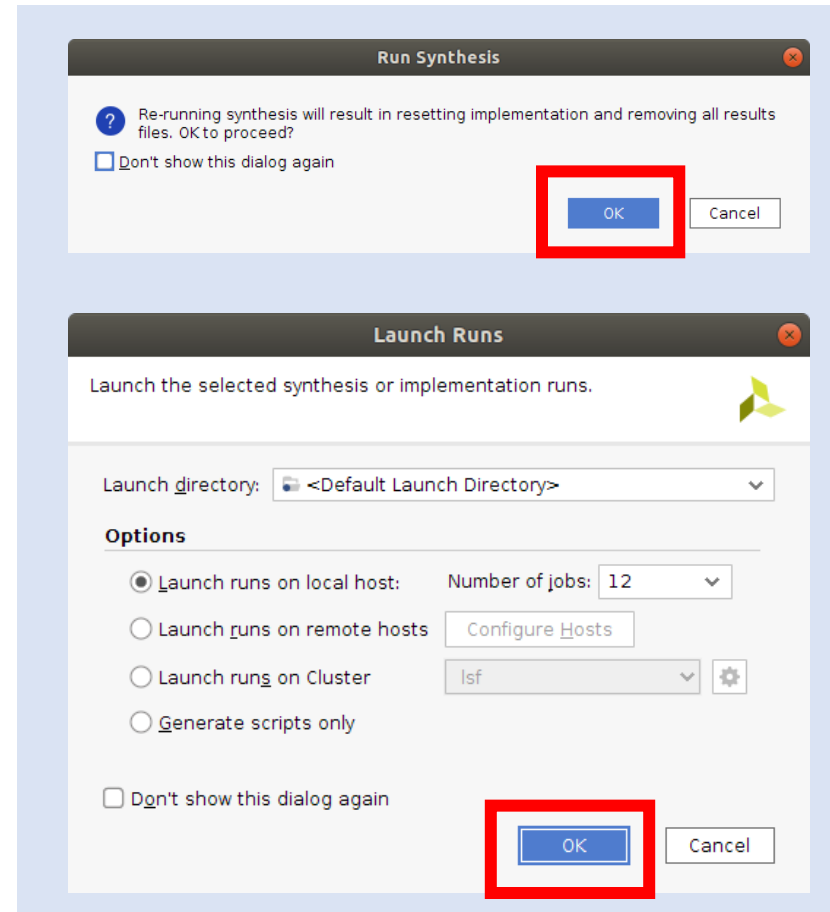
Lab 2: Intermediate Vivado

Step 25 – Run the Synthesis.



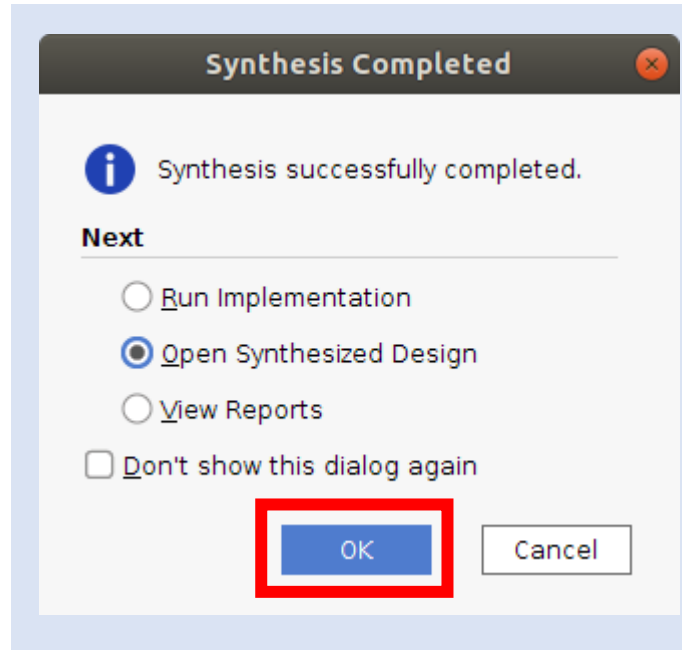
Lab 2: Intermediate Vivado

Step 26 – On both resultant dialogs click **OK** and wait for synthesis to complete.



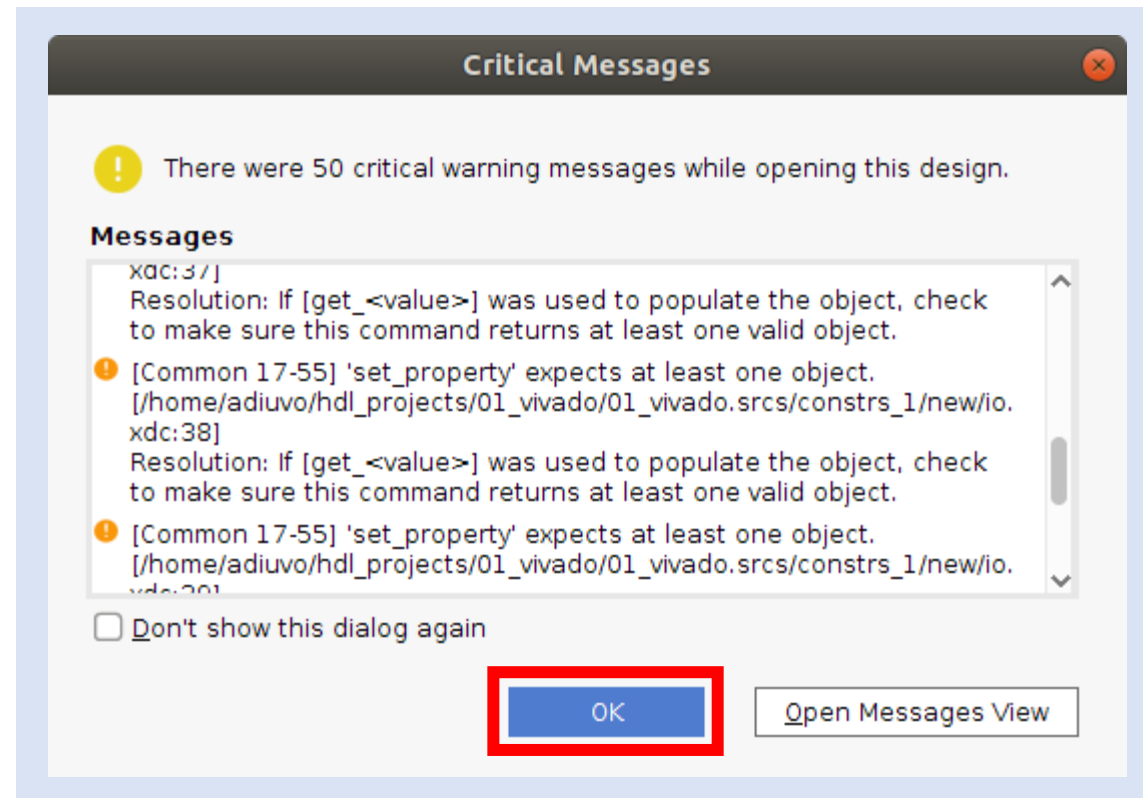
Lab 2: Intermediate Vivado

Step 27 – When synthesis completes, **Open the Synthesized Design.**



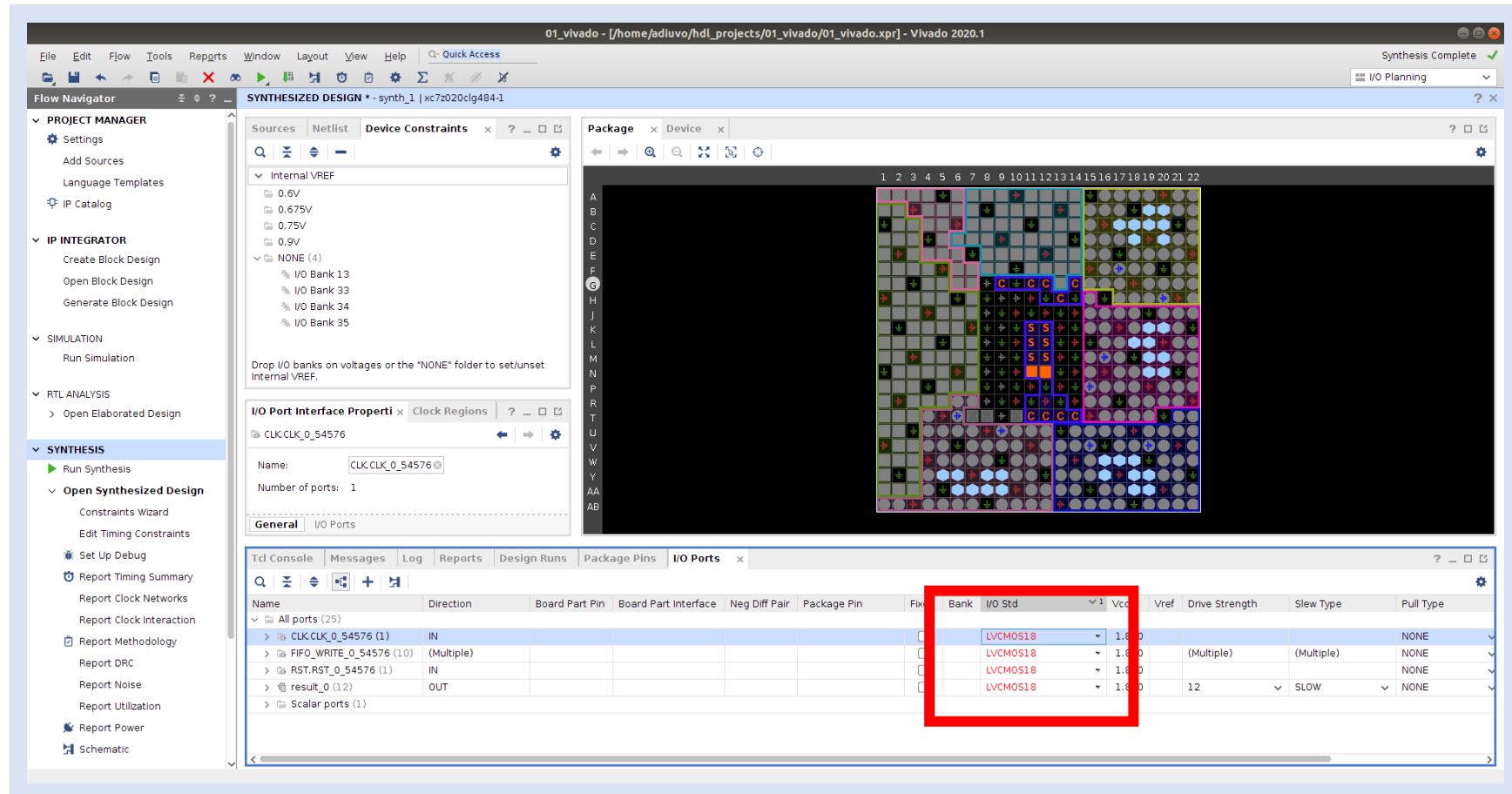
Lab 2: Intermediate Vivado

Step 28 – If any critical warnings pop up, select **OK**. This is due to out of data constraints which we are about to address.



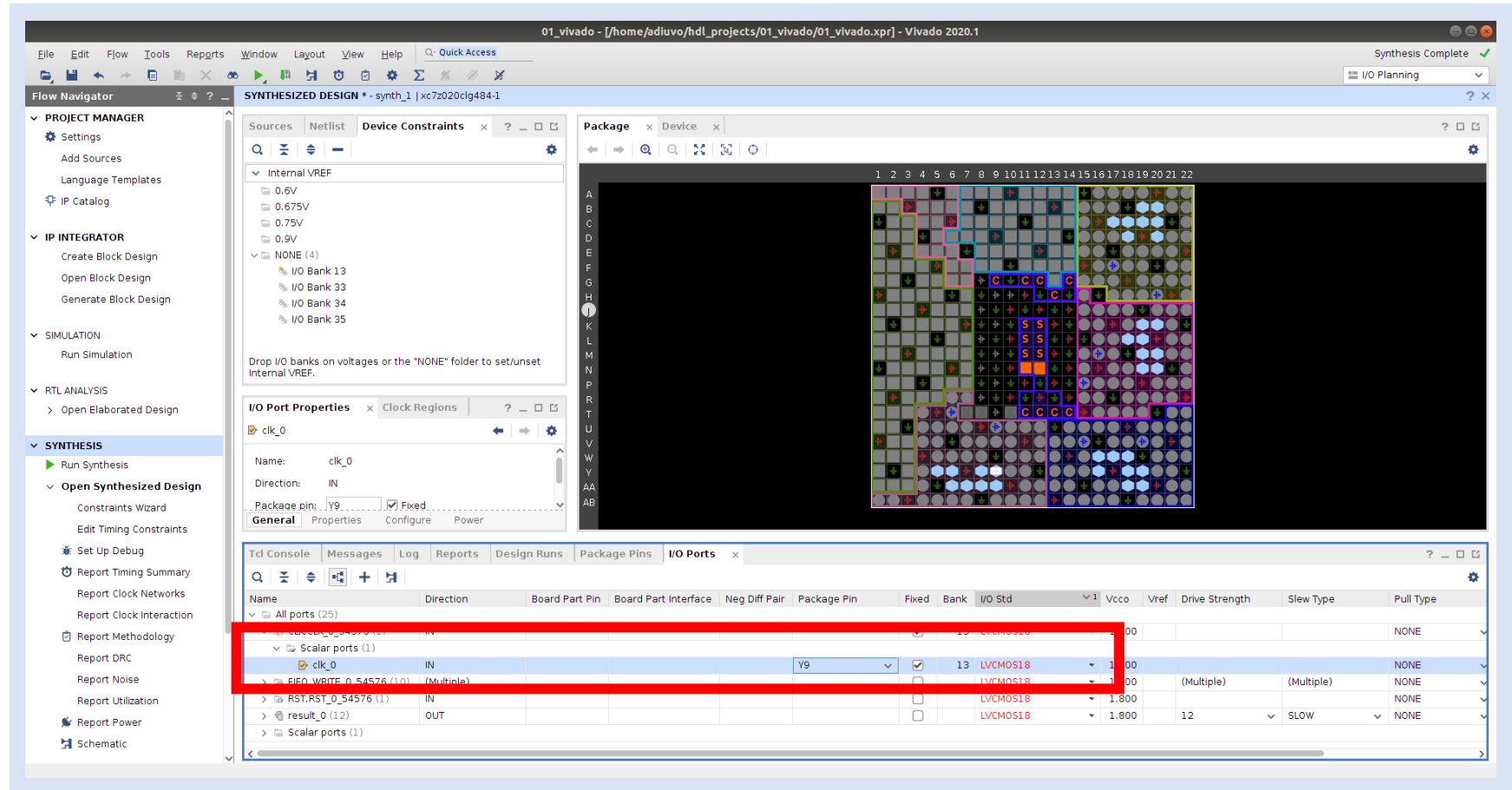
Lab 2: Intermediate Vivado

Step 29 – Change the I/O standard from default to **LVC MOS18**.



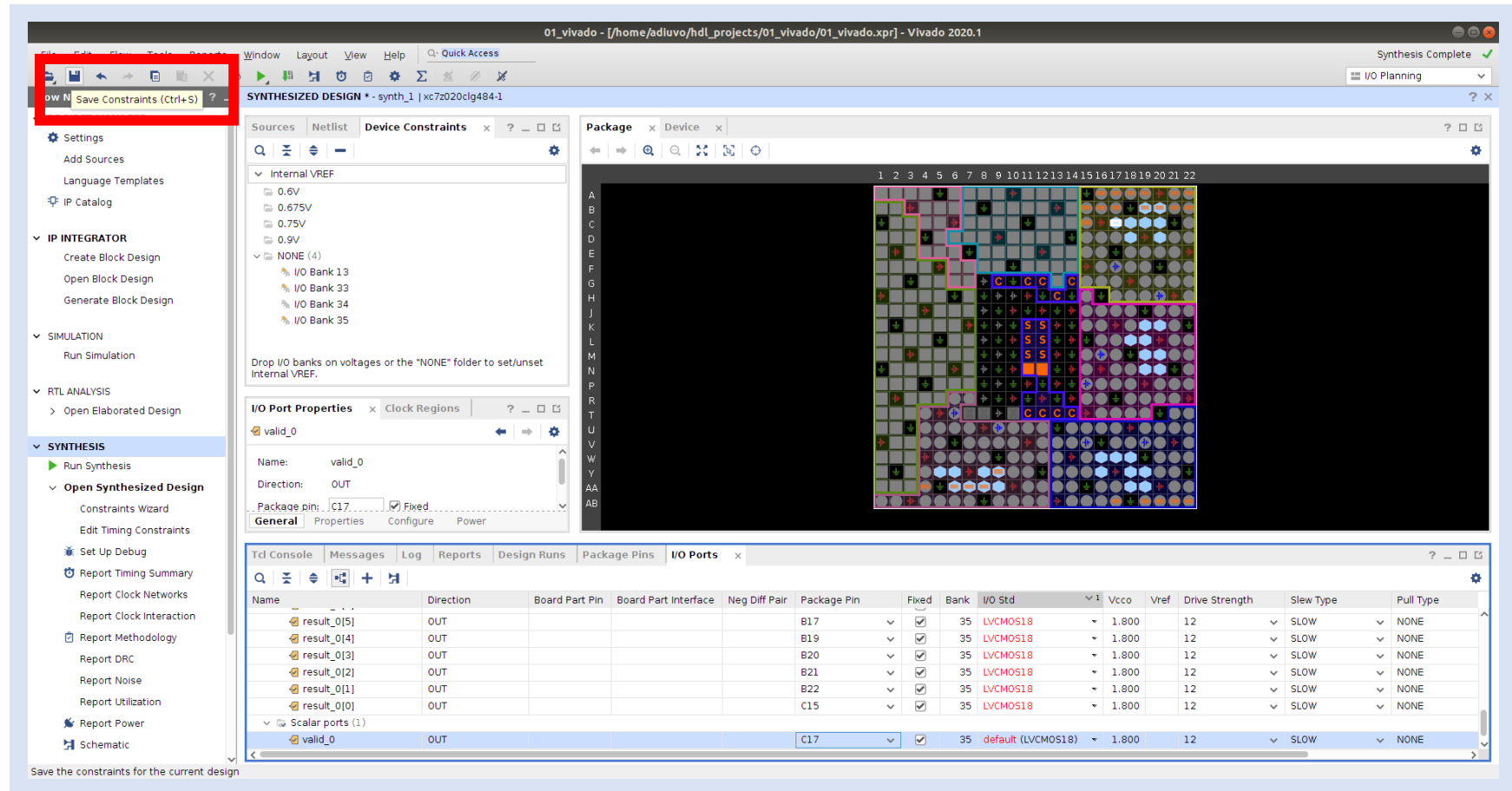
Lab 2: Intermediate Vivado

Step 30 – Assign the clock input to pin **Y9**. Assign all other IO to pins of your choice.



Lab 2: Intermediate Vivado

Step 31 – Click on Save Constraints.



Lab 2: Intermediate Vivado

Step 32 – If an out-of-date warning appears, click **OK**.



Lab 2: Intermediate Vivado

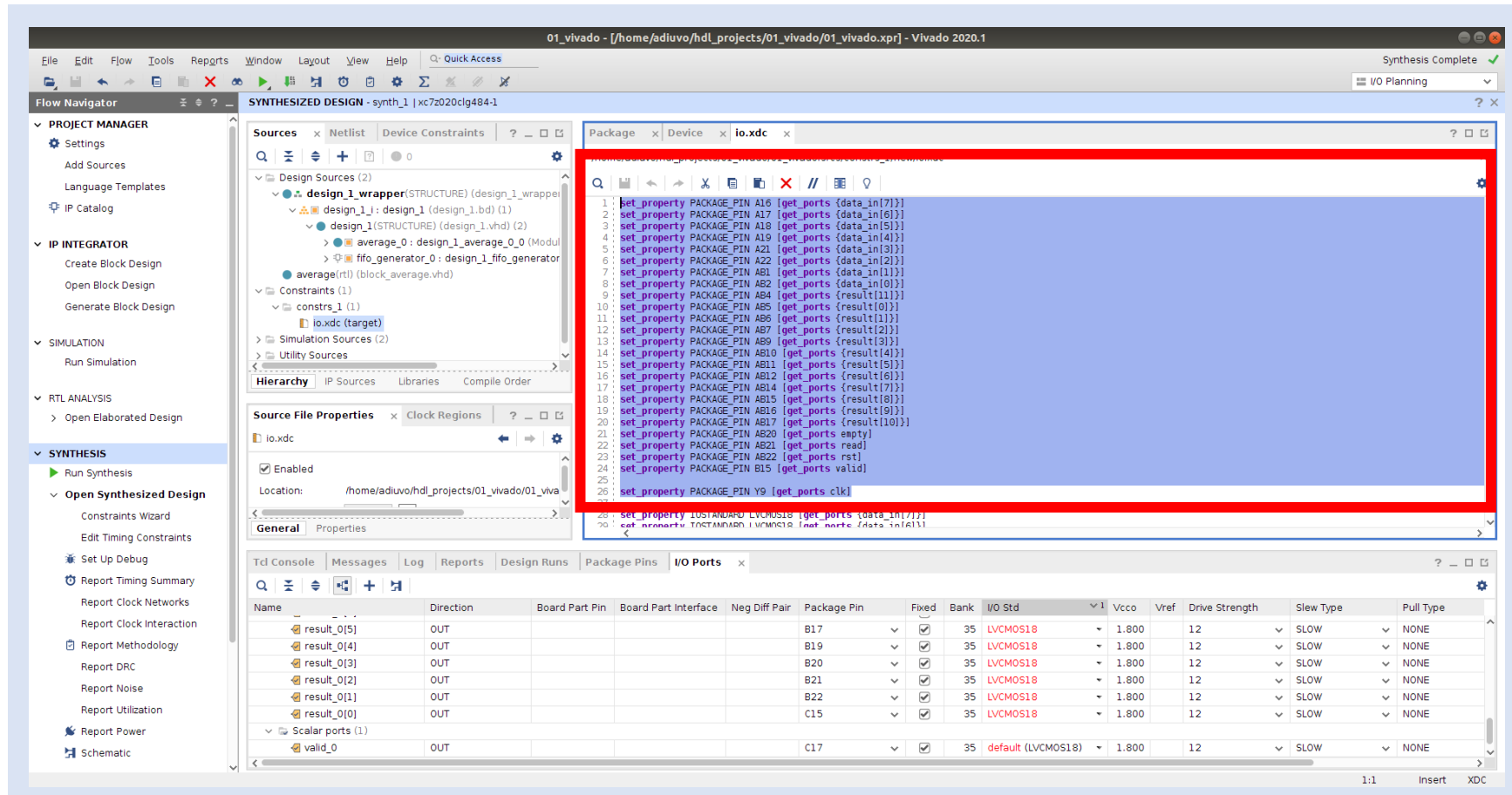
Step 33 – From the sources tab, open the **IO constraints**. You will see the old pin out for the previous project and your new project. This is because we have evolved the original project.

The screenshot shows the Vivado IDE interface. The left sidebar contains the Project Manager, IP Integrator, Simulation, RTL Analysis, and Synthesis sections. The Synthesis section is active, showing the 'Open Synthesized Design' option. The main window displays the 'io.xdc' file, which contains a list of package pins and their properties. The bottom panel shows the 'I/O Ports' table, which lists the pins and their properties.

Name	Direction	Board Part Pin	Board Part Interface	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type	Pull Type
result_0[5]	OUT				B17	✓	35	LVC MOS18	1.800	12	12	SLOW	NONE
result_0[4]	OUT				B19	✓	35	LVC MOS18	1.800	12	12	SLOW	NONE
result_0[3]	OUT				B20	✓	35	LVC MOS18	1.800	12	12	SLOW	NONE
result_0[2]	OUT				B21	✓	35	LVC MOS18	1.800	12	12	SLOW	NONE
result_0[1]	OUT				B22	✓	35	LVC MOS18	1.800	12	12	SLOW	NONE
result_0[0]	OUT				C15	✓	35	LVC MOS18	1.800	12	12	SLOW	NONE
valid_0	OUT				C17	✓	35	default (LVC MOS18)	1.800	12	12	SLOW	NONE

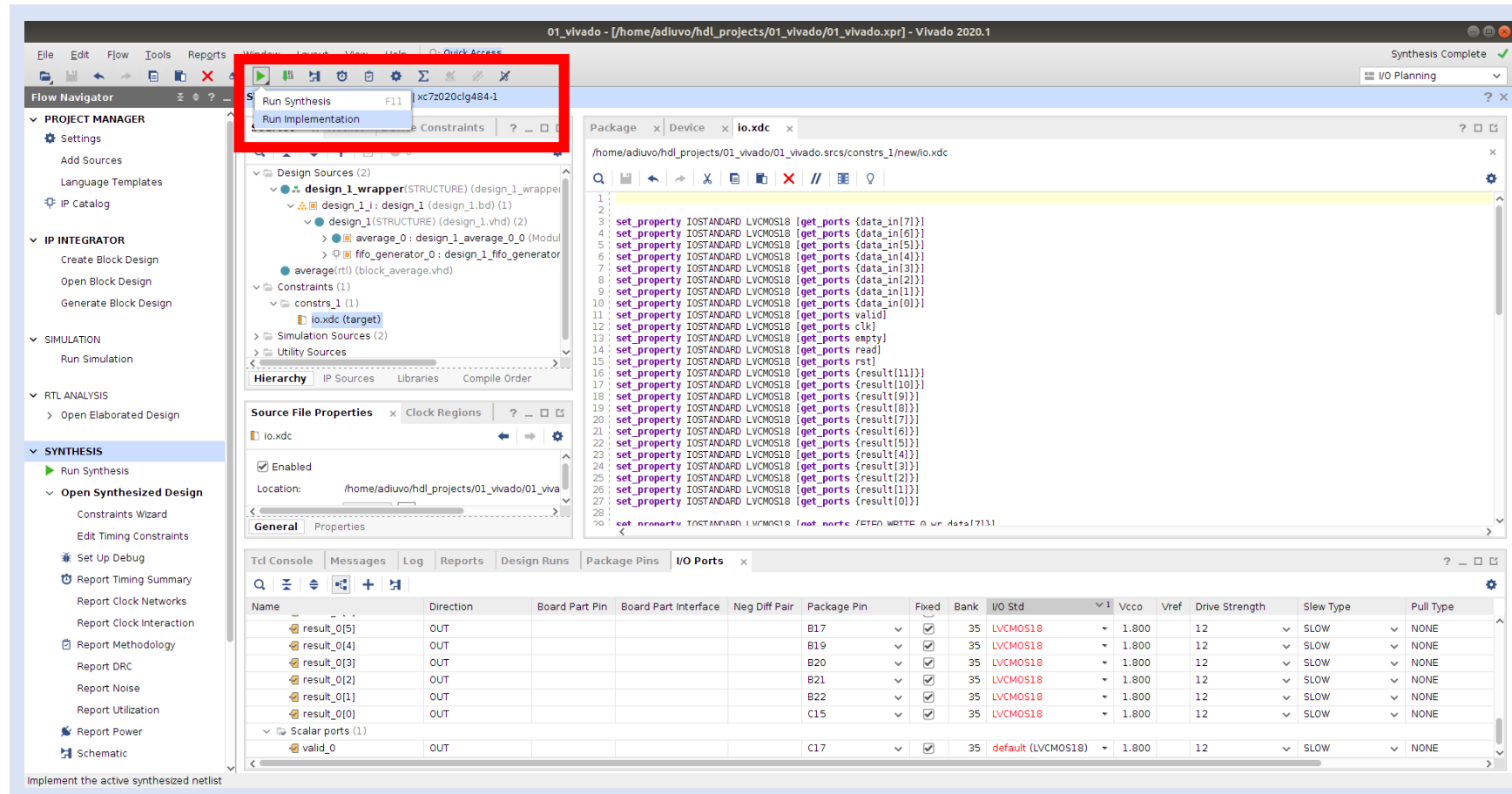
Lab 2: Intermediate Vivado

Step 34 – Select the old constraints (at the top of the file) and delete them. Save the file.



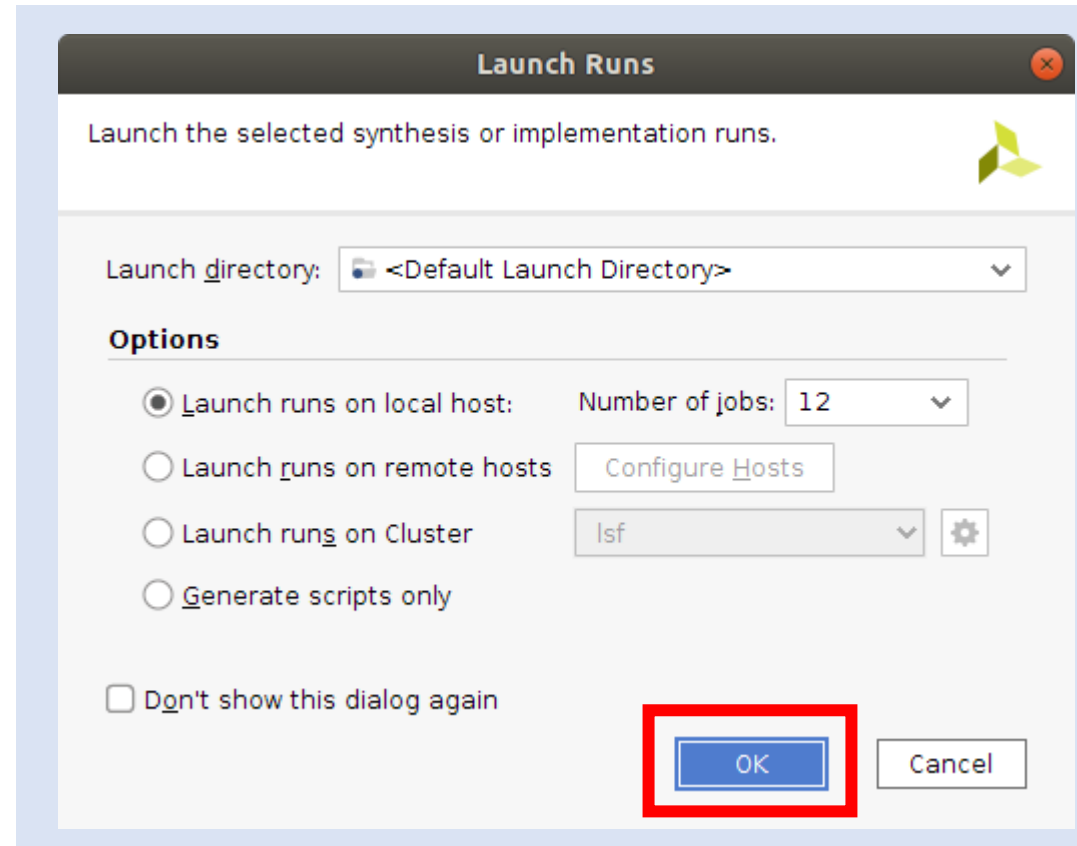
Lab 2: Intermediate Vivado

Step 35 – Run the Implementation.



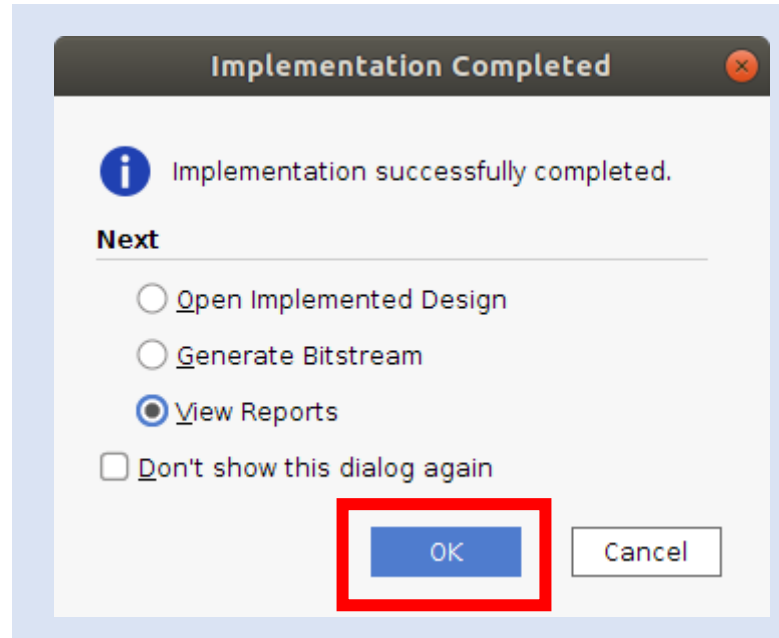
Lab 2: Intermediate Vivado

Step 36 – Click **OK** to run the implementation.



Lab 2: Intermediate Vivado

Step 37 – When the implementation completes, click **View Reports**.



Lab 2: Intermediate Vivado

Step 38 – Select Constraints under Design Sources.

The screenshot displays the Vivado 2020.1 IDE interface. The main window is titled '01_vivado - [/home/adiuvo/hdl_projects/01_vivado/01_vivado.xpr] - Vivado 2020.1'. The 'BLOCK DESIGN - design_1' window is open, showing the 'Sources' tab. The 'constrs_1 (1)' constraint set is highlighted with a red box. The 'Constraint Set Properties' window is open, showing the 'General' tab with the target constraint file 'io.xdc'. The 'Diagram' tab shows a block diagram of the design, including an 'average_0' block, an 'average_v1_0' block, and a 'fifo_generator_0' block. The 'Reports' window at the bottom shows a list of reports, with 'impl_1_route_report_timing_summary_0' selected.

Report	Type	Options	Modified	Size
impl_1_route_report_drc_0	report_drc		8/11/20, 11:10 AM	4.5 KB
impl_1_route_report_methodology_0	report_methodology		8/11/20, 11:10 AM	18.2 KB
impl_1_route_report_power_0	report_power		8/11/20, 11:10 AM	9.1 KB
impl_1_route_report_route_status_0	report_route_status		8/11/20, 11:10 AM	0.6 KB
impl_1_route_report_timing_summary_0	report_timing_summary	max_paths = 10;	8/11/20, 11:10 AM	7.4 KB
impl_1_route_report_incremental_reuse_0	report_incremental_reuse		8/11/20, 11:10 AM	10.5 KB
impl_1_route_report_clock_utilization_0	report_clock_utilization		8/11/20, 11:10 AM	0.9 KB
impl_1_route_report_bus_skew_0	report_bus_skew	warn_on_violation = true;	8/11/20, 11:10 AM	31.4 KB

Lab 2: Intermediate Vivado

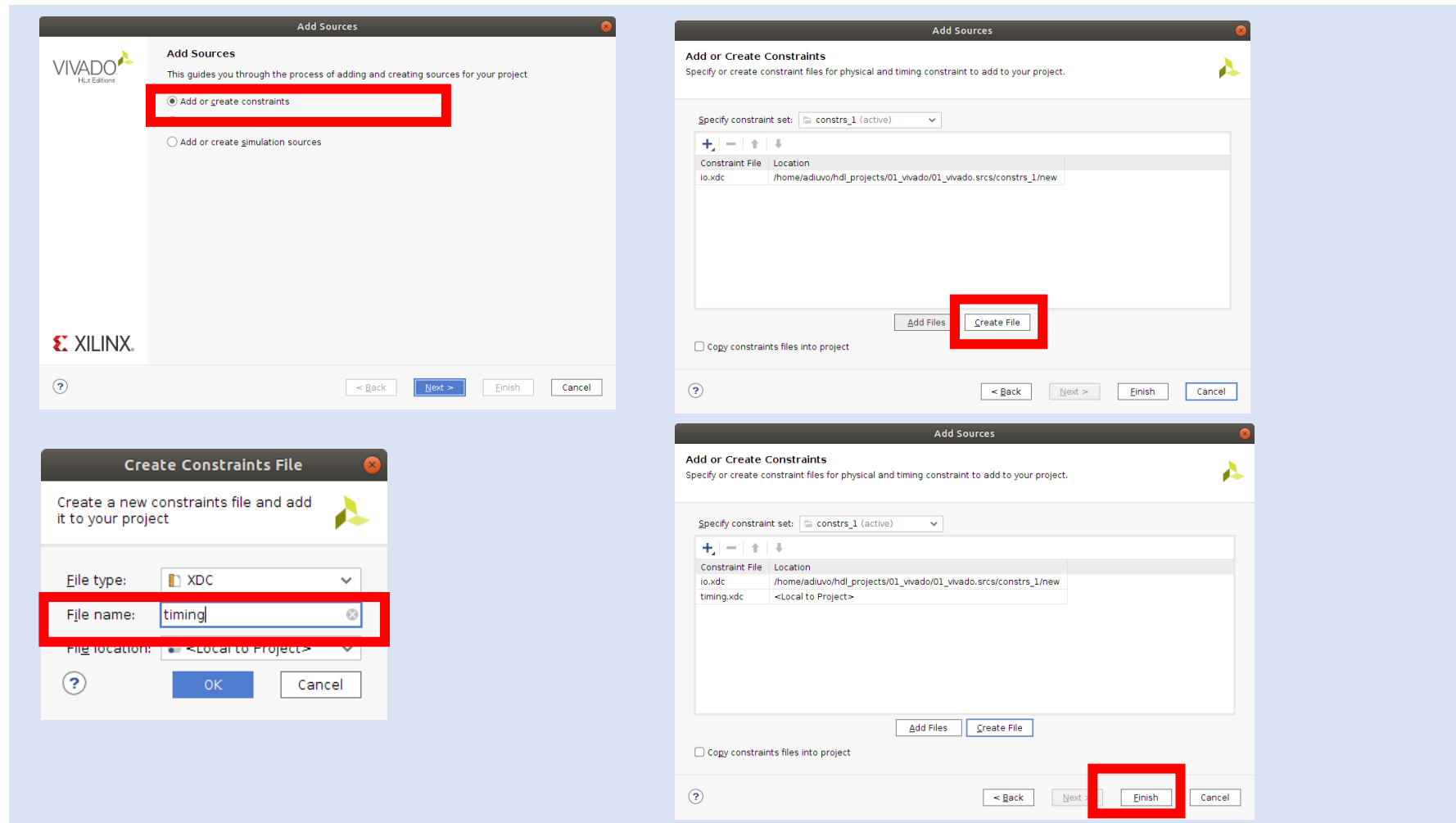
Step 39 – Click on the **Add Source** button.

The screenshot displays the Vivado IDE interface. The 'Sources' window is open, showing the project hierarchy. The 'Add Sources (Alt+A)' button is highlighted with a red box. The 'Diagram' window shows a block diagram with an 'average_0' block and a 'fifo_generator_0' block. The 'Reports' window at the bottom shows a list of reports, including 'Route Design (route_design)' and 'impl_1_route_report_timing_summary_0'.

Report	Type	Options	Modified	Size
Route Design (route_design)				
impl_1_route_report_drc_0	report_drc		8/11/20, 11:10 AM	4.5 KB
impl_1_route_report_methodology_0	report_methodology		8/11/20, 11:10 AM	18.2 KB
impl_1_route_report_power_0	report_power		8/11/20, 11:10 AM	9.1 KB
impl_1_route_report_route_status_0	report_route_status		8/11/20, 11:10 AM	0.6 KB
impl_1_route_report_timing_summary_0	report_timing_summary	max_paths = 10;	8/11/20, 11:10 AM	7.4 KB
impl_1_route_report_incremental_reuse_0	report_incremental_reuse		8/11/20, 11:10 AM	10.5 KB
impl_1_route_report_clock_utilization_0	report_clock_utilization		8/11/20, 11:10 AM	0.9 KB
impl_1_route_report_bus_skew_0	report_bus_skew	warn_on_violation = true;	8/11/20, 11:10 AM	31.4 KB
impl_1_route_implementation_log_0				

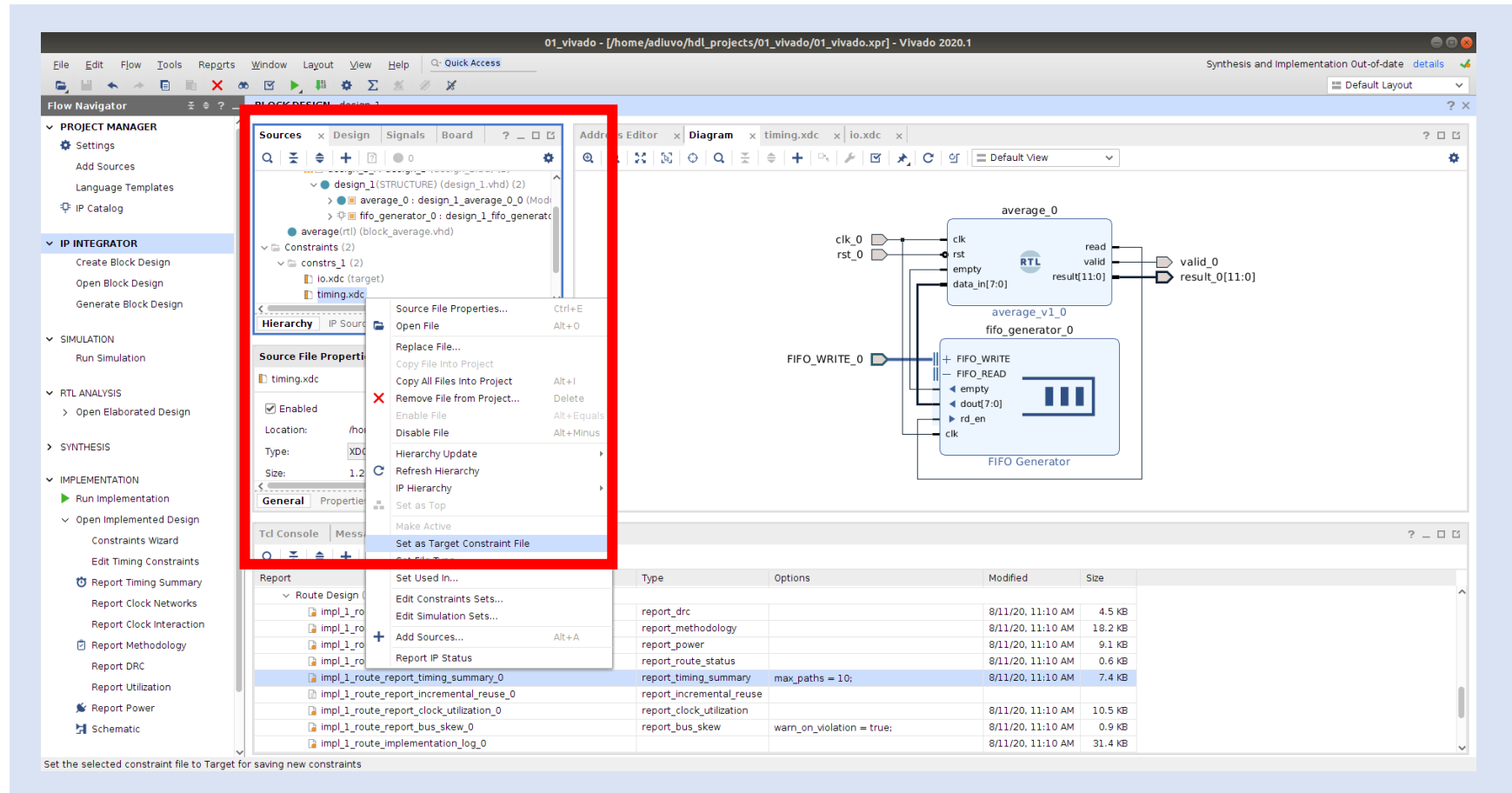
Lab 2: Intermediate Vivado

Step 40 – Select Add or Create Constraints. Then **Create File**, enter the name “**timing**” for the file, and click **Finish**.



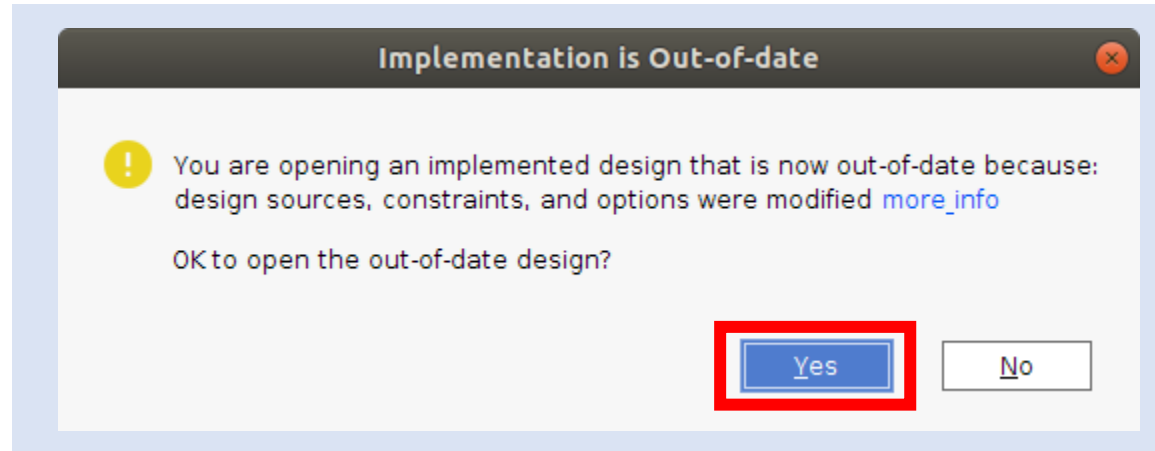
Lab 2: Intermediate Vivado

Step 41 – Right click on the newly created constraint file and select Set as Target Constraints File.



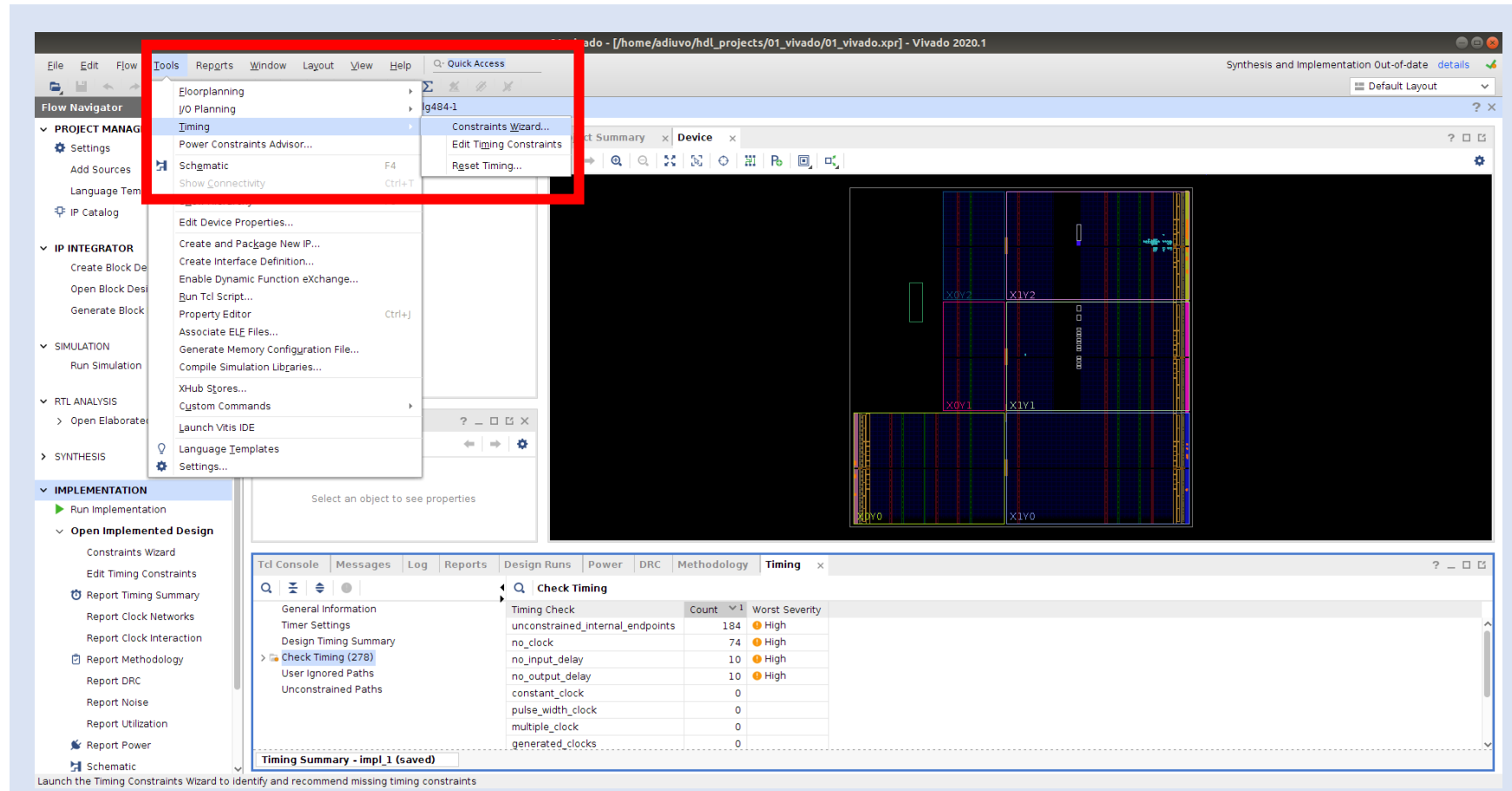
Lab 2: Intermediate Vivado

Step 42 – Open the implemented design. If you see the warning below click **OK**.



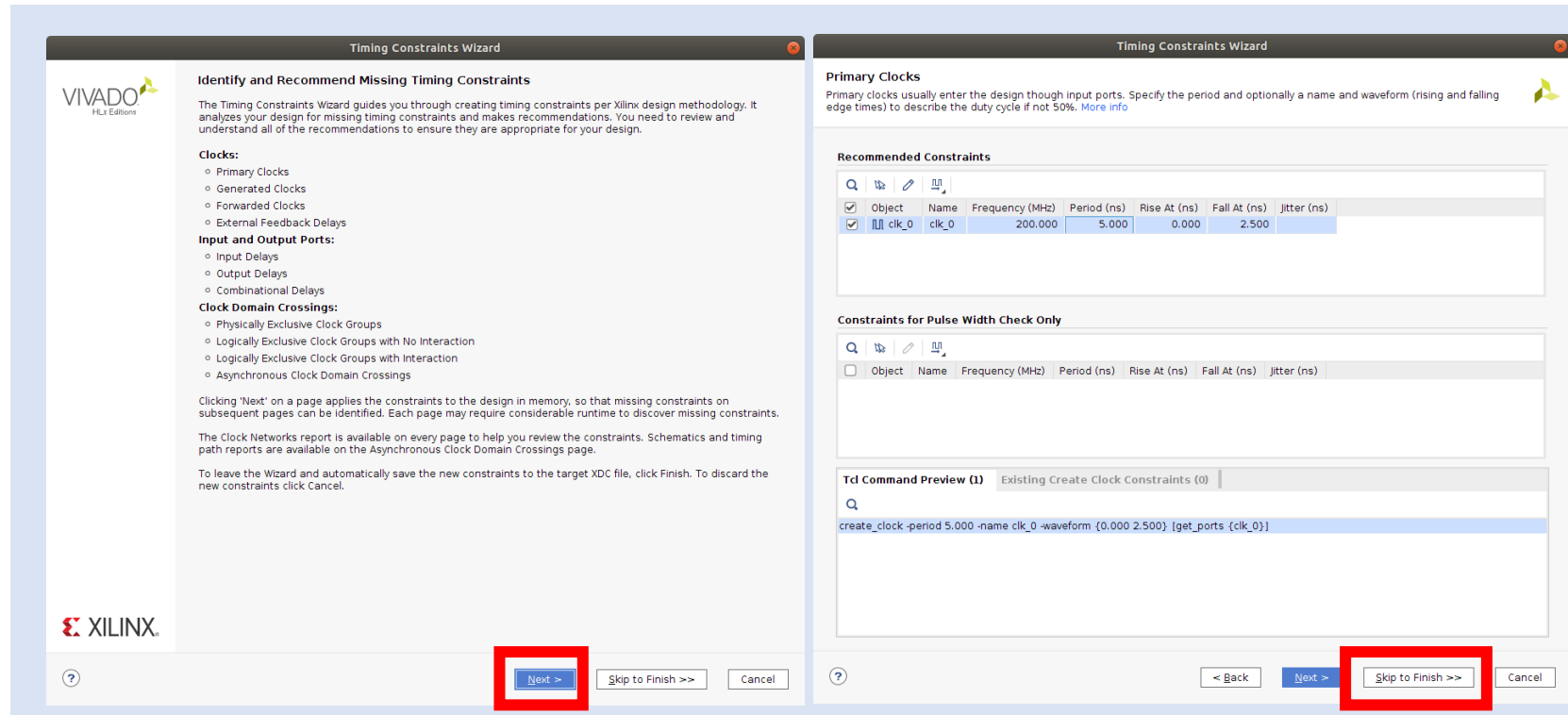
Lab 2: Intermediate Vivado

Step 43 – From the Tools menu, select Timing -> Constraints Wizard.



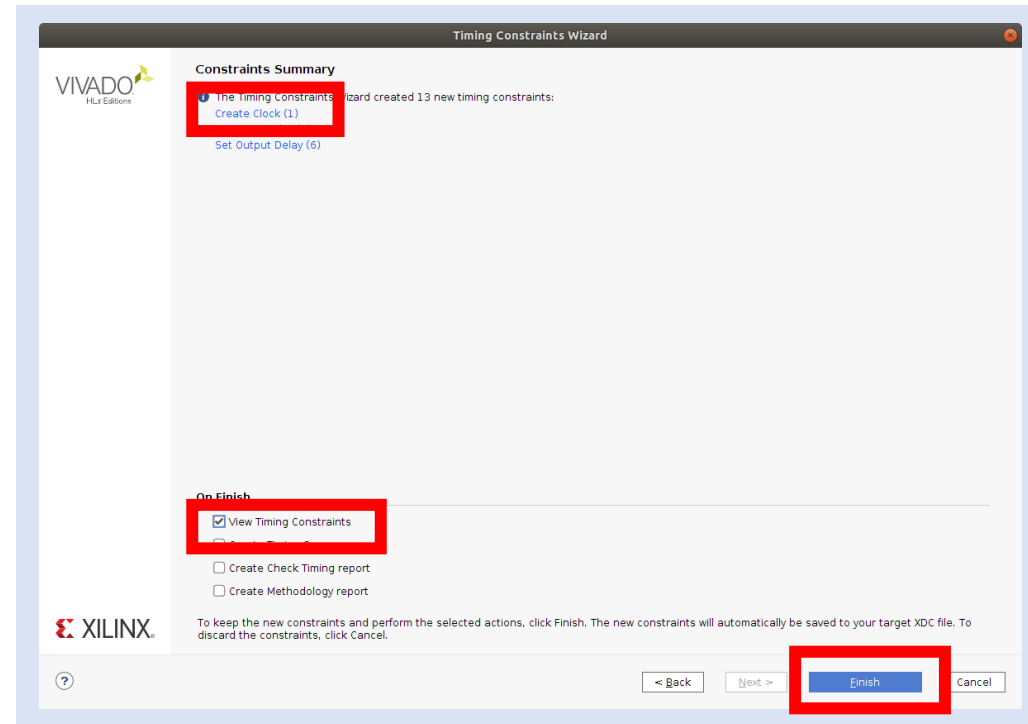
Lab 2: Intermediate Vivado

Step 44 – On the welcome screen, click **Next** and then enter **200 MHz** for the clock frequency. Once done, select **Skip to Finish**.



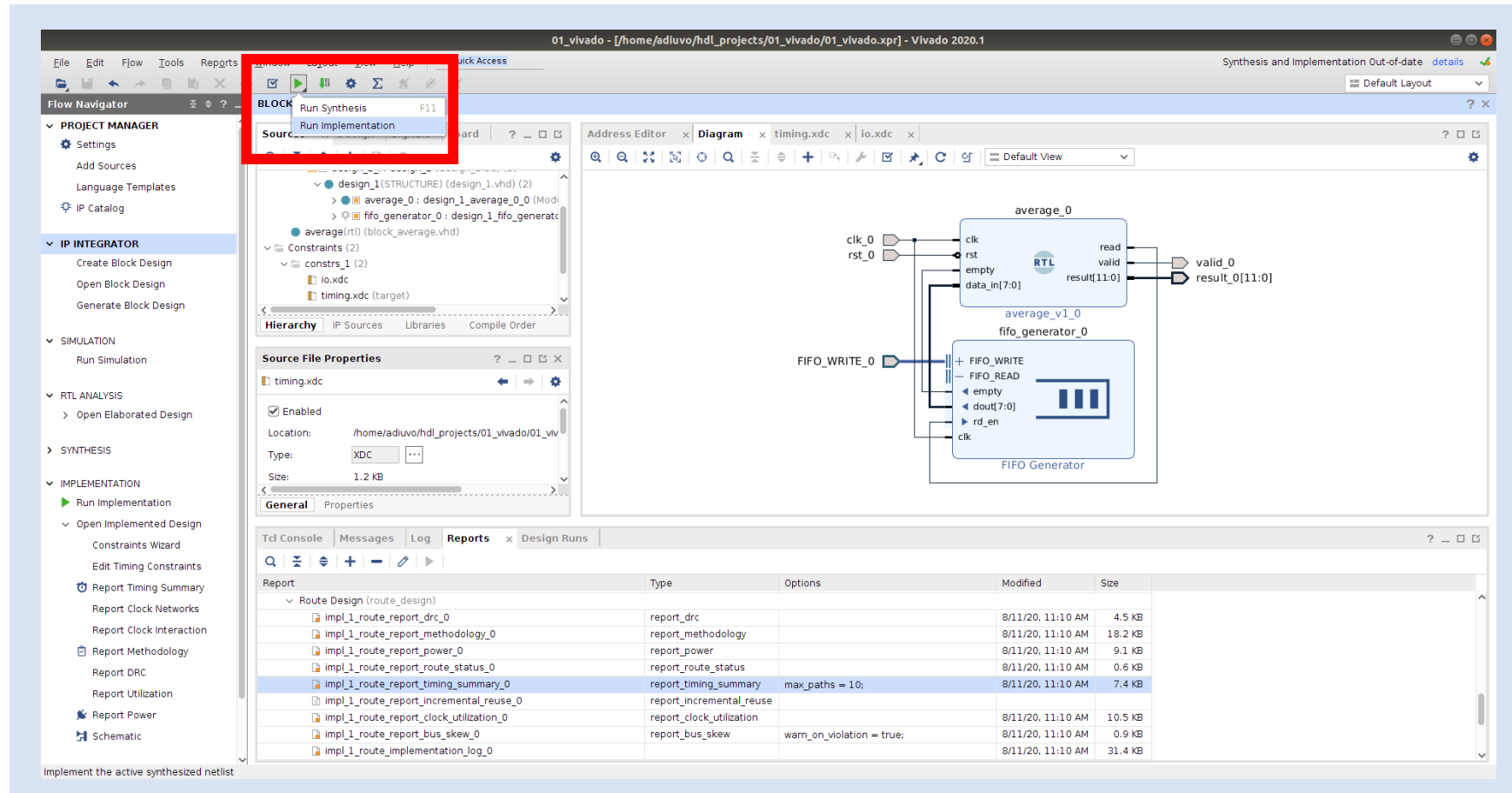
Lab 2: Intermediate Vivado

Step 45 – On the final page, check that only one constraints is being created, check the **View Timing Constraints**, and click **Finish**.



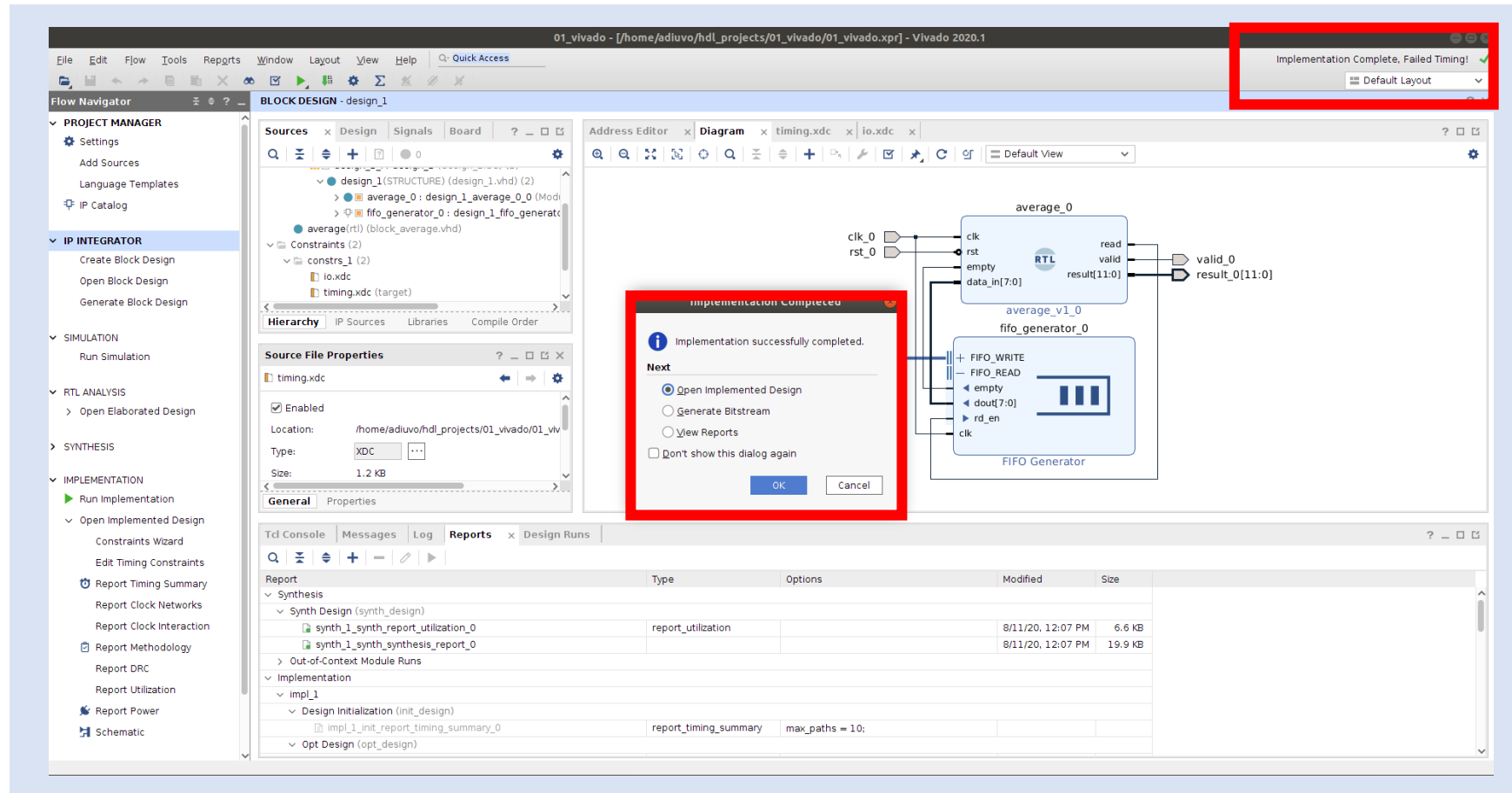
Lab 2: Intermediate Vivado

Step 46 – Close the implementation view and **rerun the implementation**. Click **OK** on any dialogs which pop up prior to implementation starting.



Lab 2: Intermediate Vivado

Step 47 – Once the implementation completes, Timing will fail. Open the **Implemented Design**.



Lab 2: Intermediate Vivado

Step 48 – In the implemented design, select the failing **Intra-Clock Paths**.

The screenshot shows the Vivado IDE interface for a project named '01_vivado'. The 'Flow Navigator' on the left is expanded to 'IMPLEMENTATION', and the 'Open Implemented Design' option is selected. The 'Sources' pane shows the project hierarchy. The 'Timing' tab is active, displaying the 'Design Timing Summary' report. A red box highlights the 'Design Timing Summary' report in the left pane. The report shows the following data:

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): -5.081 ns	Worst Hold Slack (WHS): 0.189 ns	Worst Pulse Width Slack (WPWS): 2.000 ns
Total Negative Slack (TNS): -57.392 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 25	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 194	Total Number of Endpoints: 194	Total Number of Endpoints: 75

Below the table, it states: **Timing constraints are not met.**

Lab 2: Intermediate Vivado

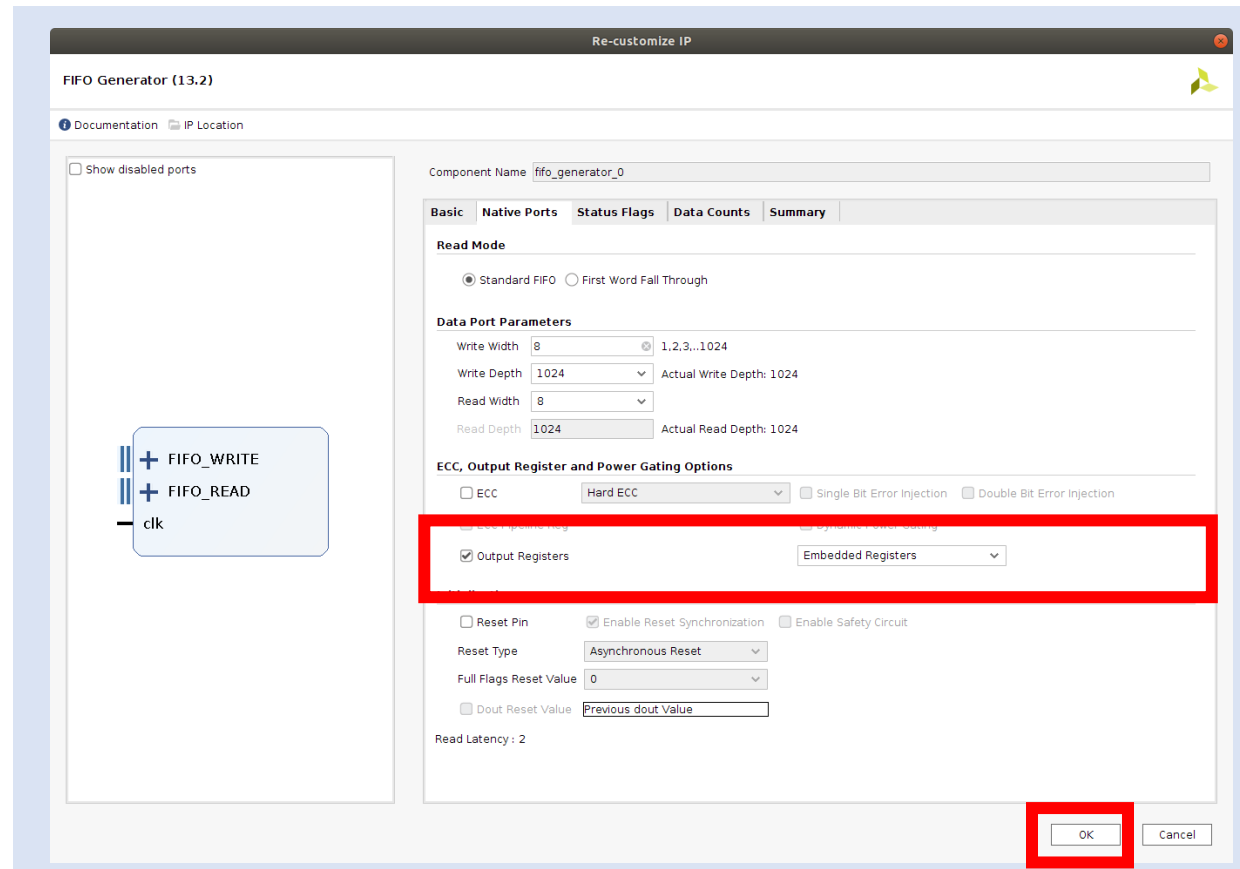
Step 49 – Select **Path 1** and zoom in. You will see that the FIFO output data passes through LUTs before finally being registered. This path is too long for timing at 200 MHz.

The screenshot shows the Vivado 2020.1 interface during the implementation phase. The top panel displays the 'Netlist' view, which is zoomed in to show the implementation grid. A red box highlights a specific path (Path 1) that passes through several LUTs before being registered. The bottom panel shows the 'Timing' report for 'Intra-Clock Paths - clk_0 - Setup', with Path 1 highlighted in red. The report indicates a setup slack of -0.249 ns, which is negative, indicating a timing violation.

Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock
Path 1	-0.249	5	1	design_1_jif...am/CLKBWRCLK	design_1_jif...or_reg[11]/D	5.264	3.887	1.377	5.0	clk_0	clk_0
Path 2	-0.127	4	1	design_1_jif...am/CLKBWRCLK	design_1_jif...ator_reg[7]/D	5.150	3.772	1.377	5.0	clk_0	clk_0
Path 3	-0.118	5	1	design_1_jif...am/CLKBWRCLK	design_1_jif...ator_reg[8]/D	5.090	3.763	1.327	5.0	clk_0	clk_0
Path 4	-0.108	5	1	design_1_jif...am/CLKBWRCLK	design_1_jif...or_reg[10]/D	5.077	3.783	1.294	5.0	clk_0	clk_0
Path 5	-0.024	5	1	design_1_jif...am/CLKBWRCLK	design_1_jif...ator_reg[9]/D	5.040	3.875	1.165	5.0	clk_0	clk_0
Path 6	-0.005	4	1	design_1_jif...am/CLKBWRCLK	design_1_jif...ator_reg[4]/D	4.976	3.649	1.327	5.0	clk_0	clk_0
Path 7	0.004	4	1	design_1_jif...am/CLKBWRCLK	design_1_jif...ator_reg[6]/D	4.963	3.669	1.294	5.0	clk_0	clk_0

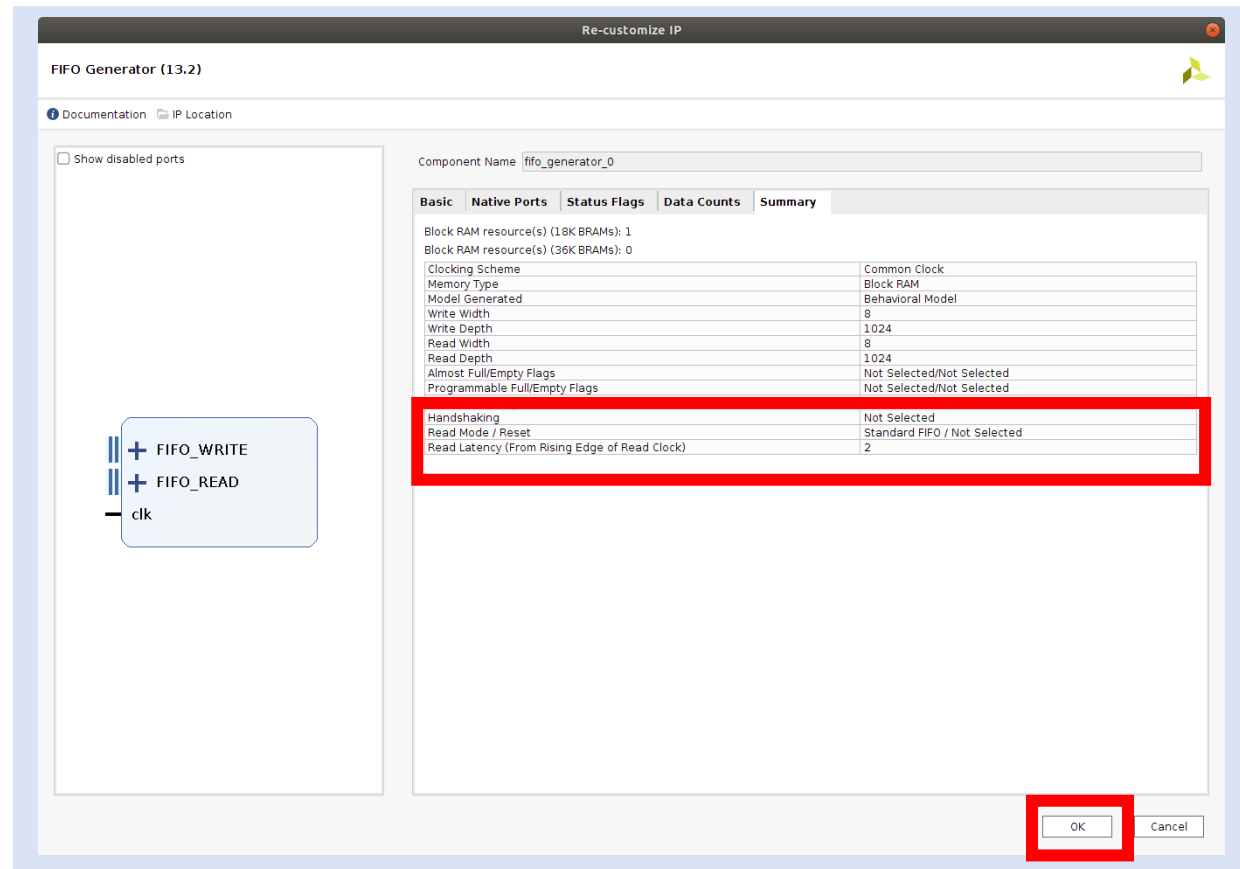
Lab 2: Intermediate Vivado

Step 50 – To fix this, we need to register the output of the FIFO. Close the implementation view and reopen the block diagram. Double click on the **FIFO** to customize.



Lab 2: Intermediate Vivado

Step 51 – Note the latency has changed from 1 to 2 clocks. We would need to correct for this in the average block, however, we proceed assuming that we have.



Lab 2: Intermediate Vivado

Step 52 – Reimplement the design. When the timing is completed, you should see that the implementation is correct and the timing is met.

The screenshot shows the Vivado 2020.1 IDE interface. The top status bar indicates "Implementation Complete" with a green checkmark. The main workspace displays the "IMPLEMENTED DESIGN" view, showing the netlist and block properties. The bottom panel shows the "Design Timing Summary" report, which confirms that all user-specified timing constraints are met.

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	0.172 ns	Worst Hold Slack (WHS):	0.185 ns	Worst Pulse Width Slack (WPWS):	2.000 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	151	Total Number of Endpoints:	151	Total Number of Endpoints:	76

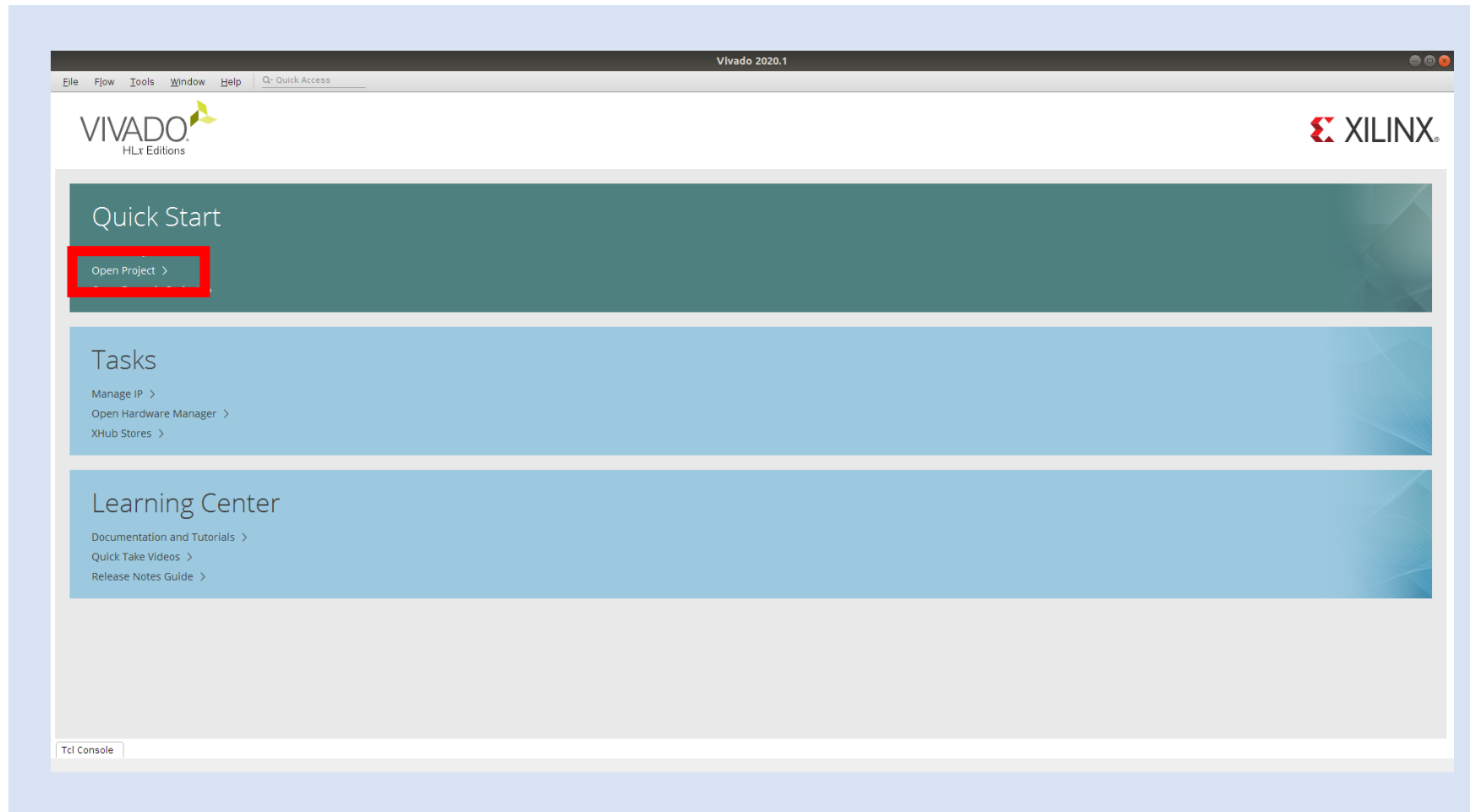
All user specified timing constraints are met.

Lab 3

Advanced Vivado

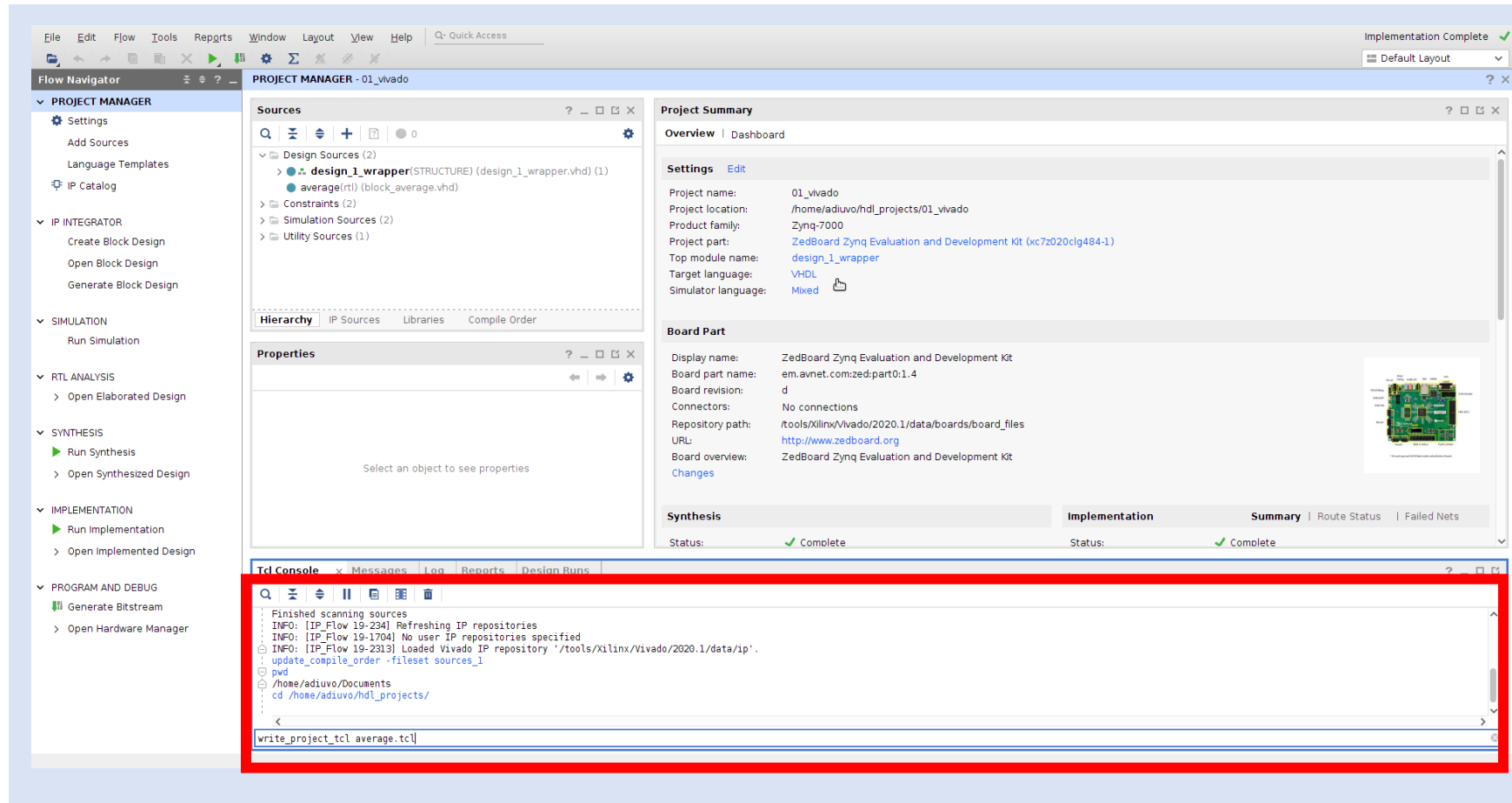
Lab 3: Advanced Vivado

Step 1 – Open the project created in part two.



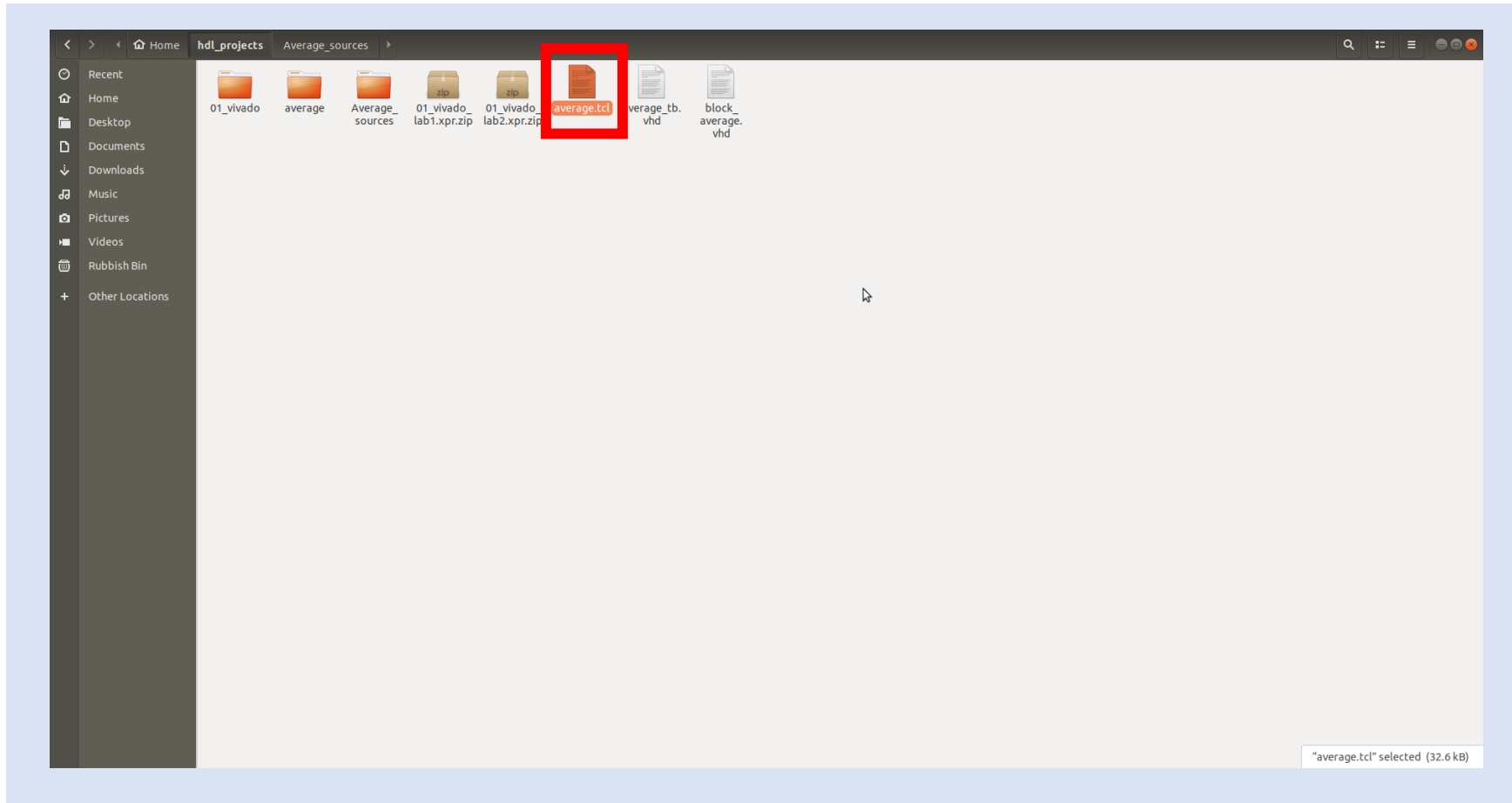
Lab 3: Advanced Vivado

Step 2 – In the TCL window, check the PWD and change directory to the location containing your project. Then enter the command `write_project_tcl average.tcl`.



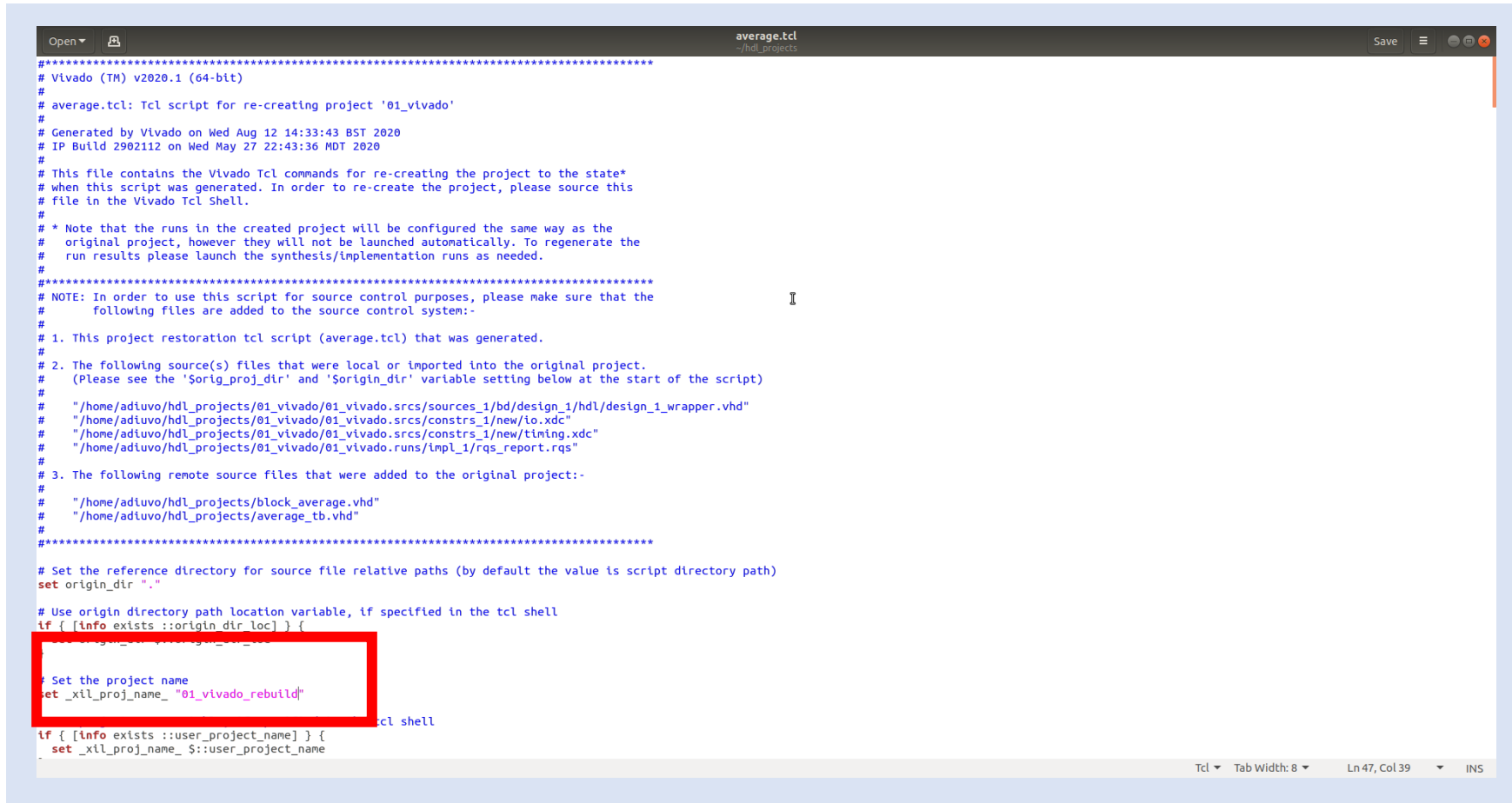
Lab 3: Advanced Vivado

Step 3 – This will write a project description in TCL to the project directory.



Lab 3: Advanced Vivado

Step 4 – Open the TCL file and change the project name from **01_Vivado to **01_Vivado_rebuild**.**



```

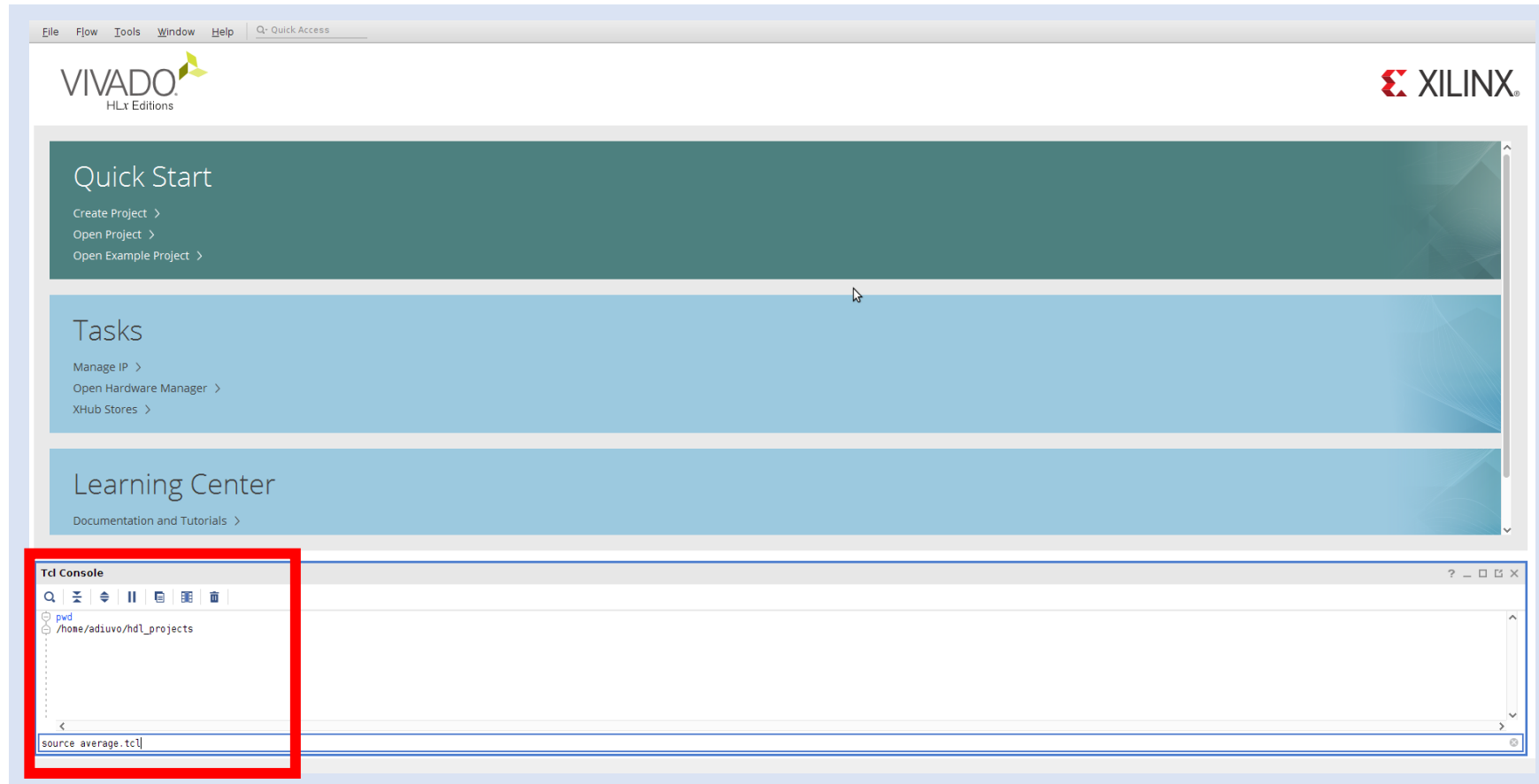
Open ▾  average.tcl
~/hdl_projects

#*****
# Vivado (TM) v2020.1 (64-bit)
#
# average.tcl: Tcl script for re-creating project '01_vivado'
#
# Generated by Vivado on Wed Aug 12 14:33:43 BST 2020
# IP Build 2902112 on Wed May 27 22:43:36 MDT 2020
#
# This file contains the Vivado Tcl commands for re-creating the project to the state*
# when this script was generated. In order to re-create the project, please source this
# file in the Vivado Tcl Shell.
#
# * Note that the runs in the created project will be configured the same way as the
# original project, however they will not be launched automatically. To regenerate the
# run results please launch the synthesis/implementation runs as needed.
#
#*****
# NOTE: In order to use this script for source control purposes, please make sure that the
# following files are added to the source control system:-
#
# 1. This project restoration tcl script (average.tcl) that was generated.
#
# 2. The following source(s) files that were local or imported into the original project.
# (Please see the 'Sorig_proj_dir' and 'Sorigin_dir' variable setting below at the start of the script)
#
# "/home/adiuvo/hdl_projects/01_vivado/01_vivado.srcs/sources_1/bd/design_1/hdl/design_1_wrapper.vhd"
# "/home/adiuvo/hdl_projects/01_vivado/01_vivado.srcs/constrs_1/new/io.xdc"
# "/home/adiuvo/hdl_projects/01_vivado/01_vivado.srcs/constrs_1/new/timing.xdc"
# "/home/adiuvo/hdl_projects/01_vivado/01_vivado.runs/impl_1/rqs_report.rqs"
#
# 3. The following remote source files that were added to the original project:-
#
# "/home/adiuvo/hdl_projects/block_average.vhd"
# "/home/adiuvo/hdl_projects/average_tb.vhd"
#
#*****
# Set the reference directory for source file relative paths (by default the value is script directory path)
set origin_dir "."
# Use origin directory path location variable, if specified in the tcl shell
if { [info exists ::origin_dir_loc] } {
    set origin_dir [::origin_dir_loc]
}
# Set the project name
set _xil_proj_name_ "01_vivado_rebuild"
#
# ***** tcl shell *****
if { [info exists ::user_project_name] } {
    set _xil_proj_name_ $::user_project_name
}

```

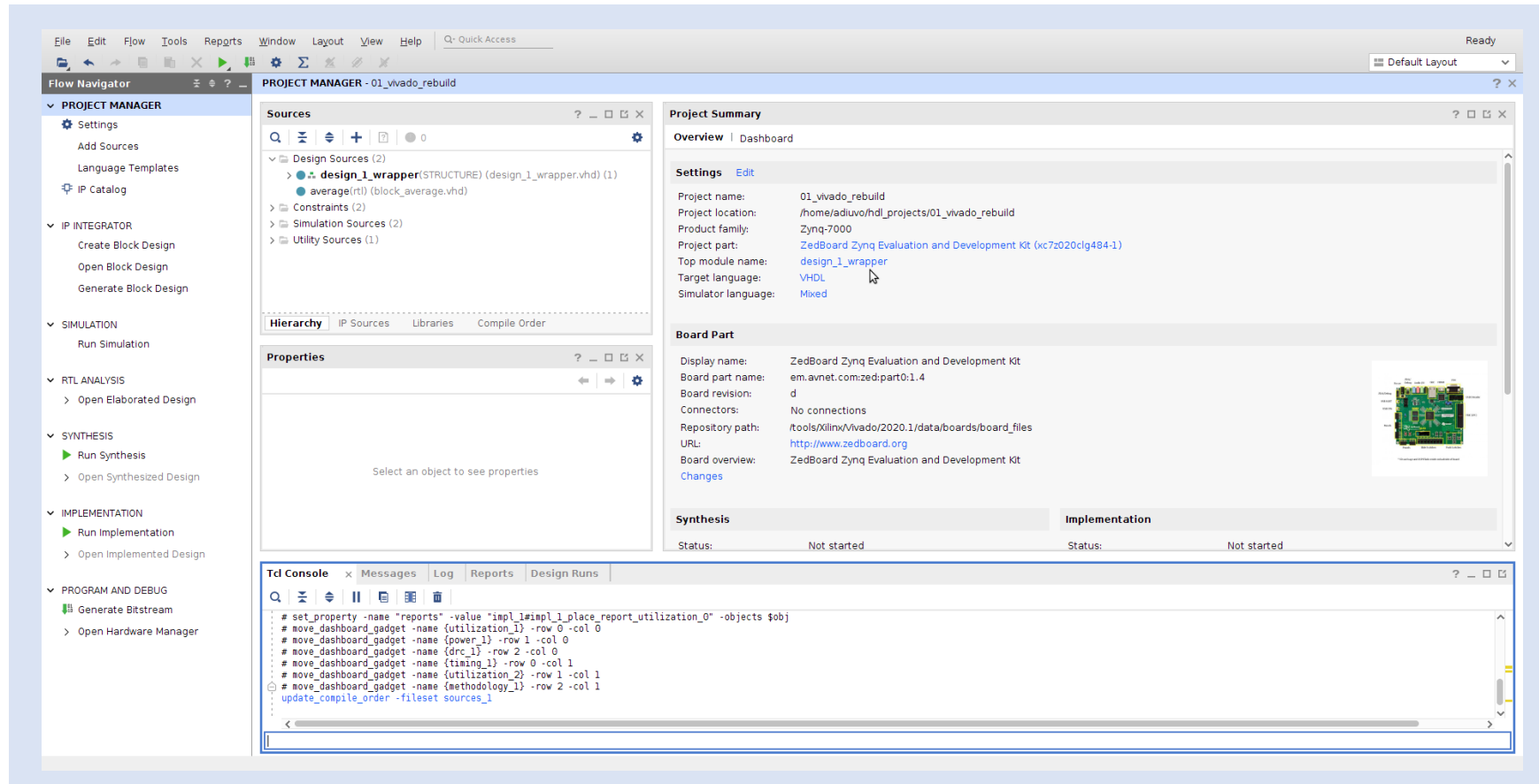
Lab 3: Advanced Vivado

Step 5 – Close the current project and type `source average.tcl` in the TCL window.



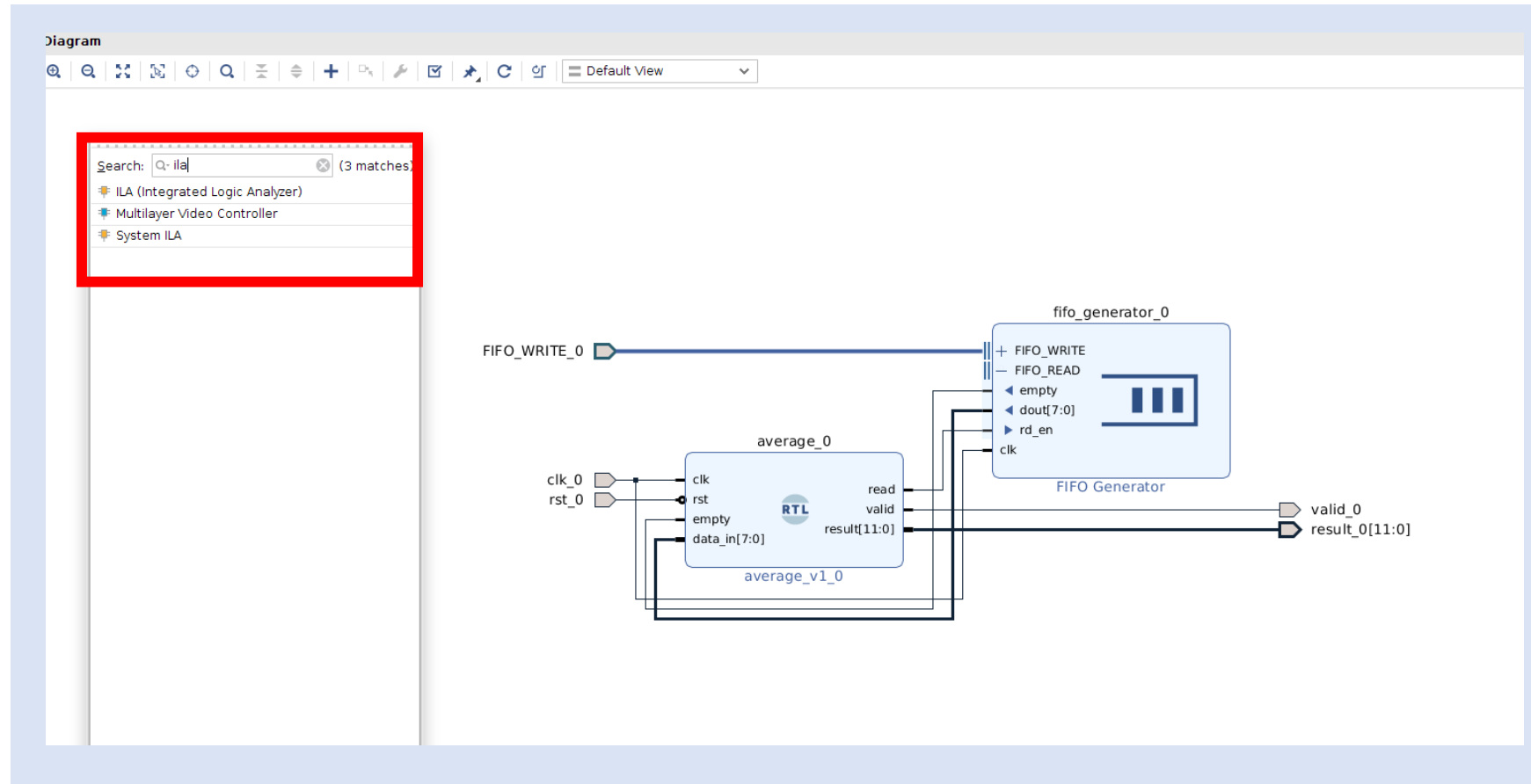
Lab 3: Advanced Vivado

Step 6 – This will rebuild the project exactly as before from the TCL file.



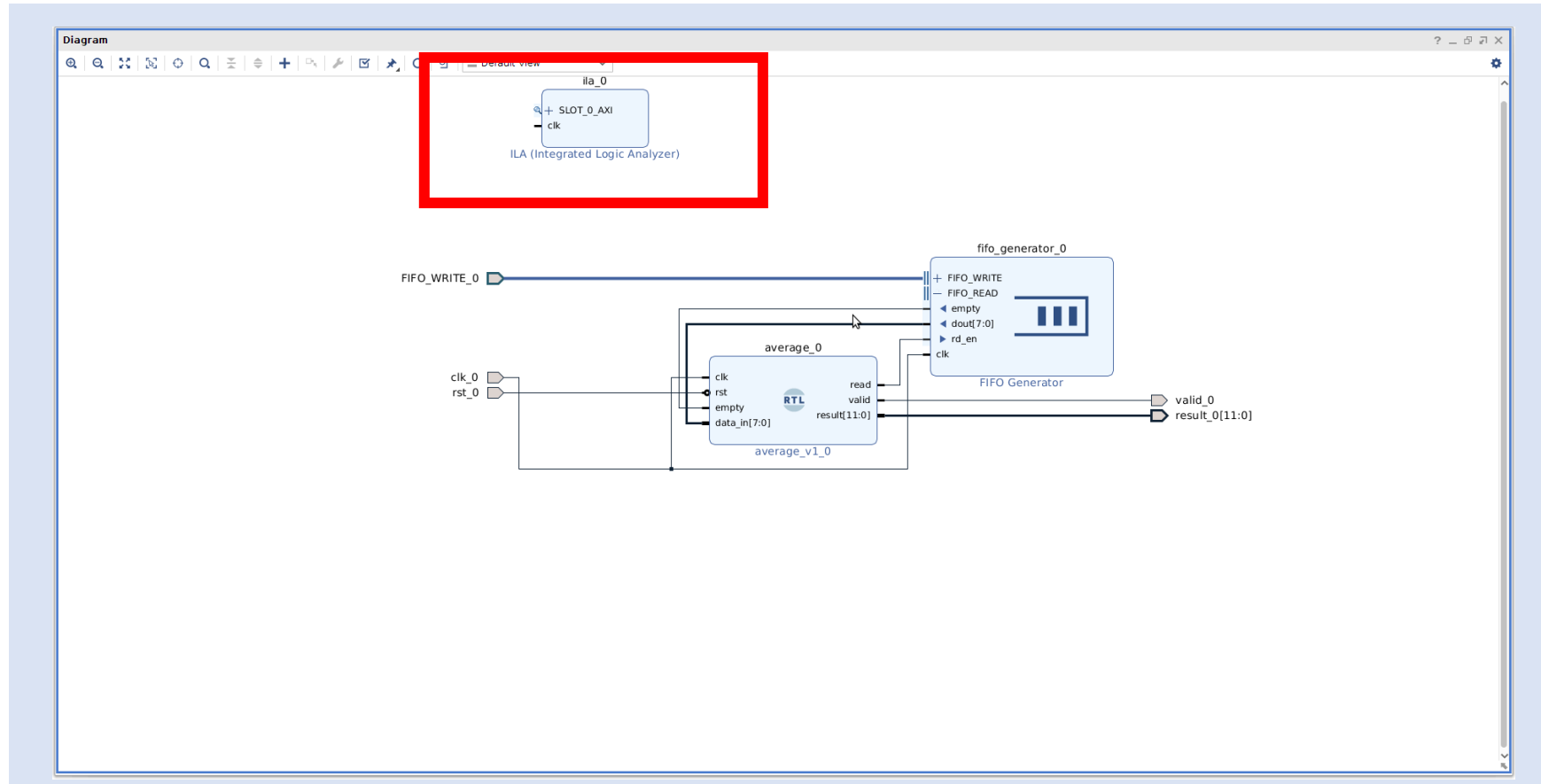
Lab 3: Advanced Vivado

Step 7 – Open the block design and click on the **+** symbol to add in IP. In the search bar, enter **ILA** and then double click on the **ILA (Integrated Logic Analyzer)** to add in the IP core.



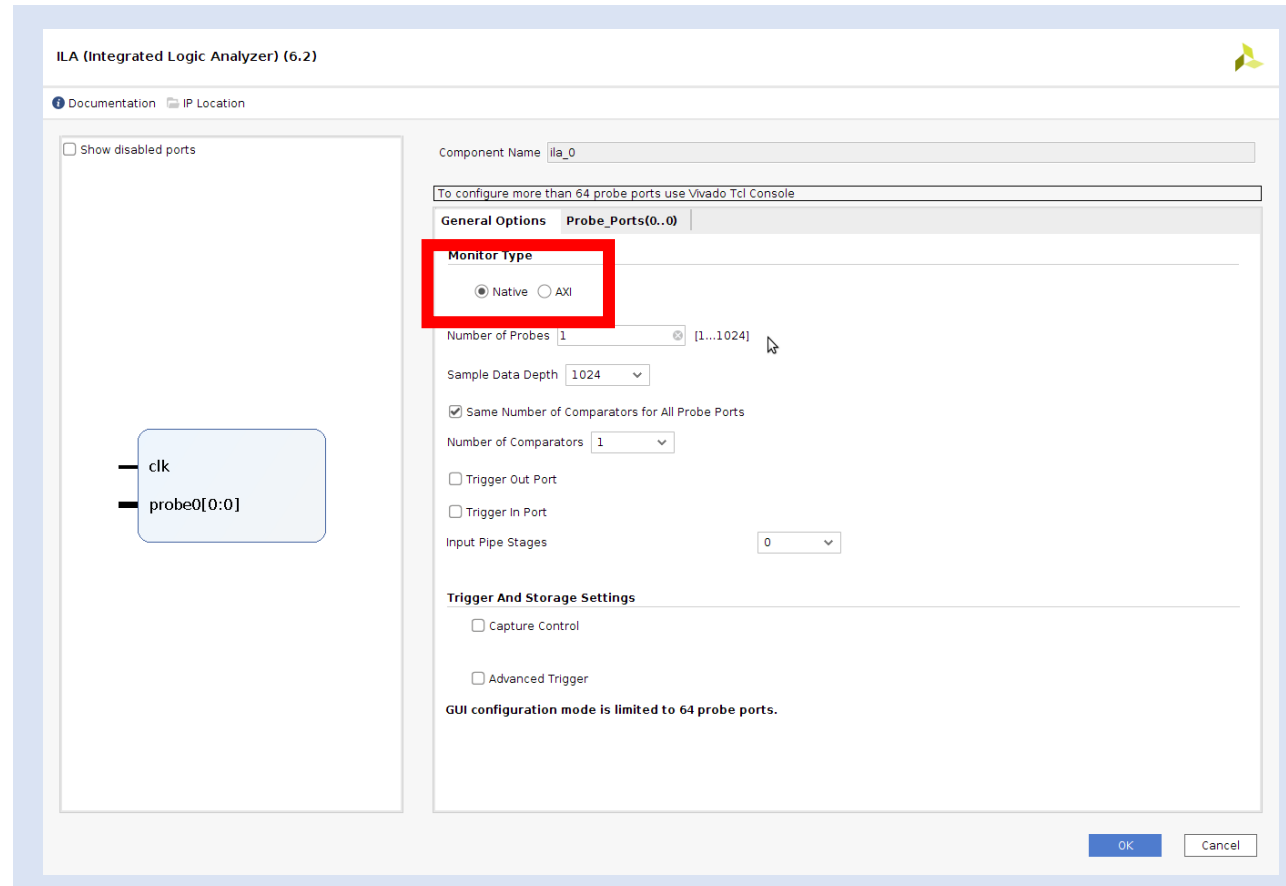
Lab 3: Advanced Vivado

Step 8 – Double click on the **ILA** to customize it.



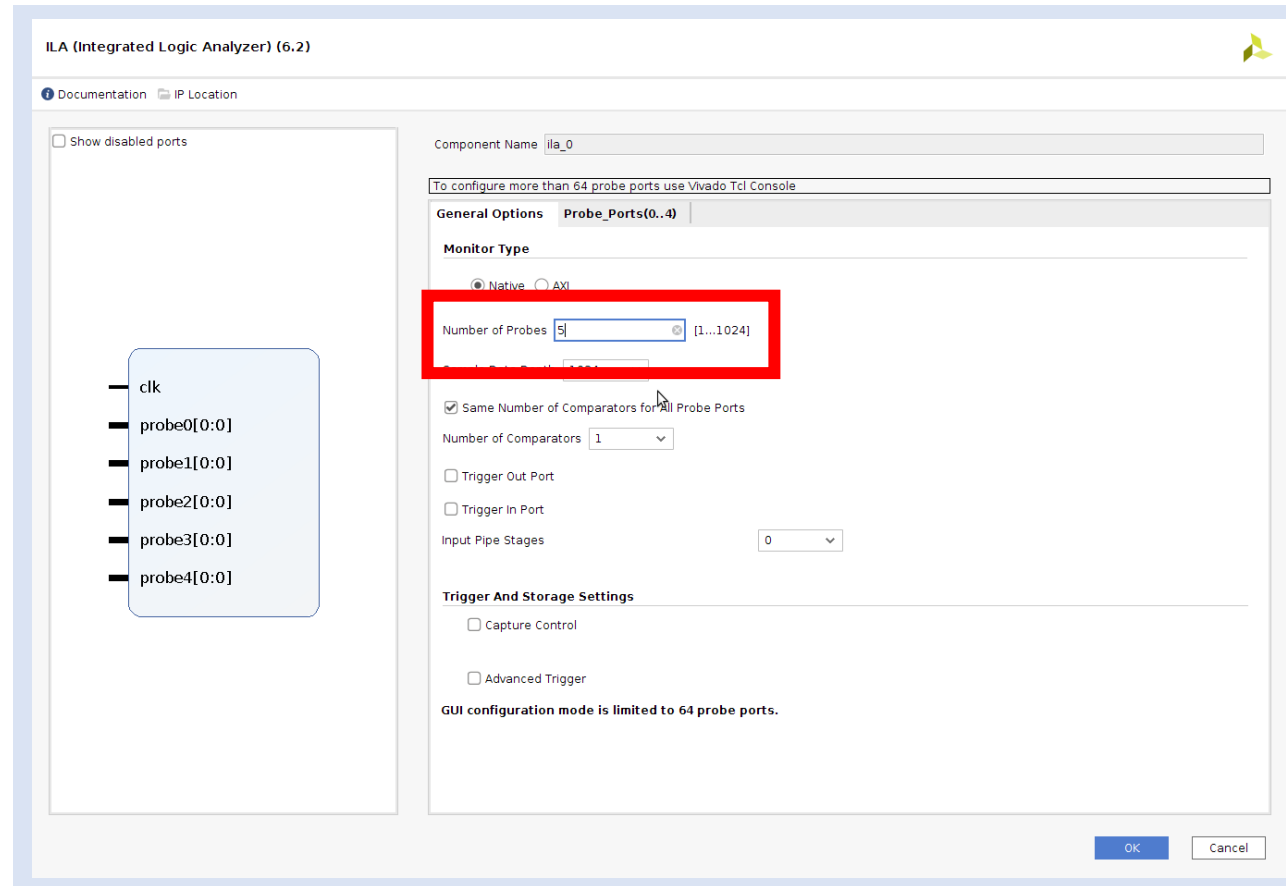
Lab 3: Advanced Vivado

Step 9 – Change the monitor type to **Native**.



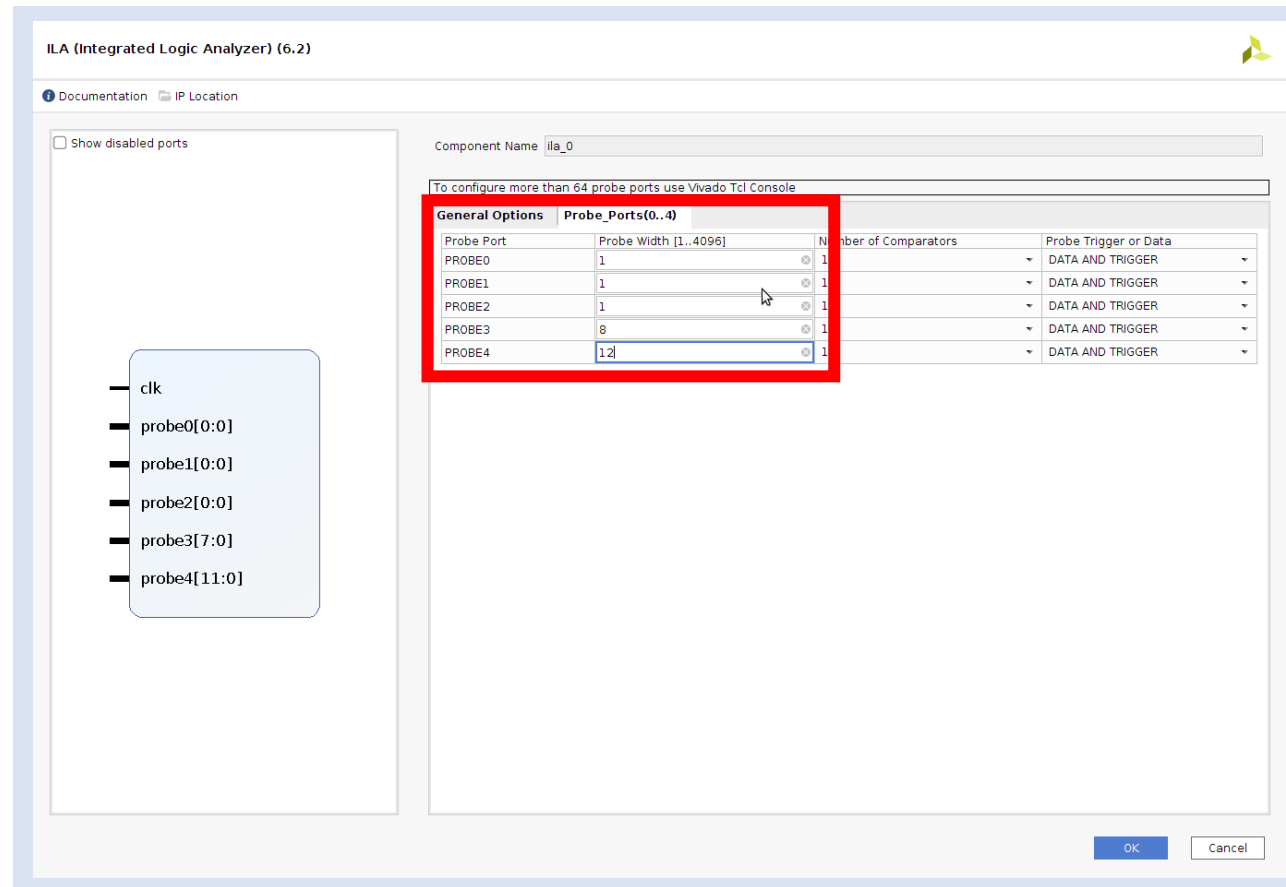
Lab 3: Advanced Vivado

Step 10 – Set the number of probes to 5.



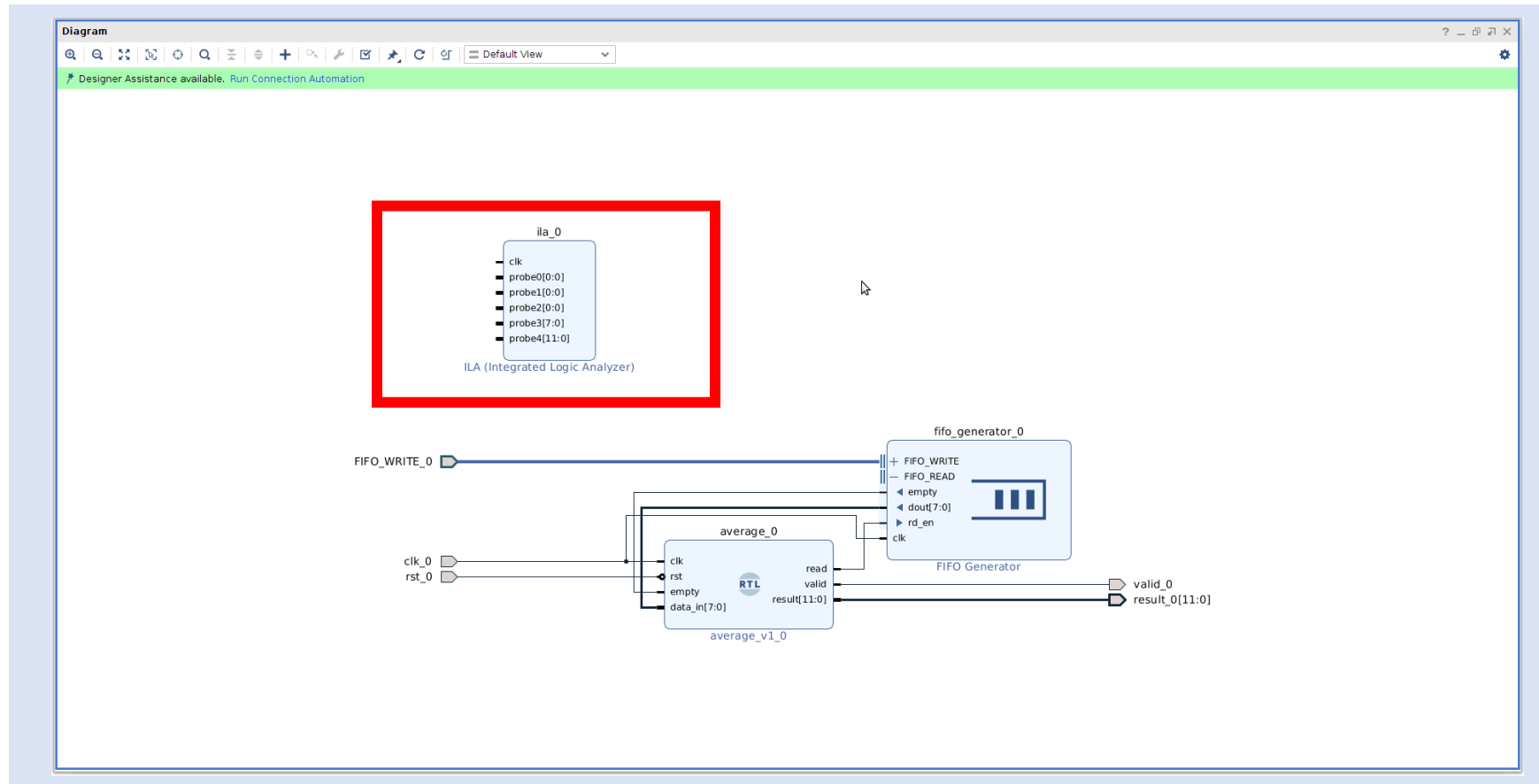
Lab 3: Advanced Vivado

Step 11 – Click on the **Probe Ports Tab** and set the widths of **PROBE3** to **8** and **PROBE4** to **12**, click **OK**.



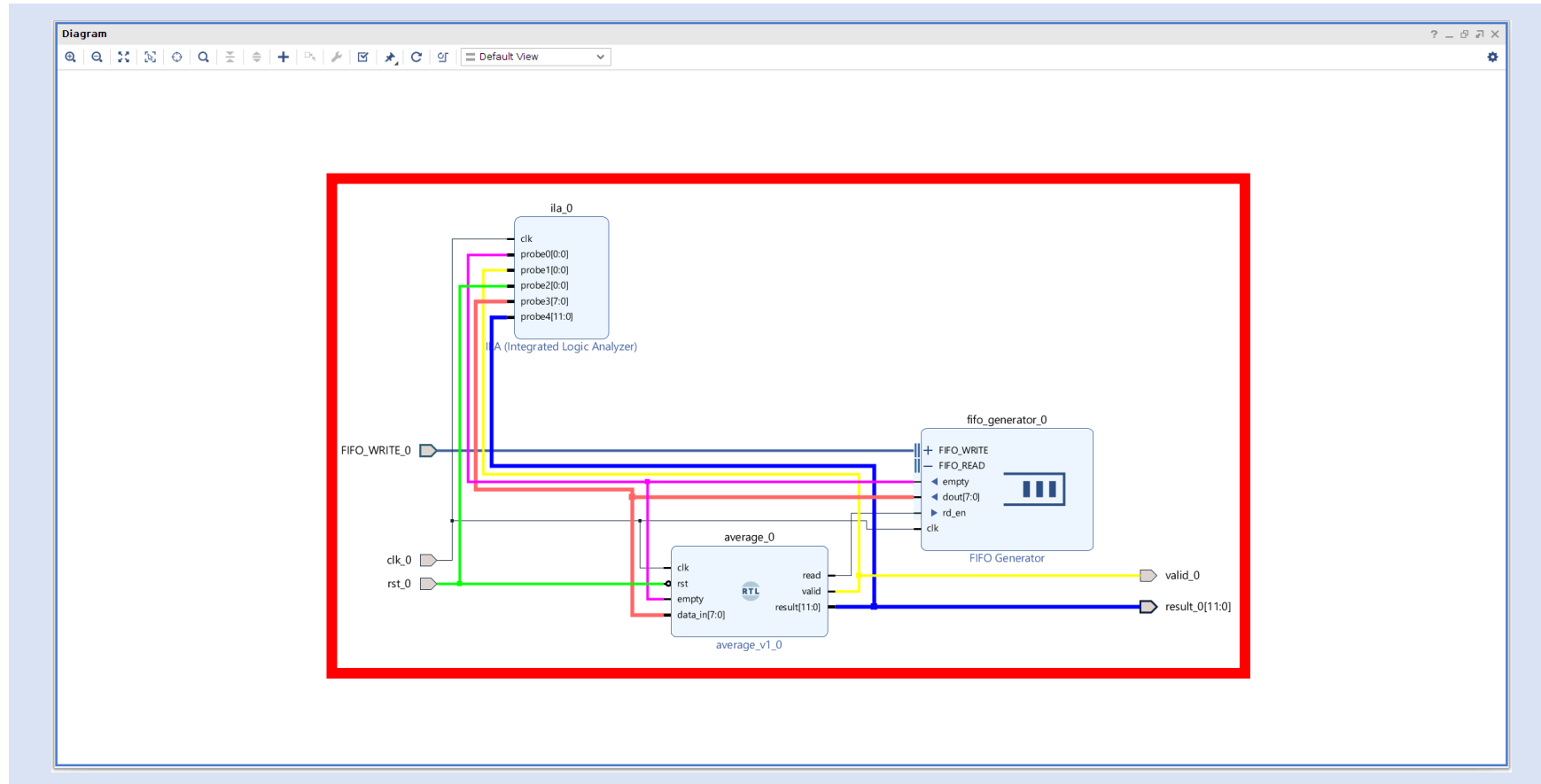
Lab 3: Advanced Vivado

Step 12 – The block diagram should now show the ILA with five ports and a clock input.



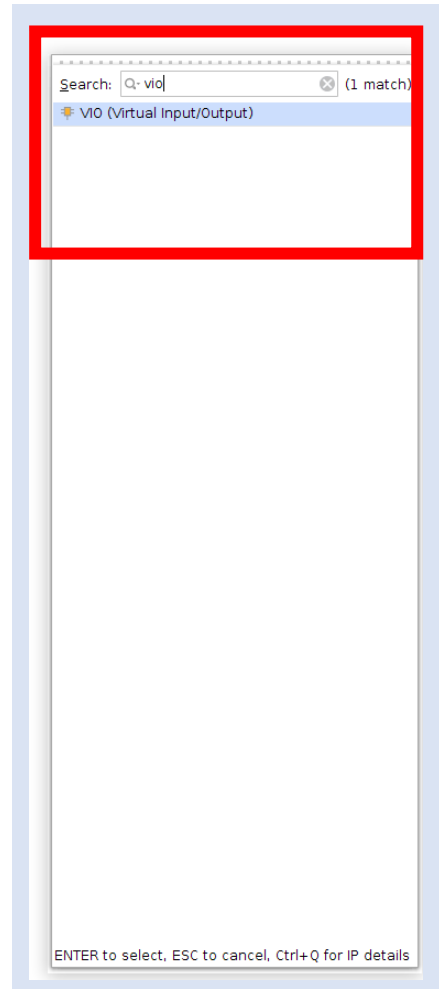
Lab 3: Advanced Vivado

Step 13 – Connect up the ILA as shown below. This will provide us visibility of all input and output of the average block.



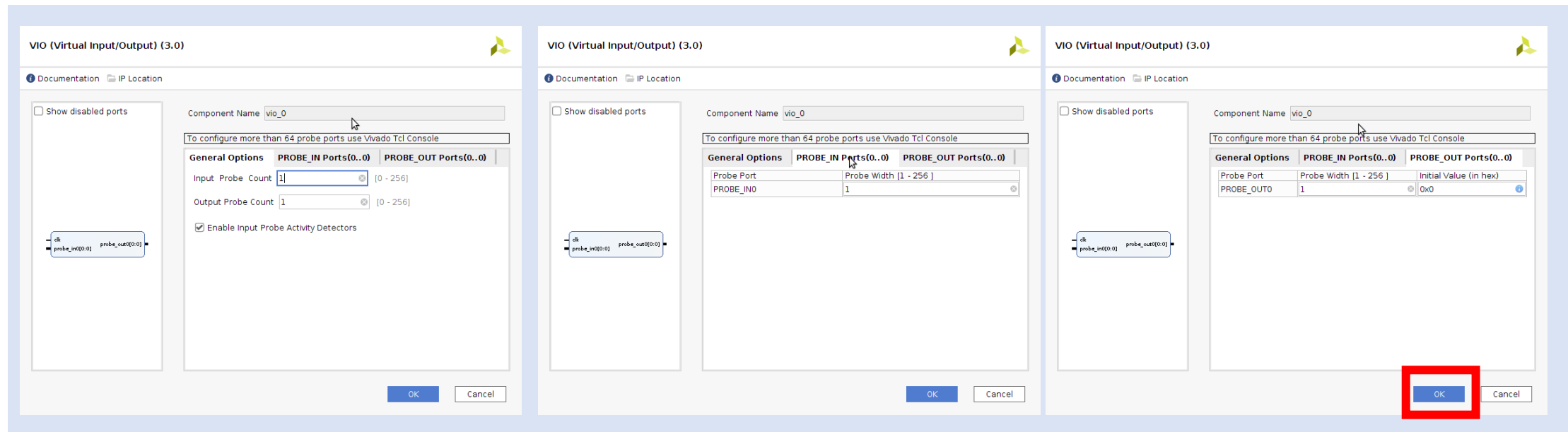
Lab 3: Advanced Vivado

Step 14 – Click on the **+** symbol to add in IP and type **VIO** in the search box. Double click on the **VIO** symbol.



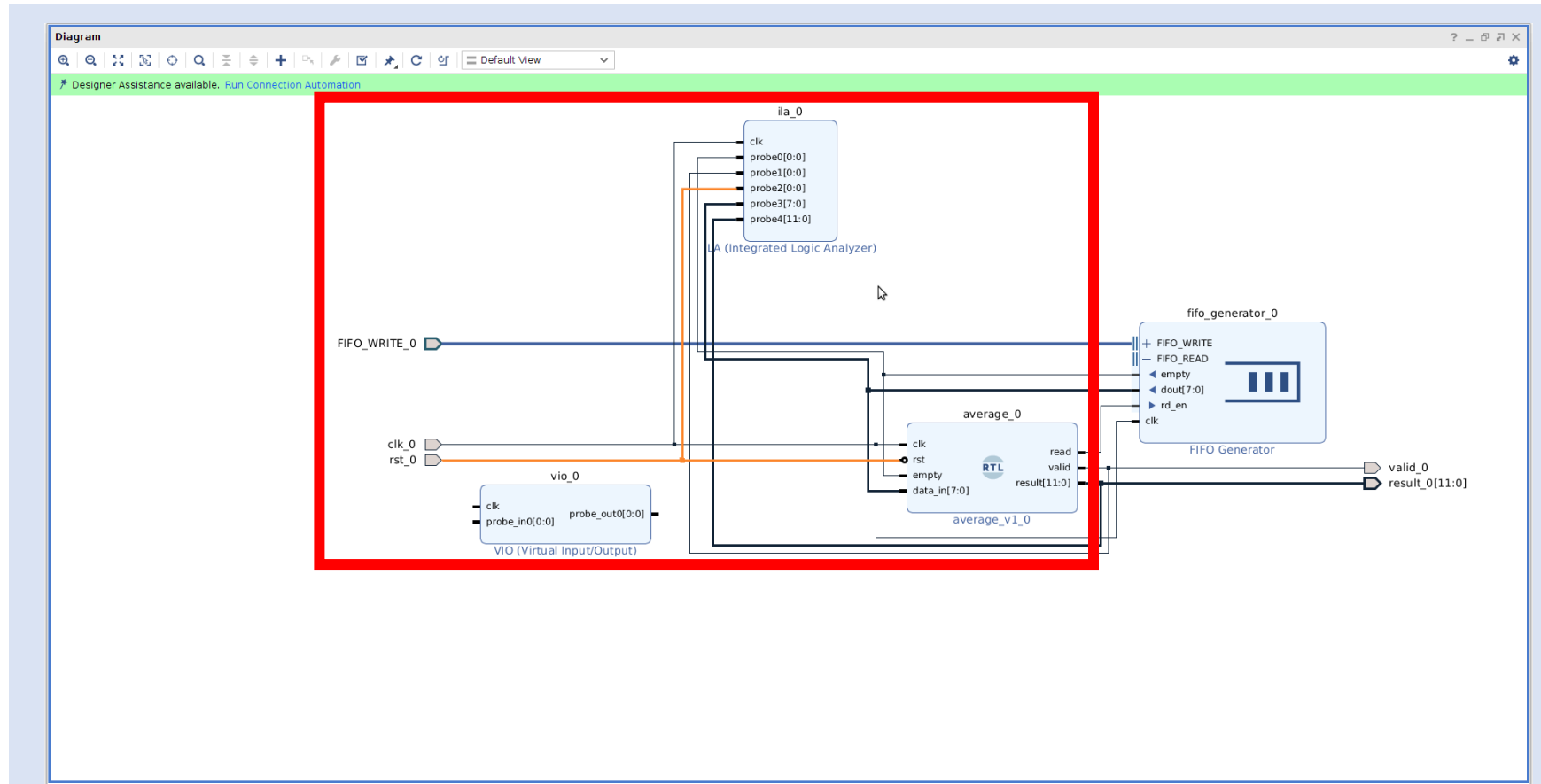
Lab 3: Advanced Vivado

Step 15 – When the VIO is added onto the block diagram, double click it to customize and explore the three tabs. Leave it all unchanged and click **OK**.



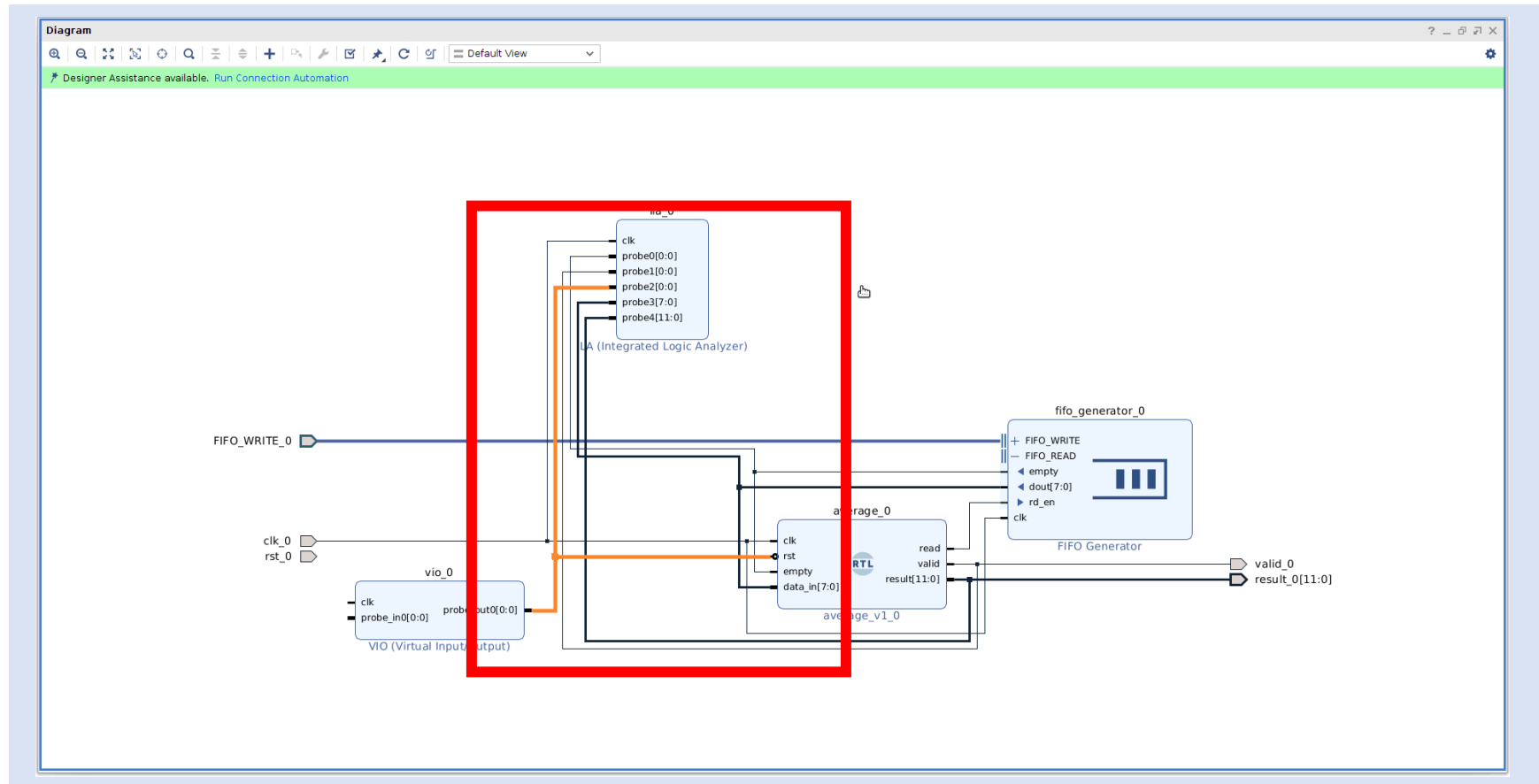
Lab 3: Advanced Vivado

Step 16 – Select the **RST_0** line and delete it.



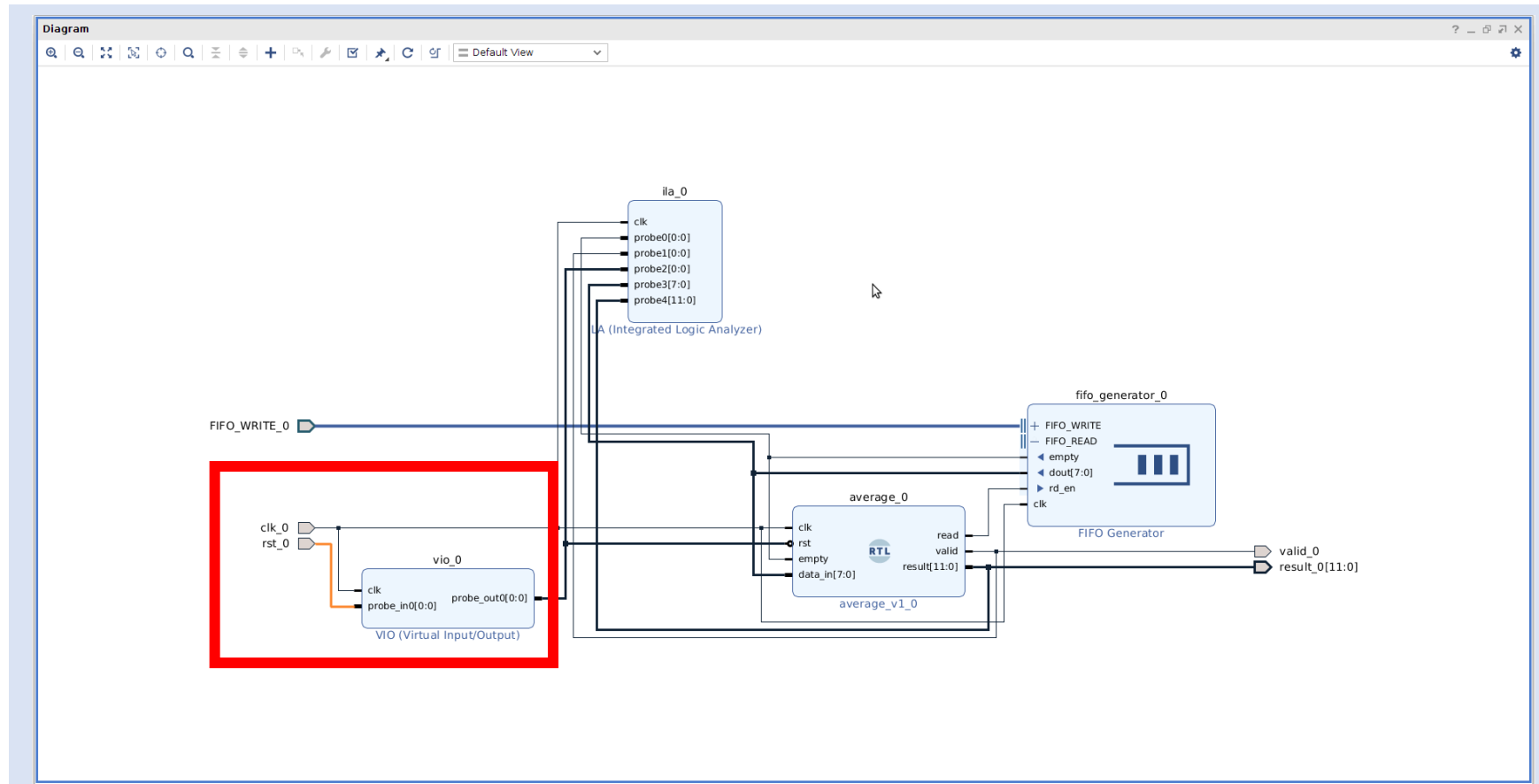
Lab 3: Advanced Vivado

Step 17 – Connect the VIO probe output to the RST and ILA.



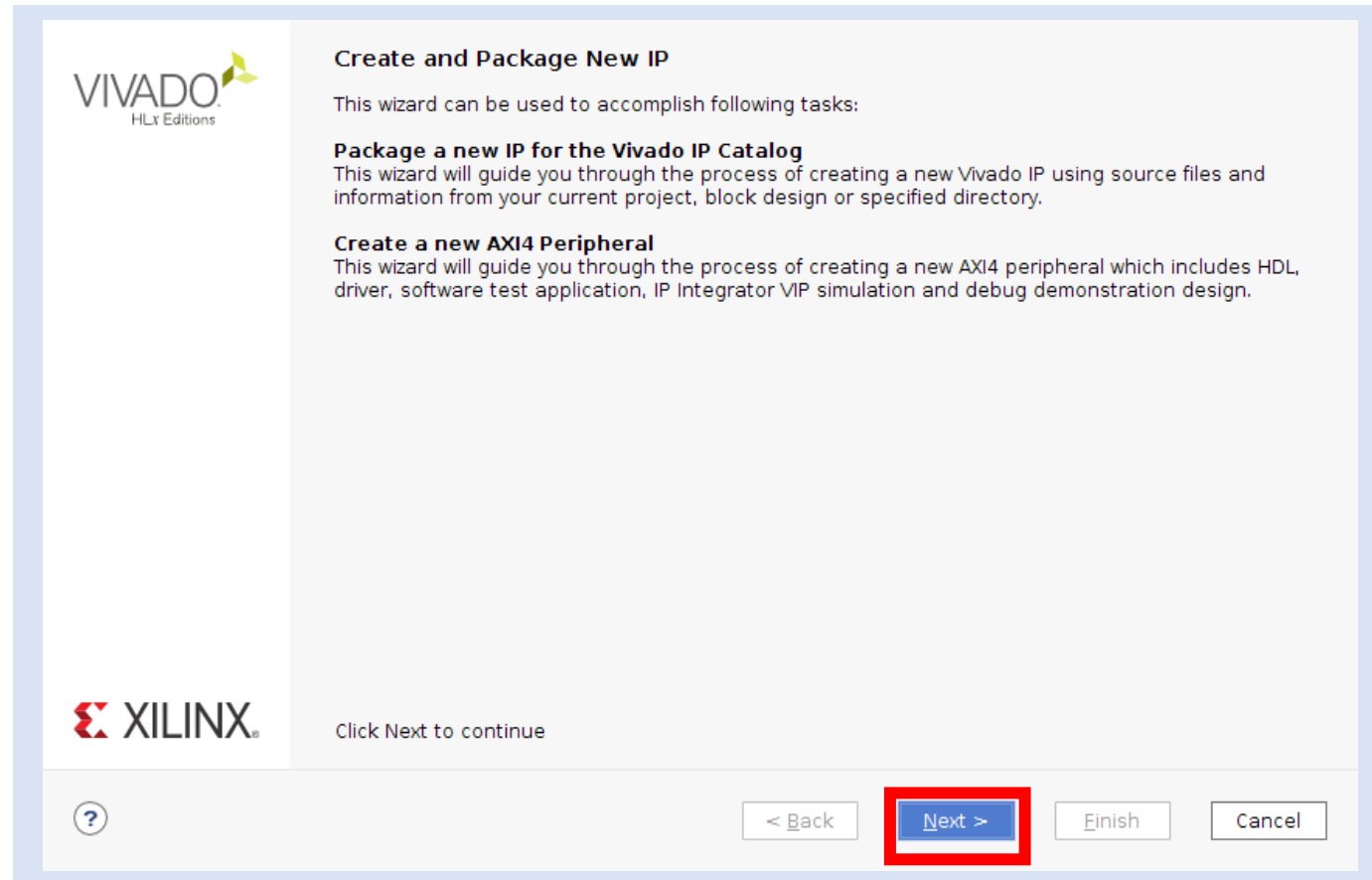
Lab 3: Advanced Vivado

Step 18 – Connect the **RST_0** input to the VIO Probe in.



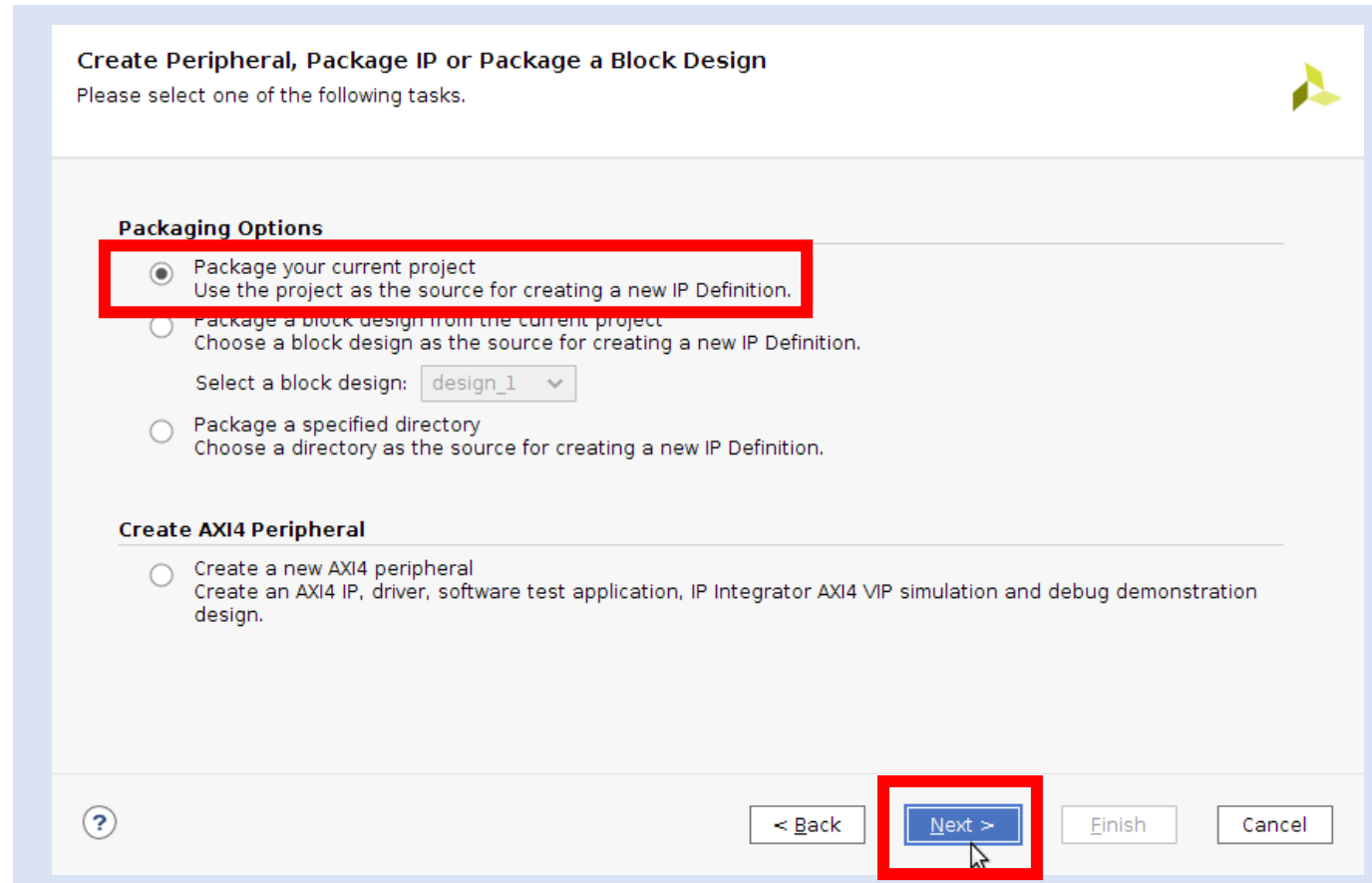
Lab 3: Advanced Vivado

Step 19 – From the Menu bar, select **Tools** → **Create and Package IP**. Select **Next** when the dialog opens.



Lab 3: Advanced Vivado

Step 20 – Select **Package your current project and then **Next**.**



Create Peripheral, Package IP or Package a Block Design
Please select one of the following tasks.

Packaging Options

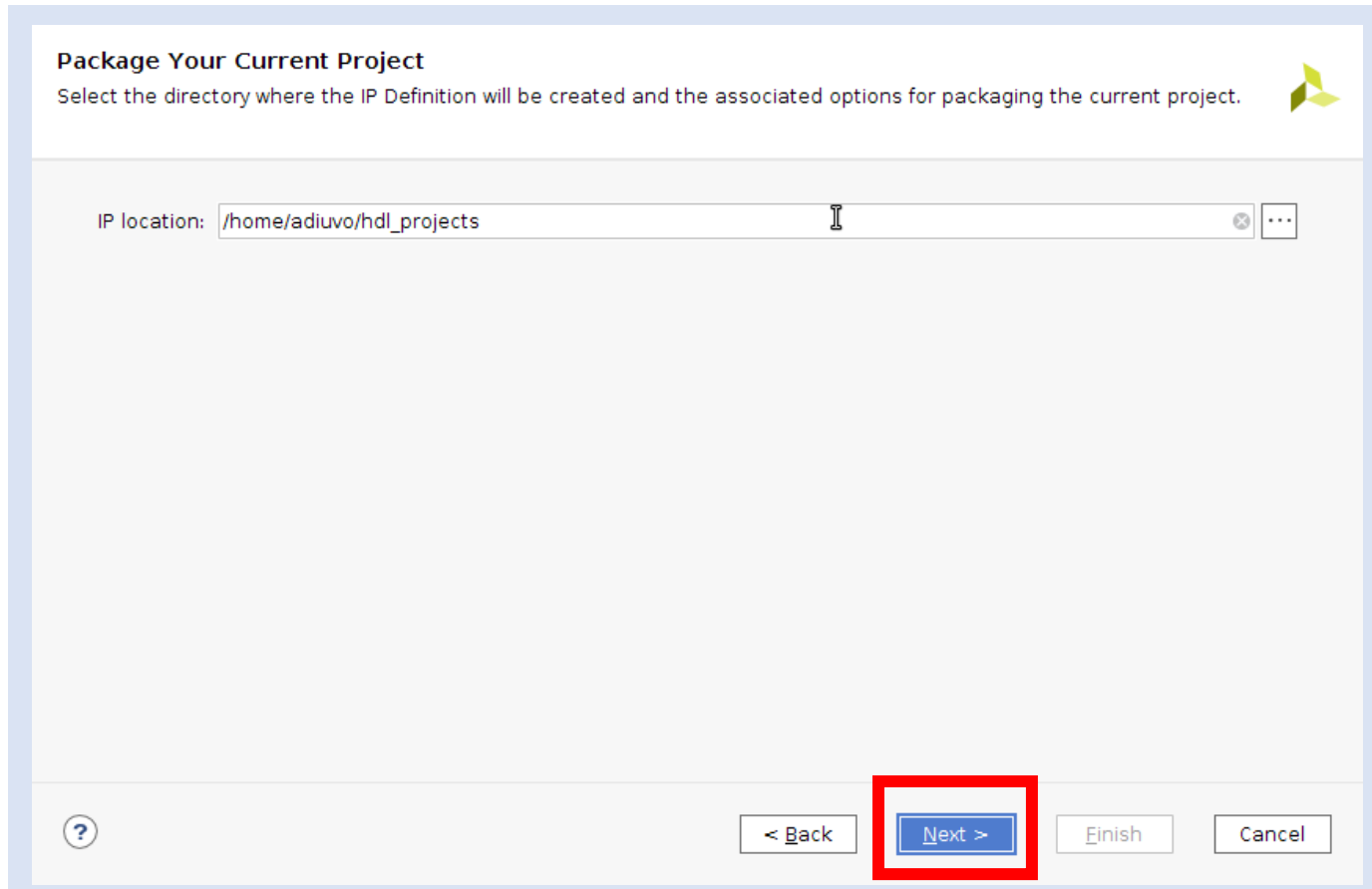
- ☒ Package your current project
Use the project as the source for creating a new IP Definition.
- ☐ Package a block design from the current project
Choose a block design as the source for creating a new IP Definition.
Select a block design:
- ☐ Package a specified directory
Choose a directory as the source for creating a new IP Definition.

Create AXI4 Peripheral

- ☐ Create a new AXI4 peripheral
Create an AXI4 IP, driver, software test application, IP Integrator AXI4 VIP simulation and debug demonstration design.

Lab 3: Advanced Vivado

Step 21 – Select the location the IP should be saved in and click **Next**.



The image shows the 'Package Your Current Project' dialog box in Vivado. The title bar is light blue. The main area has a light gray background. At the top, the title 'Package Your Current Project' is in bold, followed by the instruction 'Select the directory where the IP Definition will be created and the associated options for packaging the current project.' and the Vivado logo. Below this is a text field labeled 'IP location:' containing the path '/home/adiuvo/hdl_projects'. To the right of the text field are a close button (X) and a browse button (three dots). At the bottom, there is a row of buttons: a help button (question mark in a circle), a '< Back' button, a 'Next >' button (highlighted with a red rectangle), an 'Finish' button, and a 'Cancel' button.

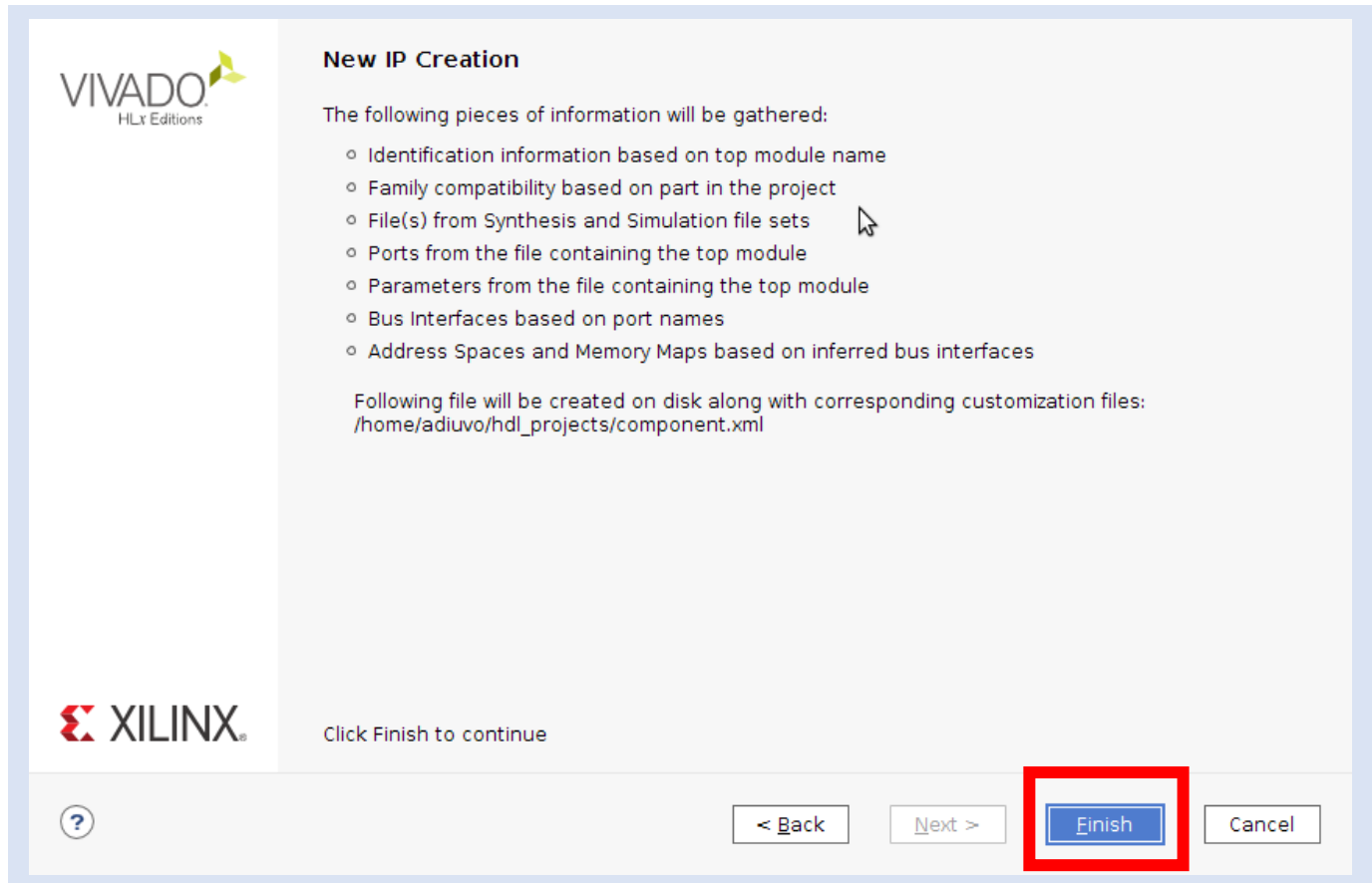
Package Your Current Project
Select the directory where the IP Definition will be created and the associated options for packaging the current project.

IP location: /home/adiuvo/hdl_projects

? < Back **Next >** Finish Cancel

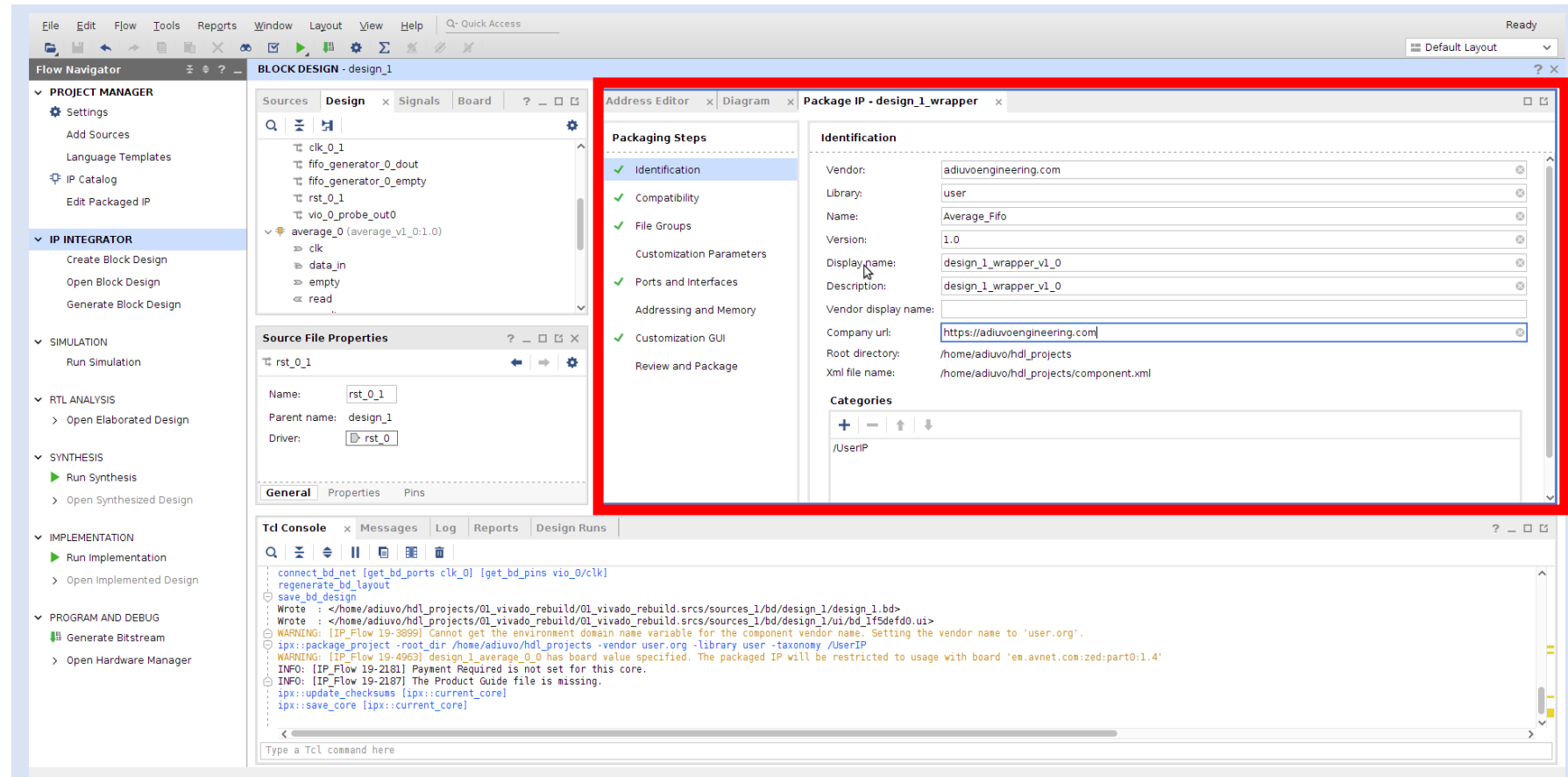
Lab 3: Advanced Vivado

Step 22 – Click on Finish.



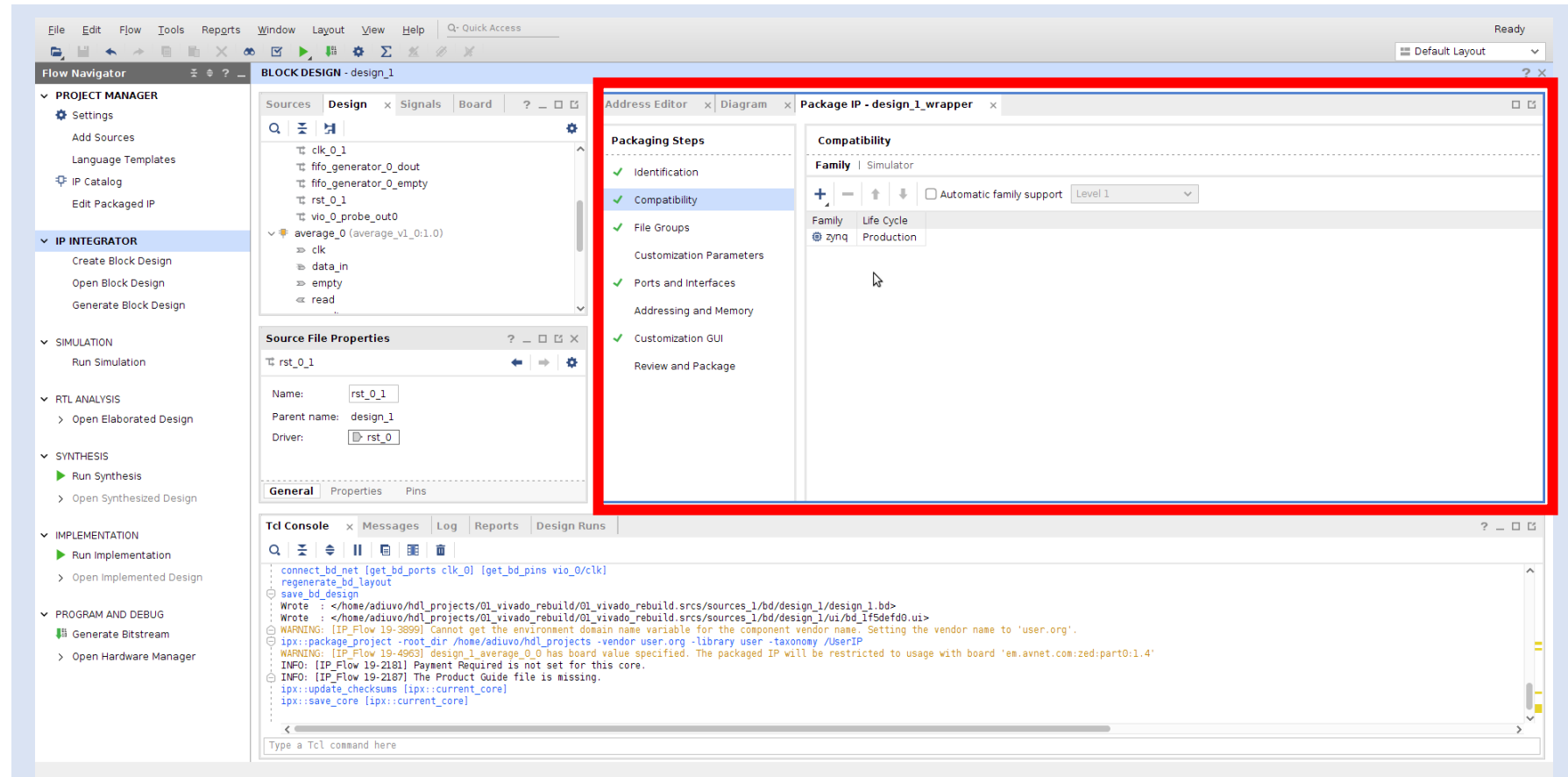
Lab 3: Advanced Vivado

Step 23 – This will open a package view in the project. Enter **Name**, **Version** and **Company URL**.



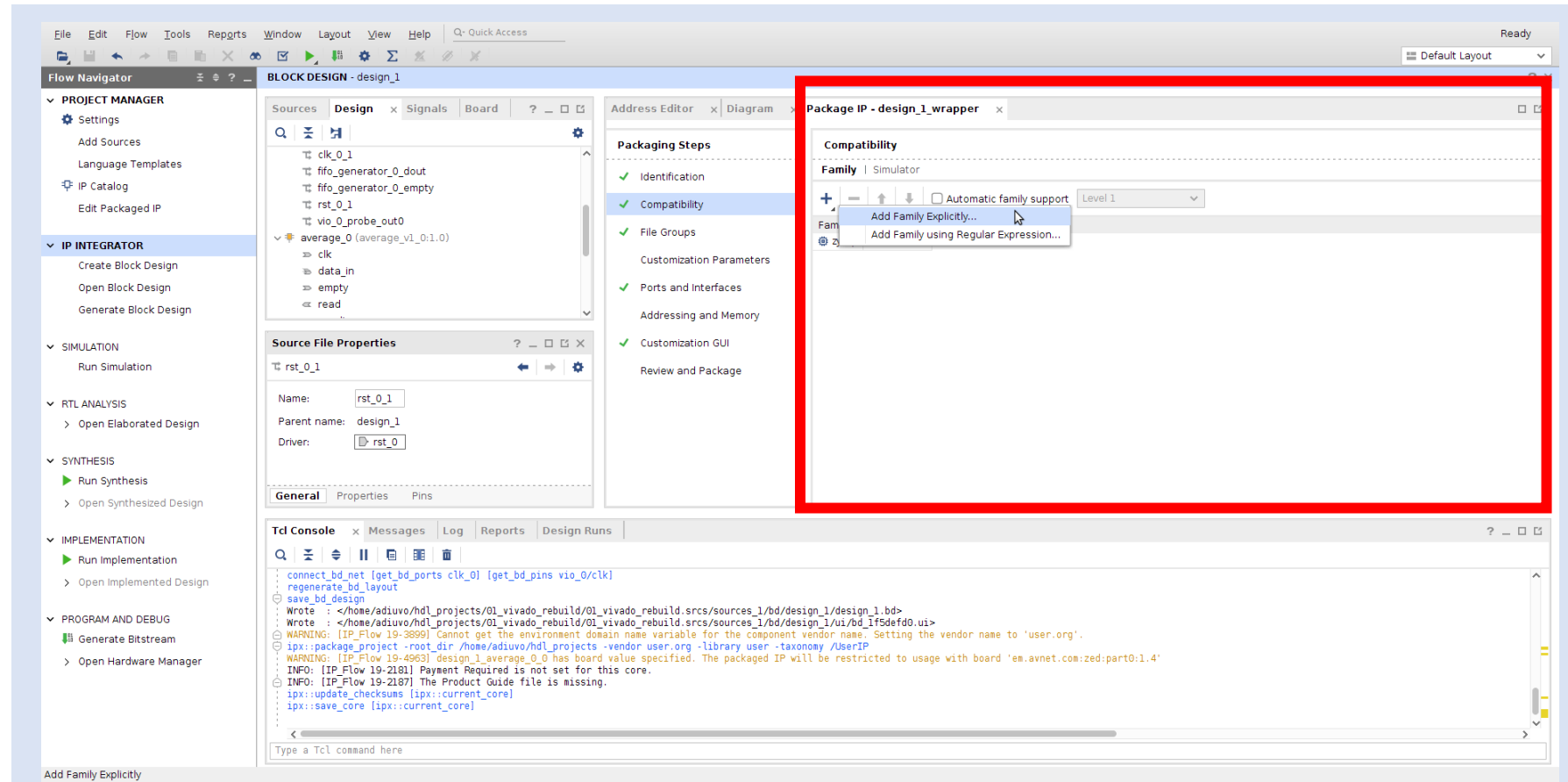
Lab 3: Advanced Vivado

Step 24 – Click on the Compatibility tab. This defines what FPGA / SoC the core will work with.



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Step 25 – Click on + and select **Add Family Explicitly**.



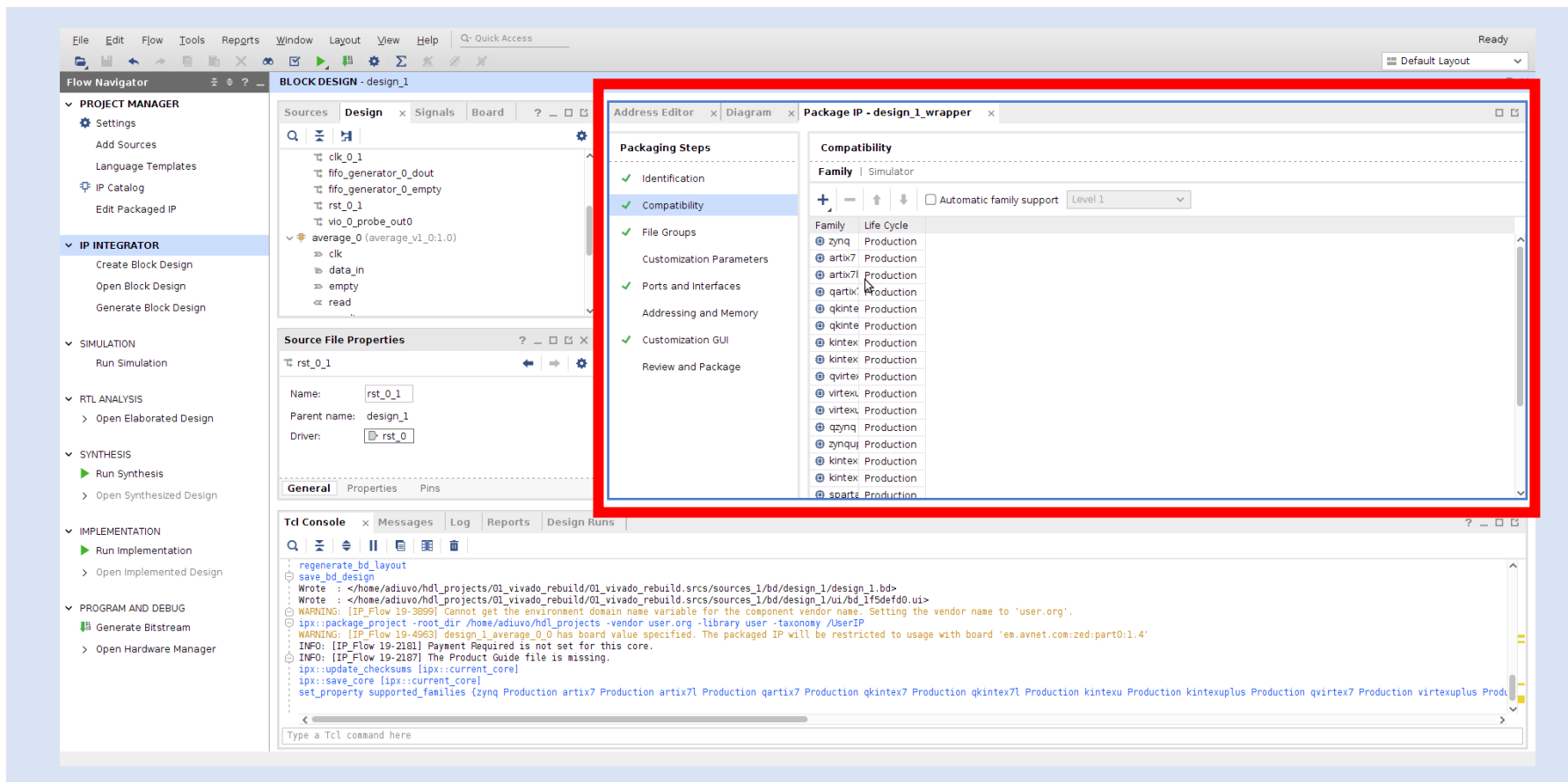
Lab 3: Advanced Vivado

Step 26 – In the resultant dialog, select **All Families** and **Parts** and **Production** from the Life-cycle.



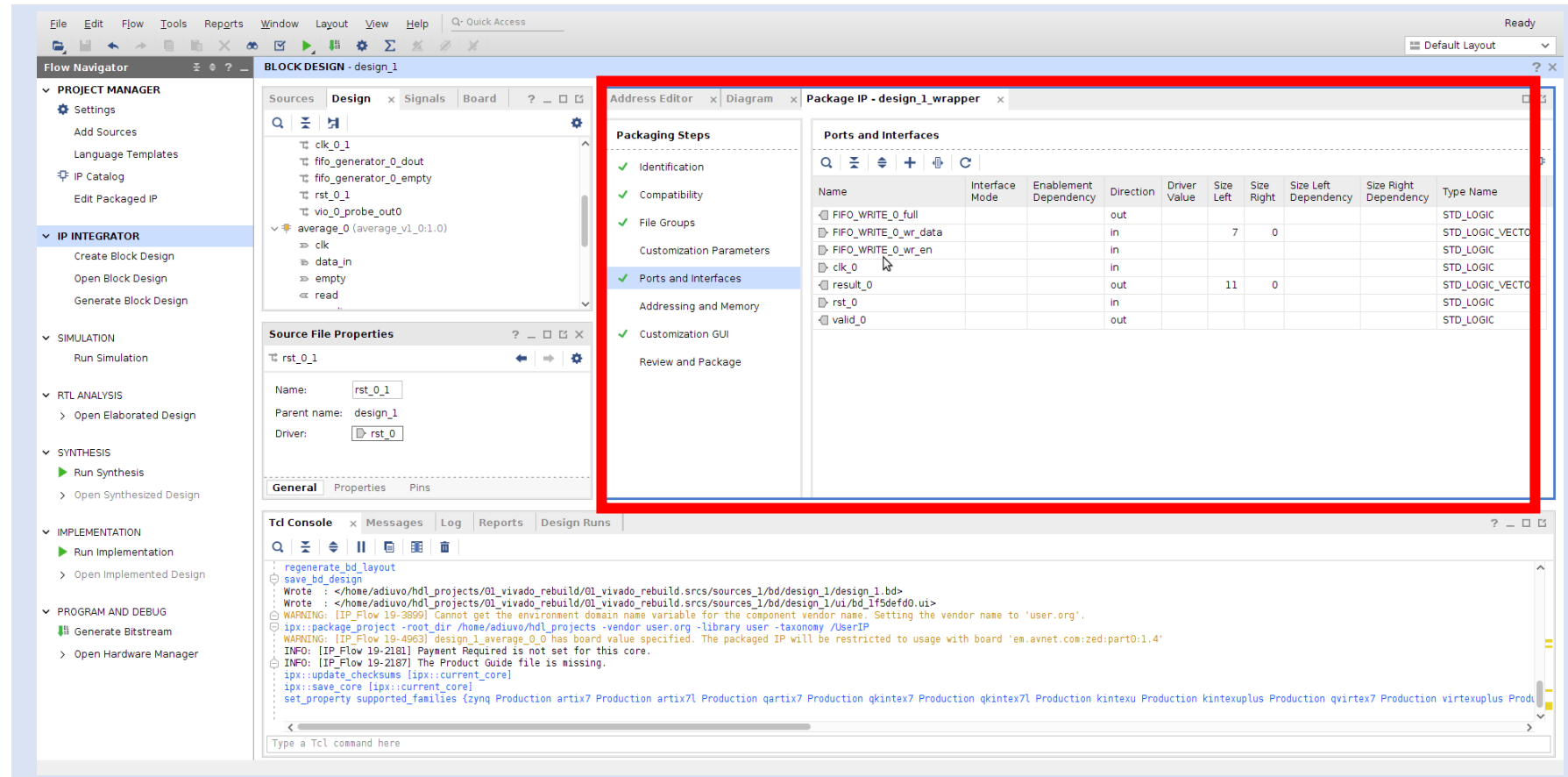
Lab 3: Advanced Vivado

Step 27 – This will make the IP core available to all families and devices.



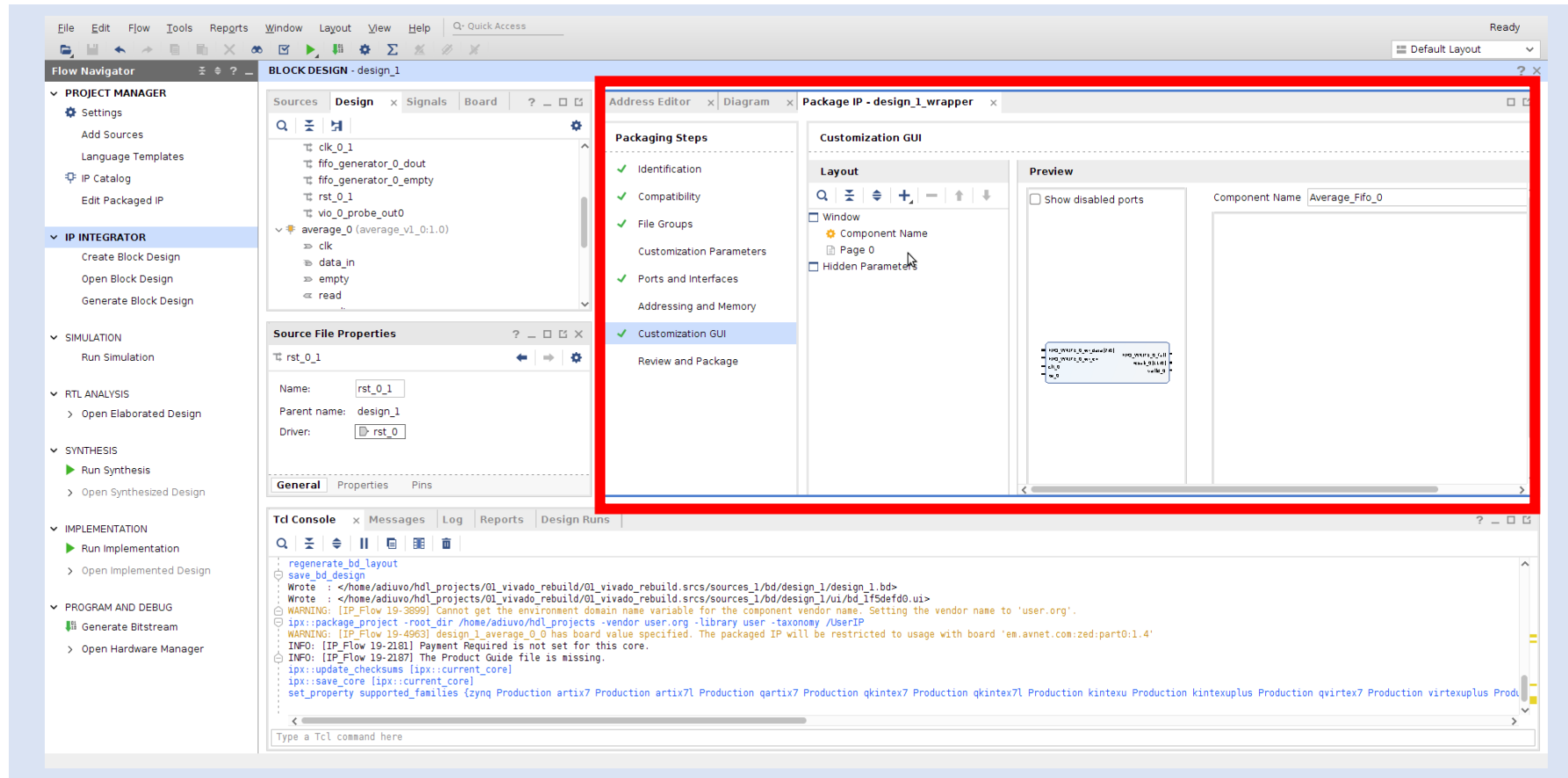
Lab 3: Advanced Vivado

Step 28 – Click on Ports and Interfaces. This should show all the blocks' inputs and outputs.



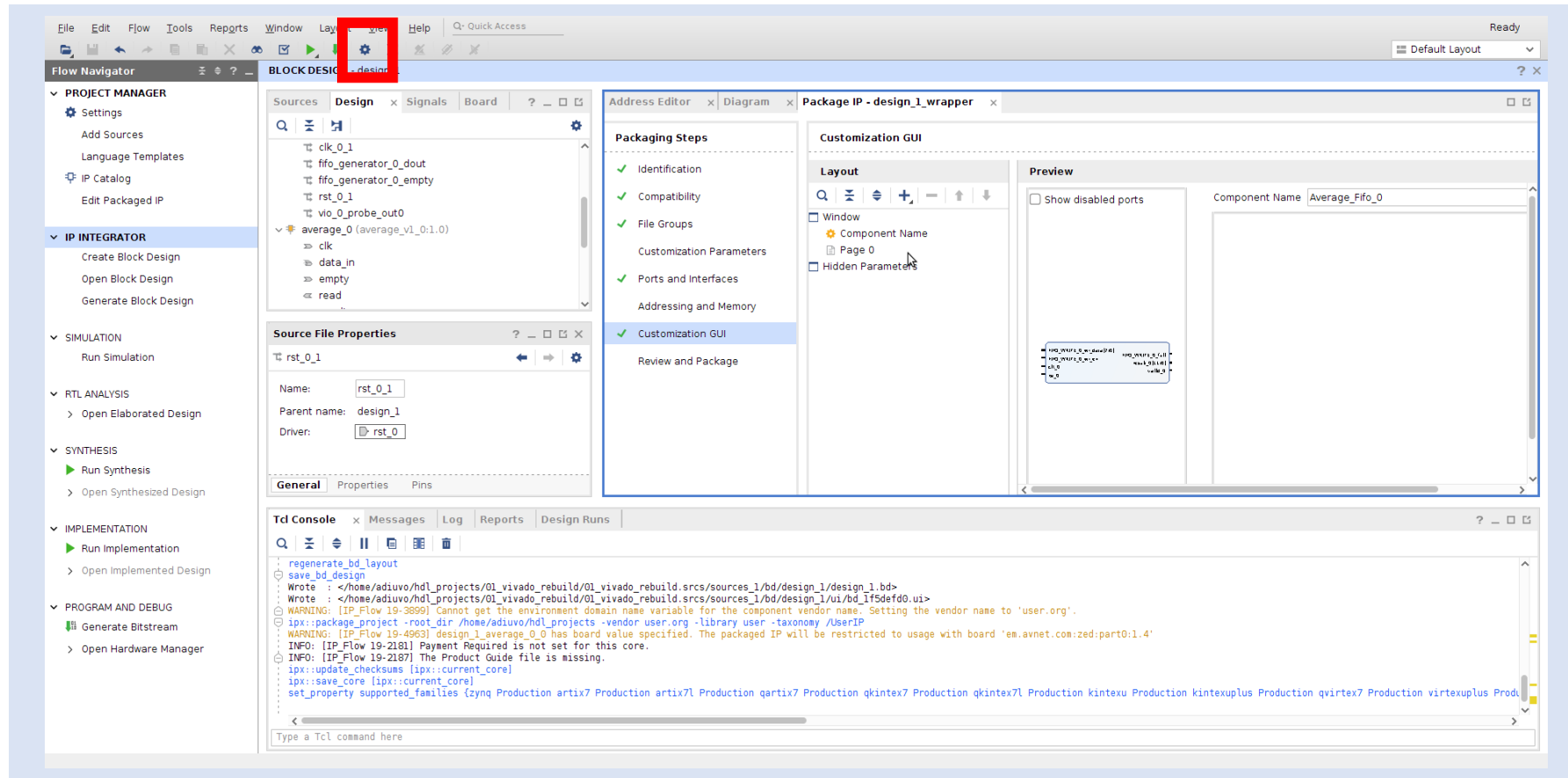
Lab 3: Advanced Vivado

Step 29 – Click on Customization. This shows the symbol.



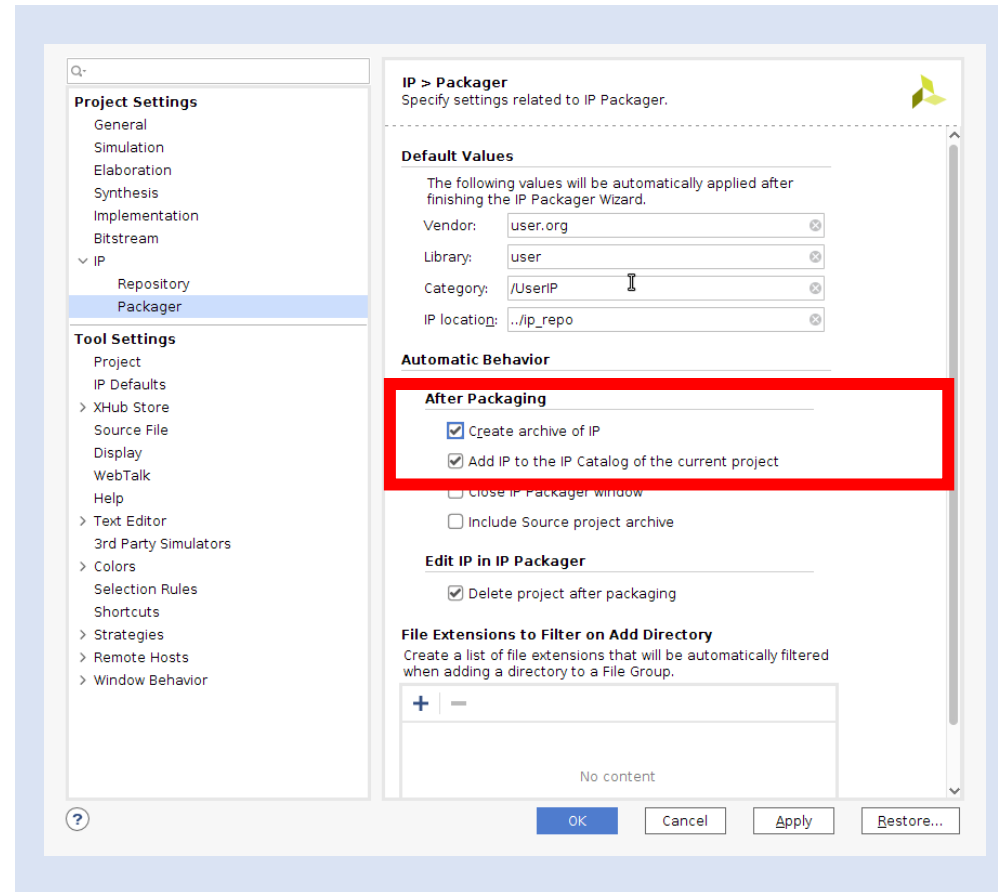
Lab 3: Advanced Vivado

Step 30 – Click on Settings.



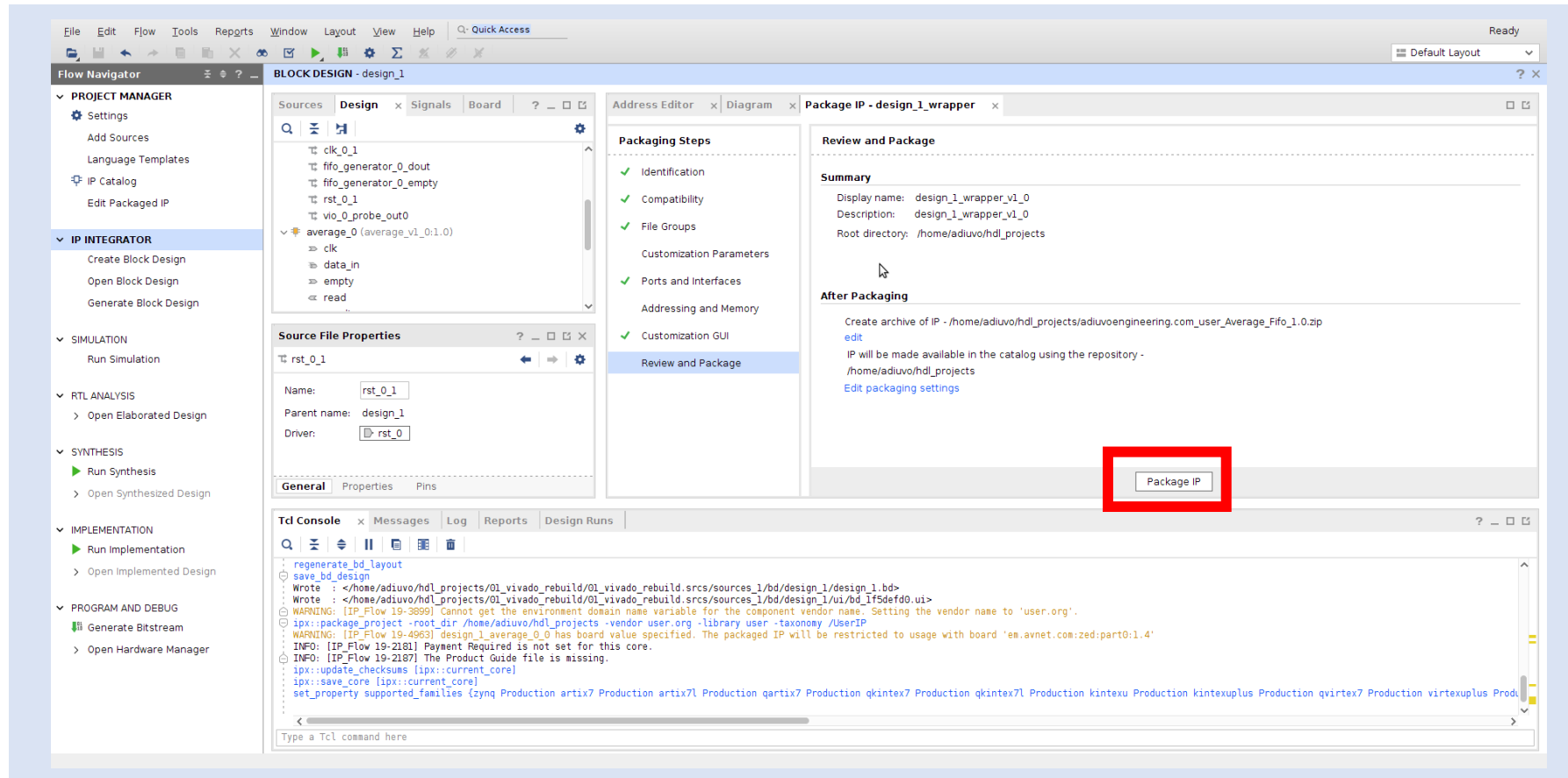
Lab 3: Advanced Vivado

Step 31 – Select Create archive of IP and click OK.



Lab 3: Advanced Vivado

Step 32 – Click on Review and Package and then Package IP.



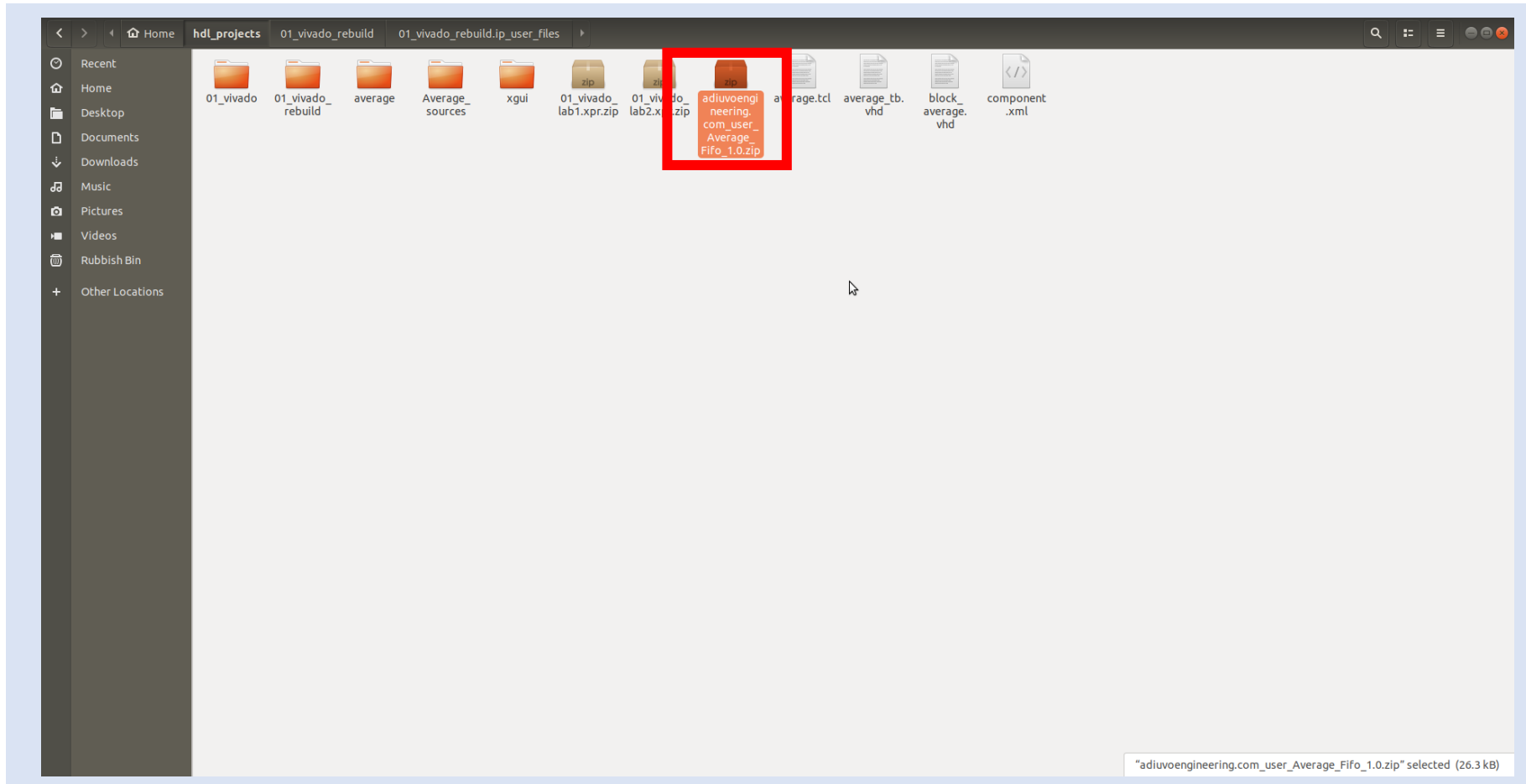
Lab 3: Advanced Vivado

Step 33 – This should show the message below. Click **OK**.



Lab 3: Advanced Vivado

Step 34 – You will see a ZIP file of the component in the directory you identified to save the IP in.



Lab 3: Advanced Vivado

Step 35 – Close the project and create a new Vivado project. In the wizard, click **Next**, then enter the **name and location** and click **Next** again.

Create a New Vivado Project

This wizard will guide you through the creation of a new project.

To create a Vivado project you will need to provide a name and a location for your project files. Next, you will specify the type of flow you'll be working with. Finally, you will specify your project sources and choose a default part.

Project Name

Enter a name for your project and specify a directory where the project data files will be stored.

Project name:

Project location:

☒ Create project subdirectory

Lab 3: Advanced Vivado

Step 36 – Select RTL Project and Next, followed by Next again.

The screenshot displays the Vivado Project Wizard interface, divided into two main sections: 'Project Type' and 'Add Sources'.

Project Type: This section prompts the user to 'Specify the type of project to create.' It lists five options:

- ☒ **RTL Project**
You will be able to add sources, create block designs in IP Integrator, generate IP, run RTL analysis, synthesis, implementation, design planning and analysis.
☐ Do not specify sources at this time
- ☐ **Post-synthesis Project**
You will be able to add sources, view device resources, run design analysis, planning and implementation.
☐ Do not specify sources at this time
- ☐ **J/O Planning Project**
Do not specify design sources. You will be able to view part/package resources.
- ☐ **Imported Project**
Create a Vivado project from a Synplify, XST or ISE Project File.
- ☐ **Example Project**
Create a new Vivado project from a predefined template.

The 'Next >' button at the bottom of this section is highlighted with a red box.

Add Sources: This section prompts the user to 'Specify HDL, netlist, Block Design, and IP files, or directories containing those files, to add to your project. Create a new source file on disk and add it to your project. You can also add and create sources later.'

It features a large text area with the instruction 'Use Add Files, Add Directories or Create File buttons below' and three buttons: 'Add Files', 'Add Directories', and 'Create File'.

Below these buttons are three checkboxes:

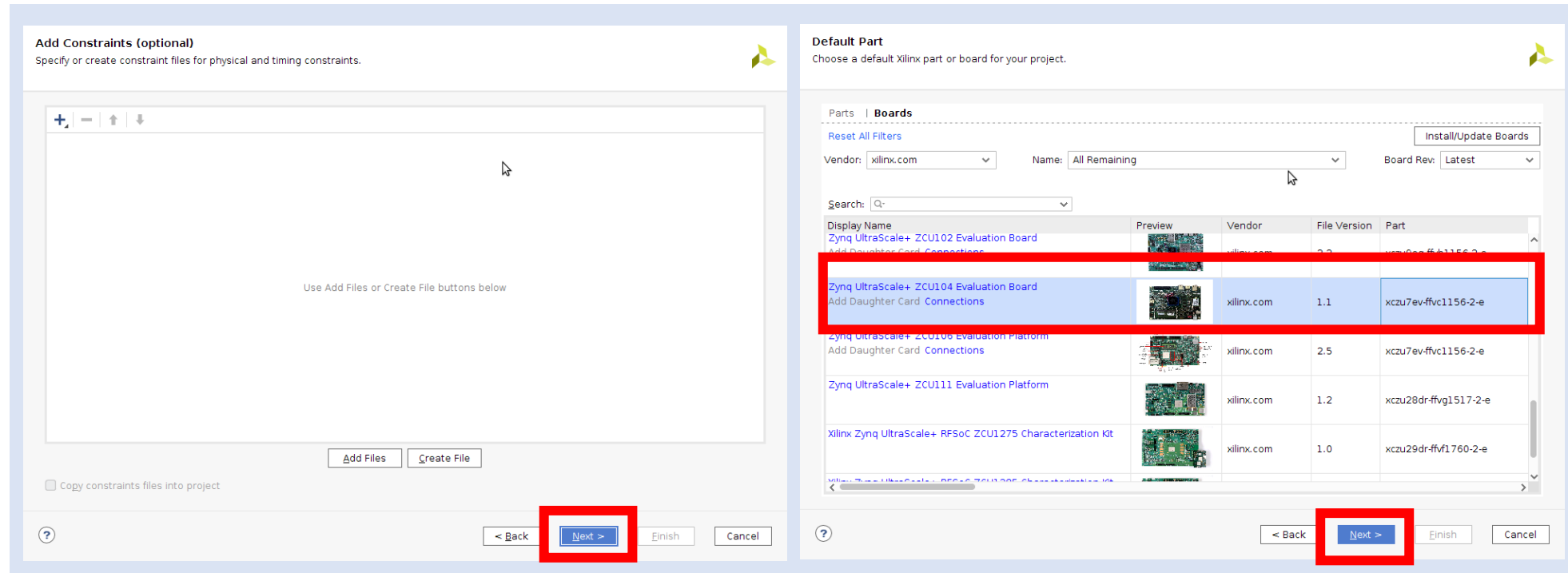
- ☐ Scan and add RTL include files into project
- ☐ Copy sources into project
- ☒ Add sources from subdirectories

At the bottom, there are two dropdown menus: 'Target language: VHDL' and 'Simulator language: Mixed'.

The 'Next >' button at the bottom of this section is also highlighted with a red box.

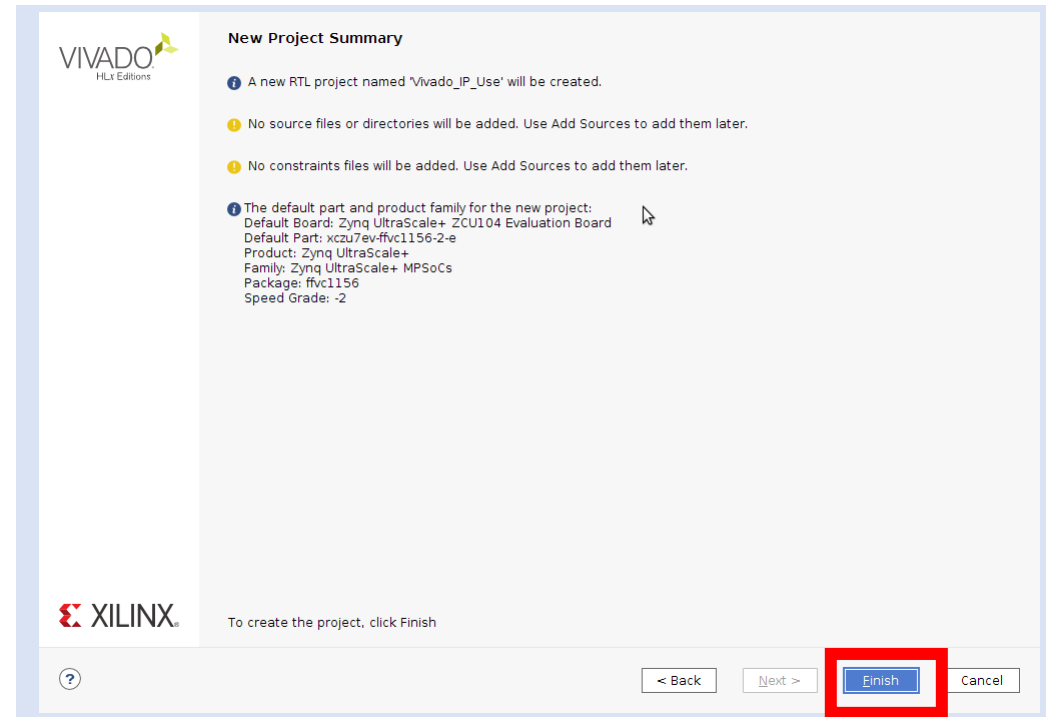
Lab 3: Advanced Vivado

Step 37 – Click **Next** on the Constraints tab. On the Default Part, select the **ZCU104 board** and click **Next**.



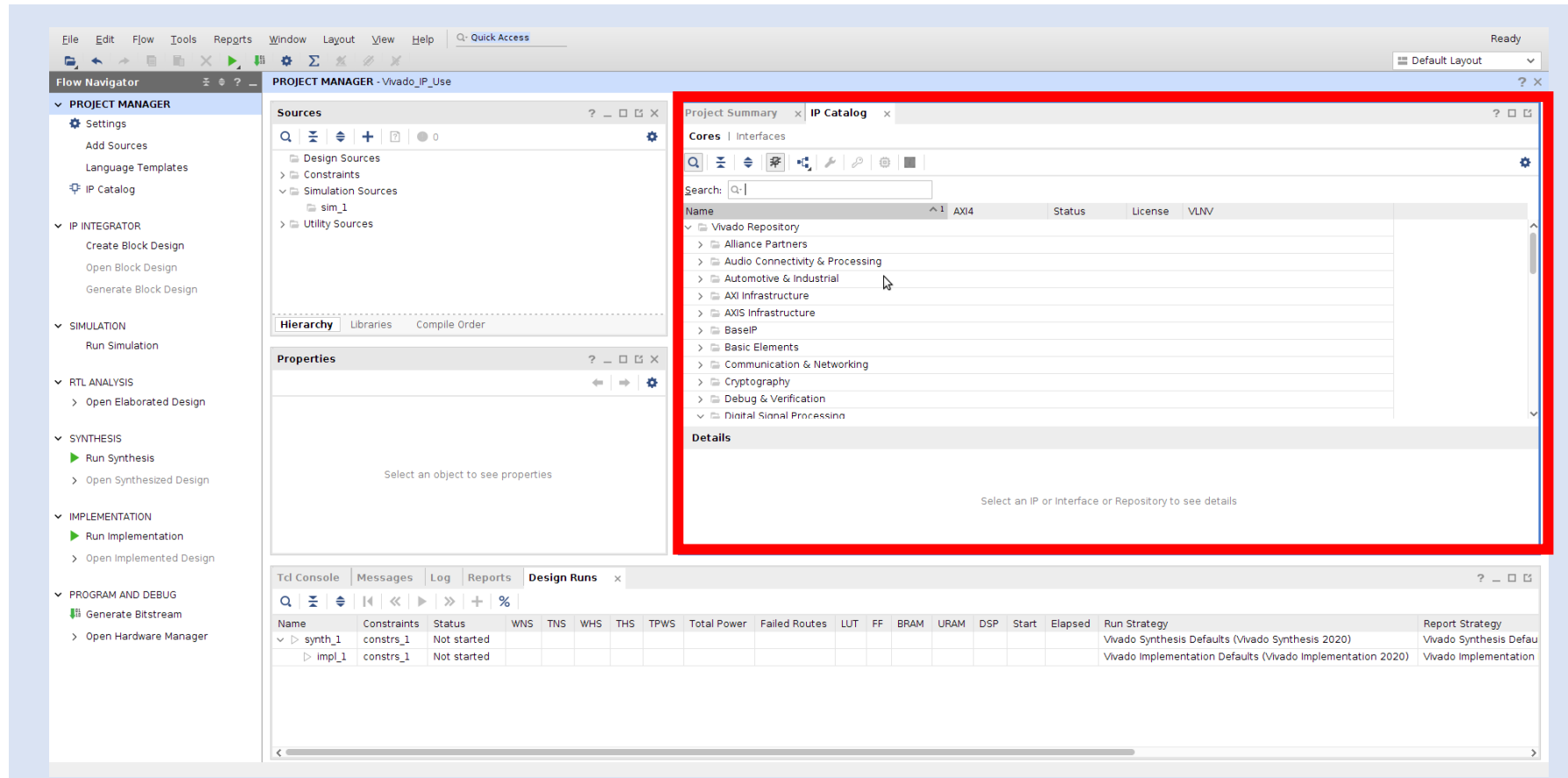
Lab 3: Advanced Vivado

Step 38 – On the final page of the Project Summary, click **OK**.



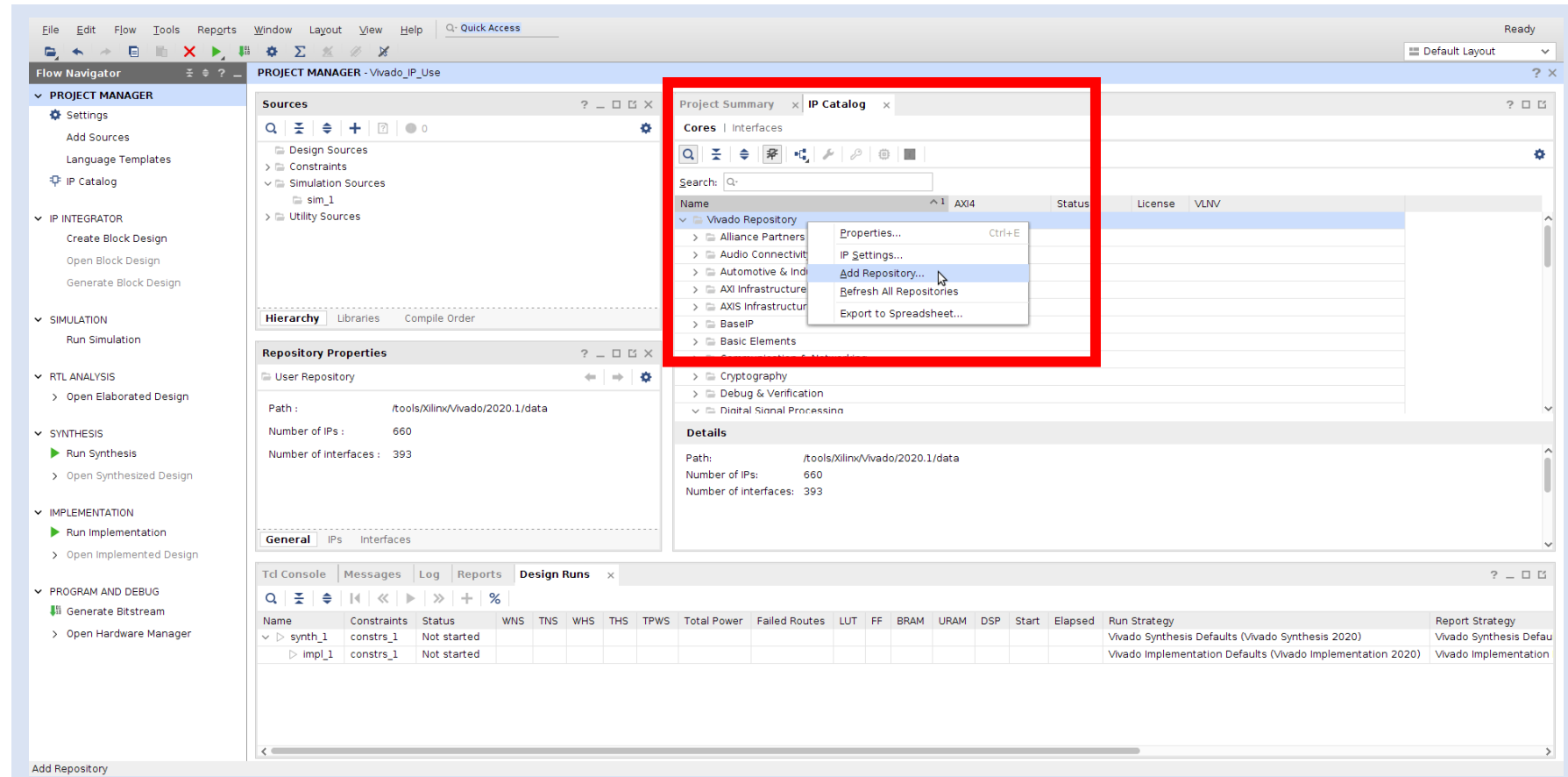
Lab 3: Advanced Vivado

Step 39 – From the menu bar, select **Window** → **IP Catalog**.



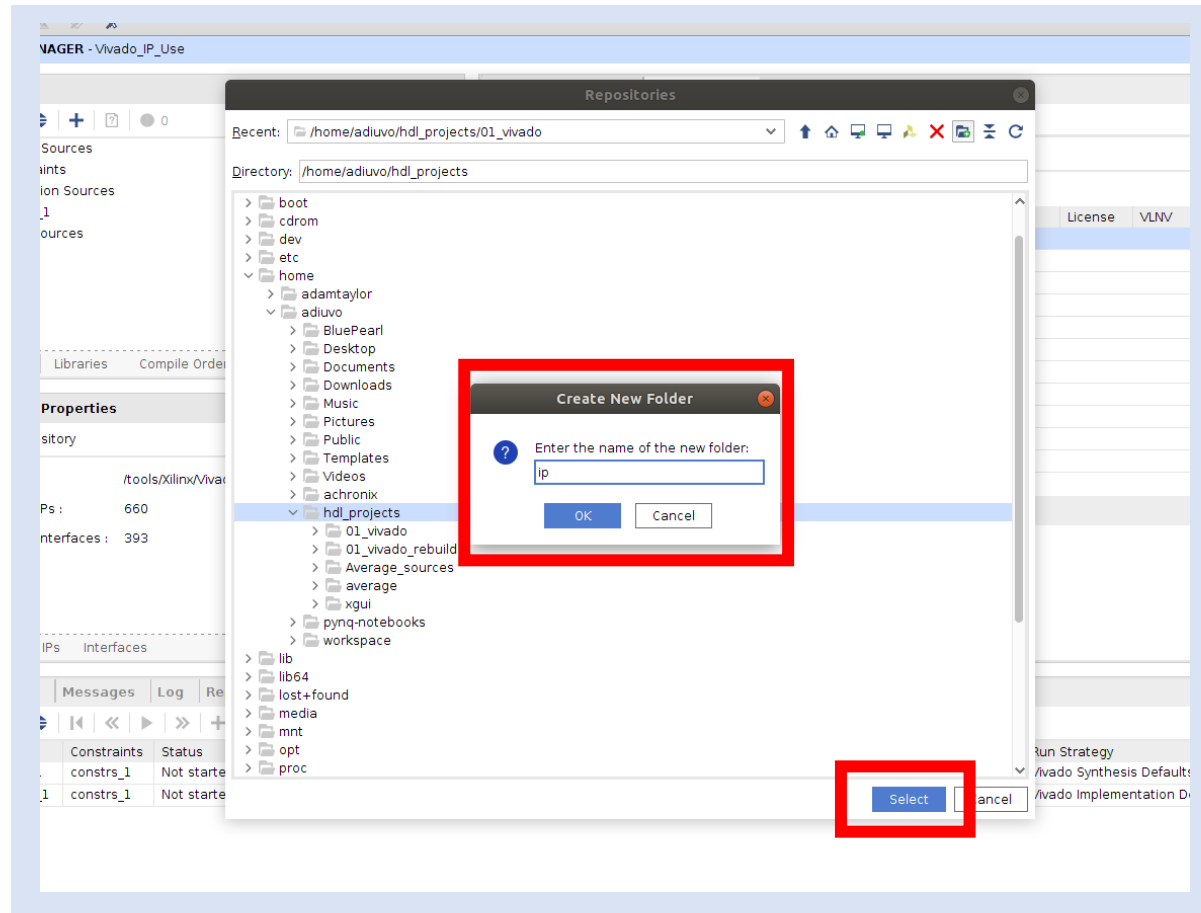
Lab 3: Advanced Vivado

Step 40 – Right click on the Vivado Repository and select Add Repository.



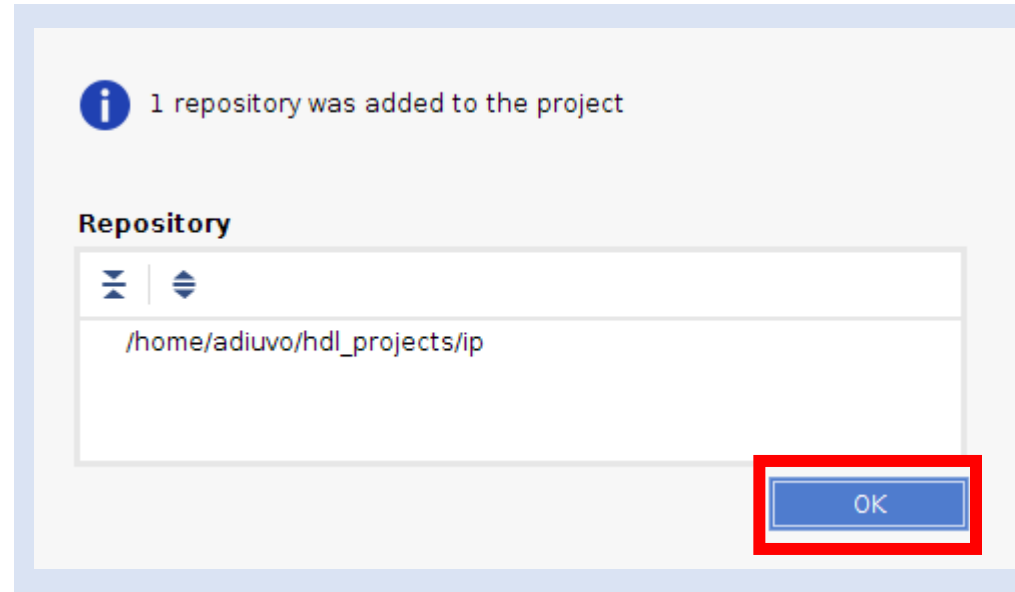
Lab 3: Advanced Vivado

Step 41 – In the open dialog, select **Create New Folder** and name it **IP** and click **Select**.



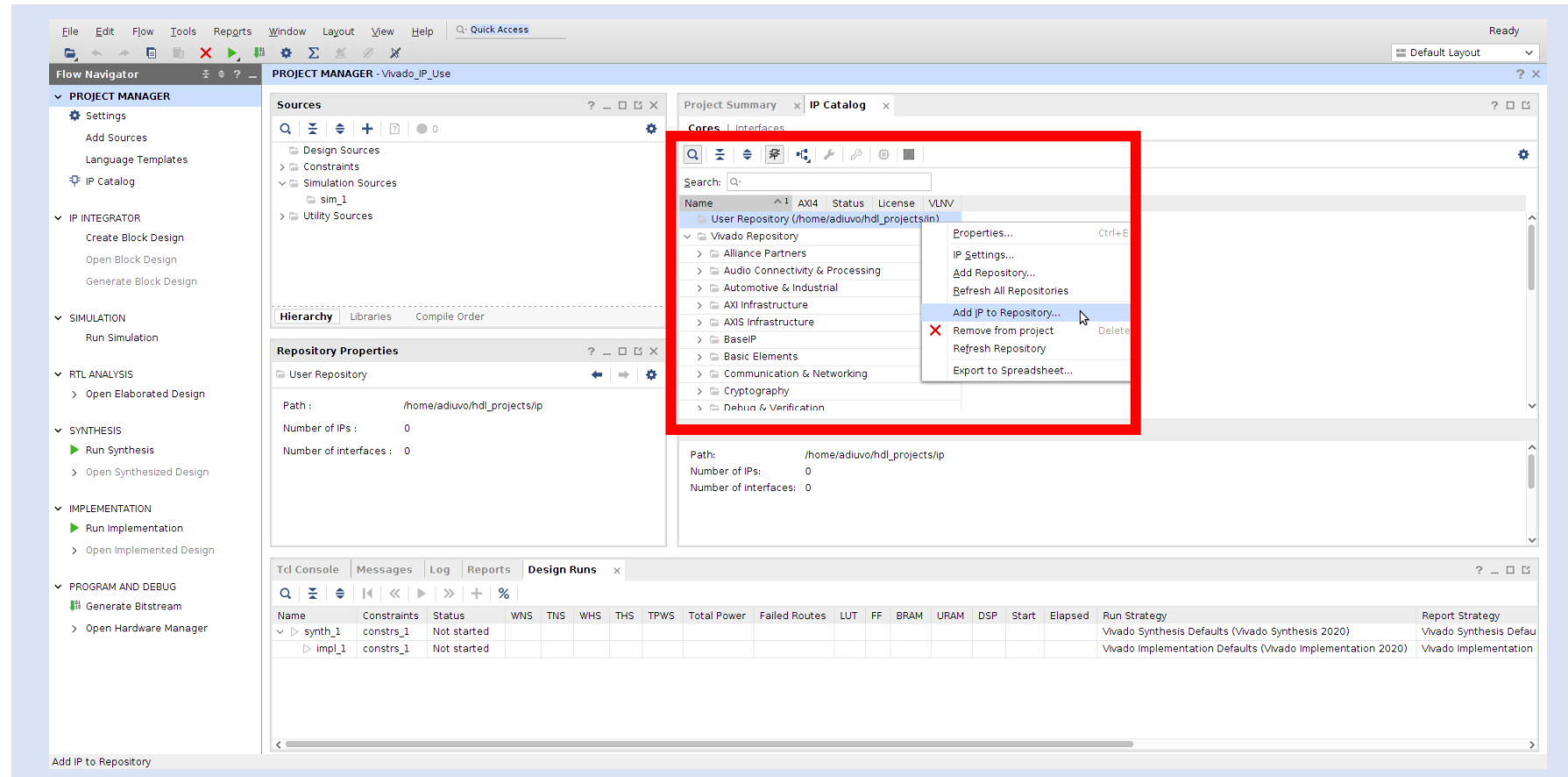
Lab 3: Advanced Vivado

Step 42 – This will show a dialog indicating that one repository was added.



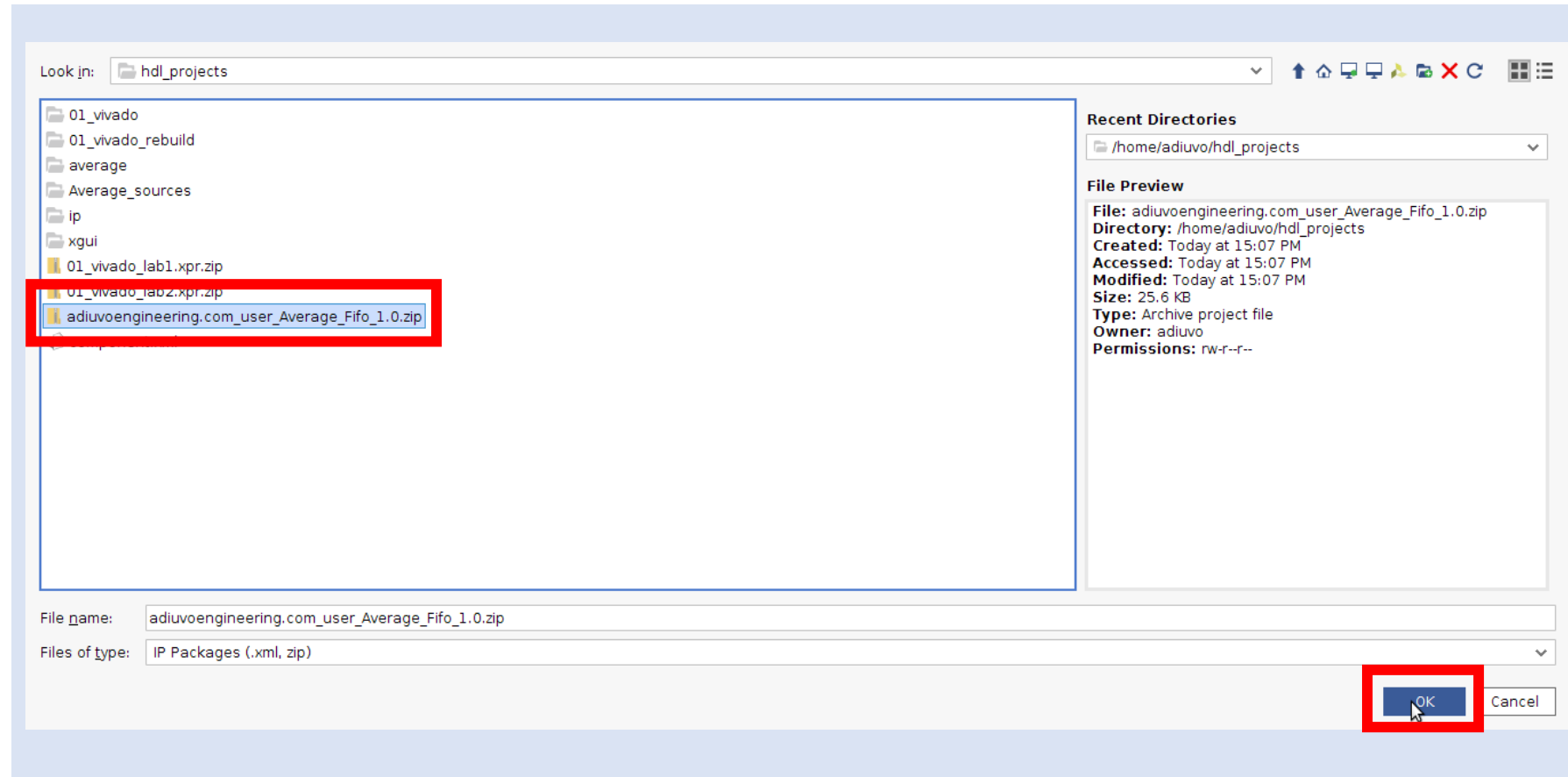
Lab 3: Advanced Vivado

Step 43 – This will add in the repository. Right click on it and select **Add IP to Repository**.



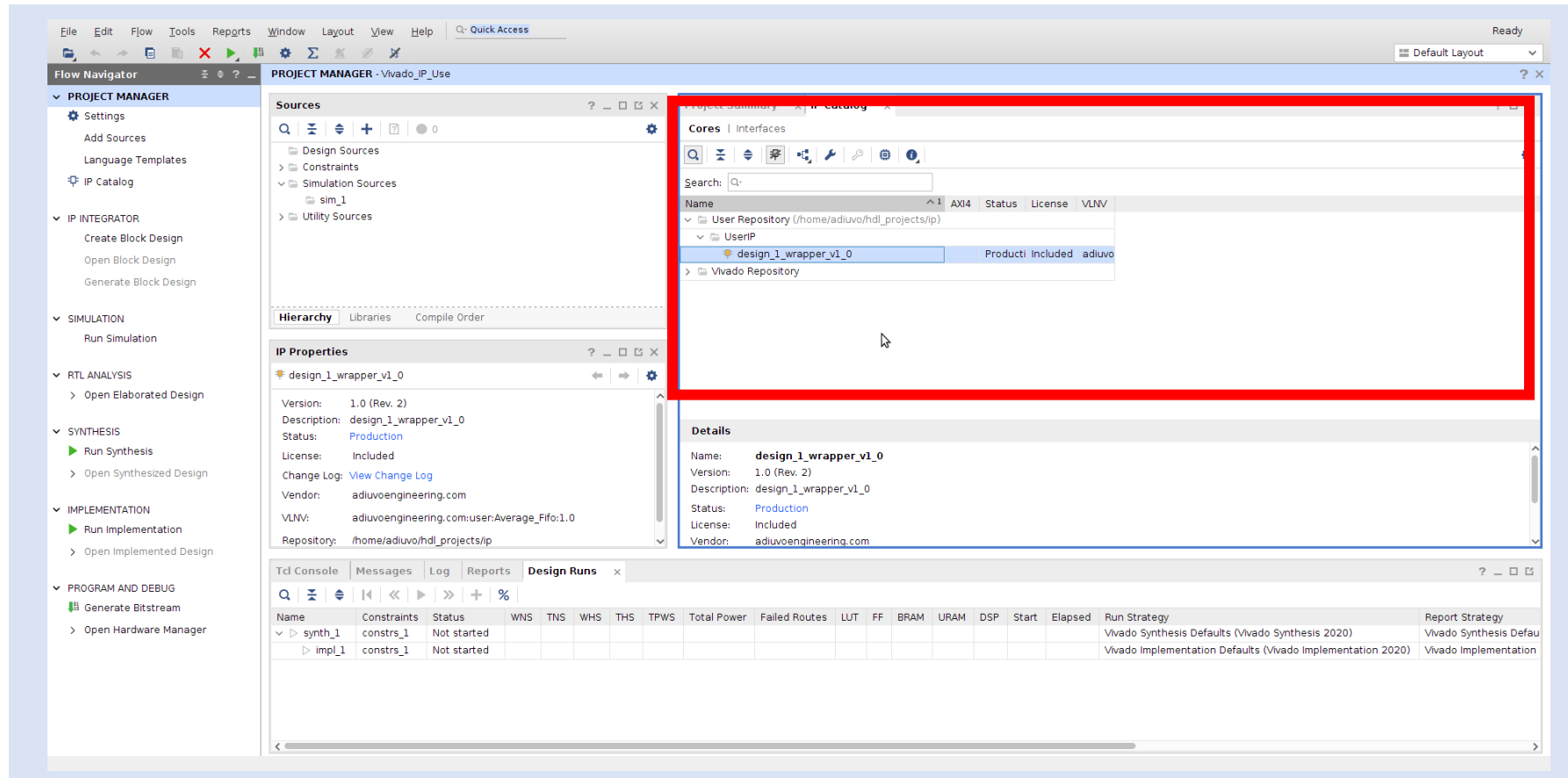
Lab 3: Advanced Vivado

Step 44 – Select the **component ZIP file** in the file dialog and click **OK**.



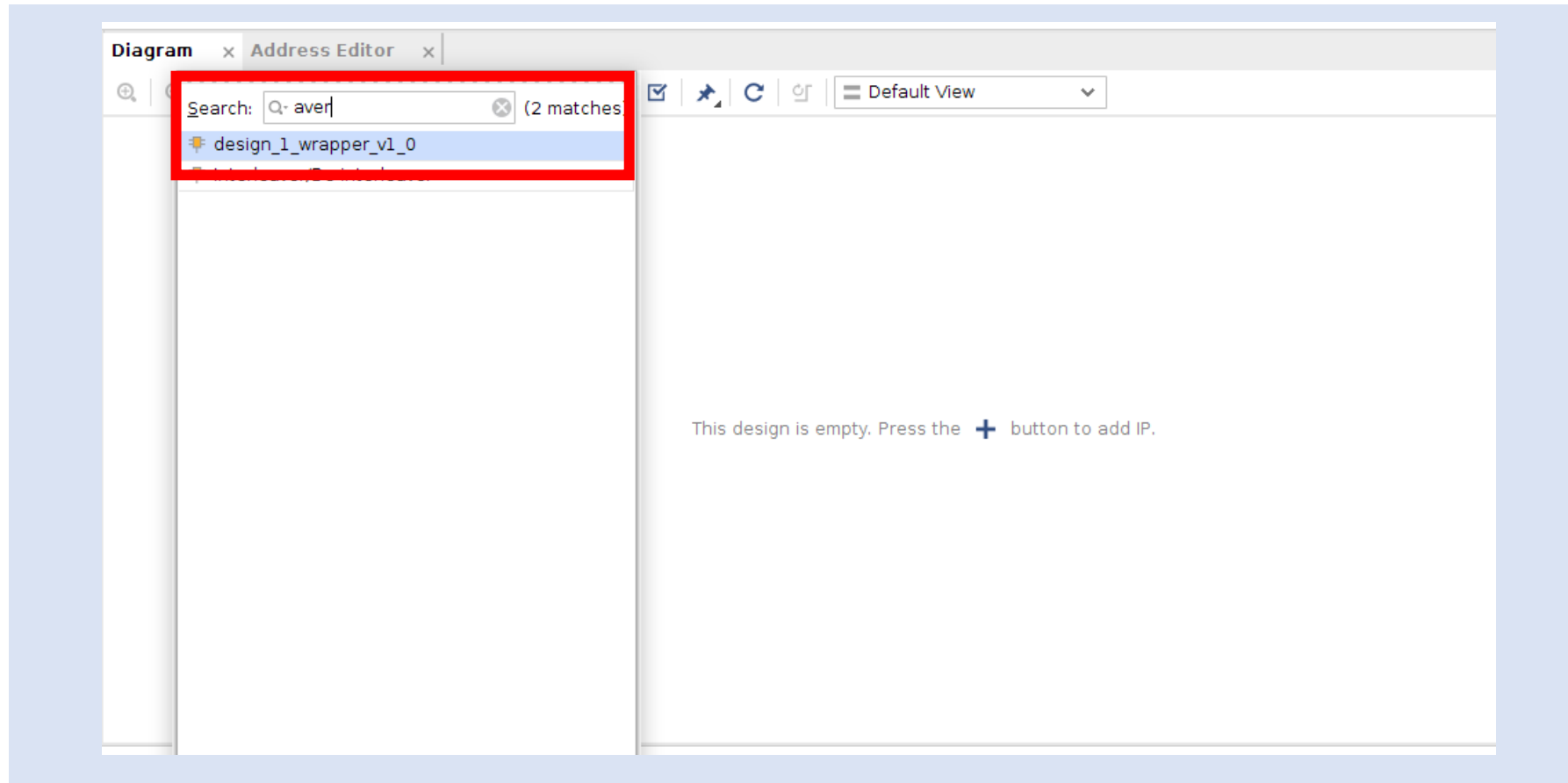
Lab 3: Advanced Vivado

Step 45 – You should now see the IP core in the library ready to go in our new design.



Lab 3: Advanced Vivado

Step 46 – Create a block design and type in the search bar average. Double click on the **design_1_wrapper_V1_0** which comes up.



Lab 3: Advanced Vivado

Step 47 – This will add in the Average block we have created and we are ready to go reusing it in our new developments.

